Claiborne EMS Protocol Manual

Vehicle and app reference
November 1, 2026

Protocols for universal patient care, adult cardiac and medical emergencies, obstetrics, airway and respiratory care, pediatric cardiac and medical emergencies, trauma and burns, and toxicology and environmental emergencies. Includes the medication reference and the retired UP-19 redirect to UP-18.

Use each quick-flow diagram together with the assessment, treatment, medication orders, provider-level care, warnings and special-population instructions. Conditional instructions appear within the flow boxes. A flow box or medication name alone is not a complete order.

Provider labels are explicit and do not depend on color. Follow the stated Medical Control requirements, provider-scope restrictions and product-specific instructions. When directed to another protocol, use its printed page reference or tap the underlined link.

Use the contents or PDF bookmarks to locate a protocol or medication. Printed page numbers match PDF page numbers.

Print on US Letter at actual size, duplex with long-edge binding. The manual uses high-contrast text, vector flow diagrams and searchable text.

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Tap any entry or use the PDF bookmarks. Printed page numbers match PDF page numbers.

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UP-01
Universal Patient Care

Revised October 31, 2026

Overview

Applies to every patient encounter and establishes the minimum assessment, treatment, reassessment, documentation, and transfer-of-care expectations for Claiborne County EMS.

Scene safety, appropriate personal protective equipment, rapid recognition of life threats, and early movement toward definitive care take priority.

Use this protocol together with the age-appropriate complaint, medical, trauma, obstetric, pediatric, operational, and procedure protocols that best fit the patient.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Scene Safe?
EMT / AEMT / Paramedic
PPE • hazards • number of patients • request resources

↓

2. Primary Survey
EMT / AEMT / Paramedic
XABCDE • control major bleeding • identify immediate threats

↓

3. Unstable / Time-Critical?
EMT / AEMT / Paramedic
Yes: treat immediate threats, notify early, move toward transport

↓

4. Focused Assessment
EMT / AEMT / Paramedic
Chief complaint • SAMPLE/OPQRST • exam • first complete vital set

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Select Protocol + Treat
EMT / AEMT / Paramedic
Use complaint-specific pathway and provider-level standing orders

↓

6. Reassess
EMT / AEMT / Paramedic
Repeat primary survey, response to treatment, and second complete vital set

↓

7. Transport + Handoff
EMT / AEMT / Paramedic
Appropriate destination • concise report • document care and response

Indications

- Every person evaluated, treated, transported, released, or otherwise encountered as a potential patient by Claiborne County EMS.
- Any call in which EMS personnel provide assessment, advice, treatment, transport, or a refusal or no-transport disposition.

Assessment

Scene approach

- Confirm scene safety before entry and reassess scene hazards throughout the incident.
- Use standard precautions and add airborne, droplet, contact, splash, or other protection when indicated.
- Determine the number of patients, mechanism of injury or nature of illness, need for triage, and need for additional resources.
- Bring the equipment reasonably anticipated for the call; at minimum, ensure immediate access to the jump bag, oxygen and airway equipment, and cardiac monitor when appropriate.
- Preserve potential evidence and document any movement of a body, clothing, or surroundings at a possible crime scene.

Primary assessment

- Form a general impression and identify the chief complaint and immediate life threats.
- Assess responsiveness and mental status using AVPU and Glasgow Coma Scale when appropriate.

- Assess and manage airway, breathing, circulation, major hemorrhage, and disability before proceeding to the secondary assessment.
- Determine whether the patient is high acuity or low acuity and choose the appropriate transport priority.

Focused history and examination

- Obtain SAMPLE history and use OPQRST for pain or other time-sensitive symptoms when applicable.
- Perform a focused medical, trauma, obstetric, behavioral, or pediatric examination based on the complaint and mechanism.
- For altered mental status, focal neurologic symptoms, seizure, syncope, suspected overdose, diabetes, or unexplained weakness, obtain blood glucose and perform the appropriate neurologic or stroke assessment.
- Obtain a 12-lead ECG when indicated by symptoms, age, risk, or protocol; transmit promptly when the tracing may alter destination or treatment.
- Screen for medications, allergies, anticoagulant or antiplatelet use, pregnancy possibility, implanted devices, and relevant advance directives or care plans.

Minimum vital signs and monitoring

- Obtain and document at least two complete sets of vital signs whenever feasible, including one early in the encounter and one before transfer of care or release.
- A complete adult vital-sign set includes systolic and diastolic blood pressure, pulse, respiratory rate, pulse oximetry, mental status, and pain score when applicable.
- Add blood glucose, temperature, end-tidal carbon dioxide, carbon monoxide level, 12-lead ECG, or other monitoring when indicated.
- Verify an automated blood-pressure value manually when it is inconsistent with the clinical presentation or differs significantly from a manual reading.
- Document why any required assessment or vital sign could not be obtained.

Ongoing reassessment

- Reassess high-acuity patients approximately every 3-5 minutes and low-acuity patients approximately every 5-15 minutes, or more frequently when clinically indicated.
- Repeat the primary assessment, vital signs, mental status, pain score, and response to each intervention.
- After medication administration, monitor for therapeutic effect, adverse effects, and any change in transport priority.
- Do not allow nonessential procedures to delay transport when the patient needs time-sensitive definitive care.

Treatment

1. Treat immediate life threats using the least invasive effective intervention within the provider's Tennessee scope of practice.
2. Apply oxygen only when clinically indicated and titrate to the condition-specific target; support ventilation when oxygenation or ventilation is inadequate.
3. Control major hemorrhage promptly and address shock, temperature exposure, pain, and patient positioning.
4. Request ALS, specialty, rescue, law-enforcement, hazardous-materials, or air-medical resources early when their response is reasonably expected to improve care or transport time.
5. Transfer care directly to an equal or higher medical authority with a concise verbal report, all available records and tracings, patient belongings, and a summary of treatment and response.
6. Complete the electronic patient-care report for every patient contact, including refusals, no-patient-found events, and nontransport dispositions.

Provider-level care

1. Initial Patient Assessment Scene safety, XABCDE, acuity, focused history and exam.

EMT: Perform scene size-up, XABCDE, major hemorrhage control, SAMPLE/OPQRST, focused exam, and initial vital signs.

- Begin BLS care immediately; do not wait for ALS for lifesaving interventions.

AEMT: Perform all EMT actions and identify early need for vascular access, fluids, or ALS intercept.

Paramedic: Perform all prior actions; lead high-acuity assessment, advanced differential, destination, and treatment decisions.

2. Airway, Oxygenation & Ventilation Use the least invasive effective airway intervention.

EMT: Position airway, suction, use basic adjuncts, oxygen when indicated, BVM ventilation, and CPAP when authorized.

AEMT: All EMT actions plus authorized supraglottic airway and advanced oxygenation support within Tennessee scope.

Paramedic: Advanced airway management, waveform capnography, ventilator strategy, and post-airway reassessment.

3. Monitoring & Vital Signs

EMT: Document BP, pulse, respirations, SpO_2, mental status, and pain score when applicable.

• Obtain glucose, temperature, ECG, or other monitoring when indicated and within scope.

AEMT: Trend vital signs and reassess after medication, fluids, procedures, or clinical change.

Paramedic: Add 12-lead ECG, $EtCO_2$, continuous monitoring, and advanced interpretation when indicated.

4. Vascular Access Do not delay urgent care or transport for repeated IV attempts.

EMT: Assist, protect existing access, and prepare equipment within local policy.

AEMT: Peripheral IV access and authorized fluid/medication administration. AEMT: eligible upper-extremity sites only, no leg or neck; no more than two attempts in one arm.

• Maximum three peripheral attempts across the entire crew; never exceed provider-specific limits or delay indicated IO access.

Paramedic: IV/IO access; proceed directly to IO when delay is unsafe or IV success is unlikely.

• Use central access only as allowed by agency policy and training.

5. Medication Administration Right patient, medication, dose, route, time, indication, and documentation.

EMT: Administer only medications authorized for EMT standing orders; verify allergies and contraindications.

AEMT: Administer authorized AEMT medications; reassess effect and adverse response.

Paramedic: Administer formulary medications within standing orders; obtain Medical Control when required.

6. Pain Management Treat pain early while preserving respiratory and hemodynamic safety.

EMT: Position, splint, ice/heat when appropriate, acetaminophen, and ibuprofen per protocol.

AEMT: All EMT care plus IV access and approved non-opioid therapy within scope.

Paramedic: Fentanyl, morphine, hydromorphone, or ketamine when indicated; monitor and document repeat pain scores.

7. Behavioral Emergency De-escalate first; use the least restrictive safe intervention.

EMT: Scene safety, verbal de-escalation, medical cause assessment, and safe restraint assistance.

AEMT: All EMT care plus monitoring, vascular access when safe, and support of authorized sedation pathway.

Paramedic: Midazolam or ketamine may be used per agitation protocol. Continuous airway and cardiac monitoring required.

8. Refusal / Non-Transport

EMT: Assess capacity, explain risks and alternatives, obtain two vital sets when permitted, and document refusal.

AEMT: Support reassessment and identify high-risk features requiring ALS or Medical Control involvement.

Paramedic: Evaluate high-risk refusals, questionable capacity, minors, or serious illness; contact Medical Control when indicated.

9. Interfacility Transfer Confirm stability, orders, equipment, medications, and receiving acceptance before departure.

EMT: Verify paperwork, patient identity, baseline status, and transport requirements within EMT scope.

AEMT: Manage authorized infusions and monitoring within scope and local transfer policy.

Paramedic: Confirm advanced monitoring, medication compatibility, airway risk, and contingency plan.

10. Documentation Record assessment, clinical reasoning, treatment, response, destination, and handoff.

EMT: Complete an accurate PCR for every patient contact, refusal, cancellation after contact, or no-transport encounter.

AEMT: Document vascular access, medication, reassessment, and any unsuccessful attempts.

Paramedic: Document advanced assessment, ECG interpretation, differential, protocol deviation, Medical Control, and destination rationale.

11. Destination Selection Choose the most appropriate facility based on capability, time, stability, and local plan.

EMT: Identify potential specialty needs and communicate early with ALS and dispatch.

AEMT: Support destination decision and early notification for time-sensitive conditions.

Paramedic: Apply specialty destination criteria, bypass policy, patient preference, and air-medical considerations.

12. Transfer of Care / Handoff Direct verbal handoff to equal or higher medical authority.

EMT: Report age, complaint, acuity, vital signs, findings, treatment, response, and belongings.

AEMT: Include access, fluids, medications, complications, and reassessment.

Paramedic: Include ECG findings, advanced procedures, diagnostic concerns, and Medical Control orders.

13. Death in the Field Follow resuscitation, termination, evidence preservation, and notification policy.

EMT: Recognize obvious death criteria only as authorized; preserve the scene and notify appropriate resources.

AEMT: Continue care within scope and assist with documentation and family support.

Paramedic: Apply termination criteria, contact Medical Control when required, and coordinate disposition under local policy.

14. Medical Control Use for protocol-required orders, uncertainty, deviation, or high-risk decisions.

EMT: Request higher-level consultation through the established chain when care exceeds standing orders.

AEMT: Provide concise clinical information and repeat back orders.

Paramedic: Contact Medical Control for required orders, unusual circumstances, protocol conflicts, or high-risk refusal and destination decisions.

Medical Control: Orders must be read back, documented, and linked to the physician or authorized clinician providing direction.

Warnings

- Perform only care permitted by the provider's current Tennessee license/scope, documented training and agency credentialing, and applicable Claiborne orders. Obtain Medical Control for a required order, uncertainty, or proposed deviation from local standing orders, but only within that authorized scope; an online order does not expand the provider's scope.
- Protocol sequence may be changed when required by patient condition, scene limitations, or resource availability, but the reason must be documented.
- A normal initial assessment or vital-sign set does not exclude serious illness; use the complete clinical picture and repeat assessment.
- Patient refusal is a high-risk disposition and requires a capacity assessment, explanation of risks and alternatives, appropriate signatures, and detailed documentation.
- Do not delay transport for documentation, nonessential IV access, repeated diagnostic testing, or procedures that can safely occur en route.

Clinical pearls

- The most qualified provider caring for the patient is responsible for ensuring that assessment and treatment information is communicated to responding units, Medical Control, and the receiving facility.
- A useful radio report includes age, chief complaint, stability, complete vital signs, level of consciousness, major findings, interventions, response, and estimated arrival time.
- When no complaint-specific protocol fits, continue supportive care, serial reassessment, and Medical Control consultation as needed.
- Document clinical reasoning, especially when deviating from protocol, selecting a specialty destination, or accepting a nontransport disposition.
- Maintain patient dignity, privacy, professionalism, and clear communication throughout the encounter.

Special populations

Pediatric patients

- Use the Pediatric Assessment Triangle early: appearance, work of breathing, and circulation to skin.
- Use a length-based pediatric resuscitation system for medication doses and equipment selection when applicable, and communicate the measured color or weight to the receiving facility.
- Avoid separating the child from the caregiver unless required for safety or treatment.
- Adapt the order and technique of examination to the child's developmental stage and clinical condition.

Patients declining care or transport

- Encourage assessment and transport when a potentially serious illness or injury exists and involve a paramedic when available for high-risk refusals.
- Assess the patient's ability to communicate a stable choice, understand relevant information, appreciate the consequences, and reason about options.
- Obtain at least two sets of vital signs and a problem-focused examination whenever the patient permits.
- Explain the suspected condition, recommended care, material risks of refusal, alternatives, and specific instructions for seeking further help.
- Consider Medical Control consultation for high-risk refusal, questionable capacity, minors, vulnerable adults, or disagreement among involved parties.

Older adults and patients with special needs

- Account for atypical presentations, polypharmacy, anticoagulant use, frailty, communication barriers, cognitive impairment, implanted devices, and baseline functional status.
- Seek caregivers, medication lists, advance directives, and individualized emergency plans when available without delaying urgent care.

UP-02 Mass-Casualty Incident (MCI) Triage

Revised August 2026

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. MCI / Resource Imbalance?
EMT / AEMT / Paramedic
Scene safety • establish command • declare MCI level • request resources • identify hazards and ingress/egress

↓

2. Global Sort
EMT / AEMT / Paramedic
Patients who can walk follow commands to the MINOR (Green) area • rapidly identify patients who are not moving or have an obvious life threat

↓

3. Adult START
EMT / AEMT / Paramedic
Breathing • respiratory rate • perfusion • follows commands

↓

4. Pediatric JumpSTART
EMT / AEMT / Paramedic
Breathing • pulse if apneic • 5 rescue breaths when pulse present • respiratory rate • peripheral pulse • age-appropriate AVPU

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Immediate Lifesaving Intervention Only
EMT / AEMT / Paramedic
Open airway • control major hemorrhage • authorized antidote • chest decompression by Paramedic when immediately indicated

↓

6. Tag / Mark + Move
EMT / AEMT / Paramedic
Immediate Red • Delayed Yellow • Minor Green • Expectant/Deceased Black • record intervention and time when feasible

↓

7. Repeat Triage
EMT / AEMT / Paramedic
Reassess in treatment and transport areas, after intervention or movement, with deterioration, and when resource availability changes

Indications

- A declared or suspected mass-casualty incident.
- Multiple patients whose immediate needs exceed, or may soon exceed, available personnel, equipment, treatment space, or transport capacity.
- An incident in which command requests standardized primary or secondary triage.

Contraindications

- Do not use MCI triage categories as a substitute for routine patient assessment when resources are adequate for individual care.

Assessment

Adult START primary triage

- Able to walk: direct to the designated MINOR (Green) area for secondary triage; reassess infants, patients with mobility limitations, and anyone unable to follow the global instruction rather than assigning a category from ambulation alone.
- Not breathing: open or reposition the airway. If spontaneous breathing begins, classify IMMEDIATE (Red). If the adult remains apneic, classify EXPECTANT/DECEASED (Black) according to the MCI plan.
- Breathing with a respiratory rate greater than 30/min: classify IMMEDIATE (Red). If 30/min or less, assess perfusion.

- Absent radial pulse or capillary refill greater than 2 seconds: control major hemorrhage and classify IMMEDIATE (Red). Use radial pulse preferentially when cold, poor lighting, skin characteristics, or other conditions make capillary refill unreliable.
- Unable to follow simple commands: classify IMMEDIATE (Red). If respiratory rate, perfusion, and command-following criteria are adequate, classify DELAYED (Yellow).

Pediatric JumpSTART primary triage

- Able to walk: direct to the MINOR (Green) area for secondary triage. Infants and developmentally nonambulatory children are not automatically Immediate; assess them through the JumpSTART sequence.
- Not breathing: open or reposition the airway. If spontaneous breathing begins, classify IMMEDIATE (Red).
- Still apneic: check for a palpable pulse. If no pulse is present, classify EXPECTANT/DECEASED (Black) according to the MCI plan. If a pulse is present, provide 5 rescue breaths. If spontaneous breathing begins, classify IMMEDIATE (Red); if apnea persists, classify EXPECTANT/DECEASED (Black).
- Spontaneously breathing with a respiratory rate less than 15/min or greater than 45/min: classify IMMEDIATE (Red).
- No palpable peripheral pulse: classify IMMEDIATE (Red).
- Mental status inappropriate for age, inappropriate response to pain, posturing, or unresponsiveness: classify IMMEDIATE (Red). If respiratory rate, pulse, and age-appropriate mental status are adequate, classify DELAYED (Yellow).

Scene and incident assessment

- Ensure responder safety, appropriate PPE, hazard-zone control, decontamination needs, and law-enforcement or rescue support before patient entry.
- Estimate patient count and injury pattern, declare or request the appropriate MCI response level, establish or join incident command, and identify triage, treatment, transport, staging, ingress, and egress functions as resources permit.
- Multiple patients with similar unexplained symptoms should prompt immediate consideration of hazardous-material, carbon-monoxide, infectious, radiologic, or intentional exposure and movement to a safe assessment area.

Treatment

1. Attach or clearly mark the triage category, record critical interventions when feasible, communicate immediate hazards or resource needs, and move to the next patient.
2. Move patients to assigned treatment areas when safe and directed. Begin complaint-specific care, monitoring, vascular access, medications, splinting, and complete assessment only after primary triage or when assigned to the treatment function.
3. Repeat triage after movement, decontamination, treatment, clinical change, arrival in the treatment or transport area, and whenever resource availability changes.
4. Coordinate transport priority, destination distribution, and patient tracking through the Transport Unit and the current regional MCI plan; prevent uncoordinated transport from overwhelming a single facility.

Provider-level care

Provider-Level Actions Every provider uses the same triage criteria; provider level determines which immediate lifesaving interventions are authorized.

EMT: Perform START or JumpSTART primary triage as assigned, attach or clearly mark the triage category, and communicate patient location and immediate needs through the incident structure.

- Perform only rapid lifesaving interventions authorized at the EMT level during primary triage: open or position the airway, control life-threatening external hemorrhage including tourniquet use, provide JumpSTART rescue breaths, and administer an authorized antidote auto-injector when indicated.
- Move to the next patient after assigning a category and completing the permitted immediate intervention; do not delay primary sorting for a complete assessment, routine vital signs, splinting, IV access, or noncritical treatment.

AEMT: Perform all EMT triage actions using the same START or JumpSTART criteria; do not change a category solely because an advanced provider is available.

- Do not initiate routine IV/IO access, fluids, or medications during primary triage. When assigned to a treatment or transport function, provide AEMT care within Tennessee scope, local credentialing, and the incident treatment plan.

Paramedic: Perform all prior triage actions and, when immediately indicated and consistent with the MCI plan, perform needle chest decompression within current Tennessee Paramedic scope.

• When assigned a clinical incident role, coordinate secondary triage, treatment priorities, destination distribution, transport sequencing, Medical Control communication, and reassessment as resources permit.

• Do not allow advanced procedures, routine monitoring, vascular access, or medication administration to delay completion of primary triage unless the intervention is an immediately lifesaving action authorized by the MCI plan.

Warnings

- Primary MCI triage is not routine Universal Patient Care. Two complete vital-sign sets, ECG acquisition, IV/IO access, routine medications, splinting, and detailed documentation must not delay primary sorting.
- Triage category is based on current physiology and available resources and may change. Repeat assessment is mandatory throughout the incident.
- A Black category communicates MCI resource priority; it does not by itself replace Tennessee and local requirements for field determination or pronouncement of death.
- Do not enter a hazardous or contaminated area without appropriate PPE, hazard control, and decontamination coordination.
- Reverse triage may be required for lightning incidents; use the applicable environmental protocol and incident plan.

Clinical pearls

- The goal of MCI triage is the greatest overall benefit with the resources available, not delivery of complete individual care during the first pass.
- Use objective criteria consistently. Do not upgrade or downgrade based only on injury appearance, age, provider intuition, or pressure from bystanders.
- Clearly separate the triage, treatment, and transport functions. Uncoordinated care or transport can create a second resource bottleneck.
- Green patients require secondary triage; walking does not guarantee absence of serious injury or delayed deterioration.

Special populations

Children

- Use JumpSTART according to the adopted agency cutoff or appearance definition. The standard JumpSTART rescue-breath sequence uses 5 breaths, not 2.

Mobility, communication, and developmental limitations

- A patient who cannot walk, hear the instruction, understand the language, follow commands at baseline, or move because of a preexisting disability requires direct physiologic assessment rather than automatic category assignment.
- Use caregivers, interpreters, baseline information, and adaptive communication when immediately available without delaying the triage pass.

Hazardous-material or infectious incidents

- Perform contamination and medical triage in coordination with the HazMat branch. Do not move contaminated patients into the cold zone, treatment area, or transport unit before required decontamination unless an immediate lifesaving exception is directed by command.

UP-03
Abdominal Pain / Vomiting / Diarrhea

Revised August 2026

Overview

Evaluate nontraumatic abdominal or genitourinary pain, nausea, vomiting, and diarrhea broadly. Prehospital symptom control must occur together with active screening for shock, hemorrhage, pregnancy-related emergencies, acute coronary syndrome, aortic disease, sepsis, metabolic disease, obstruction, perforation, and toxic exposure.

Do not withhold analgesia or antiemetic treatment solely to preserve the abdominal examination. Reassess the examination, vital signs, perfusion, pain, nausea, and response after every intervention.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Primary Survey
EMT / AEMT / Paramedic
Airway • breathing • perfusion • mental status • major GI bleeding • pregnancy emergency • peritonitis

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. Focused Assessment
EMT / AEMT / Paramedic
OPQRST/SAMPLE • abdominal and back exam • emesis/stool/urine blood • LMP/pregnancy • intake/output • surgery • medications • exposures

↓

3. Glucose + Cardiac Screen
EMT / AEMT / Paramedic
Check blood glucose • acquire and transmit 12-lead ECG for adult abdominal pain or unexplained nausea/vomiting

↓

4. Shock / High-Risk Features?
EMT / AEMT / Paramedic
Hypotension • poor perfusion • GI bleeding • rigid/guarded abdomen • pain out of proportion • suspected AAA/ACS/ectopic/sepsis

↓

5. Treat Symptoms + Cause
AEMT / Paramedic
AEMT: vascular access, fluid, ondansetron • Paramedic: ECG interpretation, analgesia, second-line antiemetic, cause-specific protocol

↓

6. Reassess + Transport
EMT / AEMT / Paramedic
Repeat vital signs, perfusion, pain, nausea, mental status, and abdominal exam • notify early for time-critical findings

Indications

- Nontraumatic abdominal, flank, pelvic, or genitourinary pain.
- Nausea, vomiting, diarrhea, dehydration, or suspected nontraumatic gastrointestinal bleeding.
- Unexplained symptoms in which a gastrointestinal, genitourinary, metabolic, cardiac, obstetric, infectious, or toxicologic cause is possible.

Contraindications

- Use the abdominal/pelvic trauma protocol when recent trauma is the likely cause.
- Do not give oral or ODT medication when the patient cannot protect the airway, has active uncontrolled vomiting that prevents administration, or has another aspiration risk.

Assessment

History

- Onset, provocation/palliation, quality, region/radiation, severity, progression, and duration of pain or nausea; timing and character of vomiting or diarrhea.
- Last oral intake, urine output, bowel movement, flatus, and emesis; blood or coffee-ground material in emesis; melena, hematochezia, hematuria, dysuria, vaginal bleeding, or discharge.
- Prior abdominal disease or surgery, aneurysm, cardiac disease, diabetes, renal disease, pregnancy history, immunocompromise, recent antibiotics or hospitalization, sick contacts, travel, unusual foods, alcohol, cannabis, medications, anticoagulants, and possible toxic exposure.
- For patients who could be pregnant: last menstrual period, pregnancy possibility, gestational age if known, prior ectopic pregnancy, fertility treatment, pain, syncope, and vaginal bleeding.

Examination

- General appearance, mental status, skin temperature/color/moisture, hydration, pulses, perfusion, orthostatic symptoms when safe, and evidence of shock.
- Inspect and gently palpate the abdomen for distention, focal or diffuse tenderness, guarding, rigidity, rebound/peritoneal signs, mass, hernia, bruising, and pulsation. Assess back and flank pain, distal pulses, and cardiopulmonary findings.
- When available without delaying care, note the appearance and approximate amount of emesis or stool and preserve objective descriptions rather than assigning a final diagnosis.

High-risk findings

- Hypotension, shock index greater than 1, altered mental status, syncope, ongoing GI bleeding, anticoagulant use with bleeding, severe dehydration, or persistent tachycardia.
- Rigid or guarded abdomen, pain out of proportion to examination, pulsatile abdominal mass, sudden tearing abdominal/back pain, absent or unequal distal pulses, or rapidly worsening pain.
- Pregnancy possibility with pain, bleeding, syncope, shoulder pain, or shock; severe pelvic or testicular pain; fever with poor perfusion; or vomiting with neurologic findings.
- Epigastric or upper abdominal symptoms with dyspnea, diaphoresis, weakness, syncope,

diabetes, known coronary disease, or ischemic ECG changes.

Treatment

1. Complete the Universal Patient Care assessment and immediately address airway compromise, hypoxemia, shock, major bleeding, severe hypoglycemia, or another life threat.
2. Obtain blood glucose and the indicated 12-lead ECG early. Transition to the diabetic, acute coronary syndrome, shock, sepsis, obstetric, toxicology, or trauma protocol as soon as a time-critical syndrome is recognized.
3. Repeat vital signs, perfusion, lung sounds after fluid, pain score, nausea/vomiting, mental status, and abdominal examination after treatment and during transport.
4. Notify the receiving facility early and minimize scene time for suspected AAA, ectopic pregnancy, ACS, major GI bleeding, bowel ischemia, obstruction/perforation, sepsis, or persistent shock.

Medication orders

Normal Saline

Adult: 500 mL IV/IO bolus; reassess and repeat to adequate perfusion or SBP ≥90 mmHg, maximum 2 L. Pediatric: 20 mL/kg IV/IO; reassess after each bolus.

- Use 250 mL increments with frequent lung and perfusion reassessment in patients with heart failure, renal failure, liver failure, known volume overload, pulmonary edema, or another high risk for fluid intolerance.

Ondansetron

Adult: 4 mg IV/IO/IM/PO/ODT. Pediatric: 0.15 mg/kg IV/IO/IM/PO/ODT, maximum 4 mg. May repeat once after 15 minutes.

- Avoid when the patient has known congenital long-QT syndrome or clinically significant QT prolongation; use caution with electrolyte loss, bradyarrhythmia, or other QT-prolonging medications.

Promethazine

Adult: 12.5 mg deep IM once for persistent nausea/vomiting after ondansetron.

- Paramedic only in this protocol.
- Do not administer IV or IO.
- Avoid with significant CNS depression, inability to protect the airway, or known hypersensitivity.

Analgesia

Age 16 years or older: use UP-11 Adult Pain Control. Age 2 through younger than 16 years: use PM-04 Pediatric Pain Control. Younger than 2 years: contact Medical Control.

- Document examination and pain score before and after medication.
- Do not use UP-11 for a patient younger than 16 years.

Provider-level care

Provider-Level Actions Higher provider levels include the actions listed for the preceding levels.

EMT: Perform the primary survey; place the patient in a position of comfort unless hypotension, respiratory compromise, or another condition requires different positioning. Administer oxygen for hypoxemia, respiratory distress, or shock.

- Obtain and document pain score, complete initial vital signs, temperature when infection is possible, blood glucose, and a focused abdominal, back, cardiopulmonary, skin, and neurologic examination. Obtain at least two complete vital-sign sets and reassess after each intervention.
- Acquire and transmit a 12-lead ECG for adult abdominal pain or otherwise unexplained nausea/vomiting. Expedite transport for hypotension, poor perfusion, GI bleeding, syncope, altered mental status, rigid or guarded abdomen, pain out of proportion, pulsatile mass, pregnancy-related pain or bleeding, or suspected ACS, AAA, sepsis, obstruction, or perforation.
- Keep the patient NPO except for an authorized oral or ODT medication. Use enteric/contact precautions when infectious vomiting or diarrhea is suspected and protect the patient from heat loss.

AEMT: Establish IV access when the patient has shock, dehydration, persistent vomiting, significant bleeding, an anticipated need for parenteral medication, or another time-critical condition. Use IO access when vascular access is urgently required and IV access cannot be obtained within the Universal Patient Care attempt limit.

- For adult hypotension or poor perfusion, administer normal saline 500 mL IV/IO, reassess, and repeat as needed to restore adequate perfusion or systolic blood pressure of at least 90 mmHg; maximum 2 L in this protocol. Use 250 mL increments with frequent lung and perfusion reassessment in patients with heart failure, renal failure, liver failure, known volume overload, pulmonary edema, or another high risk for fluid intolerance.

• For pediatric hypotension or poor perfusion, administer normal saline 20 mL/kg IV/IO and reassess after each bolus for perfusion and fluid overload; continue under the applicable pediatric shock protocol when poor perfusion persists.

• For nausea or vomiting, administer ondansetron: adult 4 mg IV/IO/IM/PO/ODT; pediatric 0.15 mg/kg IV/IO/IM/PO/ODT, maximum 4 mg. One repeat dose may be given after 15 minutes for persistent symptoms after reassessment.

• Treat hypoglycemia or hyperglycemic emergency under the age-appropriate diabetic emergency protocol. Do not delay transport for IV access or symptom-control medication.

Paramedic: Interpret the 12-lead ECG and immediately transition to the acute coronary syndrome protocol when ischemia or STEMI is suspected; obtain continuous cardiac monitoring for high-risk presentation, significant electrolyte loss, dysrhythmia, or parenteral QT-prolonging medication.

• For patients age 16 years or older, treat moderate or severe pain under UP-11 Adult Pain Control. For patients age 2 through younger than 16 years, use PM-04 Pediatric Pain Control; contact Medical Control for analgesic direction for a patient younger than 2 years. Do not withhold analgesia solely because a surgical abdomen is possible; document the examination and pain score before and after medication.

• If adult nausea or vomiting persists after ondansetron and no contraindication is present, administer promethazine 12.5 mg by deep IM injection once. Do not administer promethazine IV or IO in this protocol.

• For shock, major GI bleeding, suspected ruptured AAA or ectopic pregnancy, or another time-critical presentation, consider a second large-bore IV, provide cause-specific resuscitation, notify the receiving facility early, and minimize scene time.

• Provide advanced airway management when vomiting, aspiration, altered mental status, or clinical deterioration threatens airway protection or ventilation.

Warnings

• Abdominal pain may be an atypical presentation of acute coronary syndrome, particularly in older adults, patients with diabetes, and women. Acquire and transmit the ECG early.

• A normal blood pressure does not exclude significant hemorrhage or evolving shock. Trend mental status, pulse quality, skin, shock index, and serial vital signs.

• Do not give food or unrestricted oral fluids. An approved PO/ODT medication may be used when the airway is protected and administration is feasible.

• Promethazine can cause sedation, hypotension, respiratory depression, and severe tissue injury. Deep IM is the only authorized route in this protocol; IV and IO administration are prohibited.

• Do not use NSAID analgesia when GI bleeding, significant renal dysfunction, severe dehydration, anticoagulation, pregnancy, or another contraindication is present.

Clinical pearls

• Pain severity and abdominal tenderness do not reliably predict disease severity. Mesenteric ischemia, ectopic pregnancy, AAA, and early sepsis may initially have limited examination findings.

• Nausea without active vomiting may still benefit from antiemetic treatment.

• Diarrhea with recent antibiotic exposure or hospitalization should raise concern for C. difficile; use appropriate contact precautions and communicate the risk during handoff.

• Recurrent vomiting may reflect obstruction, increased intracranial pressure, DKA, ACS, carbon monoxide exposure, cannabinoid hyperemesis, organophosphate exposure, or another non-GI emergency.

Special populations

Pediatrics - younger than 16 years

• Use actual measured weight when reliable or the length-based tool for medication and equipment dosing. Reassess carefully after each fluid bolus.

• Bilious emesis, hematemesis, persistent focal pain, distention, guarding, altered mental status, signs of dehydration, or vomiting in a young infant requires prompt transport and early notification.

• Do not administer promethazine to a pediatric patient under this standing protocol; contact Medical Control if exceptional use is considered.

Pregnancy possibility

• Treat abdominal, pelvic, or back pain with vaginal bleeding, syncope, shoulder pain, or shock as ectopic pregnancy or obstetric hemorrhage until excluded. Expedite transport and transition to the obstetric protocol.

Fluid-intolerant patients

• Older adults may have serious disease with minimal tenderness or normal initial vital signs; reassess frequently and use clinical judgment regarding fluid tolerance.

Related pathways

- UP-11 Adult Pain Control (p. 31): Use for patients age 16 years or older.
- PM-04 Pediatric Pain Control (p. 181): Use for patients age 2 through younger than 16 years.

UP-04
Altered Mental Status

Revised October 31, 2026

Overview

Altered mental status is a symptom, not a diagnosis. Assume a time-critical medical, traumatic, toxicologic, neurologic, metabolic, environmental, or obstetric cause until reversible threats have been assessed and treated.

Prioritize oxygenation and ventilation, blood glucose, temperature, trauma assessment, opioid toxicity, seizure, stroke, shock, sepsis, toxic exposure, and dysrhythmia. Treat identified causes immediately while continuing the diagnostic assessment.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Immediate Stabilization
EMT / AEMT / Paramedic
Airway • suction • oxygenation • ventilation • circulation • severe hemorrhage • trauma precautions when indicated

↓

2. Rapid Reversible-Cause Check
EMT / AEMT / Paramedic
Glucose • temperature • pupils • medication/ingestion clues • opioid toxidrome • seizure • pregnancy possibility

↓

3. Hypoglycemia?
EMT / AEMT / Paramedic
Safe swallow: oral glucose • unsafe swallow: AEMT/Paramedic D10 or glucagon • recheck glucose and neurologic status

Continue at step 4 in the next column/page

Quick flow continued at step 4

4. Respiratory Depression + Suspected Opioid?
EMT / AEMT / Paramedic
Ventilate first • naloxone titrated to adequate ventilation • avoid abrupt full withdrawal when possible

↓

5. Stroke / Seizure / Trauma / Shock / Sepsis / Toxic Exposure?
EMT / AEMT / Paramedic
FAST then C-STAT if positive • document last known well • transition immediately to the cause-specific protocol

↓

6. Monitor + Transport
EMT / AEMT / Paramedic
Serial GCS/AVPU • repeat glucose/vital signs • cardiac monitor/ECG • $EtCO_2$ when ventilation is impaired • early notification

Indications

- Any acute or unexplained change in awareness, responsiveness, cognition, speech, behavior, orientation, or baseline neurologic function.
- Unconsciousness, unresponsiveness, delirium, confusion, lethargy, postictal state, unexplained agitation, or suspected metabolic/toxic encephalopathy.

Contraindications

- Do not administer oral glucose, food, fluid, or oral medication when the patient cannot swallow reliably or protect the airway.
- Do not use naloxone solely for altered mental status without respiratory depression or a reasonable suspicion of opioid effect.

Assessment

Immediate assessment

- Airway patency and protective reflexes; respiratory rate, depth, effort, SpO_2, and $EtCO_2$ when ventilation is impaired; pulse, perfusion, blood pressure, and major hemorrhage.
- AVPU/GCS and serial neurologic examination including pupils, gaze, facial symmetry, speech, arm drift, grip, extremity movement/sensation, seizure findings, and meningeal or infectious clues.

- Full head-to-toe examination when the patient cannot provide a reliable history, including occult trauma, infection, pressure injury, maltreatment, abuse, or neglect.

History and scene information

- Last known well and onset/progression; baseline cognition and function; seizure, stroke, diabetes, renal/hepatic disease, infection, pregnancy, psychiatric history, trauma, and recent illness.
- Prescription and nonprescription medications, insulin or oral diabetic agents, anticoagulants, opioids/sedatives, medication changes, alcohol/recreational substances, possible intentional ingestion, and access to toxins.
- Witness description, recent fall or injury, oral intake, vomiting/diarrhea, fever, environmental temperature, carbon-monoxide risk, occupational or hazardous-material exposure, and whether others have similar symptoms.

Cause-directed screening

- Hypoxia/hypercapnia; hypo- or hyperglycemia; opioid/sedative toxicity; seizure/postictal state; stroke or intracranial hemorrhage; head trauma; shock; sepsis/meningitis; hypo- or hyperthermia; dysrhythmia/ACS; electrolyte or renal/hepatic failure; pregnancy-related emergency; and toxic exposure.
- Do not stop after finding alcohol, drugs, dementia, or psychiatric illness. These conditions may coexist with hypoglycemia, trauma, infection, stroke, overdose, or another life threat.

Treatment

1. Transition to the identified cause-specific protocol without delaying transport. Continue serial GCS/AVPU, vital signs, glucose, ventilation, and response-to-treatment assessment.

Medication orders

Naloxone - intranasal

Adult: 2 mg IN; repeat every 2-3 minutes to adequate ventilation; maximum cumulative dose 8 mg. Pediatric: product/device-specific Medical Control order required pending stocked-product selection; no default numeric IN regimen.

- EMT/AEMT/Paramedic within authorized scope.
- Do not substitute a commercial 4 mg spray or carry the parenteral pediatric sequence into the IN route.
- Before pediatric intranasal naloxone, obtain a product/device-specific Medical Control order identifying the nasal product and strength, or injectable concentration and atomizer model, with dose, repeat interval and dose limits. For an atomized injectable preparation, the order must specify deliverable volume, nostril division and device dead-space handling. Do not substitute the pediatric IV sequence for an IN order or substitute a commercial 4 mg spray for the adult 2 mg order.

Oral Glucose

One packaged adult or weight-appropriate product dose PO; repeat based on glucose and clinical response.

- EMT/AEMT/Paramedic.
- Give only when the patient can follow commands, swallow reliably, and protect the airway.

Dextrose 10% (D10)

Adult: up to 250 mL IV/IO, titrated to improving mental status and glucose ≥70 mg/dL; maximum 25 g. Pediatric: initial dose 2 mL/kg IV/IO (0.2 g/kg); recheck after 5 minutes and repeat once for persistent hypoglycemia. Contact Medical Control for additional pediatric dextrose dosing or infusion.

- AEMT/Paramedic.
- Recheck glucose and neurologic status 5 minutes after administration.

Glucagon

Adult: 1 mg IM. Pediatric: <20 kg, 0.5 mg IM; ≥20 kg, 1 mg IM.

- AEMT/Paramedic when oral glucose is unsafe and vascular access cannot be obtained promptly.
- Protect against aspiration; vomiting may occur.

Naloxone - parenteral, AEMT/Paramedic

Adult: 0.4-2 mg IV/IO/IM. Pediatric: begin 0.01 mg/kg IV/IO/IM; if ventilation remains inadequate, escalate to 0.1 mg/kg, maximum 2 mg per dose. Repeat every 2-3 minutes to adequate ventilation; maximum cumulative dose 8 mg. Not an intranasal order.

- Ventilation is the treatment priority.
- If ventilation remains inadequate after 8 mg, continue airway support and reassess the diagnosis.

Normal Saline

For hypotension: adult 500 mL IV/IO; pediatric 10 mL/kg IV/IO, then reassess and use the cause-specific protocol.

- Use 250 mL adult increments when heart failure, renal failure, liver failure, pulmonary edema, known volume overload, or another high risk for fluid intolerance is present.

Provider-level care

Provider-Level Actions Higher provider levels include the actions listed for the preceding levels.

EMT: Open and suction the airway as needed; insert an OPA or NPA when indicated; provide BVM ventilation for apnea, inadequate respiratory effort, or ineffective ventilation. Administer oxygen for hypoxemia, respiratory distress, shock, or another clinical indication.

• Assess AVPU/GCS, orientation and behavior compared with baseline, pupils, speech, facial symmetry, extremity strength, sensation, skin, trauma, medical-alert information, medication access, and exposure clues. Obtain complete vital signs, SpO_2, blood glucose, and temperature early; obtain at least two complete vital-sign sets.

• Perform FAST for acute or unexplained neurologic change; if positive, complete C-STAT and document last known well, anticoagulant use, baseline function, and witness contact information. Acquire and transmit a 12-lead ECG for adult acute or unexplained altered mental status, unresponsiveness, suspected ingestion, or postictal state.

• If symptomatic hypoglycemia is present and the patient is awake, follows commands, can swallow, and protects the airway, administer oral glucose according to the packaged dose. Do not administer anything orally when airway protection or swallowing is impaired.

• When opioid toxicity is suspected with respiratory depression, support ventilation immediately. Adult: naloxone 2 mg IN, repeat every 2-3 minutes to adequate ventilation, maximum cumulative dose 8 mg. Pediatric IN administration requires a product/device-specific Medical Control order pending the medical director's stocked-product selection; do not use the parenteral sequence by the IN route. Do not delay BVM ventilation.

• Protect the patient from injury during seizure or agitation. Use the least restrictive safe approach and transition to the seizure, agitation, trauma, stroke, diabetic, sepsis, shock, environmental, or toxicology protocol as soon as the syndrome is recognized.

AEMT: Establish IV access when parenteral dextrose, naloxone, fluid, or another time-critical intervention is indicated. Use IO access when vascular access is urgently required and IV access cannot be obtained within the Universal Patient Care attempt limit.

• For glucose below 70 mg/dL with symptoms, or when symptoms cannot be assessed reliably because of altered mental status, administer D10 when oral glucose is unsafe or ineffective: adult up to 250 mL IV/IO, titrated to improving mental status and glucose at least 70 mg/dL, maximum 25 g; pediatric initial dose 2 mL/kg IV/IO (0.2 g/kg). Recheck glucose and neurologic status after 5 minutes. The pediatric dose may be repeated once for persistent hypoglycemia; contact Medical Control for additional dextrose dosing or infusion.

• If vascular access cannot be obtained promptly, administer glucagon IM: adult 1 mg; pediatric 0.5 mg when less than 20 kg or 1 mg when 20 kg or greater. Position for aspiration protection and continue airway monitoring because vomiting may occur.

• AEMT/Paramedic parenteral naloxone for suspected opioid-related respiratory depression: adult 0.4-2 mg IV/IO/IM; pediatric begin 0.01 mg/kg IV/IO/IM and, if ventilation remains inadequate, escalate to 0.1 mg/kg, maximum 2 mg per dose. Repeat every 2-3 minutes to adequate ventilation, maximum cumulative dose 8 mg. If ventilation remains inadequate after 8 mg, continue airway/ventilatory support and reassess the diagnosis. Intranasal dosing is a separate product- and age-specific pathway; do not apply this parenteral sequence to IN administration.

• For hypotension, administer normal saline: adult 500 mL IV/IO; pediatric 10 mL/kg IV/IO. Reassess blood pressure, perfusion, and lung sounds after each bolus and transition to the cause-specific shock, sepsis, diabetic, cardiac, or trauma protocol before additional fluid. Do not administer a fluid bolus for altered mental status without hypotension.

• Insert a supraglottic airway when BVM ventilation is inadequate or prolonged airway support is required and the patient meets airway-protocol indications. Use continuous waveform capnography when available after advanced airway placement or when ventilation is impaired.

Paramedic: Interpret the ECG, institute continuous cardiac monitoring, and treat a dysrhythmia, ischemia, hyperkalemia pattern, or other identified cardiac cause under the applicable cardiac protocol.

• Use waveform capnography for impaired ventilation, suspected opioid or sedative toxicity, advanced airway management, or chemical sedation. Perform endotracheal intubation when the patient cannot protect the airway or cannot be adequately oxygenated or ventilated with less invasive measures.

• Complete a focused differential for neurologic, metabolic, infectious, toxicologic, environmental, traumatic, obstetric, and behavioral causes; direct treatment and destination according to the time-critical syndrome identified.

• If agitation creates an immediate danger and verbal de-escalation and physical safety measures are insufficient, use midazolam or ketamine under the agitation protocol. Continuous airway, SpO_2, $EtCO_2$, ECG, and blood-pressure monitoring are required after chemical sedation.

• Provide early destination notification for persistent coma, airway compromise, suspected stroke, status epilepticus, sepsis/shock, severe toxic exposure, or unexplained deterioration. Minimize scene time when definitive treatment is time sensitive.

Warnings

• Suspected opioid-related respiratory depression: support ventilation immediately and titrate reversal to adequate ventilation, not complete arousal. Do not delay BVM ventilation while preparing or awaiting naloxone. Keep intranasal and parenteral dosing in separate age-specific instructions. For resuscitation immediately after birth, follow AO-02 Newly Born; do not substitute naloxone for neonatal ventilation.

• Ventilation takes priority over naloxone. The goal is adequate ventilation, not complete arousal; abrupt reversal may cause vomiting, aspiration, severe agitation, withdrawal, or sympathetic surge.

• A normal glucose, temperature, ECG, or initial neurologic examination does not exclude a serious cause. Continue reassessment and transport unexplained or persistent altered mental status.

• Do not assume intoxication, dementia, psychiatric illness, or postictal state is the sole diagnosis until trauma and medical causes have been assessed.

• Physical or chemical restraint may worsen occult hypoxia, hypercapnia, acidosis, hyperthermia, or shock. Use the agitation/restraint protocol and continuous monitoring when restraint is necessary.

• Glucagon may be ineffective in malnutrition, chronic alcohol use, severe liver disease, or depleted glycogen states.

Clinical pearls

• Collateral history, medication containers, medical-alert identification, witness contact information, and the exact scene circumstances may be diagnostically critical and should accompany the handoff.

• Persistent focal neurologic deficit after glucose correction is stroke until proven otherwise.

• Pinpoint pupils alone do not establish opioid toxicity; respiratory depression and response to ventilation/naloxone are more important.

• Multiple patients with headache, nausea, confusion, or syncope should prompt immediate consideration of carbon monoxide or another environmental exposure.

• Patients who regain consciousness after treatment still require evaluation for recurrence, long-acting agents, co-ingestion, trauma, or an alternative diagnosis.

Special populations

Pediatrics - younger than 16 years

• Use measured weight when reliable or the length-based tool for dosing. Consider ingestion, infection, seizure, trauma, hypoglycemia, metabolic disease, intussusception, and nonaccidental trauma.

• Caregiver observations of baseline behavior and the timing of change are important, but do not delay assessment of airway, glucose, temperature, perfusion, and trauma.

Pregnancy

• Consider eclampsia, hemorrhage/ectopic pregnancy, medication toxicity, hypoglycemia, stroke, and other pregnancy-related causes. Use left uterine displacement when later pregnancy and hypotension are present.

Dementia, developmental disability, or communication limitation

• Establish baseline status through caregivers or records when immediately available. An acute deviation from baseline requires the same reversible-cause assessment as any other patient.

UP-05
Back Pain

Revised October 31, 2026

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Initial Assessment
EMT / AEMT / Paramedic
Primary survey • vital signs • pain score • relevant history • traumatic or atraumatic onset

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. Focused Neurologic Assessment
EMT / AEMT / Paramedic
Bilateral leg strength and sensation • distal pulses • gait when safe • ask about saddle anesthesia and bowel/bladder dysfunction

↓

3. New Neurologic Deficit?
EMT / AEMT / Paramedic
Weakness, sensory loss, saddle anesthesia by history, urinary retention/incontinence, or bowel dysfunction

↓

4. Time-Critical Finding
EMT / AEMT / Paramedic
Prompt transport • early receiving-facility notification • repeat neurologic examination

↓

5. Treat Pain
EMT / AEMT / Paramedic
Position of comfort • adult UP-11 or pediatric PM-04 analgesia • cardiac monitoring when medication is administered

↓

6. Reassess + Transport
EMT / AEMT / Paramedic
Pain score • vital signs • neurologic findings • medication response

Indications

- Acute or chronic back pain without a more specific immediately apparent treatment pathway.

Assessment

Focused assessment

- Pain onset, location, radiation, severity, provoking or relieving factors, trauma, prior episodes, and relevant medications or anticoagulants.

Treatment

1. Also screen for sudden severe back pain with chest or abdominal pain, syncope, shock, or unequal pulses; fever or infection risk (including immunosuppression, injection drug use, or a recent spinal procedure); cancer history; and pregnancy-related symptoms. These findings require prompt assessment for vascular, infectious, spinal, or obstetric emergencies and transition to the appropriate protocol, even when the initial leg examination is normal.

Medication orders

Analgesia

Age 16 years or older: use UP-11 Adult Pain Control. Age 2 through younger than 16 years: use PM-04 Pediatric Pain Control. Younger than 2 years: contact Medical Control.

- Apply cardiac monitoring when analgesic medication is administered.
- Reassess pain, vital signs, neurologic findings, and medication response.

Provider-level care

Provider-Level Actions Higher provider levels include the actions listed for the preceding levels.

EMT: Obtain the history, pain score, complete vital signs, and a focused examination. Determine whether the pain followed trauma.

- Assess bilateral lower-extremity strength and sensation, distal pulses, and the ability to stand or walk when safe. Ask about new saddle anesthesia, urinary retention or incontinence, and bowel dysfunction. Do not perform a routine perineal examination.
- For traumatic back pain, apply spinal protection when indicated. For uncomplicated atraumatic pain, place the patient in a position of comfort and avoid unnecessary immobilization.
- Provide nonpharmacologic care. For patients age 16 years or older, use UP-11 Adult Pain Control; for patients age 2 through younger than 16 years, use PM-04 Pediatric Pain Control; for patients younger than 2 years, contact Medical Control for analgesic direction. Apply cardiac monitoring when medication is administered.
- Transport promptly and notify the receiving facility early for a new or progressive neurologic deficit, saddle anesthesia, or new bowel or bladder dysfunction.

AEMT: Establish IV access only when needed for parenteral analgesia, hemodynamic instability, or

another specific clinical indication. Do not start an IV or administer fluid routinely for isolated back pain.

• Administer analgesia under the age-appropriate pain pathway, maintain cardiac monitoring after medication administration, and reassess pain, vital signs, neurologic findings, and medication response.

Paramedic: Interpret the cardiac rhythm and administer parenteral analgesia under UP-11 for patients age 16 years or older or PM-04 for patients age 2 through younger than 16 years when indicated. Continue the medication-specific airway, respiratory, hemodynamic, and cardiac monitoring required by the applicable pathway.

• Acquire a 12-lead ECG or transition to the appropriate cardiac, abdominal, vascular, trauma, or obstetric protocol only when the history, examination, or clinical condition provides a separate indication.

Warnings

- Do not perform a routine perineal examination solely to assess saddle anesthesia; obtain this finding by history.
- Do not routinely immobilize atraumatic back pain or delay transport for nonessential procedures.

Clinical pearls

- A normal initial neurologic examination does not replace reassessment after analgesia or when symptoms change.

Related pathways

- UP-11 Adult Pain Control (p. 31): Use for patients age 16 years or older.
- PM-04 Pediatric Pain Control (p. 181): Use for patients age 2 through younger than 16 years.

UP-06
IV / IO Access

Revised October 31, 2026

Overview

Establish vascular access only when needed for medication, fluid, blood sampling, or anticipated time-critical treatment. Use the least invasive route that meets the patient's immediate clinical need.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Is Vascular Access Needed?
EMT / AEMT / Paramedic
Medication • fluid • time-critical treatment • anticipated deterioration

↓

2. Immediate IO Indication?
AEMT / Paramedic
Cardiac arrest • profound shock • critical treatment cannot wait

↓

3. Peripheral IV / Saline Lock
AEMT / Paramedic
Best conventional site • maximum 3 total attempts across the crew • do not delay transport

↓

4. Critical Access Still Needed?
AEMT / Paramedic
After unsuccessful peripheral attempts, proceed to IO

↓

5. Preferred IO Site
AEMT / Paramedic
Proximal tibia • adult humeral head only when tibial access is contraindicated

↓

6. Confirm + Secure + Reassess
AEMT / Paramedic
Patency • flush • secure • inspect for infiltration • document site and every attempt

Indications

- Medication, fluid, or blood sampling required by the applicable treatment protocol.
- Critical illness, cardiac arrest, profound shock, or anticipated deterioration for which vascular access is needed.

Contraindications

- Do not place an IO through infection, into a fractured target bone, or at a site with a relevant prosthesis or prior significant orthopedic procedure. Follow the current instructions for the agency-stocked device. For Arrow EZ-IO, do not use a target bone that has had IO access or an attempted IO insertion within the previous 48 hours; select another eligible bone.
- Do not place an IO when landmarks cannot be identified or the device needle is not appropriate for the patient's tissue depth.
- Do not place a peripheral IV in an injured extremity, an extremity with a dialysis fistula or graft, or a site with infection or severely impaired circulation when another site is available.

Assessment

Access decision

- Determine the treatment that requires access, the urgency of that treatment, expected flow needs, transport priority, and whether a nonvascular route can provide the required medication without delay.
- Before IO placement, inspect the proposed bone and overlying tissue for fracture, infection, orthopedic hardware or prosthesis, recent IO use, excessive tissue depth, and identifiable landmarks.

Treatment

1. For a conscious IO patient, administer 2% preservative-free, epinephrine-free lidocaine suitable for IV/IO administration before flushing, then flush, secure, and monitor the site continuously.
2. Document the indication, provider, site, catheter or needle size, number and location of every attempt, medications administered, patency, complications, and response.

Medication orders

Lidocaine 2% - conscious IO infusion pain

Adult: 20-40 mg IO administered slowly before the initial forceful flush. Pediatric: 0.5 mg/kg IO administered slowly before the initial forceful flush; maximum 40 mg.

- Use only 2% preservative-free, epinephrine-free lidocaine suitable for IV/IO administration. AEMT/Paramedic within device training and local authorization. This dose is for IO infusion pain, not dysrhythmia treatment.
- Confirm the IO is appropriately positioned and secure before medication administration.

Provider-level care

Provider-Level Actions Higher provider levels include the actions listed for the preceding levels.

EMT: Identify whether the patient may require medication, fluid, blood sampling, or another time-critical treatment and notify the AEMT or Paramedic.

- Prepare and assist with vascular-access equipment, position the extremity, maintain aseptic technique, monitor the patient, and document assistance as appropriate. EMT personnel do not insert IV or IO catheters under this protocol.

AEMT: Establish a peripheral IV or saline lock when clinically indicated. Use an eligible upper-extremity site; do not cannulate a leg or neck. Limit AEMT peripheral attempts to two in one arm. Select a conventional peripheral site appropriate for the treatment and avoid an injured extremity, an extremity with a dialysis fistula or graft, or a site with infection or impaired circulation.

- Limit peripheral IV attempts to three total attempts across the entire crew. Do not allow repeated attempts by successive providers to restart the count. If vascular access is critical after unsuccessful attempts, proceed to IO access.
- In cardiac arrest, profound shock, or another immediately life-threatening condition in which peripheral attempts would delay critical treatment, proceed directly to IO access without a required preceding IV attempt.
- Use the proximal tibia as the preferred IO site. In adults, the humeral head may be used as an alternative when tibial access is contraindicated and appropriate landmarks can be identified. Use proximal tibial access for pediatric patients younger than 16 years.
- For a conscious patient, administer 2% preservative-free, epinephrine-free lidocaine suitable for IV/IO administration slowly through the IO before the initial forceful flush: adult 20-40 mg; pediatric 0.5 mg/kg, maximum 40 mg. Flush and operate the device according to manufacturer instructions, using pressure-assisted infusion when needed.
- Confirm catheter stability and flow, secure the catheter, and inspect repeatedly for swelling, leakage, displacement, increasing pain, resistance, or impaired distal perfusion. Stop using the site immediately if infiltration or another complication is suspected.

Paramedic: Perform the AEMT vascular-access actions and select access according to the medication, resuscitation, anticipated flow requirement, patient anatomy, and transport priority.

• When conventional peripheral access is unsuccessful or unavailable and IV access remains clinically necessary, establish an external-jugular IV. Do not use the external jugular as a routine first site and do not place a saline lock in an external-jugular vein.

• Do not access a PICC, central venous catheter, implanted port, or dialysis catheter under UP-06. Existing central-device use during interfacility transport is governed by the separate interfacility-transfer policy.

Warnings

- AEMT peripheral IV access: use an eligible upper-extremity site; do not cannulate a leg or neck. Limit AEMT peripheral attempts to two in one arm. The maximum of three peripheral attempts across the entire crew is an overall cap, not permission to exceed a provider-specific limit or a requirement to delay indicated IO access.
- Do not delay transport, resuscitation, or a time-critical medication for repeated vascular-access attempts.
- IO infusion is painful in conscious patients. Administer lidocaine before the initial forceful flush unless an immediate life threat prevents delay.
- Stop using an IV or IO immediately for swelling, leakage, displacement, increasing pain, resistance, loss of flow, or suspected infiltration or compartment syndrome.

Clinical pearls

- A functioning IO provides access for resuscitation medications and fluids when peripheral access is not rapidly available, but pressure-assisted infusion is commonly required.
- Aspiration of marrow or blood may support placement but is not required when the catheter is stable, flushes appropriately, and there is no evidence of infiltration.
- For trauma patients, establish access during transport whenever feasible rather than extending scene time solely for an IV.

Special populations

Pediatrics - younger than 16 years

- Use the proximal tibia for IO access and select needle length according to patient size and manufacturer guidance.

UP-07
Dental Problems

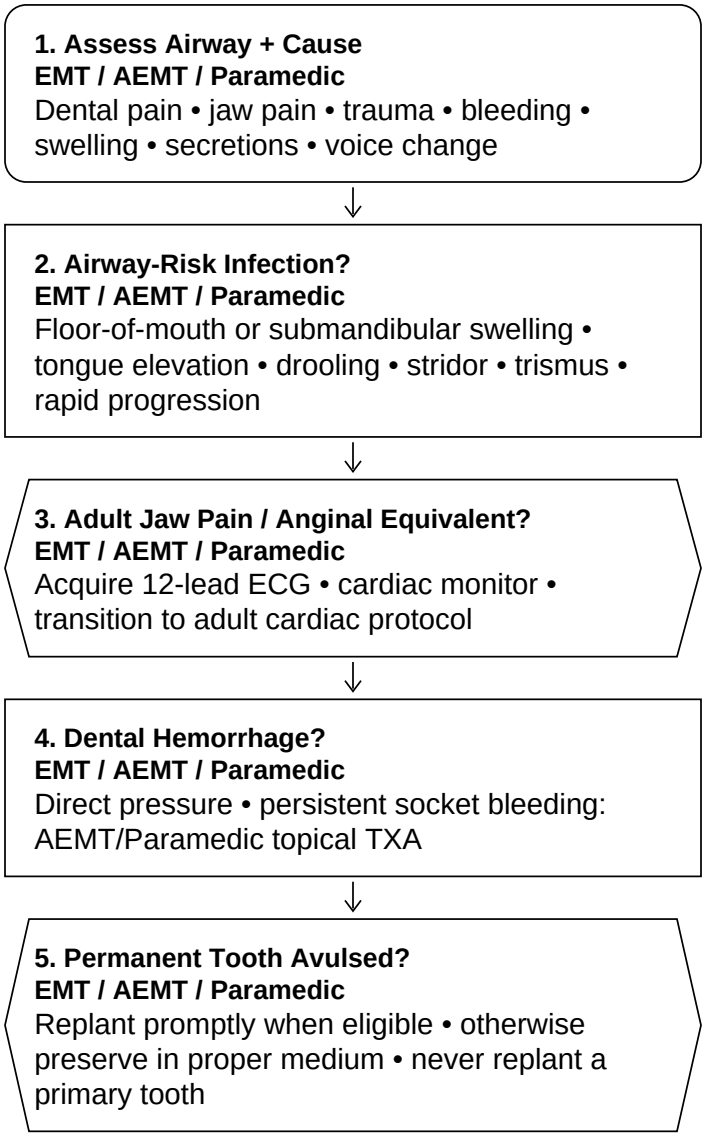
Revised August 2026

Overview

Most dental complaints require focused assessment, bleeding control, analgesia, and dental follow-up. Airway-threatening swelling, uncontrolled hemorrhage, significant facial trauma, and anginal-equivalent jaw pain require immediate transition to the appropriate emergency pathway.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.



Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Analgesia + Reassess
EMT / AEMT / Paramedic
Adult UP-11 or pediatric PM-04 pain control • cardiac monitoring with medication • repeat airway, bleeding, pain, and vital signs

Indications

- Dental or jaw pain, oral or tooth-socket bleeding, dental infection or swelling, fractured tooth, or avulsed tooth.

Contraindications

- Do not attempt field replantation when the patient cannot protect the airway, is uncooperative, has significant associated facial or jaw trauma, has a suspected socket fracture, or the tooth cannot be inserted gently without force.
- Do not administer topical TXA to a patient with a known hypersensitivity to tranexamic acid.

Assessment

Focused dental and airway assessment

- Onset and source of pain; tooth versus jaw location; trauma; bleeding; fever; swelling progression; difficulty swallowing; drooling; voice change; trismus; dyspnea; chest, arm, back, or epigastric discomfort; diaphoresis; nausea; and syncope.
- Inspect the face, jaw, mouth, floor of mouth, tongue position, visible tooth or socket, bleeding source, and ability to handle secretions without aggressive probing or manipulation.
- For tooth avulsion, determine whether the tooth is permanent, time out of the socket, dry time, storage medium, root integrity, contamination, and associated facial or neurologic injury.

Medication orders

Tranexamic Acid - topical dental hemorrhage

AEMT/Paramedic: TXA 1 g/10 mL applied topically to saturate gauze or cotton; pack the bleeding socket and maintain direct pressure for 20 minutes.

- Use only after bleeding persists despite firm direct pressure.
- EMT administration is not authorized under UP-07.
- Do not administer IV/IO TXA solely for dental bleeding.

Analgesia

Age 16 years or older: use UP-11 Adult Pain Control. Age 2 through younger than 16 years: use PM-04 Pediatric Pain Control. Younger than 2 years: contact Medical Control.

- Apply cardiac monitoring when analgesic medication is administered.
- Reassess pain, airway, bleeding, vital signs, and medication response.

Provider-level care

Provider-Level Actions Higher provider levels include the actions listed for the preceding levels.

EMT: Assess airway patency, ability to handle secretions, voice, oral and facial swelling, trismus, bleeding, trauma, pain, fever, and complete vital signs. Suction blood or secretions as needed and position the patient to protect the airway.

- For floor-of-mouth or submandibular swelling, tongue elevation, drooling, inability to handle secretions, stridor, trismus, toxic appearance, or rapidly progressive swelling, request Paramedic support, prepare airway equipment, minimize oral manipulation, and begin prompt transport.
- For adult jaw pain or another anginal-equivalent complaint, acquire and transmit a 12-lead ECG, apply cardiac monitoring, and transition to the applicable adult cardiac protocol. Do not obtain an ECG routinely for isolated tooth pain with a clear dental source.
- Control dental bleeding with suction as needed and firm direct pressure using folded gauze placed over the socket. Have the alert patient close the teeth over the gauze while maintaining airway safety. EMT personnel do not administer topical TXA under this protocol.
- For an avulsed permanent tooth, handle only the crown. If contaminated, gently rinse with milk or normal saline without rubbing or scrubbing the root. When the patient is conscious, cooperative, protects the airway, the tooth is intact, and significant jaw or socket fracture is not suspected, gently return the tooth to its socket without force and have the patient hold it in position by biting on gauze.
- Never replant a primary tooth. If replantation is unsuccessful or contraindicated, place the tooth in commercial tooth-preservation solution, milk, normal saline, or the patient's saliva in a closed container. Do not store the tooth in the patient's mouth, on ice, or dry.
- Provide nonpharmacologic care. For patients age 16 years or older, use UP-11 Adult Pain Control; for patients age 2 through younger than 16 years, use

PM-04 Pediatric Pain Control; for patients younger than 2 years, contact Medical Control for analgesic direction. Apply cardiac monitoring when medication is administered and reassess pain, airway, bleeding, vital signs, and treatment response.

AEMT: For persistent tooth-socket bleeding despite firm direct pressure, saturate gauze or cotton with TXA 1 g/10 mL, pack it into the socket, and maintain direct pressure with the patient closing the teeth for 20 minutes. Continue suction and airway protection as needed. Do not administer IV/IO TXA solely for dental bleeding.

• Establish IV access only when required for parenteral analgesia, hemodynamic instability, airway management, or another specific clinical indication. Do not routinely establish IV access or administer fluid for an isolated dental complaint.

• Administer analgesia under the age-appropriate pain pathway, maintain cardiac monitoring after medication administration, and reassess pain, bleeding, airway, vital signs, and medication response.

Paramedic: Interpret the 12-lead ECG and cardiac rhythm for adult jaw pain or another anginal-equivalent complaint, and treat identified ischemia or dysrhythmia under the applicable adult cardiac protocol.

• For progressive oral, submandibular, or floor-of-mouth swelling with airway compromise, maintain spontaneous ventilation whenever possible, prepare suction and a difficult-airway plan, and transition to the adult or pediatric airway protocol without delaying transport.

• Administer topical TXA and parenteral analgesia as indicated, using UP-11 for patients age 16 years or older and PM-04 for patients age 2 through younger than 16 years, with the medication-specific airway, respiratory, hemodynamic, and cardiac monitoring required by the applicable pathway.

Warnings

- Oral or facial swelling may progress rapidly. Drooling, inability to handle secretions, tongue elevation, stridor, trismus, voice change, or rapidly progressive submandibular swelling requires prompt airway preparation and transport.
- Do not probe or attempt to drain a dental abscess in the field.
- An avulsed or loose tooth and blood in the mouth may obstruct the airway, particularly when mental status is impaired.

Clinical pearls

- Immediate replantation offers the best chance of retaining an avulsed permanent tooth when the procedure is safe and the tooth can be inserted gently.
- If replantation cannot be performed, preventing the root from drying is time critical. Commercial preservation solution or milk is preferred when immediately available.
- Document the tooth type when known, time of avulsion, estimated dry time, storage medium, replantation attempt and result, bleeding treatment, associated injuries, and destination notification.

Special populations

Pediatrics - younger than 16 years

- Carefully distinguish a permanent tooth from a primary tooth. Never replant a primary tooth because of potential injury to the developing permanent tooth.
- Use PM-04 Pediatric Pain Control for patients age 2 through younger than 16 years. Contact Medical Control for analgesic direction for a patient younger than 2 years. Maintain close airway observation when facial swelling, bleeding, or loose teeth are present.

Related pathways

- UP-11 Adult Pain Control (p. 31): Use for patients age 16 years or older.
- PM-04 Pediatric Pain Control (p. 181): Use for patients age 2 through younger than 16 years.

UP-08
Emergencies Involving Indwelling Central Lines

Revised October 31, 2026

Overview

Recognize and stabilize complications involving PICC lines, tunneled central catheters, implanted ports, and dialysis catheters. Prevent hemorrhage, additional air entry, medication exposure, and further catheter damage.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Identify the Emergency
EMT / AEMT / Paramedic
Bleeding • damage • dislodgement • air entry • infusion reaction • infection • occlusion

↓

2. Stop Further Harm
EMT / AEMT / Paramedic
Stop infusion • close clamp • clamp damaged external tubing proximal to defect • protect open site

↓

3. Suspected Air Embolism?
EMT / AEMT / Paramedic
Stop infusion, clamp/seal • level supine or recovery position • 100% oxygen • rapid transport

↓

4. Dislodged / Damaged / Bleeding?
EMT / AEMT / Paramedic
Do not advance, remove, reinsert, or force-flush • direct pressure • sterile or air-occlusive dressing

↓

5. Monitor + Transport
EMT / AEMT / Paramedic
Cardiac rhythm • SpO_2 • $EtCO_2$ • airway • perfusion • neurologic status • early notification

Indications

- Bleeding, damage, breakage, leakage, partial or complete dislodgement, suspected venous air embolism, occlusion, infusion reaction, infection, or other emergency involving an indwelling central device.

Contraindications

- Do not use UP-08 to initiate routine access, administer medication or fluid, or manage an uncomplicated interfacility infusion.
- Do not advance, reinsert, exchange, remove, repair, or force-flush a damaged, dislodged, occluded, painful, leaking, or otherwise abnormal central device.

Assessment

Device and complication assessment

- Identify the device type when possible: PICC, tunneled catheter, implanted port, or dialysis catheter. Determine whether it was accessed before EMS arrival, what is infusing, the ordered rate, when the problem began, and what manipulation occurred immediately beforehand.
- Inspect without probing for external damage, open connections, clamp position, leakage, bleeding, catheter-length change, dressing disruption, erythema, warmth, drainage, swelling, tenderness, and impaired distal circulation.
- Assess for acute dyspnea, chest pain, hypoxemia, cough, hypotension, altered mental status, seizure, or focal neurologic findings suggesting air embolism; assess for fever, rigors, shock, urticaria, bronchospasm, or angioedema suggesting infection or infusion reaction.

Provider-level care

Provider-Level Actions Higher provider levels include the actions listed for the preceding levels.

EMT: Identify the device and problem without disconnecting, accessing, flushing, advancing, removing, or manipulating the catheter. Obtain the device and infusion history from the patient, caregiver, sending facility, labels, and pump display when immediately available.

• For line damage, dislodgement, suspected air entry, or an infusion reaction, stop the infusion. Close an existing clamp or clamp intact external tubing between the patient and the damaged segment. Use a purpose-designed line clamp when available and do not apply an unprotected metal instrument directly to the catheter.

• For bleeding at the insertion site, apply direct pressure around the catheter without removing it. If the catheter is completely dislodged, apply direct pressure until hemorrhage is controlled, then cover the site with an air-occlusive dressing. Do not reinsert the catheter.

• If the catheter is partially dislodged, do not advance or remove it. Stabilize the external segment in its found position and cover the site with a sterile dressing.

• For suspected air embolism, stop the infusion, clamp the line, seal any open insertion site, and administer 100% oxygen. Position supine on a level surface; if mental status is impaired but breathing is adequate and no advanced airway is present, use the recovery position on a level surface to protect the airway. Do not routinely use head-down positioning. If

cardiac arrest occurs, provide standard CPR supine. Transport rapidly and notify Medical Control and the receiving facility early so hyperbaric consultation can be considered without delaying stabilization.

• Apply cardiac, SpO_2, and $EtCO_2$ monitoring for suspected air embolism. Repeat airway, respiratory, perfusion, neurologic, and complete vital-sign assessments during transport.

• Do not alter an uncomplicated infusion under UP-08. Routine continuation, adjustment, or discontinuation of an interfacility infusion is governed by the separate interfacility-transfer policy.

AEMT: Perform the immediate complication-control actions above. If medication or fluid remains necessary after the central device is no longer usable, establish peripheral IV or IO access according to UP-06 without delaying transport.

• Treat hypotension or another identified complication under the applicable shock, allergic-reaction, sepsis, or complaint-specific protocol. Do not administer medication or fluid through the affected central device under UP-08.

• If parenteral nutrition is interrupted, obtain a blood glucose promptly and repeat glucose assessment for symptoms of hypoglycemia or during prolonged transport.

Paramedic: Interpret the cardiac rhythm, manage airway or ventilatory compromise, and treat shock, dysrhythmia, anaphylaxis, seizure, or another identified complication under the applicable protocol.

• For suspected air embolism, continue high-concentration oxygen and cardiac, SpO_2, and waveform $EtCO_2$ monitoring. Do not delay transport for attempts to aspirate air through the catheter.

• Do not access, repair, exchange, advance, remove, or force-flush the catheter under UP-08. Central-device use during an interfacility transfer requires the separate interfacility-transfer policy.

Warnings

• Never force-flush an occluded or resistant central line. Forced flushing may dislodge thrombus, rupture the catheter, or cause infiltration or embolization.

• An open central-line tract may permit air entry even after the catheter is completely removed. Maintain an air-occlusive dressing and monitor for delayed cardiopulmonary or neurologic symptoms.

• Do not delay rapid transport for catheter troubleshooting, repair, or air aspiration.

• A life-sustaining infusion that is not associated with line damage, dislodgement, air entry, or reaction must not be altered under UP-08; follow the separate interfacility-transfer policy and sending/receiving physician orders.

Clinical pearls

• Caregivers and sending-facility staff may provide important device-specific information, but immediate stabilization takes priority when hemorrhage, air entry, or a serious infusion reaction is suspected.

• Preserve the pump, medication container, tubing, labels, disconnected components, and the dislodged catheter when available for transfer with the patient; do not discard them at the scene.

• If parenteral nutrition is stopped, monitor for hypoglycemia, particularly in pediatric patients or during prolonged transport.

Special populations

Pediatrics - younger than 16 years

• Secure damaged or partially dislodged tubing carefully to prevent further movement and use age-appropriate positioning while maintaining airway and respiratory monitoring.

UP-09
Epistaxis

Revised August 2026

Overview

Sustained compression of the soft lower nose is the first-line treatment. Topical TXA on gauze is available to AEMTs and Paramedics for persistent atraumatic anterior bleeding; oxymetazoline is not included because it is not currently on the Claiborne formulary.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Airway + Position
EMT / AEMT / Paramedic
Sit upright and lean forward when tolerated • suction as needed • never tilt the head backward

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. Major Facial Trauma / Unprotected Airway?
EMT / AEMT / Paramedic
Do not place intranasal gauze or a clamp • protect airway • control external bleeding • transition to trauma/airway care

↓

3. Clear Clots + Compress
EMT / AEMT / Paramedic
Alert atraumatic patient gently clears clots once • compress soft lower nose continuously for 15 minutes

↓

4. Maintain Uninterrupted Pressure
EMT / AEMT / Paramedic
Manual pressure • commercial nasal clamp • or two tongue depressors taped together at one end

↓

5. Persistent Uncontrolled Bleeding?
AEMT / Paramedic
AEMT/Paramedic: TXA-saturated anterior gauze • bilateral if uncontrolled or source uncertain • pressure 20 minutes

↓

6. Hypotension?
AEMT / Paramedic
IV/IO access and fluid only for hypotension • reassess after each bolus

↓

7. Monitor After Bleeding Control + Transport
EMT / AEMT / Paramedic
Secure pressure first • then cardiac monitoring when indicated without interrupting hemorrhage control

Indications

- Active, recurrent, or recently controlled nasal bleeding requiring EMS evaluation or treatment.

Contraindications

- Do not administer topical TXA to a patient with a known hypersensitivity to tranexamic acid.

Assessment

Focused bleeding and airway assessment

- Determine onset, duration, recurrence, suspected side, trauma, prior treatment, prior nasal surgery or recurrent unilateral bleeding, estimated blood loss, anticoagulant or antiplatelet use, bleeding disorder, intranasal drug use, and symptoms of anemia or shock.
- Assess the ability to handle blood and secretions, airway patency, respiratory status, mental status, skin signs, perfusion, complete vital signs, and the posterior pharynx for ongoing blood.
- After clots are cleared once in an alert atraumatic patient, inspect only the visible anterior nasal opening without probing. Persistent blood in the posterior pharynx or continued bilateral bleeding despite correct compression suggests a posterior source.

Medication orders

Tranexamic Acid - topical epistaxis

AEMT/Paramedic: adult TXA 1 g/10 mL total; pediatric TXA 15 mg/kg, maximum 1 g total. Apply to appropriately sized anterior gauze and maintain external pressure for 20 minutes.

- Use only for persistent atraumatic anterior bleeding after 15 minutes of uninterrupted compression.
- If bleeding is uncontrolled when TXA is applied or the side is uncertain, divide the approved total dose between gauze pledgets placed gently in both anterior nostrils.
- Leave gauze tails visible externally. Do not atomize TXA and do not perform deep or blind packing.
- EMT administration is not authorized under UP-09. Do not administer IV/IO TXA solely for isolated epistaxis.

Crystalloid - hypotension only

AEMT/Paramedic: adult 500 mL; pediatric 10 mL/kg. Reassess after each bolus and use smaller aliquots for CHF, renal failure, liver failure, or known volume overload.

- Do not establish vascular access or administer fluid routinely for controlled epistaxis in a hemodynamically stable patient.
- Transition to UP-04 and the applicable hemorrhagic-shock or trauma protocol when significant blood loss or shock is present.

Provider-level care

Provider-Level Actions Higher provider levels include the actions listed for the preceding levels.

EMT: Assess airway patency, ability to handle blood and secretions, respiratory status, perfusion, mental status, bleeding severity, estimated blood loss, associated facial trauma, anticoagulant or antiplatelet use, and complete vital signs. Suction blood and secretions as needed.

• Seat the patient upright and leaning forward when tolerated. Do not tilt the head backward. If the alert, cooperative patient has atraumatic bleeding and protects the airway, have the patient gently clear or blow clots from the nose once before compression.

• Compress the soft lower third of the nose continuously for 15 minutes without releasing pressure to check for bleeding. Pressure may be applied manually, with a commercial nasal clamp, or with two tongue depressors taped together at one end. Position the untaped ends externally over the soft nasal alae; never insert the tongue depressors into the nostrils.

• If major facial trauma, suspected basilar skull fracture, altered mental status with an unprotected airway, or inability to manage blood and secretions is present, do not place intranasal gauze or a nasal clamp. Suction, protect the airway, control external bleeding, and transition immediately to the applicable trauma and airway protocols.

• EMT personnel may assist with equipment and continued external pressure but may not administer topical TXA under UP-09.

• Begin prompt transport for airway compromise, hypotension or shock, suspected posterior bleeding, major facial trauma, persistent bleeding, significant estimated blood loss, or uncontrolled bleeding in a patient taking an anticoagulant or antiplatelet medication.

AEMT: For persistent atraumatic anterior bleeding after 15 minutes of uninterrupted compression, apply topical TXA to gauze. Adult: TXA 1 g/10 mL total. Pediatric: TXA 15 mg/kg, maximum 1 g total. Saturate an appropriately sized gauze pledget and place it gently in the anterior affected nostril without deep or blind insertion.

• If bleeding is uncontrolled when TXA gauze is applied or the bleeding side cannot be identified, place appropriately sized TXA-saturated gauze gently in both anterior nostrils, dividing the approved total dose between the two sides. Leave gauze tails visible externally and maintain external pressure or the nasal clamp for 20 minutes.

• Do not atomize TXA. Do not perform deep, blind, balloon, commercial, or posterior nasal packing. Do not administer IV/IO TXA solely for isolated epistaxis.

• Establish IV or IO access only for hypotension, significant hemorrhage requiring resuscitation, airway management, or another specific clinical indication. Administer crystalloid only for hypotension according to UP-04: adult 500 mL and pediatric 10 mL/kg, followed by reassessment. Use smaller aliquots and frequent reassessment for CHF, renal failure, liver failure, or known volume overload.

• Bleeding control takes priority over routine cardiac-monitor placement. Once uninterrupted pressure or the clamp is secured, apply cardiac monitoring for hypotension, significant blood loss, suspected posterior bleeding, or clinical instability without interrupting hemorrhage control.

Paramedic: Perform the approved topical TXA treatment and hypotension-directed vascular access and fluid resuscitation described above.

• For persistent blood in the posterior pharynx, bleeding from both nares despite correct anterior compression, continued brisk hemorrhage, or inability to maintain airway protection, suspect posterior bleeding, maintain suction, prepare advanced airway management, and transport promptly with early receiving-facility notification.

• After hemorrhage-control measures are secured, interpret the cardiac rhythm and manage airway compromise, respiratory failure, shock, or dysrhythmia under the applicable protocol. Do not delay transport for repeated packing attempts.

Warnings

- Airway protection and suction take priority when the patient cannot manage blood or secretions. Do not place gauze or a clamp in a patient with an unprotected airway.
- Never tilt the head backward. Blood entering the posterior pharynx may cause aspiration, vomiting, or failure to recognize ongoing hemorrhage.
- A tongue-depressor clamp is applied only to the exterior soft nasal alae. Never insert a tongue depressor into a nostril or allow rigid pressure over the nasal bridge.
- Do not delay airway care, hemorrhage control, or transport to obtain cardiac monitoring. Apply monitoring after uninterrupted pressure is secured.

Clinical pearls

- Most epistaxis is anterior. Persistent blood in the posterior pharynx, continued bilateral bleeding, or brisk bleeding despite correct anterior compression should raise concern for a posterior source.
- Do not interrupt pressure repeatedly to inspect the nose. Use the scene timer to document the full 15-minute compression period and the 20-minute TXA-gauze period.
- Document anticoagulant and antiplatelet medications, bleeding duration, estimated blood loss, compression time, clamp method, TXA total dose and laterality, response, vital-sign trends, and destination notification.

Special populations

Pediatrics - younger than 16 years

- Maintain continuous airway observation. Do not place intranasal gauze or a clamp when the child cannot cooperate, cannot protect the airway, or cannot manage blood and secretions.

Anticoagulant or antiplatelet use

- Use the same first-line compression and topical treatment. Do not delay immediate bleeding control while seeking medication-reversal information.
- Persistent or recurrent bleeding, hemodynamic change, significant estimated blood loss, or suspected posterior bleeding requires prompt transport and early notification.

UP-10
Fever / Infection Control

Revised August 2026

Overview

Fever is a clinical sign, not a diagnosis. Identify the probable source, high-risk age or condition, evidence of organ dysfunction, environmental hyperthermia, and the need for transmission-based precautions.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Standard + Transmission-Based Precautions
EMT / AEMT / Paramedic
Select contact, droplet, and/or airborne precautions by suspected route • mask respiratory-symptom patient when tolerated

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. Assess + Measure Temperature
EMT / AEMT / Paramedic
Airway • breathing • perfusion • mental status • rash • source • temperature method • last antipyretic

↓

3. Infant ≤60 Days + Temperature ≥100.4°F / 38°C?
EMT / AEMT / Paramedic
Prompt transport and early notification • do not delay for antipyretics

↓

4. Environmental Hyperthermia / Heat Stroke?
EMT / AEMT / Paramedic
Transition to heat-emergency protocol • active cooling • antipyretics are not effective

↓

5. Uncomfortable Fever ≥100.4°F / 38°C?
EMT / AEMT / Paramedic
EMT/AEMT/Paramedic: one oral agent if airway protected and no contraindication

↓

6. Suspected Infection + Organ Dysfunction?
EMT / AEMT / Paramedic
Hypotension • altered mental status • respiratory distress/hypoxemia • poor perfusion → sepsis/shock pathway

↓

7. Reassess + Transport
EMT / AEMT / Paramedic
Repeat vital signs, temperature when useful, mental status, perfusion, respiratory status, and medication response

Indications

- Measured or reliably documented temperature of at least 100.4°F/38°C, suspected infection, recent fever reduced by an antipyretic, or illness requiring infection-control precautions.

Contraindications

- Do not give oral medication to a patient who cannot protect the airway, cannot swallow safely, is actively vomiting, or has another aspiration risk.

- Do not administer ibuprofen with known NSAID hypersensitivity, age 6 months or younger, renal disease or renal transplant, active gastrointestinal bleeding, significant dehydration, hypotension, clinically significant anticoagulant-associated bleeding risk, or known or suspected pregnancy.

Assessment

Focused fever and infection assessment

- Onset, duration, highest measured temperature and method, last antipyretic medication/dose/time, recent antibiotics, immune compromise, diabetes, cancer or chemotherapy, transplant, pregnancy, recent hospitalization or procedure, indwelling device, sick contacts, travel or environmental exposure, and suspected source.
- Assess for cough, dyspnea, chest pain, headache, neck stiffness, photophobia, sore throat, dysuria, flank pain, abdominal pain, vomiting, diarrhea, myalgias, rash, focal skin infection, altered mental status, rigors, seizure, poor intake, decreased urine output, and signs of dehydration.
- Identify organ dysfunction: hypotension, altered mental status, respiratory distress, hypoxemia, poor perfusion, mottling, delayed capillary refill, or rapidly worsening clinical appearance.

Treatment

1. Repeat complete vital signs and reassess mental status, airway, respiratory status, perfusion, temperature when useful, and medication response without delaying transport.

Medication orders

Acetaminophen - oral fever treatment

EMT/AEMT/Paramedic: adult 650-1,000 mg PO; pediatric older than 3 months 15 mg/kg PO, maximum 650 mg.

- Use for uncomfortable fever of at least 100.4°F/38°C when the patient protects the airway and can swallow safely.
- Do not administer with severe liver disease or liver failure, suspected acetaminophen overdose, known hypersensitivity, a recent maximum dose, or an uncertain previous dose/time.
- Do not delay transport of a high-risk infant, toxic-appearing patient, or unstable patient to administer or assess the effect of acetaminophen.

Ibuprofen - oral fever treatment

EMT/AEMT/Paramedic: adult 400 mg PO; pediatric older than 6 months 10 mg/kg PO, maximum 400 mg.

- Use for uncomfortable fever of at least 100.4°F/38°C when the patient protects the airway and can swallow safely.
- Do not administer with NSAID hypersensitivity, age 6 months or younger, renal disease/transplant, active GI bleeding, significant dehydration, hypotension, clinically significant anticoagulant-associated bleeding risk, pregnancy, or an uncertain previous dose/time.
- Do not administer for environmental hyperthermia or heat stroke.

Crystalloid - hypotension or shock only

AEMT/Paramedic: adult 500 mL; pediatric 10 mL/kg. Reassess after each bolus and use smaller aliquots for fluid-sensitive patients.

- Do not administer crystalloid for fever alone.
- Continue treatment according to UP-04 and the applicable sepsis or shock protocol.

Provider-level care

Provider-Level Actions Higher provider levels include the actions listed for the preceding levels.

EMT: Apply standard precautions and add contact, droplet, and/or airborne precautions according to the suspected route of transmission and agency infection-control policy. Place a mask on a patient with respiratory symptoms when tolerated and when it does not interfere with oxygen delivery or ventilation.

- Assess airway, respiratory effort, SpO_2, perfusion, mental status, skin, HEENT, neck, lungs, abdomen, back, neurologic status, hydration, rash, pain, and complete vital signs. Identify possible infectious source, environmental exposure, immune compromise, pregnancy, recent procedures, indwelling devices, sick contacts, and medications.
- Measure and document temperature and the measurement method. Do not perform rectal temperatures in the field. Accept a reliable documented temperature obtained at home or by another clinician even when antipyretics lowered the temperature before EMS arrival. Document the medication, dose, and administration time when known.
- Any infant 60 days old or younger with a documented temperature of at least 100.4°F/38°C requires prompt transport and early receiving-facility notification, even when well appearing. Do not delay transport to administer or assess the effect of an antipyretic.
- For an uncomfortable patient with temperature at least 100.4°F/38°C who is alert, protects the airway, can swallow safely, and has no medication

contraindication, administer one oral antipyretic: acetaminophen or ibuprofen. Do not routinely administer both agents.

• Acetaminophen: adult 650-1,000 mg PO; pediatric older than 3 months 15 mg/kg PO, maximum 650 mg. Ibuprofen: adult 400 mg PO; pediatric older than 6 months 10 mg/kg PO, maximum 400 mg. Do not repeat an agent when the previous dose or administration time is uncertain.

• Do not administer aspirin to a patient younger than 16 years. Do not administer an antipyretic for environmental hyperthermia or heat stroke; transition immediately to the applicable heat-emergency protocol and begin active cooling.

• For seizure activity, protect the airway and transition to the seizure protocol. Do not delay transport of a toxic-appearing, unstable, immunocompromised, or high-risk patient for temperature reduction.

AEMT: Do not establish IV/IO access or administer crystalloid for fever alone. Establish vascular access for hypotension, shock, critical illness, required parenteral treatment, or another specific clinical indication.

• For hypotension or shock, administer crystalloid according to UP-04: adult 500 mL and pediatric 10 mL/kg, followed by reassessment. Use smaller aliquots and frequent reassessment for CHF, renal failure, liver failure, or known volume overload.

• Obtain blood glucose for altered mental status, seizure, a critically ill infant or child, suspected sepsis with organ dysfunction, diabetes-related concern, or another specific indication. Routine glucose measurement is not required for uncomplicated fever.

• Administer the same oral acetaminophen or ibuprofen doses authorized above when indicated. UP-10 does not authorize IV acetaminophen, ketorolac, or antibiotic administration solely for fever.

Paramedic: Fever alone is not sepsis. When suspected infection is accompanied by hypotension, altered mental status, respiratory distress or hypoxemia, poor skin perfusion, or another sign of organ dysfunction, transition immediately to the applicable sepsis and shock protocols.

• Cardiac monitoring is not required for uncomplicated fever. Apply and interpret cardiac monitoring for suspected sepsis with organ dysfunction, hypotension, significant respiratory distress, dysrhythmia symptoms, or other clinical instability.

• Manage airway or ventilatory failure, shock, seizure, dysrhythmia, or another identified complication under the applicable protocol. Notify the receiving facility early for sepsis, a high-risk infant, immunocompromise, meningismus, petechial or purpuric rash, or other time-sensitive findings.

Warnings

• Infants 60 days old or younger with a documented temperature of at least 100.4°F/38°C require prompt evaluation even when currently afebrile or well appearing.

• Petechial or purpuric rash, meningismus, altered mental status, hypotension, hypoxemia, poor perfusion, toxic appearance, or immune compromise requires prompt transport and early notification.

• Antipyretics do not treat heat stroke and must never delay active cooling. Do not give NSAIDs in environmental heat emergencies.

• Do not delay transport of an unstable or high-risk patient to obtain a temperature, vascular access, cardiac monitoring, or antipyretic response.

Clinical pearls

• Treat the patient rather than the temperature. The purpose of an antipyretic is relief of discomfort; normalization of temperature is not required before transport.

• A normal temperature after acetaminophen or ibuprofen does not exclude serious infection. Document the highest reliable temperature and the medication, dose, and time given before EMS arrival.

• Antipyretics do not reliably prevent recurrence of febrile seizures. Treat active seizure activity under the seizure protocol.

• The absence of fever does not exclude infection or sepsis, especially in older, immunocompromised, or critically ill patients.

• Document suspected source, immune status, PPE used, temperature and method, medication contraindication screening, administered dose, response, vital-sign trends, and receiving-facility notification.

Special populations

Infants 60 days old or younger

• A reliably documented temperature of at least 100.4°F/38°C within the current illness is high risk even when the infant appears well or is afebrile after medication.

Immunocompromised patients

- Maintain a low threshold for prompt transport and early notification in patients with chemotherapy, cancer, transplant, HIV with immune suppression, chronic immunosuppressive medication, or another significant immune deficit.
- Serious infection may be present without marked fever or classic inflammatory signs.

Pediatrics - younger than 16 years

- Use actual or length-based weight for medication dosing. Acetaminophen requires age older than 3 months; ibuprofen requires age older than 6 months.
- Do not administer aspirin. Assess hydration, urine output, interaction, work of breathing, capillary refill, rash, and caregiver-reported behavior.

UP-11
Pain Control - Adult

Revised October 31, 2026

Overview

Assess and treat acute pain early while preserving airway protection, ventilation, perfusion, and neurologic assessment. Use the least invasive effective option and tailor treatment to pain severity, clinical condition, contraindications, and patient preference.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Assess Pain + Cause
EMT / AEMT / Paramedic
Pain score • location • quality • onset • cause • allergies • airway • respirations • perfusion • mental status

↓

2. Immediate Cause-Specific Care
EMT / AEMT / Paramedic
Airway care • hemorrhage control • splinting • positioning • ice when appropriate • calm reassurance

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. Mild Pain / Oral Route Appropriate?
EMT / AEMT / Paramedic
EMT/AEMT/Paramedic: acetaminophen 650-1,000 mg PO OR ibuprofen 400 mg PO

↓

4. Parenteral Nonopioid Appropriate?
Paramedic
Paramedic: acetaminophen ≥50 kg: 1,000 mg; <50 kg: 15 mg/kg, max 750 mg, over 15 minutes IV (IO only if separately authorized) OR ketorolac 15 mg IV once or 30 mg IM once (maximum). Expanded AEMT parenteral analgesia is not enabled while authorization is unverified.

↓

5. AEMT Authorization Check
AEMT
Expanded parenteral analgesia is disabled by default while individual authorization is unverified. Exact Tennessee drug/route authority, documented approved training, agency credentialing/competency, and medical-director standing orders are all required. Request Paramedic care when needed.

↓

6. Moderate-Severe Pain?
Paramedic
Paramedic: select one opioid OR analgesic-dose ketamine • titrate to meaningful improvement

↓

7. Monitor + Reassess
EMT / AEMT / Paramedic
Cardiac monitor for systemic nonoral analgesia • continuous SpO_2/$EtCO_2$ for opioid or ketamine • reassess every 5-10 minutes

↓

8. Transport + Handoff
EMT / AEMT / Paramedic
Do not delay airway care, hemorrhage control, splinting, or transport • document medication response and adverse effects

Indications

- Acute pain in a patient 16 years of age or older when treatment can be provided without delaying management of immediate threats or time-sensitive transport.
- Painful procedures or movement when analgesia is appropriate and the patient can be safely monitored.

Contraindications

- Do not give oral medication when the patient cannot protect the airway, cannot swallow safely, is actively vomiting, or has another aspiration risk.
- Do not administer acetaminophen with known hypersensitivity, severe liver disease or liver failure, suspected acetaminophen overdose, a recent maximum dose, or an uncertain prior acetaminophen dose or administration time.
- Do not administer ibuprofen or ketorolac with NSAID hypersensitivity, renal disease or renal transplant, significant dehydration, hypotension, active gastrointestinal bleeding, clinically significant anticoagulant-associated bleeding risk, known or suspected pregnancy, or recent NSAID administration.
- Do not administer an opioid with respiratory depression, inability to protect the airway, severe hypotension, or significant uncorrected hypoxemia.
- Do not administer analgesic-dose ketamine with known hypersensitivity, uncontrolled hypertension, active severe cardiac ischemia, or active psychosis.

Assessment

Pain and medication-safety assessment

- Use patient self-report whenever possible. Record pain severity before medication, after each intervention, during transport, and at handoff. Pain score alone does not determine medication selection.
- Assess airway protection, respiratory rate and effort, SpO_2, perfusion, BP, heart rate and rhythm, mental status, sedation level, injury or illness severity, and whether pain may represent a time-sensitive cardiac, vascular, neurologic, surgical, obstetric, or traumatic emergency.
- Review allergies, analgesics already taken or administered, alcohol or sedative use, opioid tolerance, obstructive sleep apnea, COPD, renal or hepatic impairment, anticoagulant use, GI bleeding, pregnancy, and prior adverse responses to opioids, NSAIDs, acetaminophen, or ketamine.

Treatment

1. Reassess pain, sedation, airway, respiratory rate and effort, SpO_2, $EtCO_2$ when used, BP, cardiac rhythm, and adverse effects at least every 5-10 minutes after parenteral medication and at handoff.
2. Do not delay airway treatment, hemorrhage control, splinting, or transport to complete pain treatment or reach a specific pain score.

Medication orders

Acetaminophen - oral

EMT/AEMT/Paramedic: 650-1,000 mg PO once.

- Use when the patient protects the airway and can swallow safely.
- Do not administer with severe liver disease or liver failure, suspected acetaminophen overdose, a recent maximum dose, or an uncertain prior dose/time.

Ibuprofen - oral

EMT/AEMT/Paramedic: 400 mg PO once.

- Do not administer with NSAID hypersensitivity, renal disease/transplant, significant dehydration, hypotension, active GI bleeding, clinically significant anticoagulant-associated bleeding risk, pregnancy, or recent NSAID administration.
- Do not combine with ketorolac.

Acetaminophen - IV/IO

Paramedic, age 16 or older: acetaminophen 1,000 mg at weight ≥50 kg or 15 mg/kg (maximum 750 mg) below 50 kg, infused over 15 minutes once; never rapid push. IV route; IO requires separate verified authorization and must never be established solely for pain. Expanded AEMT use remains disabled pending verification.

- Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing and competency verification, and medical-director standing orders; training alone does not grant permission.
- Verify all prior acetaminophen sources and dose/time. At least 6 hours must have elapsed since the previous dose; no EMS repeat. All-route 24-hour maximum: ≥50 kg, 4,000 mg; <50 kg, 75 mg/kg, not exceeding 3,750 mg.
- Do not administer an additional oral acetaminophen dose routinely.

Ketorolac

Paramedic: 15 mg IV once or 30 mg IM once. Expanded AEMT use is not enabled by default while individual authorization is unverified.

- Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing and competency verification, and medical-director standing orders; training alone does not grant permission.
- Do not administer with the NSAID contraindications listed above.
- Do not combine with ibuprofen or another NSAID.

Fentanyl

Paramedic: 50 mcg slow IV/IO; repeat 25-50 mcg every 5 minutes as needed; maximum total 100 mcg.

- Use continuous SpO_2 and waveform $EtCO_2$ and monitor airway, ventilation, sedation, and hemodynamics.
- Do not combine with morphine or hydromorphone.

Morphine

Paramedic: 5 mg slow IV/IO; repeat 5 mg after 10 minutes as needed; maximum total 10 mg. If no vascular access, 5-10 mg IM once.

- Use continuous SpO_2 and waveform $EtCO_2$ and monitor airway, ventilation, sedation, and hemodynamics.
- Do not combine with fentanyl or hydromorphone.

Hydromorphone

Paramedic: 0.5 mg slow IV/IO; repeat 0.5 mg after 10 minutes as needed; maximum total 1 mg. If no vascular access, 0.5-1 mg IM once.

- Use continuous SpO_2 and waveform $EtCO_2$ and monitor airway, ventilation, sedation, and hemodynamics.
- Do not combine with fentanyl or morphine.

Ketamine - analgesic dose

Paramedic: 0.3 mg/kg IV/IO slowly over 10 minutes, maximum 30 mg per dose; repeat every 20 minutes as needed, maximum three total doses. If no vascular access, 0.5-1 mg/kg IN once, maximum 100 mg.

- Do not administer by rapid IV push.
- For suspected penetrating or open-globe eye trauma, ketamine requires express Medical Control authorization; follow TB-11.
- Do not routinely initiate ketamine with an opioid. Fully reassess before changing agents.
- Use continuous SpO_2 and waveform $EtCO_2$ and monitor airway, ventilation, sedation, cardiac rhythm, and hemodynamics.

Ondansetron - analgesia-associated nausea

AEMT/Paramedic: 4 mg IV/IO, IM, or PO.

- Use when clinically indicated for nausea or vomiting associated with pain or analgesic administration.
- Continue cardiac and respiratory monitoring required by the analgesic medication.

Naloxone - opioid-induced hypoventilation

Paramedic: after airway support and ventilation, administer 0.4-2 mg IV/IO/IM and titrate to adequate ventilation. If vascular access is unavailable, administer 2 mg IN. Repeat every 2-3 minutes as needed; maximum cumulative dose 8 mg.

- Ventilate first and titrate to adequate spontaneous ventilation rather than complete reversal of analgesia when clinically safe.
- Transition to the naloxone/overdose protocol if overdose rather than an analgesic adverse effect is suspected.

Provider-level care

Provider-Level Actions Higher provider levels include the actions listed for the preceding levels.

EMT: Identify and treat immediate threats before analgesia. Provide airway support, hemorrhage control, wound coverage, splinting, positioning, ice when appropriate, and reassurance without delaying transport.

- Assess onset, location, quality, radiation, severity, aggravating or relieving factors, cause, associated symptoms, allergies, current medications, previous analgesics, substance or opioid tolerance, and relevant renal, hepatic, respiratory, cardiovascular, bleeding, and pregnancy history.
- Record a numeric 0-10 pain score when the patient can self-report. Use a descriptive severity assessment when reliable numeric reporting is not possible. Obtain airway, respiratory, perfusion, mental-status, and complete vital-sign assessments before medication.
- For mild pain when the patient is alert, protects the airway, can swallow safely, and has no contraindication, administer one oral medication: acetaminophen 650-1,000 mg PO once or ibuprofen 400 mg PO once. Do not routinely administer both.
- Do not administer aspirin for pain under UP-11. Aspirin remains limited to its approved cardiac indication. Nitrous oxide is not authorized because it is not on the Claiborne formulary.

• Reassess pain, airway, respiratory status, perfusion, mental status, and vital signs after treatment and at handoff. Request the higher provider level when pain remains moderate to severe or oral medication is inappropriate or ineffective.

AEMT: Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing and competency verification, and medical-director standing orders; training alone does not grant permission. Do not administer IV/IO acetaminophen or IV/IM ketorolac from this unverified AEMT view; request Paramedic care or the separately verified medication-specific standing order. Do not routinely combine parenteral acetaminophen with an additional acetaminophen dose.

• Do not administer ketorolac when ibuprofen or another NSAID has already been given during the current treatment interval. Screen for NSAID hypersensitivity, renal disease, renal transplant, significant dehydration, hypotension, active gastrointestinal bleeding, clinically significant anticoagulant-associated bleeding risk, and known or suspected pregnancy.

• Establish IV access when needed for parenteral treatment or another clinical indication. Do not establish IO access solely to treat pain in an otherwise stable patient; use an approved oral or nonvascular option or request Paramedic treatment when appropriate.

• Apply cardiac monitoring before IV/IO/IM analgesia and continue through reassessment and transport. Reassess pain, BP, heart rate, respiratory rate, SpO_2, mental status, and adverse effects after medication.

Paramedic: For parenteral nonopioid treatment, a Paramedic may administer weight-based acetaminophen infused over 15 minutes once (≥50 kg: 1,000 mg; <50 kg: 15 mg/kg, maximum 750 mg; IV, or IO only with separately verified authorization) or ketorolac 15 mg IV once or 30 mg IM once after the medication-specific contraindication screen. Do not routinely combine IV acetaminophen with an additional acetaminophen dose or ketorolac with another NSAID.

• For moderate to severe pain, select one opioid or analgesic-dose ketamine based on the clinical situation, contraindications, prior response, and patient preference. Titrate to meaningful improvement rather than complete elimination of pain.

• Fentanyl: 50 mcg slow IV/IO; repeat 25-50 mcg every 5 minutes as needed; maximum total 100 mcg.

• Morphine: 5 mg slow IV/IO; repeat 5 mg after 10 minutes as needed; maximum total 10 mg. If vascular access is unavailable, administer 5-10 mg IM once.

• Hydromorphone: 0.5 mg slow IV/IO; repeat 0.5 mg after 10 minutes as needed; maximum total 1 mg. If vascular access is unavailable, administer 0.5-1 mg IM once.

• Select one opioid. Do not combine fentanyl, morphine, and hydromorphone. Additional opioid dosing beyond the approved total limits requires medical-control authorization.

• Ketamine: 0.3 mg/kg IV/IO slowly over 10 minutes, maximum 30 mg per dose. May repeat every 20 minutes as needed for a maximum of three total doses. Do not administer by rapid IV push. For suspected penetrating or open-globe eye trauma, do not use ketamine for analgesia unless Medical Control expressly authorizes it; follow TB-11.

• If vascular access is unavailable, administer ketamine 0.5-1 mg/kg IN once, maximum 100 mg. Intranasal ketamine is a single-dose option under UP-11.

• Do not routinely initiate weight-based ketamine and an opioid together. If the initial medication is inadequate, fully reassess airway, ventilation, hemodynamics, sedation, pain, and adverse effects before changing agents.

• Apply cardiac monitoring before systemic nonoral analgesia. Maintain continuous SpO_2 and waveform $EtCO_2$ with any opioid or ketamine, keep suction and ventilation equipment immediately available, and reassess at least every 5-10 minutes.

Warnings

- Analgesia must not delay airway management, hemorrhage control, shock treatment, splinting, or time-sensitive transport.
- A fall in respiratory rate, increasing sedation, abnormal $EtCO_2$, hypoxemia, or loss of airway protection after an opioid requires immediate airway support and assisted ventilation; administer naloxone when indicated.
- Use the lower end of the approved opioid dose range and careful titration in older or frail patients, obstructive sleep apnea, severe COPD, renal or hepatic impairment, or concurrent alcohol, opioid, benzodiazepine, or other sedative exposure.
- Do not routinely combine ketamine with a benzodiazepine. Treat a clinically significant emergence reaction or other adverse effect

according to the applicable protocol and medical-control direction.
- Analgesic response does not exclude a serious underlying cause. Continue complaint-specific evaluation and transport.
- The ketorolac maximum is 15 mg IV once or 30 mg IM once for all adults, including patients age 65 years or older or weighing less than 50 kg. Do not administer 60 mg IM; expanded AEMT authority is not established by this dose limit.

Clinical pearls

- Pain is subjective. Patient self-report should guide assessment when reliable; appearance, diagnosis, or concern for drug-seeking behavior must not be used as the sole reason to withhold appropriate analgesia.
- Meaningful improvement in pain and function is the goal; complete elimination of pain is not required.
- Do not withhold indicated analgesia solely because abdominal pain is undifferentiated, the patient has opioid tolerance, the injury is a burn, or the patient has sickle-cell disease.
- Nonopioid options may provide analgesia comparable to opioids for selected acute pain while avoiding opioid-associated respiratory depression. Medication selection should match the patient and clinical circumstance.
- Document the pain score or descriptive assessment, medication choice, dose, route, time, monitoring, reassessment findings, adverse effects, and response.

Special populations

Older, frail, respiratory-risk, or organ-impaired patients

- Use the lower end of the approved dose range, allow adequate time for effect, and reassess before repeating medication.

Pregnancy

- Avoid ibuprofen and ketorolac. Use acetaminophen or carefully titrated Paramedic analgesia when indicated and transition to the applicable obstetric or complaint-specific protocol.

Pediatrics - younger than 16 years

- Do not use UP-11 for patients younger than 16 years. Open PM-04 Pediatric Pain Control and use actual or length-based weight.

Related pathways

- TB-11 Eye Trauma (p. 198): Protect suspected open globe; ketamine for analgesia requires express Medical Control authorization.
- PM-04 Pediatric Pain Control (p. 181): Open for patients age 2 through younger than 16 years; use actual or length-based weight.

UP-12
Police Custody

Revised October 31, 2026

Overview

Persons in law-enforcement custody receive the same clinical assessment, treatment, dignity, privacy, and access to emergency care as every other patient.

Clinical needs determine EMS assessment, treatment, transport recommendation, and destination. Law enforcement retains responsibility for custody and scene security.

This protocol applies to custody encounters, law-enforcement restraints, chemical-irritant exposure, and conducted-energy weapon exposure. Use the appropriate complaint, trauma, airway, behavioral, or refusal protocol concurrently.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Scene Safe + Medical Assessment
EMT / AEMT / Paramedic
Obtain law-enforcement report • address life threats • determine illness, injury, exposure, and use of force

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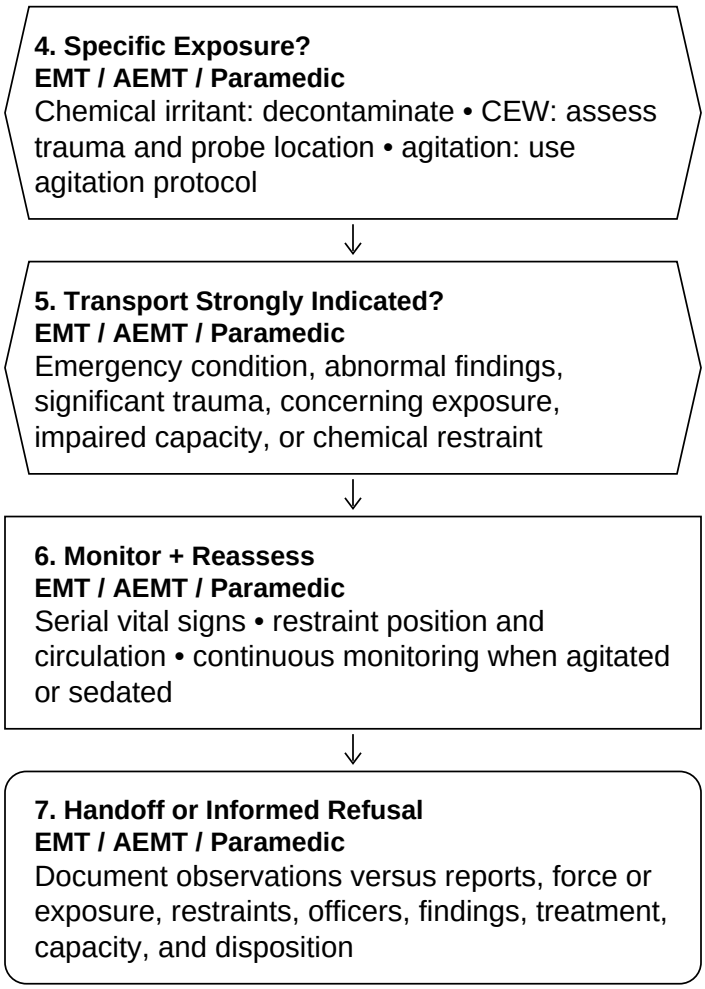
2. Capacity + Consent
EMT / AEMT / Paramedic
Custody alone does not remove capacity • capable adults retain the right to accept or refuse care

↓

3. Restraint Safety
EMT / AEMT / Paramedic
Least restrictive • never prone or hog-tied • protect airway, breathing, circulation, and access to treatment

Continue at step 4 in the next column/page

Quick flow continued at step 4



Indications

- Any patient detained, arrested, incarcerated, or physically restrained by law enforcement.
- Any patient evaluated after chemical-irritant spray, conducted-energy weapon exposure, or force used during apprehension or custody.
- Any request for EMS assessment before admission to a jail or detention facility.

Assessment

Medical assessment

- Obtain the patient's complaint directly whenever possible and obtain a separate law-enforcement report describing the event, use of force, chemical exposure, CEW exposure duration or number of cycles, fall, struggle, and observed behavior.
- Perform a primary assessment, focused medical and trauma examination, and at least two complete vital-sign sets when feasible.
- Evaluate for hypoxia, hypoglycemia, hyperthermia, head injury, neck compression, intoxication or withdrawal, overdose, psychiatric emergency, severe agitation, respiratory disease, and cardiac symptoms.
- Do not attribute abnormal behavior solely to intoxication, psychiatric illness, resistance, or custody status.

Restraint assessment

- Ensure restraints do not compromise airway access, ventilation, circulation, neurologic assessment, patient positioning, or treatment.
- EMS personnel will not apply handcuffs or attach locking law-enforcement restraints to the stretcher or side rails.
- When locking law-enforcement restraints remain during transport, a law-enforcement officer capable of immediately releasing them and possessing the key must accompany the patient in the ambulance.
- Request immediate adjustment or removal of any restraint that interferes with necessary medical assessment or care.

Transport indicators

- Transport for an emergency condition, abnormal vital signs, altered mental status or impaired capacity, significant force or trauma, head injury, neck compression, hyperthermia, suspected ingestion, respiratory or cardiac symptoms, persistent chemical-irritant symptoms, high-risk CEW findings, severe agitation, or chemical restraint.
- Persistent eye pain, abnormal vision, corneal concern, wheezing, dyspnea, chest pain, syncope, focal neurologic findings, or worsening symptoms require transport.
- Do not delay care or transport to complete law-enforcement questioning, evidence collection, charging, or confinement paperwork.

Required custody documentation

- Document custody status; chief complaint; patient statements; capacity assessment; vital signs; examination; treatment; response; and disposition.
- Document the reported mechanism, force, restraint, chemical agent, CEW mode, number or duration of cycles when known, falls, and probe locations.
- Record restraint type, position, application time when known, serial airway and circulation checks, adjustments, and the officer accompanying the patient with immediate access to the key.
- Record names or identifying numbers of involved officers when available, receiving personnel, refusal discussion, signatures, Medical Control contact, and any assessment or treatment the patient declined.

• Clearly identify which information EMS directly observed and which information was reported by the patient, law enforcement, or witnesses.

Treatment

1. Treat immediate life threats and follow the applicable complaint-specific, trauma, airway, cardiac, toxicology, behavioral, pediatric, or obstetric protocol.
2. Do not use this protocol as a fit-for-confinement examination. When a potentially serious injury or emergency condition exists, transport for appropriate medical evaluation.
3. A capable adult who declines care may remain in law-enforcement custody only after the standard informed-refusal process is completed.
4. A patient restrained because of agitation requires continuous direct observation during transport and reassessment after every intervention or change in condition.
5. Release after chemical-irritant or brief CEW exposure is appropriate only when the patient is alert, has decision-making capacity, has normal vital signs and examination, has no transport indicator, and completes the applicable refusal or non-transport documentation.
6. Provide an early receiving-facility report describing custody status, restraints, significant force, exposure, findings, treatment, and safety considerations.

Medication orders

Midazolam

Per Agitation protocol

• Paramedic only for agitation when indicated.

• Continuous airway and cardiac monitoring required.

Ketamine

Per Agitation protocol

• Paramedic only for agitation when indicated.

• Continuous airway and cardiac monitoring required.

Provider-level care

Provider Actions

EMT: Perform complete BLS assessment and immediate lifesaving care. Treat illness or injury under the applicable protocol.

• Provide chemical-irritant decontamination and remove uncomplicated superficial CEW probes using the approved procedure.

• Assess restraint position, airway, breathing, circulation, distal neurovascular status, and decision-making capacity.

AEMT: Perform all EMT care. Establish IV/IO access and administer fluids or medications only when indicated by the patient's clinical condition and the applicable protocol.

• Assist with continuous monitoring, serial reassessment, and treatment of respiratory or hemodynamic complications.

Paramedic: Perform all prior care. Lead evaluation of significant agitation, abnormal vital signs, altered mental status, respiratory compromise, cardiac symptoms, severe injury, or high-risk CEW exposure.

• Provide advanced airway, cardiac, and pharmacologic treatment under the applicable complaint or agitation protocol.

Consent, Capacity & Refusal

EMT: Determine whether the adult patient can understand the condition, proposed care, material risks, alternatives, and consequences of refusal.

• A capable adult may refuse assessment, treatment, or transport under the Refusal / Non-Transport protocol. Obtain two vital-sign sets when permitted and document any assessment the patient declines.

AEMT: Reassess for intoxication, hypoxia, hypoglycemia, head injury, medication effects, psychiatric emergency, or another condition that may impair capacity.

Paramedic: Evaluate questionable or high-risk capacity and contact Medical Control when uncertainty, disagreement, or a serious suspected condition remains.

• When capacity is absent and an emergency condition exists, provide necessary care and transport under implied consent with law-enforcement assistance when required for safety.

Restraint Safety & Monitoring Use restraint only for immediate safety or medically necessary care, never for punishment or convenience.

EMT: Use the least restrictive safe method. Never transport a patient prone, hog-tied, with wrists connected directly or indirectly to ankles, or with pressure on the neck, chest, back, or abdomen.

• Position supine with the head elevated or lateral when clinically appropriate. Do not use a long backboard solely as a restraint device.

• Reassess airway, breathing, circulation, restraint position, skin, and distal neurovascular status at least every 5 minutes and document findings.

AEMT: Add continuous SpO_2 and serial vital signs for patients restrained because of agitation. Obtain glucose and temperature when clinically indicated.

Paramedic: Use continuous cardiac monitoring and SpO_2 for significant agitation or chemical restraint.

Add waveform $EtCO_2$ after chemical restraint or when significant agitation or respiratory risk is present.

• Midazolam or ketamine may be used per agitation protocol. Continuous airway and cardiac monitoring required.

Chemical-Irritant Exposure Prevent cross-contamination and provide prompt decontamination.

EMT: Use appropriate PPE, move the patient to fresh air, remove contaminated clothing, and contain contaminated items when feasible.

• Remove contact lenses when easily accomplished. Irrigate burning or visually impaired eyes with water or normal saline for 10-15 minutes. Wash exposed skin with soap and water without scrubbing.

• Do not apply chemical neutralizers, creams, oils, or eyedrops.

AEMT: Treat wheezing, bronchospasm, or respiratory distress under the applicable respiratory protocol and obtain vascular access when clinically indicated.

Paramedic: Manage persistent respiratory, cardiac, ophthalmic, or neurologic complications and determine need for specialty consultation or destination.

Conducted-Energy Weapon Exposure

EMT: Confirm the device has been rendered safe by law enforcement. Assess for fall, head, neck, penetrating, and other traumatic injury before probe removal.

• Remove only an uncomplicated superficial probe in soft tissue using the approved procedure. Clean and dress the wound and dispose of the probe as a contaminated sharp.

• Leave probes involving the eye or face, neck, genitalia, breast, joint, bone, or suspected major vessel in place; stabilize and transport.

AEMT: Perform all EMT care and treat associated injury, dehydration, or other clinical findings under the applicable protocol.

Paramedic: Obtain ECG and provide complaint-specific evaluation when the patient has chest pain, dyspnea, abnormal vital signs, altered mental status, significant intoxication or agitation, prolonged or repeated exposure, pregnancy, significant trauma, or another concerning finding.

• An awake, alert, asymptomatic patient with normal findings after a brief exposure does not require routine ECG, laboratory testing, prolonged observation, or automatic transport solely because of the electrical exposure.

Warnings

• Transport is strongly indicated for the listed high-risk features; those criteria alone do not establish incapacity or authority for involuntary transport. If a legally authorized decision-maker refuses, assess and document decision-specific capacity, explain material risks and alternatives, contact Medical Control for high-risk refusal, and follow the approved refusal/safeguarding procedure. Use an authorized emergency or protective-custody pathway when its legal conditions apply. A patient aged 16 or 17 remains a minor for consent purposes unless a legally recognized exception applies; the clinical age-16 dosing boundary does not establish legal adulthood.

• Never use EMS restraints, equipment, medications, or procedures solely for punishment, law-enforcement convenience, interrogation, or evidence collection.

• Never transport a patient prone, hog-tied, sandwiched between backboards or mattresses, or positioned in a manner that compromises airway, breathing, circulation, or access to care.

• Do not place a surgical mask, oxygen mask, or spit-protection device in a manner that obstructs the airway, impairs ventilation, or prevents continuous observation.

• A badge, handcuffs, intoxication, or apparent behavioral disturbance does not exclude serious illness or injury.

Clinical pearls

• Use calm communication, explain each medical action, preserve dignity, and minimize the number of personnel speaking to an agitated patient.

• Simple brief CEW exposure in an otherwise normal patient rarely causes a clinically important electrical injury; associated trauma, struggle, intoxication, hyperthermia, and agitation often determine risk.

• Chemical-irritant symptoms usually improve after removal from exposure and decontamination, but persistent ocular or respiratory symptoms require further evaluation.

Special populations

Pediatrics

• Use pediatric protocols for patients younger than 16 years. A parent or legal guardian ordinarily provides consent or refusal unless another legal exception applies; custody status does not change pediatric clinical care.

Pregnancy

- For a known pregnant inmate in correctional custody, use the least restrictive restraint and request removal whenever the restraint interferes with medical assessment, treatment, positioning, or transport.
- Do not restrain the patient's hands behind her back. During labor or delivery, do not use restraints around the ankles, legs, or waist.
- Under Tennessee Code § 41-51-202, a healthcare professional responsible for the inmate's health and safety may request that correctional restraints not be used or be removed.

UP-13
Seizure

Revised October 31, 2026

Overview

Prioritize patient protection, airway and ventilation support, rapid recognition of convulsive status epilepticus, timely anticonvulsant administration, and identification of reversible causes.

Treat the seizure while evaluating for hypoxia, hypoglycemia, cardiac arrest, trauma, stroke, pregnancy or postpartum eclampsia, infection, toxins, withdrawal, and medication nonadherence.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Protect + Assess
EMT / AEMT / Paramedic
Prevent injury • check pulse • airway position • suction • oxygen or BVM when indicated

↓

2. Status Epilepticus?
EMT / AEMT / Paramedic
Generalized seizure ≥5 minutes • unknown prolonged duration • recurrent without recovery

↓

3. Glucose + Reversible Causes
EMT / AEMT / Paramedic
Check glucose without delaying anticonvulsant • treat hypoglycemia under UP-4

Continue at step 4 in the next column/page

Quick flow continued at step 4

4. Midazolam
Paramedic
Adult: 10 mg IM/IN or 5 mg IV/IO • Pediatric: weight-based • maximum total 10 mg

↓

5. Still Seizing?
Paramedic / Medical Control
Adult: ketamine 1 mg/kg IV/IO/IM, max 100 mg • Pediatric: contact Medical Control

↓

6. Continuous Monitoring
EMT / AEMT / Paramedic
Cardiac monitor • SpO_2 • waveform $EtCO_2$ after medication • airway and BVM readiness

↓

7. Reassess + Transport
EMT / AEMT / Paramedic
Neurologic recovery • trauma • pregnancy/postpartum • fever • toxin • stroke • document times and doses

Indications

- Active generalized or focal seizure.
- Convulsive seizure lasting 5 minutes or longer, unknown prolonged duration, or recurrent seizures without return to baseline.
- Postictal altered mental status requiring evaluation for injury, reversible cause, or incomplete neurologic recovery.

Assessment

History

- Document exact seizure onset time, duration, type, number of seizures, recovery between events, and baseline neurologic status.
- Ask about previous seizures, prescribed antiseizure medications, adherence, recent dose changes, prescribed rescue medication, and all medications administered before EMS arrival.
- Assess for pregnancy and postpartum status, trauma, diabetes, fever or infection, alcohol or sedative withdrawal, medication or toxin exposure, overdose, headache, stroke symptoms, and recent illness.

Examination

- Assess airway, ventilation, oxygenation, circulation, pulse, mental status, pupils, gaze or eye deviation, focal movement, symmetry, and postictal neurologic recovery.
- Examine for head or neck injury, tongue or oral trauma, active bleeding, incontinence, hyperthermia, meningismus, rash, and evidence of ingestion or injection.
- Obtain glucose in every patient when feasible. Obtain temperature when fever, infection, hyperthermia, or pediatric febrile seizure is possible.
- Persistent unilateral weakness, aphasia, gaze deviation, or another focal deficit beyond the expected postictal period requires stroke and intracranial-emergency evaluation.

Monitoring

- Use continuous cardiac monitoring and SpO_2 for status epilepticus, recurrent seizure, abnormal vital signs, or medication administration.
- Apply waveform $EtCO_2$ after midazolam or ketamine and whenever ventilation is impaired or assisted.
- Obtain a 12-lead ECG when a cardiac cause, syncope, toxin, overdose, electrolyte disturbance, or abnormal rhythm is suspected; it is not required for every uncomplicated seizure.
- Repeat complete vital signs, respiratory assessment, mental status, and neurologic examination after seizure cessation and every medication dose.

Treatment

1. Protect the patient from injury without forcibly restraining convulsions. Never place an object, medication, finger, or bite block in the mouth during active seizure.
2. Administer the age-appropriate midazolam regimen for seizure lasting 5 minutes or longer, unknown prolonged duration, recurrent seizure without recovery, or earlier when unstable airway, breathing, or circulation makes continued seizure immediately dangerous.
3. Treat pregnancy or postpartum seizure under the obstetric/eclampsia protocol, trauma under the appropriate trauma protocol, and suspected toxin or withdrawal under the applicable toxicology protocol.
4. Transport promptly with continuous reassessment and early receiving-facility notification for ongoing or recurrent seizure, incomplete recovery, or any high-risk feature.

Medication orders

Midazolam - Adult

10 mg IM preferred without IV/IO; 10 mg IN alternative; or 5 mg IV/IO when access is established

- For IV/IO administration, repeat 5 mg once after 5 minutes if seizure continues.
- Maximum total EMS midazolam dose: 10 mg.
- Paramedic only.

Midazolam - Pediatric

0.2 mg/kg IM/IN, max 10 mg; or 0.1 mg/kg IV/IO, max 5 mg

- May repeat once after 5 minutes if seizure continues.
- Maximum total: 0.4 mg/kg, not to exceed 10 mg.
- Count prescribed rescue benzodiazepine given before EMS arrival toward the two-dose maximum.
- Paramedic only.

Ketamine - Adult Refractory Seizure

1 mg/kg IV/IO or IM once; maximum 100 mg

- Use for seizure persisting after the adequate midazolam regimen.
- Continuous cardiac monitoring, SpO_2, waveform $EtCO_2$, and airway and ventilation readiness are required.
- Paramedic only.

Ketamine - Pediatric

Contact Medical Control for authorization and dosing

- Use only for seizure persisting after the adequate benzodiazepine regimen.
- Continuous cardiac monitoring, SpO_2, waveform $EtCO_2$, and airway and ventilation readiness are required.

Provider-level care

Provider Actions Treat status epilepticus promptly while supporting oxygenation and ventilation.

EMT: Protect the patient from injury, loosen restrictive clothing, position laterally when feasible, suction as needed, and provide oxygen or BVM ventilation when clinically indicated.

- Check pulse and rhythm promptly because brief seizure-like activity may occur at the onset of cardiac arrest.
- Obtain glucose without delaying indicated anticonvulsant administration; treat hypoglycemia under UP-4.
- Determine seizure onset, duration, number of events, recovery between events, and medications given before EMS arrival.

AEMT: Perform all EMT care. Establish IV/IO access when clinically useful, but do not delay IM or IN midazolam while attempting vascular access.

• Treat hypoglycemia under UP-4, assist ventilation, trend vital signs, and support continuous monitoring and rapid transport.

Paramedic: Perform all prior care. Administer midazolam for convulsive status epilepticus or recurrent seizures without recovery.

• For persistent adult seizure after the adequate midazolam regimen, administer ketamine 1 mg/kg IV/IO or IM, maximum 100 mg.

• For pediatric ketamine, contact Medical Control for authorization and dosing.

• Prepare for respiratory depression, assisted ventilation, and advanced airway management after anticonvulsant medication.

Adult Anticonvulsant Pathway Use an adequate early dose and do not delay treatment for vascular access.

EMT: Continue patient protection, airway positioning, suction, oxygenation, ventilation, glucose assessment, and preparation for medication administration.

AEMT: Obtain IV/IO access when it can be accomplished without delaying anticonvulsant treatment.

Paramedic: Without established IV/IO access: midazolam 10 mg IM is preferred. Midazolam 10 mg IN, divided between nostrils, is an alternative when IM administration is impractical.

• With established IV/IO access: midazolam 5 mg IV/IO; repeat 5 mg once after 5 minutes if seizure continues. Maximum total EMS midazolam dose is 10 mg.

• If seizure persists after the adequate midazolam regimen: ketamine 1 mg/kg IV/IO or IM once, maximum 100 mg. Continue cardiac monitoring, SpO_2, waveform $EtCO_2$, and immediate airway and ventilation readiness.

Pediatric Anticonvulsant Pathway Pediatric patients are younger than 16 years; use actual or length-based weight.

EMT: Use pediatric airway equipment and length-based resources. Identify prescribed rescue medication and the exact dose, route, and time given before EMS arrival.

AEMT: Perform all EMT care and obtain IV/IO access when it will not delay anticonvulsant treatment.

Paramedic: Midazolam IM or IN: 0.2 mg/kg, maximum single dose 10 mg.

• Midazolam IV/IO: 0.1 mg/kg, maximum single dose 5 mg.

• May repeat once after 5 minutes if seizure continues. Maximum total midazolam is 0.4 mg/kg, not to exceed 10 mg.

• Count prescribed rescue benzodiazepine administered before EMS arrival toward the maximum of two total benzodiazepine doses during the episode.

• For seizure persisting after the adequate benzodiazepine regimen, contact Medical Control before pediatric ketamine administration.

Warnings

- Transport is strongly indicated for the listed high-risk features; those criteria alone do not establish incapacity or authority for involuntary transport. If a legally authorized decision-maker refuses, assess and document decision-specific capacity, explain material risks and alternatives, contact Medical Control for high-risk refusal, and follow the approved refusal/safeguarding procedure. Use an authorized emergency or protective-custody pathway when its legal conditions apply. A patient aged 16 or 17 remains a minor for consent purposes unless a legally recognized exception applies; the clinical age-16 dosing boundary does not establish legal adulthood.
- Do not delay an indicated full anticonvulsant dose while attempting IV or IO access or obtaining glucose.
- Benzodiazepines and ketamine may produce apnea, hypoventilation, or airway obstruction; maintain immediate BVM, suction, and advanced-airway readiness.
- Brief seizure-like activity may occur at the onset of ventricular fibrillation or ventricular tachycardia; confirm pulse and rhythm.
- Postictal altered mental status does not exclude hypoglycemia, hypoxia, head injury, stroke, intoxication, infection, or eclampsia.

Clinical pearls

- Most isolated seizures stop within 1-2 minutes. A seizure lasting 5 minutes or recurrent seizures without recovery should be treated as status epilepticus.
- An adequate early benzodiazepine dose is more effective than several delayed, partial doses.
- Document the exact time seizure activity began, medication administration times, observed cessation, recurrence, and return to baseline.

- Postictal agitation should prompt reassessment for hypoxia, hypoglycemia, injury, hyperthermia, intoxication, and other medical causes before behavioral treatment.
- A focal deficit may be postictal, but persistent or unexplained focal findings require stroke or intracranial-emergency evaluation.

Special populations

Pregnancy and postpartum

- Treat seizure after 20 weeks of pregnancy or within 6 weeks postpartum as eclampsia until proven otherwise.
- Follow the obstetric/eclampsia protocol; magnesium sulfate is the primary anticonvulsant for suspected eclampsia.
- Do not allow benzodiazepine or ketamine administration to replace indicated magnesium therapy.

Pediatrics

- A first febrile seizure, prolonged or recurrent febrile seizure, incomplete recovery, abnormal examination, or age outside the expected simple-febrile-seizure range requires transport.
- Remove excess clothing and treat fever under the fever protocol after stabilization; do not use ice-water immersion or delay seizure treatment for cooling.

Refusal after a seizure

- Transport is strongly indicated for first seizure, status or recurrent seizure, incomplete return to baseline, abnormal vital signs or glucose, injury, fever, pregnancy or postpartum state, persistent neurologic deficit, suspected overdose or withdrawal, respiratory compromise, or any EMS midazolam or ketamine administration.
- A capable adult with known epilepsy may refuse only after one typical seizure, complete return to neurologic baseline, normal examination, vital signs and glucose, no injury or concerning cause, no EMS anticonvulsant administration, and confirmation of responsible adult supervision.
- Complete the Refusal / Non-Transport protocol and document the patient's capacity, risks explained, responsible adult, and return precautions.

UP-14
Suspected Stroke

Revised October 31, 2026

Overview

Treat every acute focal neurologic deficit or sudden unexplained neurologic change as a time-critical stroke until proven otherwise.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Recognize Stroke
EMT / AEMT / Paramedic
Cincinnati screen • posterior-circulation warning signs • exact last known well

↓

2. Complete C-STAT
EMT / AEMT / Paramedic
Gaze 2 • LOC questions + commands 1 • arm weakness 1 • score ≥2 suggests LVO

↓

3. Glucose + Mimics
EMT / AEMT / Paramedic
Check glucose • treat hypoglycemia under UP-4 • repeat neurologic examination

↓

4. Protect Brain + Airway
EMT / AEMT / Paramedic
NPO • manage secretions • protect affected side • oxygen only for hypoxemia

↓

5. Stroke Alert + Destination
EMT / AEMT / Paramedic
Follow current Claiborne Stroke Destination Plan • consider C-STAT and transport time

↓

6. Monitor + Access
AEMT / Paramedic
Cardiac monitor • 12-lead if no delay • one IV preferred • NS TKO

Continue at step 7 in the next column/page

Quick flow continued at step 7

7. Rapid Transport
EMT / AEMT / Paramedic
Scene target ≤15 minutes • serial neurologic exams • transmit stroke report

Indications

- Acute facial droop, unilateral weakness or numbness, speech or language disturbance, gaze deviation, visual loss, or another focal neurologic deficit.
- Sudden unexplained ataxia, severe vertigo, diplopia, dysarthria, unilateral hearing loss, persistent vomiting, or other possible posterior-circulation symptom.
- Resolved focal neurologic symptoms or suspected transient ischemic attack.
- Pediatric patient with a sudden focal neurologic deficit or unexplained acute neurologic change.

Assessment

Time and Baseline

- Record exact last known well, exact symptom-discovery time, and the time EMS first assessed the patient.
- For a wake-up stroke, last known well is the last time the patient was observed neurologically normal-not the time symptoms were discovered.
- Document baseline speech, cognition, mobility, dominant hand, functional status, and preexisting neurologic deficits.
- Obtain the witness or caregiver name and telephone number. Bring the witness when practical without delaying transport.
- Do not exclude stroke activation solely because the last-known-well time exceeds 6 hours or is unknown.

Neurologic Examination

- Perform and document the Cincinnati screen and C-STAT. Repeat after glucose correction and with any clinical change.
- Assess pupils, gaze, facial symmetry, speech, language, arm and leg strength, sensation, coordination, gait only when safe, and level of consciousness.
- A negative Cincinnati screen does not exclude posterior-circulation stroke. Evaluate sudden severe dizziness or vertigo, ataxia, diplopia, dysarthria, visual loss, unilateral hearing loss, and persistent vomiting.
- Consider seizure or Todd paralysis, hypoglycemia, migraine, intoxication or overdose, trauma, infection, and other stroke mimics without delaying stroke-system transport for persistent focal deficits.

Medication and Bleeding History

- Document all current medications, with special attention to anticoagulants and antiplatelet medications.
- Ask about prior intracranial hemorrhage, recent surgery or invasive procedure, recent trauma, active bleeding, known bleeding disorder, and recent anticoagulant reversal.

Monitoring

- Obtain glucose and at least two complete sets of vital signs when feasible without delaying transport.
- Use continuous cardiac monitoring and obtain a 12-lead ECG when it will not delay transport.
- Use continuous SpO_2. Apply waveform $EtCO_2$ whenever ventilation is impaired or assisted or an advanced airway is placed.
- Repeat the neurologic examination and C-STAT during transport and after every significant clinical change.

Treatment

1. Support airway and ventilation, suction as needed, and keep the patient NPO. Elevate the head approximately 30 degrees when tolerated and when hypotension or another contraindication is absent.
2. Provide oxygen only for hypoxemia and titrate to maintain SpO_2 at or above 94%. Avoid unnecessary hyperoxia.
3. Check glucose immediately. Treat confirmed hypoglycemia under UP-4 and repeat the neurologic examination; persistent focal findings remain a stroke alert.
4. Activate the stroke alert early and transport according to the current Claiborne Stroke Destination Plan, considering C-STAT severity and estimated transport time.
5. Target a scene time of 15 minutes or less. Do not delay transport for nonessential IV attempts, ECG acquisition, blood collection, or an exhaustive mimic investigation.
6. Adult pathway only, age 16 or older. Do not give labetalol with bradycardia, advanced heart block, cardiogenic shock, acute decompensated heart failure/pulmonary edema, active bronchospasm, bronchial asthma, or known drug hypersensitivity.

Pediatric stroke requires Medical Control for an age-specific plan; do not apply the fixed adult order. Do not routinely lower blood pressure in suspected stroke. For undifferentiated suspected CVA with manually confirmed SBP above 220 mm Hg or DBP above 110 mm Hg, give labetalol 10 mg slow IV over 2 minutes. Reassess BP, HR, rhythm, perfusion, and neurologic status every 5 minutes; repeat 10 mg once after at least 10 minutes only if still above target. Maximum total dose is 20 mg. Target SBP is 220 mm Hg or less and DBP is 110 mm Hg or less. Do not use this target after thrombolysis; follow AM-06 and written facility orders.

7. Transmit glucose, exact last known well, symptom-discovery time, Cincinnati findings, individual C-STAT findings and total score, baseline function, anticoagulant and antiplatelet use with last dose, and witness contact information.
8. Transport patients with resolved symptoms or suspected transient ischemic attack because symptom resolution does not eliminate the risk of completed or recurrent stroke.

Medication orders

Labetalol

Adult age ≥16 only. For undifferentiated suspected CVA with SBP >220 or DBP >110: 10 mg slow IV over 2 minutes; repeat 10 mg once after at least 10 minutes only if still above target; maximum total 20 mg

- Adult pathway only, age 16 or older. Do not give labetalol with bradycardia, advanced heart block, cardiogenic shock, acute decompensated heart failure/pulmonary edema, active bronchospasm, bronchial asthma, or known drug hypersensitivity. Pediatric stroke requires Medical Control for an age-specific plan; do not apply the fixed adult order. Paramedic standing order after manual confirmation.
- Goal BP is SBP ≤220 and DBP ≤110; do not use this target after thrombolysis or solely to normalize BP.

Provider-level care

Provider Actions

EMT: Perform the Cincinnati Prehospital Stroke Scale and document facial droop, arm drift, and speech findings.

- Complete and document C-STAT when stroke is suspected. Check glucose and treat hypoglycemia under UP-4.
- Determine exact last known well, symptom-discovery time, baseline function, witness contact information, and current anticoagulant or antiplatelet use.
- Keep the patient NPO, protect the affected side, manage secretions, provide oxygen only for hypoxemia, and ventilate with BVM when indicated.
- Activate the stroke alert and begin transport according to the current Claiborne Stroke Destination Plan.

AEMT: Perform all EMT care. Establish one IV when it can be accomplished without delaying transport.

- Administer normal saline at TKO. Give fluid only for hypotension and reassess for pulmonary edema or volume overload.
- Apply continuous cardiac monitoring when available and support serial vital signs and neurologic examinations.

Paramedic: Perform all prior care. Obtain a 12-lead ECG when it will not delay transport, particularly when atrial fibrillation or another cardiac cause is suspected.

- Manage airway or ventilation failure and use waveform $EtCO_2$ whenever ventilation is impaired, assisted, or an advanced airway is placed.
- Adult pathway only, age 16 or older. Do not give labetalol with bradycardia, advanced heart block, cardiogenic shock, acute decompensated heart failure/pulmonary edema, active bronchospasm, bronchial asthma, or known drug hypersensitivity. Pediatric stroke requires Medical Control for an age-specific plan; do not apply the fixed adult order. Repeat an extreme blood pressure manually. For undifferentiated suspected CVA with SBP above 220 mm Hg or DBP above 110 mm Hg, give labetalol 10 mg slow IV over 2 minutes; reassess every 5 minutes and repeat 10 mg once after at least 10 minutes only if still above the target. Maximum total dose is 20 mg. Target SBP is 220 mm Hg or less and DBP is 110 mm Hg or less. Do not routinely lower blood pressure in the field.
- Coordinate early destination notification and transmit the complete stroke report, including C-STAT score and anticoagulant or antiplatelet last-dose information.

C-STAT Stroke Severity Maximum score 4; a score of 2 or greater suggests possible large-vessel occlusion.

EMT: Conjugate gaze deviation: 2 points.

- Incorrectly answers at least one orientation question-age or current month-and fails at least one

command-close eyes or open and close hand: 1 point.

• Cannot hold either arm up for 10 seconds: 1 point.

• Document the individual findings and total C-STAT score. A low score does not exclude stroke or eliminate the need for stroke-system transport.

AEMT: Confirm the documented C-STAT findings during reassessment and report any improvement or deterioration.

Paramedic: Use C-STAT severity with the current Claiborne Stroke Destination Plan and estimated transport time; do not use the score alone to exclude a patient from stroke activation.

Medication and Bleeding-Risk History

EMT: Anticoagulants: warfarin (Coumadin), apixaban (Eliquis), rivaroxaban (Xarelto), dabigatran (Pradaxa), edoxaban (Savaysa), enoxaparin (Lovenox), and heparin injections or infusion.

• Antiplatelet medications: aspirin, clopidogrel (Plavix), ticagrelor (Brilinta), prasugrel (Effient), and aspirin/dipyridamole (Aggrenox).

• Bring medication containers or an accurate medication list when available, but do not delay transport.

AEMT: Confirm medication name, dose, indication, exact last dose time, missed doses, and any recent bleeding, surgery, trauma, or anticoagulant reversal.

Paramedic: Include anticoagulant and antiplatelet details in the stroke alert and receiving-facility report because they may affect thrombolysis and hemorrhage treatment decisions.

Warnings

• Do not administer aspirin, an anticoagulant, or a thrombolytic for undifferentiated suspected stroke. Do not administer an antihypertensive solely to normalize blood pressure.

• Do not administer nitroglycerin solely to lower blood pressure in suspected stroke.

• Naloxone is indicated only when opioid exposure and clinically significant respiratory depression are suspected; do not use it empirically for an isolated focal neurologic deficit.

• Do not delay transport for nonessential procedures, a second IV, blood collection, or complete resolution of diagnostic uncertainty.

• A normal Cincinnati screen, low C-STAT score, symptom improvement, young age, or last-known-well time beyond 6 hours does not exclude a treatment-eligible stroke.

Clinical pearls

• Time is brain, but modern stroke treatment decisions are not limited to the traditional 6-hour window; accurate timing and rapid transport remain essential.

• Posterior-circulation stroke may present without facial droop or arm weakness and may be missed by common anterior-circulation screens.

• Avoid hypotension. Routine prehospital blood-pressure reduction may reduce cerebral perfusion and should not be performed for uncomplicated suspected stroke.

• Medication names alone are insufficient: the exact last dose time of an anticoagulant may directly affect receiving-facility treatment.

Special populations

Pediatrics

• Use pediatric protocols for patients younger than 16 years and dose any required supportive medication by actual or length-based weight.

• Stroke occurs in children. Sudden focal weakness, facial asymmetry, speech change, seizure with persistent focal deficit, severe headache, or unexplained acute neurologic change requires stroke-capable evaluation.

• Do not delay transport while attempting to prove an alternative pediatric diagnosis.

Posterior-circulation stroke

• Maintain suspicion with sudden severe vertigo or dizziness, inability to walk, marked ataxia, diplopia, dysarthria, visual loss, unilateral hearing loss, persistent vomiting, or crossed neurologic findings.

• A negative Cincinnati screen does not exclude posterior stroke. Activate the stroke system when the overall examination and onset are concerning.

UP-15
Suspected Sepsis

Revised October 31, 2026

Overview

Recognize suspected infection with evolving organ dysfunction, hypoperfusion, or shock and begin time-sensitive supportive care.

Use a standardized adult or pediatric sepsis screen, but do not use a single vital sign, fever, qSOFA, or $EtCO_2$ threshold to rule sepsis in or out.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Suspect Infection
EMT / AEMT / Paramedic
History • source clues • temperature may be high, normal, or low

↓

2. Screen for Sepsis
EMT / AEMT / Paramedic
Abnormal physiology • altered mental status • hypotension • hypoxia • poor perfusion

↓

3. Sepsis Alert?
EMT / AEMT / Paramedic
Suspected infection plus organ dysfunction, hypotension, or hypoperfusion

↓

4. Support + Monitor
EMT / AEMT / Paramedic
Infection precautions • glucose • temperature • SpO_2 • cardiac monitor • ECG if no delay

↓

5. Crystalloid + Reassess
AEMT / Paramedic
Adult 500 mL • high overload risk 250 mL • pediatric 10-20 mL/kg

Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Persistent Shock?
Paramedic / Medical Control
Adult norepinephrine • pediatric epinephrine or norepinephrine after Medical Control

↓

7. Rapid Transport
EMT / AEMT / Paramedic
Early notification • reassess unstable patient every 3-5 minutes • document response

Indications

- Suspected or confirmed infection with abnormal vital signs, new altered mental status, hypoxemia, abnormal perfusion, hypotension, or other organ dysfunction.
- Suspected septic shock with hypotension or hypoperfusion requiring fluid or vasopressor support.
- High-risk infection in an older adult, pediatric patient, immunocompromised patient, recently hospitalized patient, or patient with an indwelling vascular, urinary, or prosthetic device.

Assessment

History and Source

- Ask about onset and progression, fever or chills, cough, dyspnea, chest pain, dysuria, abdominal or flank pain, vomiting or diarrhea, headache or neck stiffness, rash, wounds, pressure injuries, recent surgery, and indwelling devices.
- Identify recent antibiotics, hospitalization, healthcare exposure, known resistant organisms, chemotherapy, transplant, immune suppression, diabetes, pregnancy or postpartum status, and sick contacts.
- Review medications that may blunt tachycardia or fever, including beta-blockers, antipyretics, corticosteroids, and immunosuppressants.

Examination

- Assess mental status, airway, work of breathing, lung sounds, pulse quality, skin temperature and color, capillary refill, mottling, edema, and urine output when known.
- Examine for pulmonary, urinary, abdominal, skin or soft-tissue, neurologic, device-related, postoperative, obstetric, or other potential infection sources.

- Obtain temperature, glucose, SpO_2, and complete vital signs. Calculate MAP when hypotension or shock is present.
- Obtain at least two complete vital-sign sets when feasible and reassess an unstable patient every 3-5 minutes.

Monitoring and Diagnostics

- Use continuous cardiac monitoring for a Sepsis Alert, shock, vasopressor administration, abnormal rhythm, or significant physiologic instability.
- Obtain a 12-lead ECG when clinically indicated and when it will not delay resuscitation or transport.
- Use waveform $EtCO_2$ only for impaired or assisted ventilation or advanced-airway monitoring; no $EtCO_2$ value is required for the sepsis screen.
- Blood tubes may be collected when available and coordinated with the receiving facility, but collection must not delay transport.

Treatment

1. Support airway and ventilation. Titrate oxygen to SpO_2 92-96%; target 88-92% in a patient with known chronic hypercapnic respiratory failure unless other clinical circumstances require a higher target.
2. Check glucose and treat hypoglycemia under UP-4. Obtain temperature, but do not exclude sepsis when fever is absent.
3. Activate a Sepsis Alert for suspected infection with organ dysfunction, hypotension, or hypoperfusion and notify the receiving facility early.
4. Do not administer a prehospital antibiotic under this protocol. Piperacillin/tazobactam has been removed from the Claiborne formulary and no replacement sepsis antibiotic is approved.
5. Transport rapidly, continue resuscitation en route, and reassess an unstable patient every 3-5 minutes.

Medication orders

Norepinephrine - Adult Septic Shock

Start 0.1 mcg/kg/min IV/IO; titrate to MAP ≥65 mm Hg or SBP ≥90 mm Hg; maximum 2 mcg/kg/min

- Start after an initial crystalloid bolus when hypotension persists, or concurrently with crystalloid in profound unstable shock.
- A proximal peripheral IV may be used with frequent patency and site assessment.
- Continuous cardiac monitoring and frequent blood-pressure measurement are required.
- Paramedic only.

Epinephrine - Pediatric Septic Shock, Medical Control selection

0.05-0.1 mcg/kg/min IV/IO infusion after Medical Control contact

- Medical Control must specify initial rate, titration increment, interval, ceiling and perfusion target. Do not select an agent solely from a warm/cold label. Follow mental status, pulses, capillary refill and age-appropriate blood pressure; do not apply the adult ceiling.
- Continuous cardiac monitoring, SpO_2, frequent blood-pressure measurement, and IV/IO-site assessment are required.
- Paramedic only.

Norepinephrine - Pediatric Septic Shock, Medical Control selection

0.05-0.1 mcg/kg/min IV/IO infusion after Medical Control contact

- Medical Control must specify initial rate, titration increment, interval, ceiling and perfusion target. Do not select an agent solely from a warm/cold label. Follow mental status, pulses, capillary refill and age-appropriate blood pressure; do not apply the adult ceiling.
- Continuous cardiac monitoring, SpO_2, frequent blood-pressure measurement, and IV/IO-site assessment are required.
- Paramedic only.

Provider-level care

Provider Actions

EMT: Use appropriate standard, contact, droplet, or airborne precautions based on the suspected infection and exposure risk.

- Assess airway, breathing, circulation, mental status, skin temperature and perfusion, and possible source of infection.
- Obtain temperature, glucose, SpO_2, and complete vital signs. Apply oxygen only when indicated and assist ventilation when needed.
- Identify adult or pediatric sepsis-screen findings, activate a Sepsis Alert when criteria are met, and initiate rapid transport.

AEMT: Perform all EMT care. Establish IV access; use IO access when shock is present and IV access cannot be obtained promptly.

- For adult hypotension or hypoperfusion, administer LR or normal saline in 500 mL boluses, or 250 mL increments when volume-overload risk is present.
- For pediatric shock, administer LR or normal saline 10-20 mL/kg per bolus and reassess after every bolus.

• Apply continuous cardiac monitoring when available and repeat complete vital signs and perfusion assessment after each intervention.

Paramedic: Perform all prior care. Obtain a 12-lead ECG when it will not delay resuscitation or transport.

• For persistent adult septic shock after an initial crystalloid bolus, or concurrently with fluid in profound unstable shock, start norepinephrine.

• For persistent pediatric septic shock, contact Medical Control to select epinephrine or norepinephrine and specify the initial rate, titration increment, interval, ceiling and perfusion target. Do not select the agent solely from a warm/cold label. Use the approved 0.05-0.1 mcg/kg/min initial range, continuous monitoring, frequent vascular-site checks and individualized fluid reassessment.

• Use waveform $EtCO_2$ when ventilation is impaired or assisted or an advanced airway is placed; do not use an $EtCO_2$ threshold as a sepsis-screen criterion.

Adult Sepsis Screen Suspected infection plus abnormal physiology or organ dysfunction; clinical concern may override the screen.

EMT: Suspected infection plus two or more screening abnormalities: temperature ≥38°C or ≤36°C, heart rate >90/min, or respiratory rate >20/min.

• Alternatively, suspected infection plus any organ dysfunction: new altered mental status, SBP ≤100 mm Hg, MAP <65 mm Hg, SpO_2 <92% or new oxygen requirement, delayed capillary refill, mottling, or other abnormal perfusion.

• Fever is not required. Older adults and immunocompromised patients may be normothermic or hypothermic.

• Do not use qSOFA alone and do not use $EtCO_2$ <25 mm Hg as a sepsis-screen criterion.

AEMT: A positive physiology screen without organ dysfunction identifies possible infection and requires close reassessment. Activate the Sepsis Alert when organ dysfunction, hypotension, or hypoperfusion is present.

Paramedic: Evaluate for alternative causes of shock or abnormal vital signs, including hemorrhage, cardiogenic shock, pulmonary embolism, anaphylaxis, toxicologic causes, heat illness, adrenal crisis, and medication effects, without delaying resuscitation.

Adult Hemodynamic Support

EMT: Position for perfusion and respiratory comfort, prevent heat loss, and reassess mental status, skin perfusion, and blood pressure frequently.

AEMT: LR is preferred when available; normal saline is acceptable.

• Administer 500 mL IV/IO and reassess blood pressure, MAP, mental status, capillary refill, lung sounds, and oxygen requirement.

• Additional crystalloid may be given toward 30 mL/kg for continuing hypotension or hypoperfusion.

• Use 250 mL increments for heart failure, renal failure, liver failure, pulmonary edema, or existing volume overload. Stop or reduce fluid for worsening respiratory status or pulmonary edema.

Paramedic: Norepinephrine: start 0.1 mcg/kg/min IV/IO and titrate to MAP ≥65 mm Hg or SBP ≥90 mm Hg; maximum 2 mcg/kg/min.

• Norepinephrine may be started after an initial crystalloid bolus when hypotension persists or concurrently with crystalloid in profound unstable shock.

• A proximal peripheral IV may be used. Confirm patency, inspect the site frequently, and do not delay norepinephrine solely to obtain central access.

• Continuous cardiac monitoring and frequent blood-pressure measurement are required during vasopressor infusion.

Pediatric Sepsis and Shock Pediatric patients are younger than 16 years; recognize organ dysfunction because hypotension is a late finding.

EMT: Suspect pediatric sepsis with infection plus altered mental status, capillary refill >3 seconds, weak pulses, mottling, abnormal temperature, tachycardia or bradycardia, respiratory distress, hypoxemia, or hypotension.

• Pediatric hypotension: birth-28 days, SBP <60 mm Hg; 1-12 months, SBP <70 mm Hg; 1-10 years, SBP <70 plus twice the age in years; older than 10 years, SBP <90 mm Hg.

• Use pediatric airway equipment and actual or length-based weight. Do not apply the adult SIRS screen to pediatric patients.

AEMT: Administer LR or normal saline 10-20 mL/kg IV/IO and reassess after every bolus.

• Repeat as needed for ongoing shock, generally not exceeding 40 mL/kg prehospital.

• Stop fluid boluses for resolved shock, pulmonary edema, new hepatomegaly, worsening oxygenation, or other evidence of fluid overload.

Paramedic: For persistent pediatric septic shock, contact Medical Control to select epinephrine or norepinephrine and specify the initial rate, titration increment, interval, ceiling and perfusion target. Do not select the agent solely from a warm/cold label. Use the approved 0.05-0.1 mcg/kg/min initial range, continuous monitoring, frequent vascular-site checks and individualized fluid reassessment.

• Titrate the selected infusion to improving mental status, pulses, capillary refill, age-appropriate blood pressure, and other perfusion findings.

• Continuous cardiac monitoring, SpO_2, frequent blood-pressure measurement, and IV/IO-site assessment are required.

Warnings

• Do not delay transport for IV attempts, ECG acquisition, blood collection, or completion of every possible diagnostic evaluation.

• Do not use qSOFA alone, fever alone, or an $EtCO_2$ threshold to exclude or confirm sepsis.

• Hypotension is a late and ominous finding in pediatric sepsis; treat abnormal perfusion before hypotension develops.

• Stop or reduce fluid administration for pulmonary edema, worsening oxygenation, new hepatomegaly, or other evidence of fluid overload.

Clinical pearls

• Sepsis is infection associated with life-threatening organ dysfunction. The term severe sepsis is not used in this protocol.

• Older adults, immunocompromised patients, and patients taking antipyretics or beta-blockers may lack fever or marked tachycardia.

• Capillary refill, mental status, pulse quality, skin findings, blood pressure, and response to a fluid bolus should be interpreted together.

• Balanced crystalloid is preferred when available, but appropriate resuscitation should not be delayed when only normal saline is available.

• Early norepinephrine may be safer than repeated large fluid boluses in patients at risk for pulmonary edema or volume overload.

• A positive screening tool supports recognition but does not replace clinical judgment or evaluation for alternative causes of shock.

Special populations

Volume-overload risk

• Reassess lung sounds, SpO_2, respiratory effort, edema, blood pressure, and perfusion after every bolus.

• Consider earlier norepinephrine rather than repeated crystalloid when hypotension persists and fluid tolerance is limited.

Pediatrics

• Use pediatric protocols for patients younger than 16 years and actual or length-based weight for all weight-based treatment.

• Do not rely on hypotension alone. Altered mental status, weak pulses, delayed capillary refill, mottling, and respiratory abnormalities may identify shock earlier.

Refusal / Non-Transport

• Rapid transport is indicated for any Sepsis Alert or suspected septic shock.

• A patient with suspected infection plus organ dysfunction, hypotension, or abnormal perfusion should not be managed as a routine refusal.

• If a capable patient continues to refuse after risks and alternatives are explained, contact Medical Control and complete the Refusal / Non-Transport protocol with detailed documentation.

UP-16
Syncope

Revised October 31, 2026

Overview

Syncope is abrupt, transient loss of consciousness and postural tone caused by temporary global cerebral hypoperfusion, followed by rapid spontaneous recovery.

Near-syncope may carry similar risk and receives the same initial evaluation.

Do not label persistent altered mental status, seizure, hypoglycemia, intoxication, stroke, or traumatic loss of consciousness as uncomplicated syncope.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Primary Assessment
EMT / AEMT / Paramedic
Airway • breathing • circulation • injury • hemorrhage • pregnancy • persistent instability

↓

2. Recovered to Baseline?
EMT / AEMT / Paramedic
Persistent altered mental status or focal deficit requires the appropriate emergency pathway

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. Glucose + Cardiac Evaluation
EMT / AEMT / Paramedic
Glucose on every patient • 12-lead ECG • continuous cardiac monitoring

↓

4. High-Risk Features?
EMT / AEMT / Paramedic
Exertional or supine • abrupt • cardiopulmonary symptoms • abnormal ECG • bleeding • abnormal vitals

↓

5. Treat Identified Cause
EMT / AEMT / Paramedic
Hypoglycemia • dysrhythmia • hemorrhage • shock • opioid respiratory depression • trauma

↓

6. Transport + Reassess
EMT / AEMT / Paramedic
Repeat vitals and ECG when indicated • monitor for recurrent symptoms • early notification

Indications

- Syncope with transient loss of consciousness and postural tone followed by rapid spontaneous recovery.
- Near-syncope with transient lightheadedness, weakness, or impending loss of consciousness without complete loss of consciousness.
- Unexplained collapse with apparent spontaneous recovery when syncope remains possible.

Assessment

Event History

- Determine body position and activity at onset, prodrome, precipitating trigger, abruptness, duration, witnessed movements, color change, breathing, injuries, and time to complete recovery.
- Ask about exertion, standing, heat exposure, pain, emotional stress, urination, defecation, coughing, swallowing, dehydration, vomiting, diarrhea, and recent illness.
- Ask about chest pain, dyspnea, palpitations, headache, neck pain, abdominal or back pain, GI or vaginal bleeding, pregnancy possibility, and neurologic symptoms.
- Obtain cardiac, neurologic, seizure, bleeding, and thromboembolic history; family history of sudden death; and all medications, including antihypertensives, diuretics, QT-prolonging medications, and anticoagulants.

Examination

- Assess mental status and confirm complete return to neurologic baseline. Examine pupils, speech, gaze, facial symmetry, strength, sensation, and coordination when appropriate.
- Evaluate pulse rate and regularity, blood pressure, heart and lung findings, perfusion, hydration, abdominal or back tenderness, evidence of bleeding, and injury from the fall.
- Obtain glucose on every patient. Obtain temperature when infection or environmental illness is suspected.
- Evaluate pregnancy and postpartum status when applicable. Abdominal, pelvic, or back pain with syncope in a patient who may be pregnant requires urgent ectopic-pregnancy evaluation.

Cardiac Monitoring

- A single normal ECG or brief period of normal monitoring does not exclude intermittent dysrhythmia or another cardiac cause.
- Document monitor rhythm, ECG interpretation, recurrent symptoms, and any rhythm associated with those symptoms.

Treatment

1. Support airway and ventilation and provide oxygen only for hypoxemia or respiratory distress.
2. Check glucose immediately and treat hypoglycemia under UP-4.
3. Assess and treat injury, hemorrhage, pregnancy-related emergency, dysrhythmia, ACS, pulmonary embolism, aortic emergency, seizure, stroke, sepsis, shock, or toxicologic cause under the applicable protocol.
4. Administer naloxone only when opioid exposure and clinically significant respiratory depression are suspected; do not use naloxone empirically for isolated syncope.
5. Transport every high-risk, first unexplained, recurrent, injured, pregnant, anticoagulated, abnormal-vital-sign, abnormal-glucose, or abnormal-ECG patient with continuous reassessment.

Provider-level care

Provider Actions

EMT: Assess airway, breathing, circulation, mental status, injury from the fall, possible hemorrhage, pregnancy, and persistent instability.

• Obtain glucose, SpO_2, two complete vital-sign sets when feasible, and a focused neurologic examination.

• Acquire and transmit a 12-lead ECG and apply continuous cardiac monitoring for every syncope or near-syncope patient.

• Give oxygen only for hypoxemia or respiratory distress and initiate rapid transport when any high-risk feature is present.

AEMT: Perform all EMT care. Establish IV access when hypotension, poor perfusion, clear volume depletion, recurrent symptoms, or another high-risk finding is present.

• Administer LR or normal saline only for hypotension, poor perfusion, or clear volume depletion, using the approved adult or pediatric bolus.

• Monitor the rhythm continuously, repeat complete vital signs after intervention, and prepare for cause-specific treatment.

Paramedic: Perform all prior care. Interpret the initial 12-lead ECG and repeat it for recurrent symptoms, rhythm change, chest discomfort, or a concerning initial tracing.

• Treat bradycardia, tachycardia, ischemia, shock, hypoglycemia, respiratory failure, or another identified cause under the applicable Claiborne protocol.

• Evaluate persistent altered mental status, focal neurologic findings, suspected seizure, bleeding, ectopic pregnancy, pulmonary embolism, aortic emergency, and toxicologic causes.

High-Risk Syncope Any high-risk feature requires continuous monitoring and prompt emergency-department evaluation.

EMT: Syncope during exertion or while supine, or abrupt syncope without warning.

• Chest pain, dyspnea, palpitations, persistent headache, or recurrent symptoms.

• Persistent hypotension, hypoxia, bradycardia, tachycardia, abnormal perfusion, or failure to return completely to baseline.

• Known coronary disease, heart failure, cardiomyopathy, significant valvular disease, pacemaker, or implanted cardioverter-defibrillator.

• Family history of sudden unexplained death before age 50.

• GI bleeding, vaginal bleeding, abdominal or back pain, pregnancy, anticoagulant use, significant trauma, or focal neurologic deficit.

AEMT: Establish vascular access when clinically indicated and transport without delaying for orthostatic vital signs or nonessential procedures.

Paramedic: Treat an abnormal rhythm or identified life threat under the applicable protocol and provide early receiving-facility notification.

• Concerning ECG findings include acute ischemia, significant bradycardia or tachycardia, high-grade AV block, ventricular dysrhythmia, prolonged QT, pre-excitation, Brugada pattern, or other new conduction abnormality.

Orthostatic Vital Signs

EMT: Consider only when the patient is stable, completely recovered, able to stand safely, and has no injury, pregnancy-related concern, or high-risk feature.

• A positive test is recurrent symptoms or a fall in SBP of at least 20 mm Hg or DBP of at least 10 mm Hg after standing.

• Stop immediately for dizziness, weakness, recurrent near-syncope, syncope, chest pain, dyspnea, or instability.

• Do not use normal orthostatic vital signs to exclude a serious cause and do not delay transport to obtain them.

AEMT: When testing is appropriate, ensure monitoring and immediate assistance are available before standing the patient.

Paramedic: Interpret orthostatic findings in the context of the full history, medication list, examination, ECG, glucose, and volume status.

Fluid Support

EMT: Position the patient safely, prevent recurrent falls, and reassess perfusion while vascular access is prepared when indicated.

AEMT: Adult: LR or normal saline 500 mL IV/IO, then reassess.

• Heart failure, renal failure, liver failure, pulmonary edema, or volume-overload risk: use 250 mL increments.

• Pediatric patient younger than 16 years: LR or normal saline 10-20 mL/kg IV/IO, then reassess.

• Stop or reduce fluid for pulmonary edema, worsening oxygenation, or resolved hypotension and poor perfusion.

Paramedic: If hypotension or poor perfusion persists, reassess for hemorrhagic, cardiogenic, distributive,

obstructive, or toxicologic shock and transition to the appropriate shock pathway.

Warnings

- Transport is strongly indicated for the listed high-risk features; those criteria alone do not establish incapacity or authority for involuntary transport. If a legally authorized decision-maker refuses, assess and document decision-specific capacity, explain material risks and alternatives, contact Medical Control for high-risk refusal, and follow the approved refusal/safeguarding procedure. Use an authorized emergency or protective-custody pathway when its legal conditions apply. A patient aged 16 or 17 remains a minor for consent purposes unless a legally recognized exception applies; the clinical age-16 dosing boundary does not establish legal adulthood.
- Do not assume benign vasovagal syncope until cardiac, hemorrhagic, obstetric, neurologic, metabolic, toxicologic, and traumatic causes have been considered.
- Do not stand an unstable, injured, pregnant, anticoagulated, or high-risk patient for orthostatic vital signs.
- Do not give routine oxygen, IV fluid, aspirin, or naloxone solely because syncope occurred.
- Syncope during exertion or while supine is high risk and requires transport for cardiac evaluation.
- Persistent altered mental status or focal neurologic deficit is not uncomplicated syncope.

Clinical pearls

- A detailed event history, examination, glucose, ECG, and risk assessment are more useful than routine indiscriminate testing.
- Brief myoclonic movements can occur during cerebral hypoperfusion. Persistent postictal confusion, prolonged convulsions, or lateral tongue injury increases concern for seizure.
- Isolated syncope without focal neurologic findings is not usually a stroke presentation; persistent focal findings require stroke evaluation.
- Syncope associated with dyspnea, unexplained hypoxemia, pleuritic pain, tachycardia, or thromboembolic risk requires consideration of pulmonary embolism.
- Abdominal or back pain, diminished pulses, bleeding, hypotension, or pregnancy may identify life-threatening occult hemorrhage or an aortic emergency.

Special populations

Pediatrics and athletes

- Pediatric patients are younger than 16 years and medication or fluid doses use actual or length-based weight.
- Exertional or supine syncope, chest pain, palpitations, abnormal ECG, congenital heart disease, or family history of sudden death is high risk.
- Do not permit return to sports or exertion. Transport for cardiac evaluation.

Pregnancy and bleeding risk

- Syncope with abdominal, pelvic, or back pain or vaginal bleeding in a patient who may be pregnant requires urgent evaluation for ectopic pregnancy or other hemorrhage.
- Anticoagulant use increases concern for occult bleeding and intracranial injury after a fall.
- Do not delay transport for orthostatic vital signs when bleeding, pregnancy complication, or significant trauma is possible.

Refusal / Non-Transport

- Transport is strongly indicated for any high-risk feature, first unexplained or recurrent event, abnormal ECG, glucose or vital signs, injury, pregnancy, anticoagulant use, persistent symptoms, or incomplete return to baseline.
- A capable adult with a classic vasovagal trigger and prodrome, complete recovery, normal examination, glucose, vital signs, and ECG, and no injury or high-risk feature may refuse only after Medical Control consultation.
- Complete the Refusal / Non-Transport protocol and document capacity, event history, examination, ECG and glucose results, risks explained, Medical Control consultation, responsible adult supervision, and return precautions.

UP-17
Behavioral Health Crisis

Revised October 31, 2026

Overview

Protect the patient, public, and responders while preserving dignity and using the least restrictive effective intervention.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Scene Safe?
EMT / AEMT / Paramedic
Stage or withdraw when unsafe • request law enforcement and sufficient resources • identify weapons or hazards

↓

2. Medical Emergency?
EMT / AEMT / Paramedic
Airway • breathing • circulation • vitals • SpO_2 • glucose • temperature • trauma • neurologic and toxicologic causes

↓

3. Assign BARS
EMT / AEMT / Paramedic
1-3: altered mental status/overdose • 4: cooperative • 5: de-escalate • 6-7: UP-18 combined agitation pathway

↓

4. Assess Safety Risk
EMT / AEMT / Paramedic
Suicide • homicide • plan • intent • access to means • prior attempts • command hallucinations • ability to protect self

↓

5. De-escalate
EMT / AEMT / Paramedic
One calm speaker • reduce stimulation • maintain personal space and exit • acknowledge emotions • offer safe choices

Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Immediate Danger?
EMT / AEMT / Paramedic
Direct observation • remove accessible hazards when safe • request authorized custody assistance • do not rely on a safety contract

↓

7. Safe Disposition
EMT / AEMT / Paramedic
Emergency department for medical risk, self-harm, restraint, or sedation • alternative destination only under approved policy

Indications

- Suicidal thoughts, suicide attempt, self-harm, homicidal thoughts, psychosis, hallucinations, delusions, severe anxiety, mania, depression, or another behavioral-health crisis.
- Agitation, threatening behavior, impaired judgment, inability to protect oneself, or concern for danger to the patient or others.
- Behavioral change that may be caused by a medical, traumatic, toxicologic, medication-related, or environmental emergency.

Assessment

Medical and Traumatic Causes

- Assess airway, ventilation, oxygenation, circulation, complete vital signs, glucose, temperature, pupils, focused neurologic status, trauma, pain, pregnancy when applicable, medication adherence or changes, substance exposure, and withdrawal.
- Consider hypoxia, hypoglycemia or hyperglycemia, stroke, seizure or postictal state, head injury, shock, sepsis, hyperthermia, intoxication, overdose, withdrawal, medication reaction, endocrine/metabolic disease, dementia, and delirium.
- Obtain a 12-lead ECG when overdose, stimulant use, syncope, chest discomfort, dyspnea, palpitations, abnormal pulse, electrolyte disorder, QT-prolonging medication, or parenteral agitation medication is suspected or used.
- Use $EtCO_2$ for hypoventilation, suspected overdose, significant altered mental status, restraint, or pharmacologic management.

Behavior and Safety Risk

- Assign BARS from 1 through 7 and document objective observed behavior rather than conclusory labels such as uncooperative, crazy, or violent without supporting facts.
- Ask directly about suicidal and homicidal thoughts, plan, intent, access to means, prior attempts, self-harm, command hallucinations, substance use, supports, and ability to remain safe.
- Obtain collateral information from family, caregivers, witnesses, law enforcement, medication containers, and prior records when available, while maintaining patient dignity and necessary privacy.
- Identify abuse, neglect, exploitation, domestic violence, sexual assault, trafficking, or an unsafe living environment and follow the applicable reporting protocol.

Capacity and Reassessment

- Obtain two complete vital-sign sets when feasible. Reassess high-acuity, restrained, sedated, or clinically unstable patients at least every 3-5 minutes.
- Repeat BARS and the focused medical, airway, perfusion, neurologic, and safety assessment after every intervention or significant change.

Treatment

1. Perform the medical screen and immediately treat hypoxia, hypoventilation, hypoglycemia, opioid respiratory depression, shock, hyperthermia, trauma, seizure, stroke, overdose, or another identified emergency under the applicable Claiborne protocol.
2. Attempt verbal and environmental de-escalation before restraint whenever the situation permits.
3. Transport to an emergency department for medical abnormality, injury, overdose, delirium, recent self-harm, current suicide or homicide risk, dangerous psychosis, restraint, or pharmacologic management.
4. Use a psychiatric or crisis alternative destination only under a separately approved destination policy and only after required medical and safety screening criteria are met.
5. A patient with intact capacity and no identified immediate danger may use the standard refusal process. Contact Medical Control whenever capacity or risk is uncertain and document a clear safety and follow-up plan.

Medication orders

Diphenhydramine

Adult: 50 mg IV/IO/IM. Pediatric patient younger than 16 years: 1 mg/kg IV/IO/IM; maximum 50 mg.

- Paramedic.
- Indication: acute dystonic or extrapyramidal reaction.
- Assess for airway involvement and monitor after administration.

Provider-level care

Provider Actions Use explicit provider-level actions without substituting a generic scope statement.

EMT: Establish scene safety, request law enforcement or additional personnel when indicated, and do not enter or remain in an unsafe environment.

- Assess airway, breathing, circulation, mental status, trauma, oxygenation, glucose, temperature, medications, substance exposure, and focused neurologic findings.
- Assign and document BARS, complete the suicide/homicide safety assessment, begin verbal and environmental de-escalation, and maintain direct observation when risk is present.
- Provide indicated BLS treatment, obtain two complete vital-sign sets when feasible, and transport to the appropriate emergency destination.

AEMT: Perform all EMT care. Establish IV/IO access only when a medical condition, poor perfusion, significant overdose, or anticipated emergency treatment creates a clinical indication.

- Administer dextrose, naloxone, or IV fluid only when indicated under the applicable Claiborne medical, toxicology, or shock protocol; do not medicate solely for a psychiatric diagnosis.
- Continue physiologic monitoring, direct observation, and reassessment during transport.

Paramedic: Perform all prior care. Obtain and interpret a 12-lead ECG, use $EtCO_2$, and provide advanced airway or cause-specific treatment when the presentation or intervention indicates.

- For BARS 6-7, transition to UP-18 Behavioral Agitation and Severe Agitation. Use the severe-agitation branch when delirium, hyperthermia, unusual strength, or prolonged struggle is present.
- Agitation medication and dosing are contained only in the combined UP-18 protocol. UP-17 contains no standing-order sedative treatment.
- After restraint or medication, provide continuous airway, ventilation, SpO_2, $EtCO_2$, and cardiac monitoring as required by UP-18.

BARS Routing

EMT: BARS 1: difficult or unable to awaken. Treat as an airway, altered mental status, overdose, or other medical emergency.

• BARS 2: asleep but responds normally to verbal or physical stimulation. Evaluate for medication, alcohol, overdose, hypoglycemia, or another medical cause.

• BARS 3: drowsy or appears sedated. Evaluate altered mental status, monitor airway and ventilation, and consider overdose or other CNS depression.

• BARS 4: quiet and awake with normal activity. Continue the complete behavioral, suicide/homicide, capacity, and medical assessment.

• BARS 5: increased verbal or physical activity but not disruptive. Continue UP-17, reduce stimulation, and attempt verbal de-escalation.

• Document the initial score and repeat BARS after de-escalation or any change in condition.

AEMT: BARS 6: extremely or continuously active and disruptive but not violent. Request sufficient resources, continue de-escalation when safe, and initiate UP-18.

• Do not allow vascular access or nonessential procedures to provoke additional struggle or delay safe transport.

Paramedic: BARS 7: violent with immediate danger and restraint required. Initiate the UP-18 safety and restraint pathway without delaying for nonessential assessment.

• Use the UP-18 severe-agitation branch when agitation includes delirium, hyperthermia, unusual strength, pain tolerance, or continued/prolonged struggle.

• Repeat and document BARS after physical restraint, medication, and each clinically important change.

Suicide and Violence Safety Assessment

EMT: For medically able patients age 8 years and older, use the ASQ framework: recent wish to be dead; belief that self or family would be better off if dead; recent thoughts of suicide; prior suicide attempt; and current thoughts of suicide.

• Ask directly about plan, intent, timing, access to medications or weapons, preparations already made, previous attempts, recent self-harm, homicidal thoughts, command hallucinations, substance use, and ability to remain safe.

• If feasible and developmentally appropriate, interview pediatric patients privately for part of the assessment, then obtain collateral history from the parent or guardian.

• Any current suicidal intent, credible homicidal threat, recent serious attempt, dangerous command hallucination, or inability to protect oneself requires direct observation and emergency evaluation.

• Do not leave a high-risk patient unattended and do not use a verbal or written safety contract as proof of safety.

AEMT: Maintain continuous observation and repeat mental-status, vital-sign, and risk assessment after any medical treatment or change in behavior.

Paramedic: Integrate the safety assessment with medical findings and decision-making capacity. Contact Medical Control whenever risk, capacity, custody, or destination is uncertain.

De-escalation and Restraint Safety Use restraint only to prevent harm or permit medically necessary assessment, treatment, and transport.

EMT: Use one calm speaker, open-ended questions, active listening, emotional acknowledgment, nonthreatening posture, personal space, reduced noise and spectators, and reasonable choices.

• Maintain a clear exit and avoid confrontation, sudden movements, deceptive promises, crowding, or arguing about delusions.

• Do not use prone restraint, hog-tying, neck or chest compression, or any position or device that compromises airway, breathing, circulation, or rapid release.

AEMT: Assess respiratory and hemodynamic status and the neurovascular status of restrained extremities as soon as safely possible and at recurring intervals.

Paramedic: Physical or pharmacologic restraint requires the UP-18 pathway, appropriate physiologic monitoring, recurring reassessment, and transport to an emergency department.

• EMS medications must never be administered solely to facilitate arrest or law-enforcement custody.

Tennessee Emergency Detention and Capacity Coordinate emergency custody through the authorized Tennessee process while continuing necessary medical care.

EMT: Request law enforcement or another authorized professional when mental illness or serious emotional disturbance creates an imminent substantial likelihood of serious harm.

• EMS personnel do not independently initiate a Tennessee emergency mental-health detention solely by virtue of EMS licensure.

• Maintain observation and do not abandon a patient who presents an immediate safety threat or cannot make an informed decision.

AEMT: Continue medically necessary care and transport while the authorized officer, physician, psychologist, or commissioner-designated professional manages the custody process.

Paramedic: Assess whether the patient can understand relevant information, appreciate the condition and consequences, reason about options, and communicate a stable choice.

• A psychiatric diagnosis, unusual belief, intoxication, or refusal alone does not automatically establish incapacity; document the specific functional impairment when capacity is absent.

• Contact Medical Control and request authorized custody assistance whenever capacity or immediate risk is uncertain.

Acute Dystonic / Extrapyramidal Reaction
Recognize medication-induced dystonia as a medical condition that may be mistaken for behavioral illness.

EMT: Suspect an acute dystonic reaction with involuntary facial, jaw, neck, trunk, or extremity spasm after an antipsychotic or other dopamine-blocking medication.

• Assess for tongue, pharyngeal, or laryngeal involvement and support airway and ventilation immediately when present.

AEMT: Recognize dystonia, assess and support airway/ventilation, and request Paramedic care. AEMT diphenhydramine for dystonia is not enabled pending specific current Tennessee indication/route authority, competency and signed local orders.

Paramedic: Adult: diphenhydramine 50 mg IV/IO/IM.

• Pediatric patient younger than 16 years: diphenhydramine 1 mg/kg IV/IO/IM; maximum 50 mg.

• Monitor airway, SpO_2, vital signs, and cardiac rhythm after parenteral medication and reassess the muscular symptoms.

• Perform all prior care. Treat airway compromise immediately and evaluate for serotonin syndrome, neuroleptic malignant syndrome, seizure, tetany, toxic exposure, or another cause when findings are atypical.

Warnings

• Diphenhydramine - dystonia under UP-17: Paramedic-only administration until specific current Tennessee AEMT authority for this indication and route, applicable competency and signed local orders are documented. During that interval, the AEMT action is recognition, airway assessment/support and requesting Paramedic care. Do not infer dystonia permission from allergy treatment or use online Medical Control to expand scope. Use the Paramedic dose specified here.

• A calm, cooperative, or BARS 4 patient may still have imminent suicide or homicide risk.

• Never assume a psychiatric diagnosis until medical, traumatic, toxicologic, medication-related, and environmental causes have been assessed.

• Any patient receiving physical restraint or pharmacologic management requires continuous observation, physiologic monitoring, recurring reassessment, and emergency-department transport.

• Do not place a psychiatric-only patient at an alternative destination unless a separately approved destination policy is active and all criteria are met.

Clinical pearls

• Ask directly about suicide and homicide; asking does not create suicidal thoughts and may identify concealed risk.

• BARS is a rapid observational communication tool, not a diagnosis, suicide-risk scale, capacity assessment, or automatic medication order.

• Medical causes are especially important with new behavioral change, atypical age of onset, abnormal vital signs, fluctuating attention, focal neurologic findings, trauma, or no established psychiatric history.

• Current suicidal thoughts, a feasible plan with access to means, a recent attempt, dangerous command hallucinations, or inability to protect oneself requires immediate safety precautions and emergency evaluation.

• If law-enforcement restraints that require a key must remain, law enforcement and the key must accompany the patient; transition to the least restrictive safe restraint when feasible.

• Exact patient statements are more clinically useful than paraphrases when documenting threats, hallucinations, intent, or refusal.

Special populations

Pediatrics

• Pediatric patients are younger than 16 years and medication doses use actual or length-based weight.

• The ASQ framework may be used for medically able patients age 8 years and older. Suspected suicide risk in a child younger than 8 years requires comprehensive emergency mental-health evaluation rather than relying on a screening score.

• Evaluate for abuse, neglect, exploitation, bullying, trafficking, medication ingestion, developmental disability, and an unsafe home environment.

Law-Enforcement Custody

- A patient restrained by law enforcement remains an EMS patient when medical assessment or treatment is provided; EMS advocates for airway, breathing, circulation, positioning, monitoring, and dignity.
- Law enforcement should accompany a patient when law-enforcement restraints must remain during ambulance transport, and the required release key must be immediately available.
- Do not transport with hands restrained behind the back when that position interferes with assessment, monitoring, airway protection, or safe positioning.

Refusal / Non-Transport

- Do not permit refusal based solely on a calm appearance, a low BARS score, denial after a documented threat, or a verbal safety contract.
- Current suicidal intent, credible homicidal threat, recent serious attempt, dangerous command hallucinations, inability to protect oneself, impaired capacity, medical instability, restraint, or sedation requires emergency evaluation and safe transport.
- Document capacity, risk assessment, exact statements, collateral information, alternatives offered, risks explained, Medical Control or law-enforcement involvement, responsible support person, crisis resources, and return precautions.

UP-18
Behavioral Agitation and Severe Agitation

Revised October 31, 2026

Overview

Agitation is a clinical spectrum. Use BARS to select the least restrictive effective intervention while rapidly identifying medical, traumatic, toxicologic, and environmental causes.

The treatment goal is a calm, arousable patient who can be safely assessed and transported-not unconsciousness.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Scene Safe?
EMT / AEMT / Paramedic
Stage or withdraw when unsafe • request law enforcement and sufficient personnel • coordinate one team plan

↓

2. Assign BARS
EMT / AEMT / Paramedic
1-3: medical/overdose • 4: cooperative • 5: de-escalate • 6: moderate agitation • 7: severe immediate danger

↓

3. BARS 5
EMT / AEMT / Paramedic
One calm speaker • reduce stimulation • maintain personal space and exit • offer safe choices

↓

4. BARS 6
EMT / AEMT / Paramedic
Paramedic: midazolam pathway • use physical restraint only when needed for immediate safety or necessary care

↓

5. BARS 7 / Immediate Danger - All Providers
EMT / AEMT / Paramedic
Coordinate safe control, positioning, airway readiness, monitoring and Paramedic support. No EMT/AEMT standing-order chemical sedation.

↓

6. BARS 7 - Adult ≥16, Paramedic
Paramedic
Ketamine 2-4 mg/kg IM, maximum 400 mg, or age/risk-adjusted adult midazolam alternative below. Follow reassessment and Medical Control requirements before additional sedation.

Continue at step 7 in the next column/page

Quick flow continued at step 7

7. BARS 7 - Younger Than 16
Paramedic / Medical Control
Direct Medical Control medication and dose order required before chemical sedation.

↓

8. Safe Position
EMT / AEMT / Paramedic
Approved soft restraints • supine with head elevated or lateral • never prone, hog-tied, or compressed at neck/chest

↓

9. Find + Treat Cause
EMT / AEMT / Paramedic
Airway • ventilation • glucose • temperature • trauma • neurologic emergency • toxidrome • hypoxia

↓

10. Continuous Monitoring
EMT / AEMT / Paramedic
Direct observation • ECG • SpO_2 • waveform $EtCO_2$ after sedation • BP, respirations, airway, and restraint checks every 5 minutes

↓

11. ED Transport
EMT / AEMT / Paramedic
Every physical restraint or chemical sedation patient • early notification • document BARS, interventions, response, and complications

Indications

- BARS 6 agitation that remains disruptive or creates a safety risk despite verbal and environmental de-escalation.
- BARS 7 violent behavior with immediate danger to the patient, responders, or others.
- Physical restraint or pharmacologic management required to permit assessment, treatment, or safe transport of a medical patient.

Contraindications

- Do not use physical restraint or chemical sedation solely for refusal, verbal hostility, convenience, punishment, noncompliance, or to facilitate arrest or custody.
- Do not use a paralyzing medication solely for behavioral control.

Assessment

Immediate safety and medical assessment

- Determine BARS, whether an immediate threat exists, available personnel, weapons or hazards, law-enforcement involvement, and the safest approach.
- Assess airway, breathing, circulation, oxygenation, glucose, temperature, trauma, neurologic findings, medication exposure, alcohol or substance use, pregnancy, and relevant medical or psychiatric history as soon as safely possible.
- Consider hypoxia, hypoglycemia, stroke, seizure or postictal state, traumatic brain injury, infection or sepsis, medication effect, alcohol or sedative withdrawal, stimulant or sympathomimetic toxicity, anticholinergic toxicity, and heat illness.
- Do not label severe agitation as a psychiatric condition until urgent medical, traumatic, toxicologic, and environmental causes have been considered.

Post-control physiologic assessment

- Immediately reassess airway patency, respiratory rate and effort, ventilation, SpO_2, waveform $EtCO_2$, cardiac rhythm, blood pressure, perfusion, temperature, glucose, mental status, pupils, trauma, restraint position, and distal neurovascular status.
- Obtain a 12-lead ECG when feasible after control, particularly with suspected stimulant exposure, abnormal rhythm, chest symptoms, hyperthermia, significant tachycardia, or another medical concern. Do not delay immediate safety intervention or transport solely for the tracing.
- Repeat BARS and complete physiologic assessment after each medication, restraint change, or clinically important change.

Treatment

1. Once safe, immediately identify and treat hypoxia, hypoglycemia, hyperthermia, trauma, toxidrome, shock, and other reversible causes. Use active cooling for clinically significant hyperthermia. Administer fluid only for hypotension, poor perfusion, or suspected heat illness.

Medication orders

Midazolam - Adult BARS 6

2.5 mg IV/IO or 5 mg IM/IN; repeat once after 5 minutes if dangerous agitation persists

- Complete airway, ventilation, blood-pressure, and sedation reassessment before repeat dosing.

- Maximum: 5 mg IV/IO or 10 mg IM/IN.
- Age 65 or older, frail, respiratory disease, or suspected CNS-depressant intoxication: 1-2.5 mg IV/IO or 2.5 mg IM/IN; maximum cumulative dose 5 mg.
- Paramedic only.

Midazolam - Pediatric BARS 6

0.1 mg/kg IV/IO or 0.2 mg/kg IM/IN; maximum initial dose 5 mg

- Patient younger than 16 years. Use actual or length-based weight.
- Contact Medical Control when feasible before administration. A repeat dose requires Medical Control.
- Maximum cumulative dose 10 mg.
- Paramedic only.

Ketamine - Adult BARS 7 IM

2-4 mg/kg IM; maximum 400 mg

- Primary medication for adult severe agitation with immediate danger.
- Do not delay treatment attempting vascular access.
- Do not routinely administer prophylactic midazolam after ketamine.
- If dangerous agitation persists after 10 minutes, contact Medical Control before repeat ketamine or another sedative.
- Paramedic only.

Ketamine - Adult BARS 7 IV/IO

1 mg/kg IV/IO slowly; maximum 100 mg

- Use only when vascular access is already safely established.
- Do not attempt IV access in a violently struggling patient solely to use this route.
- Paramedic only.

Midazolam - Adult BARS 7 Alternative

10 mg IM preferred; 10 mg IN only when IM is not feasible

- Use when ketamine is unavailable or contraindicated.
- Age 65 or older, frail, or increased respiratory risk: 5 mg IM.
- Do not routinely combine with ketamine.
- Paramedic only.

Pediatric BARS 7 Chemical Sedation

Direct Medical Control order required; Medical Control specifies medication and dose

- Applies to every patient younger than 16 years with BARS 7 severe agitation.
- Continuous ECG, SpO_2, waveform $EtCO_2$, airway readiness, and emergency-department transport are required.

Provider-level care

Provider Actions Use explicit provider-level roles while maintaining one coordinated safety plan.

EMT: Establish scene safety, request law enforcement and additional personnel when indicated, and withdraw when responders cannot safely engage.

- Use verbal and environmental de-escalation, assess BARS, and assist with approved physical restraint only when necessary to prevent immediate harm or permit emergency care.
- Position the patient supine with the head elevated or lateral when clinically appropriate. Maintain airway positioning, oxygenation, suction, BVM readiness, direct observation, SpO_2, restraint checks, and serial vital signs.
- Assess glucose, temperature, trauma, oxygenation, neurologic findings, medication or substance exposure, and other reversible causes as soon as safely possible.

AEMT: Perform all EMT care. Establish IV/IO access only when safely achievable and clinically indicated; do not provoke additional struggle or delay rapid control attempting vascular access.

- Administer IV fluid only for hypotension, poor perfusion, or suspected heat illness. Follow the applicable shock or environmental protocol and reassess for fluid overload risk.
- Assist ventilation, continuous monitoring, restraint reassessment, active cooling, and rapid transport. AEMTs do not administer chemical sedation under UP-18.

Paramedic: Perform all prior care. Select the BARS 6 or BARS 7 medication pathway, administer the approved sedative, and direct post-sedation airway, ventilation, hemodynamic, temperature, and neurologic assessment.

- Apply continuous ECG, SpO_2, and waveform $EtCO_2$ as soon as safely possible after parenteral sedation. Keep suction, oxygen, BVM, and advanced-airway equipment immediately available.
- Do not administer a sedative for convenience, punishment, to force cooperation with a nonessential procedure, or to facilitate arrest or law-enforcement custody.

BARS 6 - Moderate Agitation Use de-escalation first when feasible; medication is intended to achieve safe calm while preserving airway and arousability.

EMT: Continue one-speaker verbal de-escalation, reduce noise and spectators, maintain personal space and a clear exit, and avoid provocative procedures.

• Prepare monitoring, airway equipment, approved restraints, and a coordinated team approach before medication when circumstances allow.

AEMT: Obtain IV/IO access only when already tolerated or clinically necessary. Do not delay IM medication attempting vascular access.

• Provide continuous physiologic monitoring and support after medication; no AEMT chemical-sedation administration.

Paramedic: Adult: midazolam 2.5 mg IV/IO or 5 mg IM/IN. If dangerous agitation persists, repeat once after 5 minutes following complete airway and hemodynamic reassessment. Maximum 5 mg IV/IO or 10 mg IM/IN.

• Adult age 65 or older, frail, significant respiratory disease, or suspected CNS-depressant intoxication: midazolam 1-2.5 mg IV/IO or 2.5 mg IM/IN. Repeat once only after reassessment; maximum cumulative dose 5 mg.

• Pediatric patient younger than 16: midazolam 0.1 mg/kg IV/IO or 0.2 mg/kg IM/IN, maximum initial dose 5 mg. Contact Medical Control when feasible before administration; a repeat dose requires Medical Control. Maximum cumulative dose 10 mg.

BARS 7 - Severe Agitation With Immediate Danger Use rapid coordinated control to end dangerous exertion and permit emergency assessment and transport.

EMT: Request adequate personnel and law enforcement, remove bystanders and hazards, prepare approved restraints and monitoring, and execute one coordinated plan.

• Do not delay an immediately necessary safety intervention to obtain routine measurements. Begin medical assessment as soon as control permits.

AEMT: Assist the coordinated restraint and immediate airway, ventilation, circulatory, temperature, glucose, trauma, and toxidrome assessment.

• Do not attempt vascular access in a violently struggling patient. Establish access after control only when a clinical indication remains.

Paramedic: Primary adult treatment: ketamine 2-4 mg/kg IM, maximum 400 mg. Do not delay treatment attempting IV access.

• If IV/IO access is already safely established: ketamine 1 mg/kg IV/IO slowly, maximum 100 mg.

• If ketamine is unavailable or contraindicated: midazolam 10 mg IM is preferred. If IM administration is not feasible, midazolam 10 mg IN may be considered. For patients age 65 or older, frail, or at increased respiratory risk, use midazolam 5 mg IM.

• Do not routinely combine ketamine and midazolam or administer prophylactic midazolam after ketamine. If dangerous agitation persists 10 minutes after ketamine, contact Medical Control before repeating ketamine or adding another sedative.

• For every pediatric BARS 7 patient, contact Medical Control and obtain a direct order before chemical sedation. Medical Control will specify the medication and dose.

Physical Restraint Safety Restraint is a medical safety intervention and must remain the least restrictive method that protects the patient and responders.

EMT: Use only approved soft restraints that can be rapidly removed. Secure the patient to the stretcher frame according to agency training, never to movable side rails.

• Position supine with the head elevated or lateral when clinically appropriate. Never restrain or transport prone, hog-tied, with hands and feet tied together behind the back, or with pressure that constricts the neck, chest, diaphragm, or airway.

• Do not place hands, knees, equipment, backboards, mattresses, or body weight on the patient's neck, chest, back, or abdomen.

• Assess airway, breathing, circulation, distal neurovascular status, restraint security, skin, and position at least every 5 minutes and whenever the patient or restraint position changes.

• Never use an oxygen mask as a spit-control device. If an agency-approved breathable spit hood is used, maintain continuous airway observation and remove it immediately for vomiting, respiratory compromise, or airway concern.

AEMT: Continue all restraint assessments and transition to the least restrictive safe method when the clinical condition permits.

Paramedic: Direct restraint positioning and physiologic reassessment. If rigid law-enforcement restraints must remain, require an accompanying officer with immediate access to the key and transition to approved soft restraints when safely possible.

Monitoring, Transport, and Documentation Every restraint or sedation encounter requires close physiologic monitoring, emergency-department transport, and quality review.

EMT: Maintain continuous direct observation. Obtain two complete vital-sign sets when feasible and repeat airway, respiratory rate and effort, blood pressure, mental status, BARS, position, and distal neurovascular checks every 5 minutes until stable.

• Document the behavior creating immediate danger, initial and repeat BARS, de-escalation attempts, personnel involved, restraint type and position, assessment findings, and response.

AEMT: Document vascular access, fluids, cooling, physiologic trends, adverse events, and every restraint reassessment.

Paramedic: After parenteral sedation, maintain continuous ECG, SpO_2, and waveform $EtCO_2$. Document medication, dose, route, exact time, indication, response, repeat assessment, and any airway or hemodynamic intervention.

• Transport every physically restrained or chemically sedated patient to an emergency department with early notification. Interfacility sedation remains governed by the separate transfer policy.

• Submit every physical-restraint or chemical-sedation encounter for medical-director or quality-assurance review.

Warnings

• Midazolam and ketamine can cause apnea, hypoventilation, airway obstruction, vomiting, hypersalivation, laryngospasm, or hemodynamic change. Maintain immediate suction, BVM, oxygen, and advanced-airway readiness.

• Do not routinely combine ketamine and midazolam. Additional sedation after the approved pathway requires Medical Control.

• Flumazenil should not be used routinely for post-midazolam respiratory depression; support airway and ventilation.

• Never use prone, hog-tie, neck/chest-compression, sandwich, or backboard/mattress restraint techniques.

• Continued physical struggle may worsen hyperthermia, acidosis, rhabdomyolysis, dysrhythmia, and sudden deterioration; coordinate rapid safe control and prompt medical assessment.

Clinical pearls

• BARS is an assessment and reassessment tool, not an automatic medication order.

• One calm speaker, fewer stimuli, personal space, and safe choices may prevent escalation when the patient is not an immediate danger.

• Lorazepam remains on the formulary but is not part of the primary UP-18 agitation algorithm. Droperidol and haloperidol are not included.

• Ketamine administration alone is not an indication for endotracheal intubation; intubate only for clinical airway or ventilation failure or another independent indication.

• Exact medication times, serial BARS scores, ventilation findings, $EtCO_2$, positioning, and adverse events are essential for safe handoff and quality review.

Special populations

Pediatrics

• Pediatric patients are younger than 16 years and medication dosing uses actual or length-based weight.

• Evaluate for ingestion, hypoglycemia, hypoxia, fever or infection, seizure, trauma, developmental disability, abuse, and an unsafe environment.

Older or Frail Adults

• Give particular attention to delirium, infection, stroke, hypoxia, medication toxicity, urinary retention, pain, and occult trauma.

Pregnancy

• Use de-escalation and the least restrictive safe intervention whenever possible. Maternal and fetal safety may require immediate control when violent agitation creates danger.

• Contact Medical Control when feasible, but do not permit prolonged dangerous struggle while awaiting consultation in an adult with immediate danger.

• Position with left uterine displacement when gestational age and circumstances make aortocaval compression possible.

Law-Enforcement Custody

• EMS sedation is a medical decision and must never be administered solely to facilitate arrest, interrogation, transport to jail, or law-enforcement compliance.

UP-19
Severe Agitation - Merged Into UP-18

Revised October 31, 2026

Retired entry - use UP-18

Overview

Use the BARS 7 severe-agitation branch in UP-18 for medication, restraint, monitoring, transport, and documentation requirements.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Use UP-18
EMT / AEMT / Paramedic
BARS 7 severe agitation and all historical UP-19 presentations now follow the combined UP-18 pathway

Indications

• Historical UP-19 link, bookmark, or reference.

Provider-level care

Compatibility Redirect

EMT: Open UP-18 Behavioral Agitation and Severe Agitation and follow the applicable BARS branch.

AEMT: Open UP-18 Behavioral Agitation and Severe Agitation and follow the applicable BARS branch.

Paramedic: Open UP-18 Behavioral Agitation and Severe Agitation. Do not use historical UP-19 medication dosing.

Warnings

• UP-19 is retired and merged into UP-18. No independent medication or fluid standing orders. Do not use the historical standalone pathway; all current agitation care and dosing are contained in UP-18.

Clinical pearls

• UP-19 is retired; use UP-18 for all agitation care.

Related pathways

• UP-18 Behavioral Agitation and Severe Agitation (p. 57): Current replacement protocol; UP-19 has no independent standing orders.

UP-20
Sickle Cell Crisis

Revised October 31, 2026

Overview

Use this protocol for a patient with known or suspected sickle cell disease and acute pain or illness. Sickle cell trait alone does not establish a vaso-occlusive crisis.

Do not assume every complaint is an uncomplicated pain crisis. Screen for acute chest syndrome, stroke, fever or sepsis, shock or severe anemia, splenic sequestration, priapism, pregnancy complications, dehydration, and pain that differs from the patient's usual crisis.

Treat pain promptly and respectfully while preserving airway, ventilation, perfusion, neurologic assessment, and rapid access to definitive care.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Confirm Sickle Cell Disease + Assess
EMT / AEMT / Paramedic
Usual crisis pattern • pain • chest symptoms • fever • neurologic findings • perfusion • pregnancy • priapism

↓

2. Life-Threatening Complication?
EMT / AEMT / Paramedic
Acute chest syndrome • stroke • sepsis • shock/severe anemia • splenic sequestration • priapism ≥4 hours

↓

3. Support Oxygenation + Temperature
EMT / AEMT / Paramedic
Oxygen for hypoxemia or respiratory distress • target SpO_2 ≥95% unless documented lower baseline • keep warm

Continue at step 4 in the next column/page

Quick flow continued at step 4

4. Treat Pain Promptly
EMT / AEMT / Paramedic
Nonpharmacologic care • adult UP-11 • approved pediatric weight-based pathway • reassess pain and sedation

↓

5. Fluids Only When Indicated
AEMT / Paramedic
No routine bolus • hypotension/poor perfusion: adult 250 mL or pediatric 10 mL/kg isotonic aliquots with reassessment

↓

6. Rapid Transport When High Risk
EMT / AEMT / Paramedic
Chest symptoms • hypoxemia • neurologic deficit • fever ≥38.5°C • shock/anemia • pregnancy • atypical or uncontrolled pain

Indications

- Known or suspected sickle cell disease with acute pain, fever, respiratory symptoms, neurologic symptoms, weakness, syncope, priapism, pregnancy concern, or other acute illness.
- Concern for a sickle cell complication even when the presenting pain resembles a prior vaso-occlusive crisis.

Assessment

Focused history

- Confirm sickle cell disease type when known, prior acute chest syndrome or stroke, baseline SpO_2, transfusion history, usual pain location and severity, and the patient's previously effective and tolerated treatment.
- Determine onset, location, severity, and character of pain and whether this episode differs from the patient's usual crisis.
- Ask about fever, chills, cough, chest pain, dyspnea, focal neurologic symptoms, syncope, reduced intake, vomiting or diarrhea, pregnancy, abdominal swelling, and priapism duration.
- Review allergies, analgesics already taken, hydroxyurea or other disease-modifying therapy, anticoagulants, recent hospitalization, and indwelling vascular access.

Focused examination and monitoring

- Obtain two complete sets of vital signs when feasible, including temperature, SpO_2, mental status, pain score, respiratory effort, and perfusion.
- Assess lung sounds, work of breathing, skin color and temperature, hydration, pulse quality, neurologic status, and the painful area for an alternate cause.
- Examine the abdomen for tenderness or enlargement when severe anemia or splenic sequestration is suspected; do not delay transport for a prolonged examination.
- Use continuous pulse oximetry for chest or respiratory symptoms. Add cardiac monitoring, 12-lead ECG, and waveform $EtCO_2$ when indicated by the presentation or treatment.

Treatment

1. Perform Universal Patient Care and immediately address airway, ventilation, hypoxemia, shock, altered mental status, seizure, stroke findings, or another life threat.
2. Do not withhold analgesia because the patient appears calm, has recurrent crises, or has received opioid medication previously. Use the patient's known effective regimen only when it is consistent with the approved protocol, formulary, and current clinical condition.
3. For suspected acute chest syndrome, support oxygenation and ventilation, apply continuous SpO_2 and cardiac monitoring, obtain a 12-lead ECG when indicated, notify early, and transport rapidly without excessive fluid.
4. For temperature 38.5°C / 101.3°F or higher, provide supportive sepsis care and prompt transport. No field antibiotic is authorized under this protocol.
5. Strongly recommend transport for acute chest symptoms, hypoxemia, neurologic deficit, fever, hypotension, poor perfusion, suspected severe anemia or sequestration, prolonged priapism, pregnancy concern, uncontrolled or atypical pain, or any opioid or ketamine administration.

Provider-level care

Provider-Level Actions Recognize complications, begin supportive care, treat pain early, and avoid unnecessary delay or excess fluid.

EMT: Perform XABCDE, obtain two complete sets of vital signs when feasible, assess pain, temperature, SpO_2, mental status, perfusion, and the patient's usual crisis pattern and prior complications.

• Ask specifically about chest pain, cough, fever, dyspnea, new weakness or speech change, syncope, pregnancy, abdominal fullness, and priapism duration when applicable.

• Keep the patient warm, position for comfort, provide oxygen for hypoxemia or respiratory distress, and support ventilation when needed.

• Use approved nonpharmacologic and EMT pain measures. Do not dismiss or delay care because the patient appears calm or has frequent EMS encounters.

AEMT: Perform all EMT care. Establish IV access when indicated without delaying transport or analgesia.

• Do not give a routine crystalloid bolus for uncomplicated vaso-occlusive pain. For hypotension or poor perfusion, administer isotonic crystalloid in reassessed aliquots: adult 250 mL or pediatric 10 mL/kg.

• For suspected dehydration without hypotension, use only KVO or maintenance-rate fluid during transport and reassess for pulmonary congestion.

• Administer approved AEMT analgesia under the applicable adult or pediatric pain pathway and monitor treatment response.

Paramedic: Perform all prior care. Treat pain promptly: use UP-11 for age 16 or older and PM-04 for age 2 through younger than 16. For a child younger than 2, provide nonpharmacologic care, obtain Medical Control direction for analgesia within scope, and do not use an adult-dose fallback. PM-04 authorizes no additional AEMT standing-order analgesic beyond its EMT measures. Follow PM-04 medication, route, dose, contraindication, monitoring, and repeat-dose limits.

• For opioid or ketamine analgesia, continuously monitor airway, respiratory effort, SpO_2, cardiac rhythm, and waveform $EtCO_2$ when feasible; keep suction and BVM immediately available.

• Obtain a 12-lead ECG for adult chest pain, dyspnea, syncope, abnormal rhythm, or another cardiac concern without delaying transport.

• Provide early receiving-facility notification for acute chest syndrome, stroke findings, fever, shock, suspected severe anemia or sequestration, prolonged priapism, pregnancy concern, or an atypical crisis.

High-Risk Complications A normal appearance or familiar pain complaint does not exclude a time-critical complication.

EMT: Acute chest syndrome: suspect with chest pain, cough, fever, dyspnea, abnormal lung sounds, or new hypoxemia. Provide oxygen and ventilation support as indicated and transport rapidly.

• Stroke: treat any new focal deficit, speech change, severe disequilibrium, vision change, or altered mental status under the suspected-stroke pathway.

• Fever: temperature 38.5°C / 101.3°F or higher is high risk for invasive infection and requires prompt emergency-department evaluation.

• Splenic sequestration or severe anemia: suspect with weakness, pallor, tachycardia, abdominal fullness or splenic enlargement, syncope, hypotension, or poor perfusion.

• Priapism: duration four hours or longer is a time-critical emergency; severe pain or a rapidly progressing episode also requires urgent evaluation.

AEMT: For hypotension or poor perfusion, provide small reassessed isotonic fluid aliquots and stop or slow fluid if respiratory distress, crackles, or worsening oxygenation develops.

Paramedic: Coordinate rapid specialty-capable transport and early notification. Provide cause-specific airway, shock, seizure, stroke, or cardiac care under the applicable Claiborne protocol.

Warnings

- Do not allow a familiar pain presentation to obscure acute chest syndrome, stroke, sepsis, shock, severe anemia, splenic sequestration, or another emergency.
- Avoid excessive crystalloid; overhydration can worsen pulmonary complications.
- Opioid or ketamine analgesia requires continuous airway, respiratory, oxygenation, cardiac, and $EtCO_2$ monitoring as specified by the applicable pain protocol.
- Do not use stigmatizing language such as 'drug-seeking' in the clinical record. Document objective findings, reported pain, treatment, response, and safety concerns.

Clinical pearls

- Acute chest syndrome may initially resemble pneumonia or an uncomplicated pain crisis and can progress rapidly.
- Pain behavior varies widely. A calm appearance does not reliably measure pain severity.

- Priapism lasting four hours or longer is an emergency because delay increases the risk of permanent dysfunction.

Special populations

Pediatric patients

- For Claiborne clinical protocols, pediatric medication dosing applies to patients younger than 16 years. Use actual or length-based weight.
- Children may deteriorate rapidly from infection, acute chest syndrome, aplastic crisis, or splenic sequestration. Fever 38.5°C / 101.3°F or higher requires prompt emergency evaluation.

Pregnancy

- Use left lateral positioning when appropriate, monitor closely for hypoxemia and shock, and coordinate obstetric-capable destination and early notification.

Related pathways

- Open UP-11 Adult Pain Control (p. 31): Use the approved adult analgesia pathway for patients 16 years or older.

UP-21
Family Violence / Abuse

Revised October 31, 2026

Overview

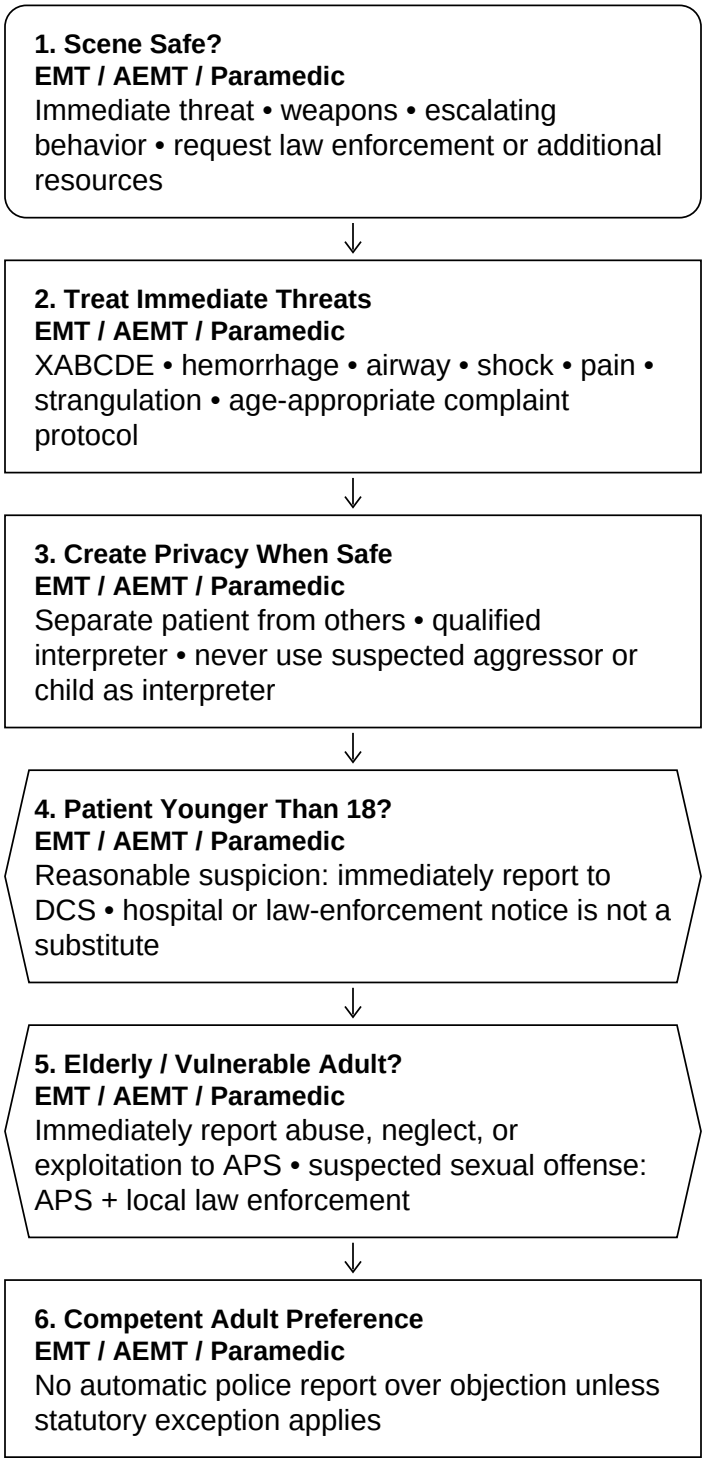
Use this protocol for suspected intimate-partner or family violence, child abuse or neglect, elderly or vulnerable-adult abuse or exploitation, caregiver abuse, or human trafficking.

Immediate patient care and scene safety come first. Provide private, trauma-informed, nonjudgmental care and use the appropriate trauma, medical, pediatric, behavioral, obstetric, or sexual-assault protocol concurrently.

EMS recognizes, treats, protects, documents, and reports. EMS does not investigate, confront a suspected offender, or attempt to prove that abuse occurred.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.



Continue at step 7 in the next column/page

Quick flow continued at step 7

7. Preserve + Document
EMT / AEMT / Paramedic
Exact words • objective findings • who was present • scene conditions • evidence handled • report details

↓

8. Safe Disposition
EMT / AEMT / Paramedic
Private handoff • required reports • do not leave child or vulnerable adult in an unsafe plan

Indications

- Patient disclosure of violence, abuse, neglect, exploitation, coercive control, trafficking, or fear of a household member, caregiver, partner, or other person.
- Injury pattern, history, behavior, living condition, caregiver interaction, or scene finding that creates reasonable suspicion for abuse, neglect, exploitation, or trafficking.
- Any child, elderly adult, or vulnerable adult for whom a mandatory safeguarding report may be required.

Contraindications

- Do not confront or accuse the suspected offender, conduct an independent investigation, or delay emergency treatment to collect evidence.
- Do not promise secrecy when a mandatory report may be required.

Assessment

Safety and immediate medical assessment

- Identify immediate threats, weapons, escalating behavior, unsafe persons, children or dependents at risk, and the need for law enforcement or additional resources.
- Perform XABCDE and assess for hemorrhage, head injury, strangulation, chest or abdominal injury, fracture, pregnancy complications, overdose, intoxication, neglect, dehydration, hypothermia, and psychiatric emergency.
- When safe, separate the patient from household members, caregivers, or a suspected aggressor and ask whether the patient feels safe returning to the current environment.

Trauma-informed inquiry

- Explain that questions are asked because violence and abuse affect health and safety.
- Use broad, non-leading prompts such as: 'Tell me what happened,' 'Has anyone hurt or threatened you?' and 'Do you feel safe where you live?'
- Do not pressure a patient to disclose, repeat a detailed account, identify an offender, or make an immediate law-enforcement decision unless a mandatory report applies.
- Use the patient's own words in quotation marks when possible and distinguish statements from EMS observations.

Possible strangulation

- Assess voice change, dysphagia, dyspnea, neck pain or tenderness, petechiae, loss of consciousness, incontinence, seizure, confusion, focal neurologic symptoms, and memory gaps.
- Visible neck injury is not required for serious airway or vascular injury. Strongly recommend emergency-department evaluation even when initial findings are minimal.
- Treat strangulation as a high-risk mechanism and follow the mandatory injury-reporting pathway.

Treatment

1. Create privacy when safe, use a qualified interpreter, and use trauma-informed, non-leading questions only to the extent needed for medical care, safety, and reporting.
2. Treat associated illness or injury under the appropriate Claiborne protocol and preserve evidence when doing so does not compromise care.
3. For a competent adult experiencing intimate-partner violence, respect the patient's decision regarding police involvement unless a statutory injury or other mandatory-reporting exception applies.
4. Strongly recommend emergency-department transport after strangulation, significant injury, altered mental status, pregnancy concern, overdose, suicidal risk, or an unsafe disposition.
5. A competent adult may refuse transport when capacity is intact, but refusal does not cancel a mandatory report. Do not leave a child or vulnerable adult solely under an unsafe caregiver refusal; request law enforcement, DCS or APS, Medical Control, and the agency safeguarding chain.
6. Provide a private handoff and document objective findings, exact statements, consent, capacity, evidence handling, safety concerns, reporting decisions, notifications, recipient, time, and reference number.

Provider-level care

Provider-Level Actions All providers share responsibility for recognition, immediate safety, compassionate care, objective documentation, and required reporting.

EMT: Treat immediate life threats and request law enforcement or additional resources when danger is present.

• When safe, speak with the patient privately. Use calm, simple, nonjudgmental, non-leading questions and avoid repeated questioning.

• Do not use the suspected aggressor or a child as interpreter. Request a qualified interpreter when needed.

• Preserve potential evidence, document objective findings and the patient's exact statements, and ensure the required reporting pathway is completed.

AEMT: Perform all EMT actions. Provide authorized vascular access, fluid, medication, and monitoring care for associated illness or injury.

• Reassess for occult injury, shock, overdose, strangulation, pregnancy-related concerns, and medical causes of altered behavior.

• Do not delay transport or a required report to obtain a detailed history or complete documentation on scene.

Paramedic: Perform all prior care. Lead assessment for serious occult injury, strangulation, head injury, cardiac or neurologic complications, pregnancy emergency, overdose, and need for specialty destination.

• Contact Medical Control for questionable capacity, high-risk refusal, conflict between patient safety and caregiver or custodial demands, or an unsafe proposed disposition.

• Provide a private receiving-facility handoff and document the agency contacted, recipient, time, report or reference number, and any safety plan.

Tennessee Reporting Pathways Reporting duties depend on patient age, vulnerability, suspected offense, injury, capacity, and transport disposition.

EMT: Child younger than 18: when reasonable suspicion exists, immediately report suspected abuse, neglect, exploitation, or child sexual abuse to Tennessee DCS at 1-877-237-0004. Ensure the report is made; hospital or law-enforcement notification alone does not replace the duty.

• Elderly or vulnerable adult: immediately report suspected abuse, neglect, or exploitation to Tennessee APS at 1-888-277-8366.

• Suspected rape, aggravated rape, sexual battery, or aggravated sexual battery of an elderly or vulnerable adult: report to both APS and local law enforcement.

• Competent adult intimate-partner violence: do not automatically release identifying information to law enforcement over the patient's objection unless a statutory exception applies.

• Life-threatening injury, strangulation, or injury involving a knife, firearm, or other deadly weapon follows the mandatory injury-reporting pathway. For an injury covered by Tennessee Code §38-1-101, complete and document the EMS notification step: when transporting, privately tell the receiving physician or emergency-department staff the suspected cause of injury; when not transporting, report the result of the call to the 911 center through the approved agency channel. This does not replace separate required child-protection, adult-protection, or other legally required reports. Follow the approved safeguarding policy when reporting duties overlap.

AEMT: Support immediate reporting and continue patient care. Do not assume another responder or the receiving hospital has completed the clinician's duty unless confirmation is obtained and documented.

Paramedic: Resolve uncertainty through the agency safeguarding chain, Medical Control, and law-enforcement or protective-service consultation without delaying an immediately required report.

Warnings

- Patient safety and emergency medical care take priority over evidence preservation.
- A patient's refusal to disclose details or involve law enforcement does not imply that abuse did not occur.
- Do not use a suspected offender, coercive caregiver, or minor child as interpreter.
- Age alone does not remove an adult patient's autonomy. Apply the statutory elderly or vulnerable-adult criteria and assess capacity individually.
- Do not leave a child or vulnerable adult in an unsafe environment solely because a caregiver refuses transport or minimizes the concern.

Clinical pearls

- EMS needs reasonable suspicion-not proof-to initiate a mandatory child or vulnerable-adult report.
- Abuse may coexist with medical illness, disability, cognitive impairment, substance use, pregnancy, or financial dependence.
- Objective documentation and the patient's exact words are more useful than conclusory labels.

- Strangulation can produce delayed airway, vascular, and neurologic complications despite few or no visible findings.

Special populations

Children and adolescents

- The mandatory child-reporting threshold is younger than 18 years, even though Claiborne pediatric medication dosing generally applies to patients younger than 16 years.
- Use developmentally appropriate, non-leading questions. Do not repeatedly interview the child or ask the child to confront the suspected offender.

Elderly and vulnerable adults

- Assess decision-making capacity individually. Age, disability, or dependence does not automatically remove the right to participate in care decisions.

Related pathways

- Call Tennessee DCS - tel:+18772370004: Child Abuse Hotline: 1-877-237-0004
- Call Tennessee APS - tel:+18882778366: Adult Protective Services: 1-888-277-8366

UP-22
Sexual Assault

Revised October 31, 2026

Overview

Provide immediate medical care, safety, privacy, patient control, and trauma-informed support after a suspected or disclosed sexual assault.

A competent adult may decline any assessment or treatment step and, unless a mandatory exception applies, may object to release of identifying information to law enforcement.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Scene Safe + Private?
EMT / AEMT / Paramedic
Separate from suspected assailant • request law enforcement for immediate danger • qualified interpreter

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. Treat Immediate Threats
EMT / AEMT / Paramedic
XABCDE • hemorrhage • head injury • strangulation • intoxication/drugging • pregnancy • suicidal risk

↓

3. Consent for Each Step
EMT / AEMT / Paramedic
Explain before touching • obtain permission • patient may pause or decline any nonemergent step

↓

4. Patient Younger Than 18?
EMT / AEMT / Paramedic
Immediately report reasonable suspicion to DCS

↓

5. Elderly / Vulnerable Adult?
EMT / AEMT / Paramedic
Report suspected sexual offense to APS and local law enforcement

↓

6. Competent Adult + Mandatory Exception?
EMT / AEMT / Paramedic
Life-threatening injury • strangulation • knife, firearm, or other deadly weapon

↓

7. Preserve Evidence Without Delaying Care
EMT / AEMT / Paramedic
Minimal handling • separate paper bags • no forensic/internal exam or swabs • necessary emergency external assessment with consent

↓

8. Appropriate SANE / Forensic Destination
EMT / AEMT / Paramedic
Medical stability first • private handoff • document consent, reports, evidence, and patient preference

Indications

- Patient disclosure of sexual assault or concern that sexual contact occurred without consent.

- Concern for sexual assault based on injury, altered consciousness, drug-facilitated assault, caregiver or witness information, or scene findings.
- Request for emergency medical evaluation after a suspected sexual assault, regardless of whether the patient wishes to involve law enforcement.

Contraindications

- Do not condition medical care or transport on the patient's willingness to speak with law enforcement or undergo a forensic examination.

Assessment

Immediate medical and safety assessment

- Assess scene safety, ongoing threat, need for law enforcement, airway, breathing, circulation, hemorrhage, head or facial injury, strangulation, chest or abdominal trauma, intoxication, overdose, and altered mental status.
- Assess for pregnancy-related emergency, suicidal thoughts, severe anxiety or dissociation, and the need for a support person or advocate chosen by the patient when safe and available.
- Consider drug-facilitated assault when there is unexpected sedation, memory loss, unexplained intoxication, or concern for covert drug administration. Provide time-sensitive receiving-facility notification.

Consent and focused history

- Ask only what is necessary to guide emergency care, evidence-preservation advice, destination, and mandatory reporting.
- Use the patient's exact words when documenting statements. Avoid judgmental language, conclusions, and repeated requests for a detailed account.

Possible strangulation

- Assess voice change, dysphagia, dyspnea, neck pain or tenderness, petechiae, loss of consciousness, incontinence, seizure, confusion, focal neurologic symptoms, and memory gaps.
- Visible injury is not required for serious airway or vascular injury. Strongly recommend emergency-department evaluation and follow the mandatory injury-reporting pathway.

Treatment

1. Ensure scene safety and treat immediate airway, breathing, circulation, hemorrhage, shock, head injury, strangulation, overdose, pregnancy emergency, or suicidal risk without delay.
2. Preserve possible evidence without delaying care. When practical, advise the patient not to wash, change clothing, eat, drink, urinate, or brush teeth, but never withhold comfort or necessary treatment.
3. If clothing is removed, minimize handling and place each item separately in a clean paper bag; do not use plastic for damp evidence. Document every item handled and transfer it according to agency evidence policy.
4. For a competent adult, respect the patient's choice about law-enforcement involvement unless a mandatory reporting exception applies. Explain that medical and forensic evaluation may still be available without an immediate police report.
5. Transport to a medically appropriate SANE or forensic-capable facility when available. Medical stability takes priority, and the final locally approved facility list remains part of the destination plan.
6. For an injury covered by Tennessee Code §38-1-101, complete and document the EMS notification step: when transporting, privately tell the receiving physician or emergency-department staff the suspected cause of injury; when not transporting, report the result of the call to the 911 center through the approved agency channel. This does not replace separate required child-protection, adult-protection, or other legally required reports. Follow the approved safeguarding policy when reporting duties overlap. Document capacity, options, warnings, reports and the patient's decision.
7. Treat pain, nausea, shock, and other medical problems under the applicable Claiborne protocol. Pregnancy prevention, STI prophylaxis, and HIV post-exposure prophylaxis are receiving-facility responsibilities.

Provider-level care

Provider-Level Actions Protect safety, consent, privacy, evidence, and access to appropriate medical and forensic care.

EMT: Treat immediate threats and obtain only the history needed for medical care, safety, destination, and mandatory reporting.

• Explain each action before touching the patient, obtain consent throughout care, and allow the patient to pause or decline nonemergent assessment or treatment.

• Do not perform a forensic genital examination, internal pelvic or rectal examination, or forensic swabs under this protocol. When uncontrolled hemorrhage, traumatic injury, or severe pain makes external assessment necessary for emergency care,

explain the need and obtain consent; expose only the area needed, preserve privacy and provide indicated trauma care. A patient with decision-making capacity may refuse all or part of the examination; honor that refusal. If capacity is absent and delaying necessary emergency care would risk serious harm, use only the applicable emergency-consent or minor-consent pathway. An emergency exception does not override a capable refusal.

• Preserve clothing or other potential evidence when feasible and transport to an appropriate medical and forensic-capable facility.

AEMT: Perform all EMT actions. Provide authorized vascular access, fluid, analgesia, antiemetic, and monitoring care for associated illness or injury.

• Do not delay transport or compromise potential evidence for nonessential procedures.

• Continue consent checks and explain that medical treatment is available regardless of whether the patient chooses immediate law-enforcement involvement.

Paramedic: Perform all prior care. Assess and treat serious injury, strangulation, airway or respiratory compromise, shock, head injury, overdose or drug-facilitated assault, pregnancy emergency, and suicidal risk.

• Select a medically appropriate SANE or forensic-capable destination when available and notify the receiving emergency department privately.

• Contact Medical Control for significant injury, questionable capacity, high-risk refusal, destination uncertainty, or conflict involving consent or mandatory reporting.

Tennessee Reporting Pathways A competent adult's law-enforcement preference is respected unless a mandatory child, vulnerable-adult, or injury-reporting exception applies.

EMT: Patient younger than 18: immediately report reasonable suspicion of sexual abuse to Tennessee DCS at 1-877-237-0004.

• Elderly or vulnerable adult: report the suspected sexual offense to both Tennessee APS at 1-888-277-8366 and local law enforcement.

• Competent adult: do not release identifying information to law enforcement over the patient's objection unless a statutory exception applies, including life-threatening injury, strangulation, or injury involving a knife, firearm, or other deadly weapon.

• For an injury covered by Tennessee Code §38-1-101, complete and document the EMS notification step: when transporting, privately tell the receiving physician or emergency-department staff the suspected cause of injury; when not transporting, report the result of the call to the 911 center through the approved agency channel. This does not replace separate required child-protection, adult-protection, or other legally required reports. Follow the approved safeguarding policy when reporting duties overlap.

AEMT: Support immediate required reports while continuing patient care and preserving privacy from bystanders, family, and nonessential personnel.

Paramedic: Resolve uncertainty through the agency safeguarding chain, Medical Control, and protective-service or law-enforcement consultation without delaying an immediately required report.

Warnings

- Medical care and patient safety take priority over evidence preservation.
- Do not promise confidentiality when a mandatory report applies. Explain what information must be shared and with whom.
- Do not pressure a competent adult to involve law enforcement, undergo a forensic examination, or accept any nonemergent procedure.
- Do not allow the suspected assailant, coercive companion, or a minor child to serve as interpreter.
- A normal external examination does not exclude assault, internal injury, strangulation injury, pregnancy risk, or infection risk.

Clinical pearls

- Trauma-informed care returns control to the patient through privacy, explanation, choice, consent, and avoidance of unnecessary repetition.
- An adult may be eligible for a medical-forensic examination and Tennessee hold-kit pathway without making an immediate police report. Do not promise exact examiner availability before confirming with the receiving facility.
- Potential DNA evidence may be affected by washing or clothing changes, but prior washing, eating, drinking, urination, or time elapsed does not eliminate the value of medical or forensic evaluation.
- Drug-facilitated assault toxicology is time-sensitive; early receiving-facility notification is important.

Special populations

Children and adolescents

- Reasonable suspicion involving a patient younger than 18 requires immediate DCS reporting. Use developmentally appropriate, non-leading questions and avoid repeated interviewing.

- Use pediatric clinical dosing for patients younger than 16 years and the appropriate adult clinical pathway for patients 16 or 17 years old while maintaining mandatory child-reporting requirements through age 17.

Elderly and vulnerable adults

- Report a suspected sexual offense to both APS and local law enforcement. Continue to involve the patient in decisions to the maximum extent supported by capacity.

Related pathways

- Call Tennessee DCS - tel:+18772370004: Child Abuse Hotline: 1-877-237-0004
- Call Tennessee APS - tel:+18882778366: Adult Protective Services: 1-888-277-8366

AC-01
Asystole / Pulseless Electrical Activity

Revised October 10, 2026

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Confirm Cardiac Arrest
EMT / AEMT / Paramedic
Unresponsive • absent or abnormal breathing • no definite pulse within 10 seconds • honor valid DNR/MOST and obvious-death criteria.

↓

2. Begin TEAM-Focused CPR
EMT / AEMT / Paramedic
Compressions 100-120/min, depth 2-2.4 in, full recoil, minimal pauses; change compressor every 2 minutes. Use 30:2 before an advanced airway.

↓

3. Confirm Nonshockable Rhythm
AEMT / Paramedic
Check rhythm every 2 minutes. Confirm asystole in 2 leads and exclude loose leads, low gain, and fine VF. Do not defibrillate or pace asystole/PEA.

↓

4. Access + Epinephrine
AEMT / Paramedic
Give epinephrine 1 mg IV/IO as soon as possible and repeat every 3-5 minutes. Make one rapid IV attempt, then proceed to IO if unsuccessful or delayed.

↓

5. Airway + Ventilation
EMT / AEMT / Paramedic
Two-person BVM first. Do not interrupt compressions for airway placement. After an advanced airway, continue compressions and give 1 breath every 6 seconds with waveform capnography.

Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Treat Reversible Causes
EMT / AEMT / Paramedic
Evaluate and treat hypovolemia, hypoxia, acidosis, potassium abnormality, hypothermia, tension pneumothorax, tamponade, toxins, and coronary or pulmonary thrombosis.

↓

7. Rhythm Changes?
EMT / AEMT / Paramedic / Medical Control
If VF/pulseless VT develops, transition immediately to AC-09. If ROSC occurs, transition to AC-10. Otherwise continue 2-minute cycles and consider AC-12 termination criteria.

Indications

- Patient 16 years of age or older in cardiac arrest with asystole or pulseless electrical activity.

Contraindications

- Valid DNR, MOST, or other recognized order directing that resuscitation not be attempted.
- Obvious-death findings or circumstances addressed by agency policy.
- A palpable pulse is present; manage the perfusing rhythm and underlying condition instead.

Assessment

Arrest Confirmation

- Assess responsiveness, breathing, and central pulse simultaneously; limit the pulse check to 10 seconds.

CPR Quality

- Keep rhythm and pulse checks under 10 seconds and change compressors approximately every 2 minutes.
- Use real-time audiovisual CPR feedback when available.

Treatment

1. Begin high-quality CPR immediately and follow the TEAM-Focused CPR pathway in AC-11.
2. Provide two-person BVM ventilation first. Maintain 30:2 before an advanced airway and avoid excessive ventilation.
3. Use changes in waveform EtCO2 as one component of assessment; an abrupt sustained rise may indicate ROSC. Never use a single EtCO2 value alone to terminate resuscitation.

4. Continue resuscitation on scene for most out-of-hospital arrests. Transport during CPR only for a defined special circumstance or after ROSC; apply AC-12 termination guidance when appropriate.
5. Use mechanical CPR only when high-quality manual compressions are unsafe or impractical and placement can occur with minimal interruption.

Medication orders

Epinephrine 1 mg/10 mL (0.1 mg/mL)

1 mg IV/IO as soon as possible; repeat every 3-5 minutes

- AEMT and Paramedic standing-order medication for adult nonshockable cardiac arrest.
- Flush the vascular line and continue CPR without interruption after administration.

Provider-level care

EMT Care Recognition, high-quality CPR, BLS airway care, AED/monitor support, and cause-directed basic care.

EMT: Confirm unresponsiveness, abnormal or absent breathing, and no definite pulse within 10 seconds; identify valid DNR/MOST or obvious-death criteria.

- Start TEAM-focused CPR immediately: 100-120 compressions/min, depth 2-2.4 inches, complete recoil, minimal interruptions, and compressor change every 2 minutes.
- Apply AED/monitor pads, follow rhythm prompts, and resume compressions immediately after every rhythm or pulse check.
- Use a two-person BVM with oxygen and an airway adjunct when available; maintain 30:2 until an advanced airway is in place.
- Assist with scene organization, medication timing, documentation, reversible-cause treatment, and safe movement only when indicated.

AEMT Care All EMT care plus vascular access, epinephrine, supraglottic airway, and quantitative capnography when available.

AEMT: Perform all EMT actions and confirm the rhythm is nonshockable; if apparent asystole, verify in 2 leads and correct lead, cable, and gain problems.

- Make one rapid IV attempt without interrupting CPR. If unsuccessful, infeasible, or delayed, establish IO access; use proximal tibia preferentially and humeral head only when tibial access is contraindicated.
- Give epinephrine 1 mg IV/IO as soon as possible and repeat every 3-5 minutes while arrest continues.
- Place a supraglottic airway without interrupting compressions when BVM ventilation is ineffective or an advanced airway will improve care.
- After advanced-airway placement, ventilate once every 6 seconds during continuous compressions and monitor waveform capnography when available.

Paramedic Care All AEMT care plus rhythm confirmation, definitive airway options, continuous waveform capnography, and advanced treatment of reversible causes.

Paramedic: Perform all EMT and AEMT actions; confirm PEA has no palpable pulse and confirm asystole in 2 leads, excluding fine VF.

- Choose supraglottic airway or endotracheal intubation according to patient and scene conditions; airway placement must not interrupt compressions.
- Use continuous waveform capnography after any advanced airway to confirm and monitor placement, assess ventilation, and follow trends in CPR effectiveness.
- Identify and treat reversible causes. Give calcium or sodium bicarbonate only for a specific suspected or confirmed indication, not routinely.
- Direct on-scene resuscitation, transition immediately to AC-09 for VF/pulseless VT or AC-10 after ROSC, and apply AC-12 when termination criteria are met.

Warnings

- Asystole and PEA are nonshockable. Do not defibrillate or pace unless the rhythm changes to an appropriate perfusing or shockable rhythm.
- Do not delay or interrupt CPR for vascular access, medication administration, airway placement, or patient movement.
- Do not routinely administer calcium, sodium bicarbonate, magnesium, or antiarrhythmics in asystole/PEA. Use only for a specific protocol-supported cause.
- Naloxone must not delay standard resuscitation in confirmed cardiac arrest.
- Do not use point-of-care ultrasound during arrest unless separately approved by the agency medical director and governed by a dedicated protocol.
- Do not use a single EtCO2 cutoff or any single finding as the sole basis for terminating resuscitation.
- For arrest-associated hypovolemia, use cause-directed fluid resuscitation; do not require a routine fixed-volume saline bolus.

Clinical pearls

- A sudden sustained increase in EtCO2 may signal ROSC; pause only for the scheduled brief rhythm and pulse check.
- Low or falling EtCO2 should prompt reassessment of compression quality, ventilation, airway placement, and reversible causes-not automatic termination.
- Moving a patient during active CPR commonly worsens compression quality and creates safety risk; favor on-scene resuscitation unless a special circumstance clearly supports transport.

Special populations

Suspected Opioid-Associated Arrest

- Prioritize CPR, ventilation, and standard cardiac-arrest care. Naloxone may be considered only when it does not delay or interrupt those interventions.

Pregnancy

- Begin standard resuscitation immediately. When the fundus is at or above the umbilicus, use continuous left uterine displacement, notify Medical Control early, and plan an obstetric-capable destination when clinically appropriate.

Hypothermia or Toxicologic Arrest

- Apply the relevant environmental or toxicology pathway because medication timing, resuscitation duration, and transport decisions may differ.

Related pathways

- AC-09 VF / Pulseless VT (p. 96): Use immediately if the rhythm becomes shockable.
- AC-10 Post Resuscitation (p. 99): Use after return of spontaneous circulation.
- AC-11 TEAM Focused CPR (p. 102): Team roles and high-performance CPR workflow.
- AC-12 Termination of CPR (p. 104): Apply when ongoing resuscitation remains unsuccessful.
- AO-03 OB-GYN Emergency (p. 140): Open for pregnancy-related causes, hemorrhage, seizure, or obstetric destination planning.

AC-02
Bradycardia With a Pulse

Revised October 10, 2026

Overview

Adult bradycardia pathway for patients 16 years of age and older. Bradycardia is typically a heart rate below 50/min, but treatment is determined by the patient's clinical condition rather than the rate alone.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Assess + Support
EMT / AEMT / Paramedic
Airway • oxygen for hypoxemia • assist ventilation as needed • monitor pulse, rhythm, SpO_2, and blood pressure • apply pacing/defibrillation pads.

↓

2. Cardiopulmonary Compromise?
EMT / AEMT / Paramedic
Hypotension • acutely altered mental status • shock • ischemic chest discomfort • acute heart failure. Confirm symptoms are caused by the bradycardia.

↓

3. No Compromise
EMT / AEMT / Paramedic
Monitor, obtain 12-lead ECG and glucose, identify and treat the cause, obtain 2 complete vital-sign sets, and transport as clinically indicated.

↓

4. Severe Instability / High-Grade Block?
Paramedic
If severe instability or high-grade AV block is present and vascular access is unavailable, begin transcutaneous pacing immediately while access is pursued.

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Atropine
Paramedic
Paramedic: atropine 1 mg IV/IO. Repeat every 3-5 minutes to a maximum total dose of 3 mg. Do not delay pacing when atropine is unlikely to be effective.

↓

6. Persistent Instability
Paramedic
Transcutaneous pacing and/or epinephrine infusion 2-10 mcg/min or dopamine infusion 5-20 mcg/kg/min; titrate to clinical response.

↓

7. Reassess + Transport
EMT / AEMT / Paramedic
Confirm mechanical capture, repeat ECG and vital signs, continue cause-directed care, and notify the receiving facility early.

Indications

- High-grade AV block or another bradyarrhythmia at risk for rapid deterioration.

Contraindications

- Do not provide rate-directed treatment solely for an asymptomatic low heart rate that is appropriate for the clinical condition.
- Do not use this pathway when the patient has no pulse; use the appropriate cardiac-arrest protocol.

Assessment

Clinical Stability

- Assess the relationship between symptom onset and the rhythm; a low heart rate may be normal in healthy adults, athletes, or during sleep.
- Continuously monitor pulse, blood pressure, rhythm, SpO_2, respiratory status, and mental status.

Rhythm + Cardiac Evaluation

- Acquire a 12-lead ECG when it will not delay urgent pacing, medication treatment, or transport.
- Identify high-grade AV block, wide-QRS escape rhythms, ischemia/infarction, pacemaker spikes without capture, and rhythm progression.
- Obtain serial ECGs and 2 complete vital-sign sets when feasible, including reassessment after each intervention.

Reversible Causes

- Hypoxia, myocardial ischemia or infarction, hyperkalemia or another electrolyte disturbance, acidosis, hypothermia, increased vagal tone, and pacemaker malfunction.
- Medication or toxicologic causes, including beta-blockers, calcium-channel blockers, digoxin, and other rate-slowing agents.

Treatment

1. Maintain a patent airway, provide oxygen for hypoxemia, assist ventilation as needed, and attach a cardiorespiratory monitor.
2. For transcutaneous pacing, set a rate of 60-80/min and increase current until electrical capture occurs. Confirm mechanical capture with a palpable pulse, improved blood pressure, pulse-ox waveform, or $EtCO_2$ response-not ECG appearance alone.
3. Identify and treat reversible causes and use the applicable ACS, CHF, hypothermia, or toxicology pathway when identified.
4. Begin transport after immediate stabilization measures, continue serial ECGs and vital signs, and notify the receiving facility early.

Medication orders

Atropine

1 mg IV/IO; repeat every 3-5 minutes to a maximum total dose of 3 mg

- Paramedic standing order for bradycardia causing cardiopulmonary compromise.
- Do not delay pacing when atropine is unlikely to be effective or the patient is severely unstable.

Epinephrine Infusion

2-10 mcg/min IV/IO infusion; titrate to clinical response

- Paramedic standing order for persistent unstable bradycardia when atropine is ineffective.
- Push-dose epinephrine is not included in this protocol.

Dopamine Infusion

5-20 mcg/kg/min IV/IO infusion; titrate to clinical response

- Paramedic standing order for persistent unstable bradycardia when atropine is ineffective.
- Taper slowly after clinical stabilization when appropriate.

Isotonic Crystalloid

250-500 mL IV/IO aliquots for hypotension when clinically appropriate; reassess after each aliquot

• Use smaller volumes or avoid additional fluid in pulmonary edema, CHF, renal failure, liver failure, or suspected volume overload.

Provider-level care

EMT Care BLS stabilization, monitoring, ECG acquisition, pacing-pad placement, and rapid transport.

EMT: Assess airway, breathing, circulation, mental status, perfusion, and whether the bradycardia is causing the patient's symptoms.

• Provide oxygen for hypoxemia and assist ventilation with BVM when respirations are inadequate.

• Apply cardiac monitor and pacing/defibrillation pads, monitor pulse, blood pressure, and SpO_2, and obtain blood glucose.

• Acquire and transmit a 12-lead ECG when it will not delay urgent treatment or transport.

• Obtain 2 complete vital-sign sets when feasible, assist the ALS provider, and initiate prompt transport for cardiopulmonary compromise or high-grade block.

AEMT Care All EMT care plus IV/IO access and cautious isotonic fluid support for appropriate hypotension.

AEMT: Perform all EMT actions and establish IV access without delaying pacing, ALS intercept, or transport.

• For severe instability, make one rapid IV attempt and proceed to IO if unsuccessful or access is otherwise infeasible or delayed.

• For hypotension when volume administration is clinically appropriate, give isotonic fluid in 250-500 mL aliquots with reassessment.

• Use smaller fluid volumes or avoid additional fluid when pulmonary edema, CHF, renal failure, liver failure, or volume overload is suspected.

• Prepare and assist with transcutaneous pacing, medication administration, serial ECGs, and continuous reassessment.

Paramedic Care All AEMT care plus rhythm interpretation, atropine, transcutaneous pacing, pacing analgesia/sedation, and vasopressor infusions.

Paramedic: Interpret the rhythm and identify sinus bradycardia, junctional escape, Mobitz I, Mobitz II, complete heart block, or pacemaker malfunction.

• Give atropine 1 mg IV/IO every 3-5 minutes to a maximum total dose of 3 mg when bradycardia is producing cardiopulmonary compromise.

• Do not delay transcutaneous pacing for atropine in severe instability, Mobitz II, complete heart block, or a new wide-QRS escape rhythm.

• If atropine is ineffective, use transcutaneous pacing and/or an epinephrine or dopamine infusion titrated to clinical response.

• Provide analgesia and sedation for pacing when the patient's condition permits, without delaying lifesaving pacing; use continuous cardiac, blood-pressure, respiratory, SpO_2, and $EtCO_2$ monitoring after sedation.

• Confirm and continually reassess mechanical capture, treat reversible causes, and coordinate early receiving-facility notification.

Warnings

• Treat the patient, not the monitor or a heart-rate number. Asymptomatic bradycardia generally does not require rate-directed treatment.

• Transcutaneous pacing is painful in a conscious patient; provide analgesia and sedation when feasible, but never delay lifesaving pacing.

Clinical pearls

• Correction of hypoxia, hyperkalemia, hypothermia, ischemia, or toxicologic causes may be more important than simply increasing the heart rate.

• Prepare for clinical deterioration and cardiac arrest whenever high-grade AV block, a wide escape rhythm, or recurrent loss of capture is present.

Special populations

Acute Coronary Syndrome

• Obtain serial 12-lead ECGs, avoid delaying stabilization, and use AC-04 when ischemia or infarction is suspected.

Medication or Toxicologic Bradycardia

• Use TE-07 and contact Poison Control for suspected beta-blocker, calcium-channel blocker, digoxin, or other toxic exposure; antidotal treatment may be required in addition to pacing and pressor support.

Hypothermia

• Use TE-05 because medication response, pacing response, handling, and transport priorities may differ in significant hypothermia.

Related pathways

• AC-04 Acute Coronary Syndrome / STEMI (p. 80): Use when myocardial ischemia or infarction is suspected.

• AC-05 CHF / Acute Pulmonary Edema (p. 83): Use when acute heart failure or pulmonary edema is present.

• TE-05 Hypothermia / Frostbite (p. 210): Use for significant hypothermia-associated bradycardia.

- TE-07 Overdose / Toxic Ingestion (p. 211): Use for medication- or toxin-associated bradycardia.

AC-03
Adult Cardiac Arrest

Revised October 10, 2026

Overview

Master adult cardiac-arrest pathway for patients 16 years of age and older. Use this protocol to begin coordinated resuscitation and route immediately to the appropriate rhythm-specific pathway.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Confirm Cardiac Arrest
EMT / AEMT / Paramedic
Unresponsive • absent or abnormal breathing • no definite pulse within 10 seconds • honor valid DNR/MOST and obvious-death criteria.

↓

2. Start TEAM-Focused CPR
EMT / AEMT / Paramedic
Compressions 100-120/min • depth 2-2.4 in • full recoil • minimal pauses • change compressor every 2 min • chest-compression fraction goal >80%.

↓

3. Attach AED / Monitor
EMT / AEMT / Paramedic
Analyze rhythm immediately. Resume CPR after every rhythm check or shock. Check a pulse only with an organized rhythm and limit the check to 10 seconds.

↓

4. Shockable Rhythm?
EMT / AEMT / Paramedic
VF/pulseless VT → open AC-09. Asystole/PEA → open AC-01. Repeat rhythm analysis every 2 minutes.

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Access + Medication
AEMT / Paramedic
One rapid IV attempt, then IO if unsuccessful or delayed. Epinephrine 1 mg IV/IO every 3-5 min; timing follows the rhythm pathway.

↓

6. Airway + Capnography
EMT / AEMT / Paramedic
Two-person BVM first. Do not interrupt compressions for airway placement. After an advanced airway, give 1 breath every 6 sec with continuous compressions and waveform capnography.

↓

7. Treat Reversible Causes
EMT / AEMT / Paramedic
Actively evaluate and treat the Hs and Ts. Use cause-specific therapy rather than routine empiric arrest medications.

↓

8. ROSC or Termination
EMT / AEMT / Paramedic / Medical Control
ROSC → open AC-10. No ROSC → continue rhythm pathway and evaluate AC-12 when appropriate. Favor on-scene resuscitation unless a defined special circumstance supports transport.

Contraindications

- Valid DNR, MOST, or other recognized order directing that resuscitation not be attempted.
- Obvious-death findings or circumstances addressed by agency policy.
- A definite pulse is present; use the appropriate perfusing-rhythm or medical protocol instead.

Assessment

Arrest Confirmation

- Assess responsiveness, breathing, and central pulse simultaneously; limit the pulse check to 10 seconds.
- If there is uncertainty about the pulse, begin CPR rather than delaying treatment.
- Identify the arrest as medical, traumatic, toxicologic, environmental, pregnancy-related,

drowning-related, or associated with a mechanical circulatory device.

CPR Quality

- Target a chest-compression fraction above 80% and use real-time CPR feedback when available.

Reversible Causes - Hs and Ts

- Hypovolemia, hypoxia, hydrogen ion excess/acidosis, hypo-/hyperkalemia, and hypothermia.
- Tension pneumothorax, cardiac tamponade, toxins, coronary thrombosis, and pulmonary thrombosis.

Treatment

1. Begin high-quality CPR immediately and open AC-11 for TEAM-focused roles and performance targets.
2. For VF/pulseless VT, open AC-09 and prioritize rapid defibrillation. For asystole/PEA, open AC-01 and give epinephrine as soon as possible.
3. Perform rhythm analysis every 2 minutes. Check a pulse only with an organized rhythm or other evidence of ROSC, and limit the check to 10 seconds.
4. Provide two-person BVM ventilation first, maintaining 30:2 before an advanced airway and avoiding excessive ventilation.
5. Use changes in waveform $EtCO_2$ as one component of assessment. An abrupt sustained rise may indicate ROSC; never use a single $EtCO_2$ value alone to terminate resuscitation.
6. Continue resuscitation on scene for most arrests. Transport during CPR only for a defined special circumstance or after sustained ROSC.
7. Use mechanical CPR only when high-quality manual compressions are unsafe or impractical and device placement can occur with minimal interruption.

Medication orders

Epinephrine 1 mg/10 mL (0.1 mg/mL)

1 mg IV/IO every 3-5 minutes

- AEMT and Paramedic standing order for adult cardiac arrest.
- Give as soon as possible for asystole/PEA and after the second shock for VF/pulseless VT.

Amiodarone

300 mg IV/IO after the third shock for refractory VF/pulseless VT; may repeat 150 mg IV/IO once

- Paramedic standing order under AC-09.
- Choose amiodarone or lidocaine; do not combine them.

Lidocaine

1-1.5 mg/kg IV/IO after the third shock; then 0.5-0.75 mg/kg every 5-10 minutes; maximum total 3 mg/kg

- Paramedic alternative to amiodarone under AC-09.
- Choose lidocaine or amiodarone; do not combine them.

Provider-level care

EMT Care Arrest recognition, TEAM-focused CPR, AED care, two-person BVM ventilation, and basic reversible-cause treatment.

EMT: Confirm unresponsiveness, absent or abnormal breathing, and no definite pulse within 10 seconds; identify valid DNR/MOST or obvious-death criteria.

- Activate additional resources and begin TEAM-focused CPR immediately, assuming or assigning compressor, airway, AED/monitor, recorder/timer, and team-lead functions.
- Perform compressions at 100-120/min to a depth of 2-2.4 inches with complete recoil, minimal interruptions, compressor changes every 2 minutes, and a chest-compression fraction goal above 80%.
- Apply the AED, deliver indicated shocks, and resume CPR immediately after every shock or rhythm analysis without an immediate pulse check.
- Provide two-person BVM ventilation with oxygen and an airway adjunct when available; maintain 30:2 until an advanced airway is in place.
- Assist with medication timing, documentation, reversible-cause treatment, family support, and safe movement only when indicated.

AEMT Care All EMT care plus vascular access, epinephrine, supraglottic airway, and capnography when available.

AEMT: Perform all EMT actions and make one rapid IV attempt without interrupting CPR or delaying defibrillation.

- If IV access is unsuccessful, infeasible, or delayed, establish IO access; use proximal tibia preferentially and humeral head only when tibial access is contraindicated.
- Give epinephrine 1 mg IV/IO every 3-5 minutes: as soon as possible for asystole/PEA and after the second shock for VF/pulseless VT.
- Place a supraglottic airway without interrupting compressions when BVM ventilation is ineffective or an advanced airway will improve care.
- After advanced-airway placement, ventilate once every 6 seconds during continuous compressions and use waveform capnography when available.

Paramedic Care All AEMT care plus manual rhythm interpretation and defibrillation, definitive airway options, antiarrhythmics, and advanced reversible-cause treatment.

Paramedic: Perform all EMT and AEMT actions and direct the rhythm-specific pathway, TEAM-focused resuscitation, scene strategy, and transition of care.

• For VF/pulseless VT, deliver an initial 200 J biphasic shock, verify the manufacturer-recommended setting, and use the same or higher energy for subsequent shocks; follow AC-09.

• After the third shock for persistent VF/pulseless VT, give either amiodarone or lidocaine according to AC-09; do not combine them.

• Choose a supraglottic airway or endotracheal intubation according to patient and scene conditions; airway placement must not interrupt compressions.

• Use continuous waveform capnography after any advanced airway and treat reversible causes with indication-specific therapy.

• Transition immediately to AC-10 after ROSC or apply AC-12 when termination criteria are met.

Warnings

• Do not interrupt CPR for IV/IO access, medication preparation, airway attempts, device placement, or prolonged rhythm checks.

• Do not transport routinely during active CPR. Movement commonly worsens compression quality and increases risk to the crew and public.

• Do not routinely administer calcium, sodium bicarbonate, magnesium, fluids, antidotes, or other cause-specific treatments without an appropriate indication.

• Naloxone must not delay CPR, defibrillation, airway support, or standard cardiac-arrest medications.

• Do not use routine head-up CPR, double-sequential defibrillation, vector-change defibrillation, or point-of-care ultrasound under this protocol.

• Do not use a single $EtCO_2$ cutoff or any single finding as the sole basis for terminating resuscitation.

Clinical pearls

• High-quality compressions and rapid defibrillation for a shockable rhythm provide the greatest immediate opportunity to improve outcome.

• Give epinephrine early for nonshockable arrest, but do not allow it to delay initial defibrillation attempts in VF/pulseless VT.

• A sudden sustained increase in $EtCO_2$ may signal ROSC; pause only for the scheduled brief rhythm and pulse check.

• Clear role assignment, closed-loop communication, visible medication and event timing, and structured debriefing improve team reliability.

Special populations

Pregnancy

• Begin standard resuscitation immediately. When the fundus is at or above the umbilicus, use continuous left uterine displacement, notify Medical Control early, and plan an obstetric-capable destination when clinically appropriate.

Traumatic Arrest

• Use TB-10 because hemorrhage control, oxygenation, bilateral chest decompression, and transport decisions differ from primary medical arrest.

Hypothermia, Drowning, or Toxicologic Arrest

• Use TE-05, TE-03, or TE-07 because pulse assessment, medication timing, resuscitation duration, antidotes, and transport decisions may differ.

Mechanical Circulatory Device

• Use AC-14 or the applicable mechanical-circulation protocol; continuous-flow devices may not produce a palpable pulse, so assess perfusion and device function before starting compressions.

Related pathways

• AC-09 VF / Pulseless VT (p. 96): Open the shockable-rhythm pathway.

• AC-01 Asystole / PEA (p. 72): Open the nonshockable-rhythm pathway.

• AC-11 TEAM-Focused CPR (p. 102): Open team roles and CPR performance targets.

• AC-10 Post-Resuscitation Care (p. 99): Open immediately after ROSC.

• AC-12 Termination of Resuscitation (p. 104): Open when resuscitation remains unsuccessful.

• AO-03 OB-GYN Emergency (p. 140): Open for pregnancy-related causes, hemorrhage, seizure, or obstetric destination planning.

AC-04
Acute Coronary Syndrome / STEMI

Revised October 16, 2026

Overview

Adult acute coronary syndrome pathway for patients 16 years of age and older with chest discomfort, an anginal-equivalent complaint, or ECG findings concerning for acute coronary occlusion.

Target scene time is less than 15 minutes unless essential stabilization requires additional time.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Suspected ACS
EMT / AEMT / Paramedic
Chest discomfort or anginal equivalent • ABCs • symptom-onset time • complete vital signs • SpO_2 • cardiac monitor

↓

2. 12-Lead Within 10 Minutes
EMT / AEMT / Paramedic
Acquire and transmit immediately • repeat for persistent or recurrent symptoms, nondiagnostic tracing, or deterioration

↓

3. Immediate Treatment
EMT / AEMT / Paramedic
Aspirin 324 mg chewed • oxygen only when indicated • prepare for instability

↓

4. Nitroglycerin Eligible?
EMT / AEMT / Paramedic
SBP ≥110 • no shock, RV infarction/preload dependence, or prohibited PDE-5 use • 0.4 mg SL every 5 min, max 3

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. STEMI / Occlusion Suspected?
EMT / AEMT / Paramedic
Transmit ECG • early alert • minimize scene time • choose the fastest appropriate reperfusion pathway

↓

6. Reperfusion Destination
EMT / AEMT / Paramedic
Hospital with catheterization-lab capability or Claiborne Medical Center for hospital-directed thrombolytic treatment

↓

7. Consider Covenant Health Air
EMT / AEMT / Paramedic
Activate when appropriate and expected to reduce time to reperfusion-capable care • do not delay ground movement or rendezvous

↓

8. Reassess En Route
EMT / AEMT / Paramedic
Symptoms • ECG • blood pressure • perfusion • respiratory status • response to treatment

Indications

- Chest discomfort, pressure, heaviness, tightness, or pain concerning for myocardial ischemia.
- Anginal-equivalent symptoms including dyspnea, diaphoresis, nausea, vomiting, syncope, weakness, or jaw, arm, back, shoulder, or epigastric discomfort.
- Diagnostic, dynamic, or evolving ECG changes concerning for STEMI or acute coronary occlusion.
- Unexplained high-risk symptoms in an older adult or a patient with diabetes, known coronary disease, or another significant cardiovascular risk factor.

Contraindications

- No exclusion should prevent appropriate assessment and ECG acquisition when acute coronary syndrome is reasonably suspected.
- Withhold aspirin for true aspirin anaphylaxis, active major bleeding, or strong suspicion of acute aortic dissection.
- Withhold nitroglycerin when SBP is below 110 mmHg, shock or preload dependence is present,

right-ventricular infarction is suspected, or prohibited recent PDE-5 inhibitor use is identified.

Assessment

Immediate assessment

- Perform the primary assessment and treat immediate life threats.
- Obtain continuous cardiac monitoring, SpO_2, blood pressure, and two complete vital-sign sets when feasible.
- Determine exact symptom onset, progression, provoking or relieving factors, and associated symptoms.
- Assess for prior myocardial infarction, PCI, CABG, heart failure, dysrhythmia, and implanted cardiac devices.
- Review aspirin already taken, antiplatelet or anticoagulant therapy, active bleeding, medication allergies, and recent PDE-5 inhibitor use.
- Consider other time-critical causes of chest or upper-body discomfort, including aortic dissection, pulmonary embolism, tension pneumothorax, esophageal rupture, and hemorrhage.

12-lead ECG

- Obtain a diagnostic-quality 12-lead ECG as early as possible, with a goal of within 10 minutes of first medical contact.
- Transmit the ECG to the anticipated receiving facility immediately and provide early notification when STEMI or acute coronary occlusion is suspected.
- Do not delay transport solely for ECG transmission, repeat ECGs, or additional lead placement.

STEMI identification

- Use the clinical presentation together with ST elevation in at least two anatomically contiguous leads.
- Measure ST elevation at the J point relative to the isoelectric baseline.
- In leads other than V2-V3: at least 1 mm of ST elevation in two contiguous leads.
- In V2-V3 for women: at least 1.5 mm.
- In V2-V3 for men age 40 or older: at least 2 mm.
- In V2-V3 for men younger than 40: at least 2.5 mm.
- The monitor's automated interpretation does not independently confirm or exclude STEMI.

Contiguous lead groups

- Inferior: II, III, aVF.
- Lateral: I, aVL, V5, V6.
- Anterior/septal: V1-V4.
- Posterior: V7-V9.
- Right ventricular: V3R-V4R.

STEMI equivalents and high-risk occlusion patterns

- Posterior MI: horizontal ST depression in V1-V3 with prominent R waves or upright T waves in the appropriate clinical setting; obtain V7-V9 when feasible.
- Right-ventricular involvement: in inferior STEMI, obtain right-sided leads when feasible; ST elevation in V4R supports RV involvement.
- New or presumed-new left bundle branch block alone is not an automatic STEMI criterion.
- In left bundle branch block or paced rhythm, immediately transmit tracings with concordant ST elevation, concordant ST depression in V1-V3, excessive discordance, dynamic ischemia, or another pattern strongly concerning for acute occlusion.
- Other high-risk patterns include diffuse ST depression with ST elevation in aVR, de Winter-type changes, hyperacute T waves, and rapidly progressive ST-segment changes.

Treatment

1. Provide oxygen only for SpO_2 below 90%, respiratory distress, shock, or another clinical indication; do not routinely administer oxygen when SpO_2 is at least 90%.
2. Withhold nitroglycerin when SBP is below 110 mmHg, shock or preload dependence is present, right-ventricular infarction is suspected, or the patient used avanafil within 12 hours, sildenafil or vardenafil within 24 hours, or tadalafil within 48 hours.
3. Establish IV access without delaying transport. Do not routinely administer fluids; for hypotension without pulmonary edema, use cautious 250 mL isotonic-crystalloid aliquots with reassessment.
4. Transport to a hospital with cardiac catheterization laboratory capability or to Claiborne Medical Center for hospital-directed thrombolytic treatment under the Claiborne destination plan.
5. Air activation must not delay assessment, aspirin, ECG acquisition, stabilization, or movement toward ground transport or rendezvous.
6. Target scene time is less than 15 minutes. Do not delay transport for repeat ECGs, IV access, additional leads, or nonessential procedures.

Medication orders

Aspirin

324 mg PO chewed

- EMT, AEMT, and Paramedic standing order for suspected acute coronary syndrome.
- If aspirin was taken before EMS arrival, give enough to complete a total loading dose of 324 mg.
- Withhold for true aspirin anaphylaxis, active major bleeding, or strong suspicion of acute aortic dissection.

Nitroglycerin

0.4 mg SL every 5 minutes as needed; maximum 3 doses

- EMT may assist with the patient's prescribed nitroglycerin; AEMT and Paramedic may administer agency nitroglycerin.
- Withhold when SBP is below 110 mmHg, shock or preload dependence is present, right-ventricular infarction is suspected, or prohibited recent PDE-5 inhibitor use is identified.
- Reassess symptoms, blood pressure, and perfusion before and after every dose.

Fentanyl

25-50 mcg IV/IO; repeat as needed to a maximum total dose of 100 mcg

- Paramedic standing order for persistent significant ischemic discomfort after aspirin and appropriate nitroglycerin.
- Use continuous cardiac, respiratory, SpO_2, and blood-pressure monitoring.
- Avoid routine opioid administration; opioids may delay absorption of oral antiplatelet medications.

Isotonic crystalloid

250 mL IV/IO aliquots with reassessment

- AEMT and Paramedic for hypotension without pulmonary edema.
- Do not administer routine fluid boluses to a normotensive patient.

Provider-level care

EMT Care Rapid recognition, aspirin, prescribed-nitroglycerin assistance, ECG acquisition, early notification, and time-sensitive transport.

EMT: Perform the primary assessment, obtain symptom-onset time and focused history, apply the cardiac monitor/AED, obtain SpO_2 and complete vital signs, and place defibrillation pads when instability is present or anticipated.

- Obtain a 12-lead ECG within 10 minutes of first medical contact when equipment and training are available; transmit the tracing immediately without delaying transport.
- Give aspirin 324 mg chewed unless contraindicated.
- Assist with the patient's prescribed nitroglycerin only when SBP is at least 110 mmHg and no contraindication is present.
- Provide oxygen only for SpO_2 below 90%, respiratory distress, shock, or another clinical indication.
- Initiate early destination notification and transport under the Claiborne reperfusion plan.

AEMT Care All EMT care plus vascular access, agency nitroglycerin, cautious fluid treatment for hypotension, and continued ECG surveillance.

AEMT: Perform all EMT actions and establish IV access without delaying transport.

- Give nitroglycerin 0.4 mg SL every 5 minutes for continuing ischemic discomfort, to a maximum of three doses, only when SBP is at least 110 mmHg and no contraindication is present.
- For hypotension without pulmonary edema, give isotonic crystalloid in cautious 250 mL aliquots with reassessment after each aliquot.
- Repeat 12-lead ECGs for persistent or recurrent symptoms, a nondiagnostic initial tracing, or clinical deterioration.
- Immediately report hypotension, dysrhythmia, worsening respiratory status, altered mental status, or other evidence of instability.

Paramedic Care All AEMT care plus ECG interpretation, high-risk occlusion recognition, refractory-pain treatment, instability management, and reperfusion destination decisions.

Paramedic: Perform all EMT and AEMT actions; interpret serial ECGs and identify STEMI, posterior or right-ventricular involvement, and other patterns concerning for acute coronary occlusion.

- Obtain V4R for suspected right-ventricular involvement or V7-V9 for suspected posterior infarction when indicated and feasible, without delaying transport.
- For persistent significant ischemic discomfort after aspirin and appropriate nitroglycerin, give fentanyl 25-50 mcg IV/IO; repeat as needed to a maximum total dose of 100 mcg.
- Use continuous cardiac, respiratory, SpO_2, and blood-pressure monitoring after fentanyl administration.
- Treat dysrhythmia, acute pulmonary edema, shock, or cardiac arrest under the corresponding linked protocol.

• Choose the fastest appropriate reperfusion pathway, initiate early facility notification, and consider Covenant Health Air when appropriate.

Warnings

- A normal or nondiagnostic initial ECG does not exclude acute myocardial infarction.
- Strong suspicion of acute aortic dissection is a reason to withhold aspirin and nitroglycerin and pursue immediate time-critical transport.
- Do not delay transport while waiting for air-medical arrival when ground transport or rendezvous is faster.
- Claiborne EMS does not administer prehospital thrombolytics, heparin, or a P2Y12 inhibitor under AC-04.
- Transport to Claiborne Medical Center for thrombolytic treatment means hospital evaluation and hospital-directed treatment; EMS does not independently determine thrombolytic eligibility.

Clinical pearls

- Transmit the first ECG as soon as possible, even when the automated interpretation is nondiagnostic.
- Early notification allows the receiving team to review the tracing and prepare the catheterization laboratory or thrombolytic pathway.
- Serial ECGs are particularly valuable when symptoms continue or the initial tracing is nondiagnostic.
- Borderline ST elevation with strong reciprocal changes may still represent acute coronary occlusion.
- Older adults, women, and patients with diabetes may present without classic chest pain.

Special populations

Atypical presentations

- Consider acute coronary syndrome in unexplained dyspnea, weakness, diaphoresis, nausea, syncope, or jaw, arm, shoulder, back, or epigastric discomfort.
- Maintain a low threshold for ECG acquisition in older adults and patients with diabetes or known coronary disease.

Right-ventricular infarction

- Inferior STEMI with hypotension, clear lungs, jugular venous distention, or disproportionate preload dependence should prompt right-sided lead acquisition when feasible.

Related pathways

- AC-02 Bradycardia With a Pulse (p. 74): Open for symptomatic bradycardia or poor perfusion.
- AC-05 Acute Pulmonary Edema (p. 83): Open for cardiogenic pulmonary edema or decompensated heart failure.
- AC-03 Adult Cardiac Arrest (p. 77): Open immediately if the patient becomes pulseless.

AC-05
CHF / Acute Pulmonary Edema

Revised October 16, 2026

Overview

Prioritize early noninvasive positive-pressure ventilation and blood-pressure-guided vasodilation while identifying acute coronary syndrome, dysrhythmia, cardiogenic shock, and alternative causes of respiratory distress.

Target scene time is less than 15 minutes unless essential stabilization requires additional time.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Suspected Pulmonary Edema
EMT / AEMT / Paramedic
Acute dyspnea • orthopnea • rales • diaphoresis • hypertension • JVD/edema • frothy sputum

↓

2. Position + Monitor
EMT / AEMT / Paramedic
Upright unless hypotensive • SpO_2 • frequent BP • cardiac monitor • IV • 12-lead without delaying support

↓

3. Moderate / Severe Distress?
EMT / AEMT / Paramedic
Spontaneously breathing • airway protected • cooperative enough for mask • SBP ≥100

Continue at step 4 in the next column/page

Quick flow continued at step 4

4. Start CPAP Early
EMT / AEMT / Paramedic
If adjustable: start 5 cm H_2O and titrate to 10 • fixed device: manufacturer setting • reassess continuously

↓

5. Nitroglycerin Eligible?
EMT / AEMT / Paramedic
SBP ≥110 before every dose • no shock, RV infarction/preload dependence, or prohibited PDE-5 use

↓

6. Nitroglycerin
EMT / AEMT / Paramedic
0.4 mg SL every 5 min • maximum 3 • EMT assists prescribed medication • AEMT/Paramedic agency medication

↓

7. Shock or Ventilatory Failure?
EMT / AEMT / Paramedic
Stop nitrates • reduce/remove CPAP if worsening hypotension • BVM support and request Paramedic airway/shock care

↓

8. Paramedic - Cardiogenic Shock
Paramedic
Norepinephrine per the AC-05 dose and perfusion targets below; continuous cardiac monitoring and frequent blood-pressure and vascular-site checks

↓

9. Reassess + Transport
EMT / AEMT / Paramedic
BP • work of breathing • mental status • SpO_2/$EtCO_2$ • lung sounds • early alert • scene target <15 min

Indications

- Suspected cardiogenic pulmonary edema or acute decompensated heart failure in a patient 16 years of age or older.
- Acute dyspnea with orthopnea, rales, diaphoresis, hypertension, jugular venous distention, peripheral edema, frothy sputum, or a history strongly suggestive of heart failure.
- Respiratory distress associated with suspected acute cardiac ischemia, dysrhythmia, valvular disease, renal failure, or medication nonadherence when pulmonary edema is clinically suspected.

Contraindications

- CPAP is contraindicated for apnea, inadequate spontaneous ventilation, active vomiting, inability to protect the airway, significant facial trauma preventing a seal, suspected untreated pneumothorax, severe agitation preventing safe application, or SBP below 100 mmHg.
- Nitroglycerin is contraindicated when SBP is below 110 mmHg, shock or preload dependence is present, right-ventricular infarction is suspected, or prohibited recent PDE-5 inhibitor use is identified.
- Do not administer routine large-volume crystalloid when pulmonary edema is present.

Assessment

Focused history

- Determine onset and progression, orthopnea, paroxysmal nocturnal dyspnea, chest discomfort, palpitations, syncope, fever, cough, sputum, medication adherence, dialysis schedule, recent weight gain, and baseline oxygen or CPAP use.
- Identify prior heart failure, myocardial infarction, PCI/CABG, valvular disease, renal failure, dysrhythmia, COPD/asthma, pulmonary embolism, and implanted cardiac devices.
- Review nitrates already taken, diuretics and antihypertensives, antiplatelet or anticoagulant therapy, and recent PDE-5 inhibitor use.

Examination and monitoring

- Assess respiratory rate and pattern, accessory-muscle use, speech, lung sounds, diaphoresis, cyanosis, frothy sputum, JVD, edema, pulse quality, capillary refill, and mental status.
- Obtain continuous cardiac monitoring, SpO_2, frequent blood pressures, and two complete vital-sign sets when feasible.
- Obtain a 12-lead ECG without delaying CPAP, ventilation, or transport; use AC-04 when ACS or STEMI is suspected.
- Use waveform $EtCO_2$ when available and technically compatible with CPAP or assisted ventilation.

Alternative and precipitating diagnoses

- Consider acute coronary syndrome, hypertensive emergency, dysrhythmia, acute valvular disease, renal failure, medication nonadherence, infection, and dietary or fluid excess as precipitants.
- Consider COPD/asthma, pneumonia, pulmonary embolism, pneumothorax, aspiration, anaphylaxis, metabolic acidosis, and other causes of respiratory distress.
- Wheezing may occur with pulmonary edema; do not assume asthma or COPD solely because wheezing is present.

Treatment

1. For adjustable CPAP, begin at 5 cm H_2O and titrate to 10 cm H_2O according to response and tolerance. For a fixed-pressure device, use the manufacturer-approved setting.
2. Reassess blood pressure, respiratory effort, mental status, SpO_2, lung sounds, perfusion, and mask tolerance after every nitroglycerin dose and CPAP adjustment.
3. Do not routinely administer furosemide, another diuretic, morphine, another opioid, or an IV fluid bolus under this protocol.
4. For hypotension or cardiogenic shock, withhold nitroglycerin, reduce or discontinue CPAP if it worsens perfusion, support ventilation, and avoid routine crystalloid when pulmonary edema is present.
5. Notify the receiving facility early, treat while moving toward transport, and target scene time under 15 minutes.

Medication orders

Nitroglycerin

0.4 mg SL every 5 minutes as needed; maximum 3 doses

- EMT may assist with the patient's prescribed nitroglycerin; AEMT and Paramedic may administer agency nitroglycerin.
- Confirm SBP of at least 110 mmHg before every dose.
- Withhold for shock, suspected right-ventricular infarction or preload dependence, avanafil within 12 hours, sildenafil or vardenafil within 24 hours, or tadalafil within 48 hours.
- IV bolus, infusion, and other high-dose nitroglycerin regimens are not authorized under AC-05.

Norepinephrine

Start 0.1 mcg/kg/min IV/IO; titrate to MAP ≥65 mmHg or SBP ≥90 mmHg; maximum 2 mcg/kg/min

- Paramedic standing order for persistent cardiogenic shock.
- Continuous cardiac monitoring, frequent blood-pressure assessment, and careful IV/IO-site surveillance are required.
- Use a proximal, well-functioning IV or IO and reassess perfusion frequently.

Albuterol

Use the approved respiratory dose only when true bronchospasm is suspected

- Do not administer solely because wheezing is present; pulmonary edema may produce cardiac wheezing.

Provider-level care

EMT Care Rapid recognition, positioning, oxygenation, CPAP, prescribed-nitroglycerin assistance, monitoring, and early transport.

EMT: Perform the primary assessment; position upright unless hypotensive and assess respiratory effort, lung sounds, perfusion, mental status, SpO_2, and complete vital signs.

- Apply the cardiac monitor/AED and obtain a 12-lead ECG when trained and equipped, without delaying respiratory support.
- Start CPAP early for moderate-to-severe distress when the patient is breathing spontaneously, can protect the airway, can tolerate the mask, and has SBP of at least 100 mmHg.
- Titrate oxygen to SpO_2 92-96%; use 88-92% for known chronic hypercapnic respiratory failure unless severe hypoxemia requires a higher target.
- Assist with the patient's prescribed nitroglycerin only when SBP is at least 110 mmHg and no contraindication is present.
- Prepare for BVM ventilation, rapid transport, and ALS support if mental status, ventilation, perfusion, or CPAP tolerance worsens.

AEMT Care All EMT care plus vascular access, agency nitroglycerin, CPAP management, and recognition of cardiogenic shock.

AEMT: Perform all EMT actions and establish IV access without delaying CPAP, nitroglycerin, or transport.

- Give nitroglycerin 0.4 mg SL every 5 minutes, maximum three doses, only when SBP is at least 110 mmHg before every dose and no contraindication is present.
- Reassess blood pressure, symptoms, work of breathing, mental status, SpO_2, and perfusion after every nitroglycerin dose and CPAP adjustment.

• Do not administer a routine fluid bolus when pulmonary edema is present.

• Support ventilation with BVM and immediately request paramedic intervention when shock, ventilatory failure, or inability to protect the airway develops.

Paramedic Care All AEMT care plus advanced ECG assessment, cardiogenic-shock treatment, advanced airway management, and linked cause-specific care.

Paramedic: Perform all EMT and AEMT actions; interpret the 12-lead ECG and evaluate for ACS, dysrhythmia, acute valvular disease, hypertensive pulmonary edema, and cardiogenic shock.

• For persistent cardiogenic shock, start norepinephrine at 0.1 mcg/kg/min IV/IO and titrate to MAP of at least 65 mmHg or SBP of at least 90 mmHg; maximum 2 mcg/kg/min.

• Withhold nitroglycerin and reduce or discontinue CPAP when either intervention is worsening hypotension or perfusion.

• Use waveform $EtCO_2$ when available and technically compatible with CPAP; trend ventilation and response rather than relying on a single value.

• Prepare for drug-assisted airway management when CPAP fails, ventilation is inadequate, mental status declines, or the airway cannot be protected; anticipate peri-intubation hypotension.

• Open AC-04 for suspected ACS/STEMI, AM-05 for shock, AR-03 for drug-assisted airway management, or AC-03 for cardiac arrest.

Warnings

• Do not delay CPAP, ventilation, or transport for IV access or a 12-lead ECG.

• Do not use CPAP for apnea, inadequate spontaneous ventilation, inability to protect the airway, active vomiting, or shock.

• Positive-pressure ventilation can worsen hypotension; reassess blood pressure and perfusion continuously.

• Intubation in cardiogenic shock may precipitate profound hypotension or cardiac arrest; optimize oxygenation and hemodynamics and prepare vasopressor support.

• Use CPAP only with confirmed equipment capabilities and device-specific training. Follow the fixed or adjustable device instructions and the pressure limits specified here.

Clinical pearls

• Early CPAP and blood-pressure-guided nitroglycerin are the principal field interventions for hypertensive cardiogenic pulmonary edema.

• Acute pulmonary edema may reflect rapid fluid redistribution rather than total-body fluid overload; a single physical finding does not establish the diagnosis.

• Wheezing does not exclude pulmonary edema, and rales do not exclude COPD, pneumonia, or another respiratory diagnosis.

• A normal initial ECG does not exclude acute coronary syndrome; use AC-04 when the history or tracing is concerning.

• Improvement in speech, respiratory rate, work of breathing, mental status, SpO_2, and blood pressure provides a useful treatment-response trend.

Special populations

Renal failure or dialysis

• Determine the last completed dialysis treatment, missed treatments, access complications, baseline weight, and known potassium problems.

• Avoid routine fluid administration and consider hyperkalemia when bradycardia, weakness, peaked T waves, or QRS widening is present.

Cardiogenic shock

• Hypotension, altered mental status, cool or mottled skin, weak pulses, delayed capillary refill, and reduced urine output suggest poor perfusion.

• Withhold nitrates, limit positive pressure when it worsens perfusion, avoid routine crystalloid in pulmonary edema, and start norepinephrine when indicated.

Related pathways

• AC-04 ACS / STEMI (p. 80): Open for suspected ischemia, STEMI, or acute coronary occlusion.

• AM-05 Hypotension / Shock (p. 129): Open for persistent hypotension or shock management.

• AR-03 Drug-Assisted Airway (p. 146): Open when CPAP fails or airway protection is lost.

• AC-03 Adult Cardiac Arrest (p. 77): Open immediately if the patient becomes pulseless.

AC-06
Adult Narrow-Complex Tachycardia

Revised October 16, 2026

Overview

Adult pathway for patients 16 years of age and older with a pulse and a narrow-complex tachycardia, generally defined by QRS duration under 0.12 seconds.

A tachyarrhythmia causing instability is typically at least 150/min, but treatment is determined by the patient's clinical condition rather than rate alone.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Assess in Clinical Context
EMT / AEMT / Paramedic
Airway/ventilation • oxygen if hypoxemic • monitor/pads • BP/SpO_2 • IV • 12-lead when it will not delay urgent treatment

↓

2. Sinus Tachycardia / Secondary Cause?
EMT / AEMT / Paramedic
Treat hypoxia, pain, fever, dehydration, hemorrhage, sepsis, stimulant exposure, or another underlying cause

↓

3. Instability Caused by Rhythm?
EMT / AEMT / Paramedic
Hypotension • acute altered mental status • shock • ischemic chest discomfort • acute heart failure

↓

4. Unstable → Synchronized Cardioversion
Paramedic
Regular narrow 100 J • AF 200 J • flutter 200 J • sedate when feasible without delaying shock • resynchronize every time

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Stable + Regular
EMT / AEMT / Paramedic
Modified Valsalva • no carotid massage • if unsuccessful, request Paramedic adenosine pathway below

↓

6. Adenosine
Paramedic
Paramedic: 6 mg rapid IV/IO + immediate flush • then 12 mg after 1-2 min if needed; may repeat 12 mg once • continuous rhythm recording

↓

7. Stable AF / Flutter
Paramedic
No routine adenosine • SBP ≥110 and no AV-nodal-agent contraindication • use diltiazem when available; during approved Cardizem shortage, metoprolol 2.5-5 mg IV push over 1 min every 10 min (max 15 mg)

↓

8. Reassess + Transport
EMT / AEMT / Paramedic
Post-treatment rhythm strip/12-lead • complete vital signs • monitor recurrence • route wide or polymorphic rhythms appropriately

Indications

- Stable regular narrow-complex tachycardia potentially responsive to modified Valsalva or adenosine.
- Atrial fibrillation or atrial flutter with rapid ventricular response requiring assessment for instability or rate control.

Contraindications

- Do not administer adenosine or perform cardioversion for sinus tachycardia; treat the underlying cause.
- Do not administer adenosine for an irregular rhythm, unstable wide-complex tachycardia, polymorphic tachycardia, or active severe bronchospasm.
- Do not administer diltiazem or the shortage-only metoprolol alternative for hypotension, cardiogenic shock, acute decompensated heart failure, suspected systolic dysfunction, acute coronary

syndrome, pre-excited atrial fibrillation/flutter, bradycardia, advanced heart block, or an undifferentiated wide-complex rhythm.

Assessment

Initial assessment

- Maintain a patent airway, assist ventilation as needed, and administer oxygen only for hypoxemia.
- Obtain continuous cardiac monitoring, defibrillation pads, frequent blood pressure, SpO_2, IV access, and two complete vital-sign sets when feasible.
- Obtain a 12-lead ECG before treatment when the patient is stable enough and it will not delay cardioversion.
- Determine onset, prior episodes, medications, stimulant or toxicologic exposure, cardiac history, heart-failure history, anticoagulant use, and known pre-excitation.

Instability attributable to tachycardia

- Confirm that the rhythm is causing instability rather than representing a compensatory response to another condition such as sepsis, hemorrhage, hypoxia, or pain.

Rhythm classification

- Confirm QRS duration under 0.12 seconds and determine whether the rhythm is regular or irregular.
- Regular narrow rhythms may include AVNRT, AVRT, atrial tachycardia, atrial flutter with fixed conduction, or sinus tachycardia.
- Irregular narrow rhythms commonly include atrial fibrillation, atrial flutter with variable conduction, or multifocal atrial tachycardia.
- If the rhythm is wide, irregular with pre-excitation, or polymorphic, stop this pathway and open AC-07 or AC-08.

Treatment

1. If synchronization is delayed and the patient's condition is critical, deliver an unsynchronized high-energy shock.
2. If modified Valsalva is unsuccessful, give adenosine 6 mg rapid IV/IO push through the most proximal practical access followed immediately by a rapid normal-saline flush.
3. Obtain a post-treatment rhythm strip and 12-lead ECG, reassess complete vital signs, monitor for recurrence, and transport.

Medication orders

Adenosine

First dose 6 mg rapid IV/IO push with immediate flush; then 12 mg after 1-2 minutes if needed; may repeat the 12 mg dose once

- Paramedic standing order for stable regular narrow-complex tachycardia after unsuccessful modified Valsalva.
- Use the most proximal practical access and record the rhythm continuously during administration.
- Do not use for sinus tachycardia, an irregular rhythm, unstable wide-complex tachycardia, polymorphic tachycardia, or active severe bronchospasm.
- Contact Medical Control before using a reduced dose for a heart-transplant patient or administration through central venous access.

Diltiazem

0.25 mg/kg IV/IO (maximum 20 mg); if needed, 0.35 mg/kg IV/IO after 15 minutes (maximum 20 mg); infusion 5-10 mg/hr

- Paramedic standing order for significantly symptomatic stable atrial fibrillation/flutter when SBP is at least 110 mmHg.
- For prolonged transport, an infusion of 5-10 mg/hr requires Medical Control.
- Avoid with hypotension, acute decompensated heart failure, suspected systolic dysfunction, acute coronary syndrome, pre-excitation, or an undifferentiated wide-complex rhythm.

Metoprolol - Cardizem shortage alternative

2.5-5 mg IV push over 1 minute every 10 minutes; maximum 3 doses or 15 mg

- Paramedic shortage-only alternative for significantly symptomatic stable atrial fibrillation/flutter when diltiazem is unavailable, SBP is at least 110 mmHg, and no AV-nodal-agent contraindication is present.
- Reassess blood pressure, heart rate, rhythm, perfusion, respiratory status, and symptoms before every dose.
- Do not combine or sequence with diltiazem without Medical Control. Avoid with hypotension, bradycardia, advanced heart block, cardiogenic shock, pre-excitation, undifferentiated wide-complex rhythm, acute decompensated heart failure, or suspected systolic dysfunction.

Midazolam

2-5 mg IV/IO; use the smallest effective dose

- Paramedic procedural sedation for synchronized cardioversion when feasible.
- Do not delay cardioversion in an unstable patient.

• Continuous airway, respiratory, SpO_2, cardiac, and blood-pressure monitoring is required.

Provider-level care

EMT Care Recognition, supportive care, monitoring, pad placement, modified Valsalva, and rapid escalation for instability.

EMT: Perform the primary assessment; support airway and ventilation, give oxygen only when hypoxemic, and assess whether the tachycardia is likely causing the patient's symptoms.

• Obtain complete vital signs, apply the cardiac monitor/AED and defibrillation pads, and obtain a 12-lead ECG when trained and equipped without delaying urgent treatment.

• Identify and treat likely causes of sinus tachycardia, including hypoxia, pain, fever, dehydration, hemorrhage, sepsis, and stimulant exposure.

• For a stable, regular narrow-complex rhythm, coach a modified Valsalva maneuver. Do not perform carotid massage.

• Immediately request paramedic intervention and prepare for synchronized cardioversion when instability is attributable to the tachycardia.

AEMT Care All EMT care plus vascular access, ECG acquisition, treatment preparation, and monitoring during cardioversion.

AEMT: Perform all EMT actions and establish IV/IO access without delaying synchronized cardioversion or transport.

• Acquire a 12-lead ECG before treatment when feasible and assist with continuous rhythm recording during adenosine administration.

• Prepare medication, suction, oxygenation/ventilation equipment, and airway equipment for cardioversion and procedural sedation.

• Reassess blood pressure, mental status, perfusion, chest discomfort, respiratory status, and rhythm after every intervention.

• Do not independently administer adenosine, diltiazem, or procedural sedation under AC-06.

Paramedic Care All AEMT care plus rhythm interpretation, adenosine, diltiazem or the approved shortage-only metoprolol alternative, synchronized cardioversion, and procedural sedation.

Paramedic: Perform all EMT and AEMT actions; determine rhythm regularity, QRS duration, hemodynamic effect, and whether the tachycardia is the cause of instability.

• For unstable regular narrow-complex tachycardia, cardiovert at 100 J. For unstable atrial fibrillation or flutter, cardiovert at 200 J.

• Confirm synchronization markers before every shock and resynchronize after every cardioversion attempt; increase energy after an unsuccessful shock.

• Sedate whenever feasible with midazolam 2-5 mg IV/IO using the smallest effective dose, but do not delay cardioversion.

• For stable regular narrow-complex tachycardia after unsuccessful modified Valsalva, give adenosine 6 mg rapid IV/IO with an immediate flush, then 12 mg after 1-2 minutes if needed; may repeat the 12 mg dose once.

• For significantly symptomatic stable atrial fibrillation/flutter with SBP of at least 110 mmHg and no contraindication, give diltiazem 0.25 mg/kg IV/IO (maximum 20 mg); if needed, repeat 0.35 mg/kg IV/IO after 15 minutes (maximum 20 mg).

• During the approved Claiborne Cardizem shortage only, when diltiazem is unavailable and the same AF/flutter eligibility criteria are met, give metoprolol 2.5-5 mg IV push over 1 minute every 10 minutes; maximum three doses or 15 mg. Reassess blood pressure, heart rate, rhythm, perfusion, respiratory status, and symptoms before every dose.

• Do not independently combine or sequence diltiazem and metoprolol. Contact Medical Control for inadequate response, intolerance, or uncertainty; do not use amiodarone as an AV-nodal substitute for stable AF/flutter with RVR.

• Route suspected pre-excited, wide-complex, or polymorphic rhythms to the appropriate pathway and contact Medical Control early.

Warnings

• Instability must be caused by the tachycardia rather than merely associated with a fast heart rate.

• Do not combine or sequence AV-nodal-blocking medications without an explicit protocol or Medical Control direction.

• In suspected pre-excited atrial fibrillation/flutter, do not administer adenosine, diltiazem, metoprolol, digoxin, or amiodarone.

• Diltiazem may worsen hypotension or decompensated systolic heart failure.

• The metoprolol pathway is a Cardizem-shortage contingency only; it is not a second routine first-line branch and must be withdrawn when diltiazem availability is restored.

• Verify synchronization before every cardioversion attempt; many devices require synchronization to be reactivated after each shock.

Clinical pearls

- A properly performed modified Valsalva is more effective than a simple seated Valsalva maneuver.
- Adenosine may transiently reveal atrial activity without terminating atrial fibrillation or flutter; it is not routine therapy for an irregular rhythm.
- Capture the rhythm before, during, and after adenosine or cardioversion whenever feasible.
- Stable atrial fibrillation/flutter of unknown duration should receive rate control rather than elective prehospital rhythm conversion.

Special populations

Heart transplant or central venous access

- Adenosine may produce an exaggerated response in a heart-transplant patient or when administered centrally.
- Contact Medical Control before using a reduced dose; do not automatically give the standard 6 mg dose through central access.

Pre-excited atrial fibrillation/flutter

- An irregular rapid rhythm with variable QRS morphology or known Wolff-Parkinson-White may represent pre-excitation.
- Avoid AV-nodal blockers and amiodarone; perform synchronized cardioversion if unstable and contact Medical Control early if stable.

Related pathways

- AC-04 ACS / STEMI (p. 80): Open for ischemic chest discomfort or suspected acute coronary syndrome.
- AC-05 Acute Pulmonary Edema (p. 83): Open when acute heart failure or pulmonary edema is present.
- AC-07 Monomorphic Wide-Complex Tachycardia (p. 90): Open when the QRS is wide and the rhythm is monomorphic.
- AC-08 Polymorphic Wide-Complex Tachycardia (p. 93): Open for polymorphic or torsades-pattern tachycardia.
- AM-05 Hypotension / Shock (p. 129): Open for persistent hypotension or shock.
- AC-03 Adult Cardiac Arrest (p. 77): Open immediately if the patient becomes pulseless.

AC-07
Adult Monomorphic Wide-Complex Tachycardia

Revised October 16, 2026

Overview

Adult pathway for patients 16 years of age and older with a pulse and a regular monomorphic wide-complex tachycardia, generally QRS duration of at least 0.12 seconds.

When the diagnosis is uncertain, presume ventricular tachycardia and avoid medications that may worsen an undifferentiated wide-complex rhythm.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Confirm Pulse + Monomorphic Wide Rhythm
EMT / AEMT / Paramedic
QRS ≥0.12 sec • assume VT when uncertain • airway/ventilation • oxygen if hypoxemic • monitor/pads • BP/SpO_2 • IV/IO

↓

2. Instability Caused by Tachycardia?
EMT / AEMT / Paramedic
Hypotension • acute altered mental status • shock • ischemic chest discomfort • acute heart failure

↓

3. Unstable → Synchronized Cardioversion
Paramedic
100 J biphasic • increase if unsuccessful • confirm sync before every shock • sedate when feasible

↓

4. Stable + Uncertain Origin
Paramedic
Only if regular and monomorphic: adenosine 6 mg rapid IV/IO + flush, then 12 mg after 1-2 min if needed; may repeat 12 mg once

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Stable Monomorphic VT
Paramedic
Amiodarone 150 mg IV/IO over 10 min • stop and cardiovert immediately if instability develops

↓

6. Persistent After Infusion?
Paramedic
Synchronized cardioversion 100 J biphasic • do not stack multiple antiarrhythmics

↓

7. Converted / Recurrence
Paramedic
If VT recurs: repeat amiodarone 150 mg over 10 min once • then infusion 1 mg/min

↓

8. Reassess + Transport
EMT / AEMT / Paramedic
Post-treatment rhythm strip/12-lead • complete vitals • early notification • prepare for polymorphic rhythm or pulseless arrest

Indications

- Sustained monomorphic ventricular tachycardia requiring synchronized cardioversion or antiarrhythmic treatment.
- Stable regular monomorphic wide-complex tachycardia of uncertain origin in which adenosine may be diagnostic or therapeutic.

Contraindications

- Do not use adenosine for unstable, irregular, or polymorphic wide-complex tachycardia or active severe bronchospasm.
- Do not administer diltiazem, verapamil, metoprolol, or another AV-nodal blocker for an undifferentiated wide-complex rhythm.
- Do not administer amiodarone for suspected torsades de pointes, a clearly prolonged-QT polymorphic rhythm, or suspected pre-excited atrial fibrillation/flutter.

Assessment

Initial assessment

- Obtain continuous cardiac monitoring, pads, frequent blood pressure, SpO_2, IV/IO access, and two complete vital-sign sets when feasible.
- Obtain a 12-lead ECG before treatment when stable enough and it will not delay cardioversion.
- Determine onset, prior dysrhythmia, cardiac history, implanted devices, medications, stimulant or toxicologic exposure, and anticoagulant therapy.

Instability attributable to tachycardia

- Hypotension, acutely altered mental status, shock or poor perfusion, ischemic chest discomfort, or acute heart failure.
- Confirm that the rhythm is driving instability rather than representing a secondary response to another critical condition.

Reversible causes and rhythm hazards

- Evaluate for ACS, structural heart disease, electrolyte abnormality, hyperkalemia, hypomagnesemia, stimulant exposure, and medication toxicity.
- Consider sodium-channel-blocker toxicity with wide QRS, hypotension, seizures, or a compatible ingestion.
- If the rhythm becomes irregular, polymorphic, or pulseless, leave AC-07 immediately and open the linked pathway.

Treatment

1. Treat an uncertain regular monomorphic wide-complex tachycardia as ventricular tachycardia and apply pads before deterioration.
2. For instability attributable to monomorphic VT with a pulse, perform immediate synchronized cardioversion at 100 J biphasic and increase energy if unsuccessful.
3. If synchronization is delayed and the condition is critical, deliver an unsynchronized high-energy shock.
4. If stable VT persists after the infusion, perform synchronized cardioversion at 100 J rather than stacking antiarrhythmics.
5. Do not combine amiodarone with lidocaine or another antiarrhythmic under AC-07 unless directed by Medical Control.
6. Obtain post-treatment rhythm strips and a 12-lead ECG, reassess complete vital signs, notify early, and transport.

Medication orders

Adenosine

6 mg rapid IV/IO with immediate flush; then 12 mg after 1-2 minutes if needed; may repeat the 12 mg dose once

- Paramedic standing order only for stable regular monomorphic wide-complex tachycardia of uncertain origin.
- Use proximal access and record the rhythm continuously.
- Do not use for unstable, irregular, or polymorphic rhythm or active severe bronchospasm.

Amiodarone

150 mg IV/IO over 10 minutes

- Paramedic standing order for stable monomorphic VT with a pulse.
- Stop the infusion and cardiovert immediately if instability develops.
- After conversion, if VT recurs, repeat 150 mg over 10 minutes once and begin maintenance infusion at 1 mg/min.
- Do not combine with another antiarrhythmic without Medical Control.

Midazolam

2-5 mg IV/IO; use the smallest effective dose

- Paramedic sedation for synchronized cardioversion when feasible.
- Do not delay cardioversion in an unstable patient.
- Continuous airway, respiratory, SpO_2, cardiac, and blood-pressure monitoring is required.

Provider-level care

EMT Care Recognition, supportive care, pad placement, reversible-cause assessment, and immediate escalation for instability.

EMT: Perform the primary assessment, confirm a pulse, support airway and ventilation, and give oxygen only when hypoxemic.

- Obtain complete vital signs, apply the cardiac monitor/AED and defibrillation pads, and obtain a 12-lead ECG when trained and equipped without delaying urgent treatment.
- Treat an uncertain wide-complex tachycardia as ventricular tachycardia and immediately request paramedic intervention.
- Identify possible ischemia, electrolyte abnormality, hyperkalemia, toxicologic exposure, stimulant use, or medication effect.
- Prepare for immediate CPR and defibrillation if the patient becomes pulseless or the rhythm becomes polymorphic.

AEMT Care All EMT care plus vascular access, treatment preparation, and continuous reassessment.

AEMT: Perform all EMT actions and establish IV/IO access without delaying synchronized cardioversion or transport.

- Acquire a 12-lead ECG before treatment when the patient is stable and assist with continuous rhythm recording during adenosine.
- Prepare medication, suction, ventilation equipment, and airway equipment for cardioversion and sedation.
- Reassess blood pressure, mental status, perfusion, chest discomfort, respiratory status, and rhythm after every intervention.
- Do not independently administer adenosine, amiodarone, or procedural sedation under AC-07.

Paramedic Care All AEMT care plus rhythm interpretation, synchronized cardioversion, adenosine, amiodarone, and procedural sedation.

Paramedic: Perform all EMT and AEMT actions; confirm rhythm regularity, monomorphic morphology, QRS duration, pulse, and whether the tachycardia is causing instability.

- For unstable monomorphic VT with a pulse, perform synchronized cardioversion at 100 J biphasic and increase energy if unsuccessful.
- Confirm synchronization markers before every shock and resynchronize after every attempt.
- Sedate whenever feasible with midazolam 2-5 mg IV/IO using the smallest effective dose, but do not delay cardioversion.
- For a stable regular monomorphic wide-complex rhythm of uncertain origin, consider adenosine 6 mg rapid IV/IO with immediate flush, followed by 12 mg after 1-2 minutes if needed; may repeat the 12 mg dose once.
- For stable monomorphic VT, give amiodarone 150 mg IV/IO over 10 minutes; stop the infusion and cardiovert immediately if instability develops.
- After conversion, treat one recurrence with amiodarone 150 mg over 10 minutes, then begin maintenance infusion at 1 mg/min.

Warnings

- Do not use adenosine, amiodarone, diltiazem, metoprolol, digoxin, or another AV-nodal blocker for suspected pre-excited atrial fibrillation/flutter.
- Verify synchronization before every cardioversion attempt; many devices require reactivation after each shock.
- Sedation must not delay cardioversion. Loss of pulse requires immediate transition to AC-03 and AC-09.

Clinical pearls

- In an older adult or patient with structural heart disease, regular wide-complex tachycardia is VT until proven otherwise.
- Apply pads early because a stable patient may deteriorate rapidly.
- Adenosine may terminate SVT with aberrancy or reveal atrial activity; it does not treat ventricular tachycardia.
- A change to polymorphic morphology requires immediate transition to AC-08.
- Capture the rhythm before, during, and after medication or cardioversion whenever feasible.

Special populations

Pre-excited atrial fibrillation/flutter

- An irregular rapid wide-complex rhythm with variable morphology or known Wolff-Parkinson-White may represent pre-excitation.
- Avoid adenosine, amiodarone, and AV-nodal blockers; cardiovert if unstable and contact Medical Control early if stable.

Toxicologic or metabolic wide-complex rhythm

- Hyperkalemia and sodium-channel-blocker toxicity require immediate cause-specific treatment in addition to rhythm and perfusion support.
- Open the linked renal/hyperkalemia or toxic-ingestion protocol rather than relying on amiodarone alone.

Related pathways

- AC-04 ACS / STEMI (p. 80): Open for ischemic chest discomfort or suspected ACS.
- AC-08 Polymorphic Wide-Complex Tachycardia (p. 93): Open immediately if the rhythm becomes polymorphic.
- AC-09 VF / Pulseless VT (p. 96): Open immediately if ventricular tachycardia becomes pulseless.
- AC-03 Adult Cardiac Arrest (p. 77): Open the master cardiac-arrest pathway after loss of pulse.
- AM-03 Dialysis / Renal Failure (p. 123): Open for suspected hyperkalemia or dialysis-related emergency.
- TE-07 Overdose / Toxic Ingestion (p. 211): Open for suspected sodium-channel-blocker or toxicologic dysrhythmia.

AC-08
Adult Polymorphic Wide-Complex Tachycardia

Revised October 16, 2026

Overview

Adult pathway for patients 16 years of age and older with sustained or recurrent polymorphic wide-complex tachycardia and a pulse.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Confirm Pulse + Polymorphic Wide Rhythm
EMT / AEMT / Paramedic
Beat-to-beat QRS variation • pads immediately • airway/ventilation • oxygen if hypoxemic • do not delay shock

↓

2. Sustained Polymorphic VT
EMT / AEMT / Paramedic
Treat as unstable regardless of apparent perfusion

↓

3. Immediate Unsynchronized Shock
Paramedic
200 J biphasic • verify manufacturer setting • escalate according to device • never attempt synchronization

↓

4. Pulse Lost?
EMT / AEMT / Paramedic
Begin CPR immediately and open AC-09 VF / Pulseless VT

↓

5. Rhythm Terminates
EMT / AEMT / Paramedic
Obtain rhythm strip/12-lead • determine whether baseline QT is prolonged

Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Long QT / Torsades Recurrence
Paramedic
Magnesium sulfate 1-2 g IV over 15 min • may repeat once

↓

7. Normal QT Recurrence
Paramedic
Treat suspected myocardial ischemia • lidocaine 100 mg IV/IO initial dose

↓

8. Reassess + Transport
EMT / AEMT / Paramedic
Pads remain attached • continuous monitoring • early notification • anticipate recurrence or VF

Indications

- Sustained polymorphic wide-complex tachycardia with a pulse in a patient 16 years of age or older.
- Recurrent self-terminating polymorphic VT requiring prevention of recurrence after pads are applied and immediate-shock readiness is established.
- Suspected torsades de pointes associated with a prolonged baseline QT interval.

Contraindications

- There is no contraindication to immediate unsynchronized defibrillation for sustained polymorphic VT.
- Do not use magnesium routinely for polymorphic VT when the baseline QT interval is normal.
- Do not use lidocaine as the torsades/long-QT treatment pathway.

Assessment

Immediate rhythm and perfusion assessment

- Apply pads immediately and do not delay defibrillation for vascular access, medication administration, sedation, or a 12-lead ECG.

QT classification after rhythm termination

- Use the preceding or post-conversion rhythm strip/12-lead ECG to determine whether the baseline QT is prolonged.
- Torsades is suggested by prolonged QT, pause-dependent onset, associated bradycardia, recurrent self-termination, congenital long-QT history, or QT-prolonging medication exposure.
- Normal-QT polymorphic VT is commonly associated with acute myocardial ischemia or infarction.

Cause assessment

- Review prescribed and nonprescribed medications for QT prolongation and discontinue suspected QT-prolonging agents.
- Assess for myocardial ischemia, renal failure, dialysis, toxicologic exposure, stimulant use, electrolyte abnormality, congenital long-QT syndrome, and digitalis toxicity.
- Do not delay recurrent shock treatment while investigating the underlying cause.

Treatment

1. If the rhythm persists or sustained polymorphic VT recurs, repeat unsynchronized defibrillation using the manufacturer-recommended setting and energy escalation.
2. Keep pads attached, reassess complete vital signs, notify early, and transport rapidly while anticipating recurrent polymorphic VT or VF.

Medication orders

Magnesium sulfate

1-2 g IV over 15 minutes; may repeat once

- Paramedic standing order for recurrent polymorphic VT associated with a prolonged QT interval/torsades de pointes.
- Defibrillation takes priority for any sustained episode.
- Do not administer routinely when the QT interval is normal.
- Monitor continuously for hypotension, bradycardia, respiratory depression, and recurrent dysrhythmia.

Lidocaine

100 mg IV/IO initial dose

- Paramedic standing order for recurrent polymorphic VT only when the baseline QT interval is normal.
- Treat associated myocardial ischemia and keep pads attached.
- Do not substitute lidocaine for immediate defibrillation of a sustained episode.

Midazolam

2-5 mg IV/IO; use the smallest effective dose

- Paramedic sedation when a conscious patient requires unsynchronized defibrillation.
- Do not delay shock delivery for sedation.
- Continuous airway, respiratory, SpO_2, cardiac, and blood-pressure monitoring is required.

Provider-level care

EMT Care Rapid recognition, immediate pad placement, supportive care, and preparation for deterioration.

EMT: Perform the primary assessment, confirm a pulse, support airway and ventilation, and give oxygen only when hypoxemic.

• Recognize polymorphic wide-complex tachycardia by beat-to-beat variation in QRS morphology and immediately request paramedic intervention.

• Apply defibrillation pads, obtain complete vital signs when feasible, and do not delay definitive treatment for a 12-lead ECG.

• Prepare for immediate CPR and AED use if the patient becomes pulseless.

• After rhythm termination, obtain a rhythm strip and 12-lead ECG when trained and equipped without delaying transport.

AEMT Care All EMT care plus vascular access, medication preparation, and continuous reassessment.

AEMT: Perform all EMT actions and establish IV/IO access without delaying defibrillation.

• Prepare magnesium sulfate, lidocaine, midazolam, suction, ventilation equipment, and airway equipment for paramedic use.

• Reassess pulse, blood pressure, mental status, perfusion, respiratory status, and rhythm after every shock or clinical change.

• Keep pads attached and prepare for recurrent polymorphic VT, ventricular fibrillation, or cardiac arrest.

• Do not independently administer magnesium sulfate, lidocaine, or procedural sedation under AC-08.

Paramedic Care All AEMT care plus immediate manual defibrillation, QT-directed medication treatment, and sedation when feasible.

Paramedic: Perform all EMT and AEMT actions and confirm a pulse with polymorphic wide-complex tachycardia.

• For sustained polymorphic VT, immediately deliver an unsynchronized biphasic shock at 200 J; confirm the manufacturer-recommended setting and escalate according to the device.

• Never attempt synchronized cardioversion for polymorphic VT because reliable synchronization is not possible.

• If the patient is conscious, sedate when immediately feasible with midazolam 2-5 mg IV/IO using the smallest effective dose, but never delay defibrillation.

• After rhythm termination, determine whether the baseline QT is prolonged using the preceding or post-conversion rhythm strip/12-lead ECG.

• For recurrent long-QT polymorphic VT/torsades, give magnesium sulfate 1-2 g IV over 15 minutes; may repeat once for continued recurrence.

• For recurrent polymorphic VT with a normal QT, treat suspected myocardial ischemia and give lidocaine 100 mg IV/IO as the initial dose.

Warnings

- Do not delay defibrillation for vascular access, medication, sedation, rhythm printing, or a 12-lead ECG.
- Do not give adenosine, diltiazem, verapamil, metoprolol, or another AV-nodal blocker for polymorphic wide-complex tachycardia.
- Avoid amiodarone and other QT-prolonging antiarrhythmics when torsades or a prolonged QT is suspected.
- Recurrent polymorphic VT may rapidly degenerate into ventricular fibrillation.

Clinical pearls

- Polymorphic VT varies in QRS shape and axis from beat to beat and may be difficult to distinguish from coarse VF.
- Torsades de pointes is polymorphic VT associated with a prolonged baseline QT; polymorphic VT with a normal QT is commonly ischemic.
- The QT interval should be assessed on the rhythm before or after the episode rather than during polymorphic VT.
- Defibrillation terminates the episode but does not prevent recurrence; subsequent therapy is directed by the baseline QT.
- Keep pads attached throughout transport because recurrence is common.

Special populations

Long-QT polymorphic VT / torsades

- Give magnesium for recurrence after immediate-shock readiness is established.
- For recurrent pause-dependent torsades or associated bradycardia, contact Medical Control for overdrive pacing.

Normal-QT polymorphic VT

- Suspect acute myocardial ischemia or infarction and open AC-04.
- Use lidocaine rather than the torsades magnesium pathway.

- Seek expert consultation for suspected catecholaminergic polymorphic VT, digitalis toxicity, or another unusual cause.

Related pathways

- AC-04 ACS / STEMI (p. 80): Open for suspected myocardial ischemia or infarction.
- AC-07 Monomorphic Wide-Complex Tachycardia (p. 90): Open if the rhythm is regular with uniform QRS morphology.
- AC-09 VF / Pulseless VT (p. 96): Open immediately if the pulse is absent or lost.
- AM-03 Dialysis / Renal Failure (p. 123): Open for suspected dialysis-related or renal emergency.
- TE-07 Overdose / Toxic Ingestion (p. 211): Open for suspected medication, stimulant, digitalis, or other toxicologic cause.

AC-09
Ventricular Fibrillation / Pulseless VT

Revised October 16, 2026

Overview

Adult pathway for patients 16 years of age and older in ventricular fibrillation or pulseless ventricular tachycardia.

High-quality CPR, immediate defibrillation, minimal peri-shock pauses, and treatment of reversible causes take priority over advanced procedures.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. VF / Pulseless VT
EMT / AEMT / Paramedic
Start CPR • bag-mask ventilation with oxygen • attach monitor/defibrillator • activate arrest team

↓

2. Shock 1
EMT / AEMT / Paramedic
200 J biphasic • verify manufacturer setting • immediately resume CPR for 2 min

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. CPR 2 min + IV/IO
EMT / AEMT / Paramedic
Rate 100-120/min • depth ≥2 in • full recoil • minimize pauses • IV preferred, IO if delayed

↓

4. Shock 2
AEMT / Paramedic
Same or higher energy per device • immediately resume CPR • epinephrine 1 mg IV/IO ASAP, then every 3-5 min

↓

5. CPR 2 min + Airway
EMT / AEMT / Paramedic
Consider advanced airway/capnography without interrupting compressions • treat reversible causes

↓

6. Shock 3
EMT / AEMT / Paramedic
Same or higher energy per device • immediately resume CPR

↓

7. CPR 2 min + Antiarrhythmic
Paramedic
Amiodarone 300 mg IV/IO; second dose 150 mg • alternative lidocaine 1-1.5 mg/kg IV/IO, then repeat per dose table (maximum total 3 mg/kg)

↓

8. Rhythm Every 2 min
EMT / AEMT / Paramedic
Shockable → shock and resume CPR • organized rhythm → pulse check • ROSC → AC-10

↓

9. Persistent Arrest
EMT / AEMT / Paramedic
Continue shock/CPR cycles • epinephrine every 3-5 min • treat Hs and Ts • consider AC-12 when criteria met

Indications

- Ventricular fibrillation without a pulse in a patient 16 years of age or older.
- Pulseless ventricular tachycardia in a patient 16 years of age or older.
- Pulseless polymorphic ventricular tachycardia managed as a shockable cardiac-arrest rhythm.

Contraindications

- There is no contraindication to immediate defibrillation for confirmed VF or pulseless VT.
- Do not defibrillate an organized rhythm unless the patient is pulseless and the rhythm is confirmed as ventricular tachycardia.
- Do not interrupt CPR or delay defibrillation for vascular access, medication administration, or advanced-airway placement.

Assessment

Immediate arrest assessment

- Begin CPR, provide bag-mask ventilation with oxygen, attach pads, and identify VF or pulseless VT.
- Establish the time last known responsive, whether the arrest was witnessed, bystander CPR/AED use, initial rhythm, downtime, and suspected cause when this does not interrupt treatment.

CPR quality

- Keep rhythm analysis, pulse checks, and peri-shock pauses under 10 seconds.

Rhythm, ventilation, and perfusion monitoring

- Use continuous waveform capnography after advanced-airway placement to confirm position and monitor ventilation and CPR trends.
- An abrupt sustained rise in $EtCO_2$ may suggest ROSC but requires rhythm and pulse confirmation; no absolute $EtCO_2$ value determines termination.

Reversible causes

- Evaluate continuously for hypovolemia, hypoxia, hydrogen ion excess/acidosis, hypo-/hyperkalemia, and hypothermia.
- Evaluate for tension pneumothorax, cardiac tamponade, toxins, pulmonary thrombosis, and coronary thrombosis.
- Use the linked renal/hyperkalemia and toxic-ingestion protocols for cause-specific treatment.

Treatment

1. During the first post-shock CPR cycle, establish IV access; use IO access if IV access is unsuccessful or would delay medication.
2. Perform a rhythm check after 2 minutes. If VF/pVT persists, deliver the second shock using the same or higher energy according to the device and immediately resume CPR.
3. After the next 2-minute CPR cycle, deliver the third shock if VF/pVT persists and immediately resume CPR.

Medication orders

Epinephrine

1 mg IV/IO after the second shock as soon as possible; repeat every 3-5 minutes

- AEMT and Paramedic standing order under AC-09; no EMT administration.
- Do not interrupt CPR or delay defibrillation for medication administration.
- Document each dose and maintain a visible 3-5-minute medication timer.

Amiodarone

First dose 300 mg IV/IO; second dose 150 mg IV/IO

- Paramedic standing order for VF/pVT refractory to the third shock.
- Administer during CPR without interrupting compressions.
- Do not combine with lidocaine without Medical Control.

Lidocaine

1-1.5 mg/kg IV/IO initial dose; then 0.5-0.75 mg/kg every 5-10 minutes; maximum total 3 mg/kg

- Paramedic alternative to amiodarone for shock-refractory VF/pVT.
- Administer during CPR without interrupting compressions.
- Do not combine with amiodarone without Medical Control.

Magnesium sulfate

1-2 g IV/IO push; may repeat once

- Paramedic standing order only for suspected pulseless torsades/long-QT polymorphic VT.
- Routine magnesium is not indicated for other VF/pVT cardiac arrests.

Provider-level care

EMT Care Immediate CPR, AED operation, ventilation, compressor rotation, timing, and documentation.

EMT: Confirm unresponsiveness, abnormal or absent breathing, and absence of a pulse for no more than 10 seconds; begin CPR immediately.

• Perform compressions at 100-120/min to a depth of at least 2 inches with complete recoil and minimal interruption.

• Apply the AED/monitor pads and deliver a shock immediately when advised; resume CPR without a post-shock pulse check.

• Provide bag-mask ventilation with oxygen at a 30:2 compression-to-ventilation ratio until an advanced airway is placed.

• Rotate compressors every 2 minutes or sooner when fatigued and announce cycle, shock, medication, and rhythm-check times.

• Check a pulse only when an organized rhythm appears and prepare for immediate re-arrest if ROSC occurs.

AEMT Care All EMT care plus vascular access, epinephrine, airway assistance, and medication preparation.

AEMT: Perform all EMT actions and establish IV access during CPR without delaying a shock; use IO access when IV access is unsuccessful or would delay medication.

• After the second shock, give epinephrine 1 mg IV/IO as soon as possible and repeat every 3-5 minutes.

• Assist with advanced-airway placement and continuous waveform capnography without interrupting compressions or delaying defibrillation.

• Prepare amiodarone, lidocaine, and magnesium sulfate for paramedic administration.

• Continuously reassess CPR quality, ventilation rate, waveform capnography, medication timing, and reversible causes.

• Do not independently administer amiodarone, lidocaine, or magnesium sulfate under AC-09.

Paramedic Care All AEMT care plus manual defibrillation, rhythm interpretation, antiarrhythmics, advanced airway/capnography, and reversible-cause treatment.

Paramedic: Perform all EMT and AEMT actions and confirm VF or pulseless VT while maintaining continuous high-quality CPR.

• Defibrillate at 200 J biphasic, confirm the manufacturer-recommended setting, and use the same or higher energy for subsequent shocks according to the device.

• Charge the defibrillator during compressions, clear briefly, deliver the shock, and resume CPR immediately.

• After the second shock, give epinephrine 1 mg IV/IO as soon as possible and repeat every 3-5 minutes.

• After the third shock, give amiodarone 300 mg IV/IO; give 150 mg IV/IO as the second dose for persistent or recurrent VF/pVT.

• As an alternative to amiodarone, give lidocaine 1-1.5 mg/kg IV/IO initially; then 0.5-0.75 mg/kg every 5-10 minutes to a maximum total of 3 mg/kg. Do not combine amiodarone and lidocaine without Medical Control.

• For suspected pulseless torsades, give magnesium sulfate 1-2 g IV/IO push; may repeat once. Do not use magnesium routinely for other VF/pVT arrests.

• Place an advanced airway and use continuous waveform capnography when this can be accomplished without interrupting compressions or delaying shocks.

Warnings

- Do not delay defibrillation for airway placement, vascular access, medication, scene movement, or a 12-lead ECG.
- Avoid excessive ventilation. After advanced-airway placement, give one breath every 6 seconds with continuous compressions.
- Routine magnesium, calcium, and sodium bicarbonate are not indicated in undifferentiated VF/pVT; use them only for a specific reversible cause or linked protocol indication.
- Routine vector-change defibrillation and double-sequential defibrillation are not included because their usefulness remains unestablished.

Clinical pearls

- Defibrillation and high-quality CPR are the highest-priority treatments for VF/pVT.
- Charging during compressions and immediate post-shock CPR reduce harmful pauses.
- An abrupt sustained $EtCO_2$ increase may indicate ROSC; confirm with an organized rhythm and pulse without an unnecessary interruption.
- VF/pVT may be the result of coronary occlusion, hyperkalemia, hypothermia, toxins, or another reversible cause that requires simultaneous treatment.

Special populations

Suspected pulseless torsades

- Defibrillate and continue standard VF/pVT arrest care.
- Open AC-08 if a pulse returns with recurrent polymorphic VT.

Suspected hyperkalemic or toxicologic arrest

- Continue standard VF/pVT care while opening the linked cause-specific protocol.

Related pathways

- AC-03 Adult Cardiac Arrest (p. 77): Open the master adult cardiac-arrest pathway.
- AC-08 Polymorphic Wide-Complex Tachycardia (p. 93): Open if a pulse returns with recurrent polymorphic VT.
- AC-10 Post-Resuscitation Care (p. 99): Open immediately after return of spontaneous circulation.
- AC-11 TEAM-Focused CPR (p. 102): Open for team roles, timing, communication, and CPR quality.
- AC-12 Termination of Resuscitation (p. 104): Open for persistent arrest when termination criteria may be met.
- AM-03 Dialysis / Renal Failure (p. 123): Open for suspected hyperkalemic or dialysis-related arrest.
- TE-07 Overdose / Toxic Ingestion (p. 211): Open for suspected sodium-channel blocker, opioid, or other toxicologic arrest.

AC-10 Post-Resuscitation Care

Revised October 16, 2026

Overview

Adult pathway for patients 16 years of age and older with sustained return of spontaneous circulation after cardiac arrest.

Prioritize controlled oxygenation and ventilation, avoidance of hypotension, rapid 12-lead ECG acquisition, cause-specific treatment, appropriate destination, and preparation for re-arrest.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Confirm Sustained ROSC
EMT / AEMT / Paramedic
Pulse • BP/MAP • organized rhythm • perfusion • keep pads attached • prepare for re-arrest

↓

2. Airway + Oxygenation
EMT / AEMT / Paramedic
100% oxygen until reliable SpO_2 • then titrate to 90-98% • waveform capnography for advanced airway

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. Control Ventilation
EMT / AEMT / Paramedic
Begin near 10 breaths/min • target $EtCO_2$ approximately 35-45 mm Hg • avoid hyperventilation

↓

4. MAP ≥65 mm Hg?
AEMT / Paramedic
If low: NS 250-500 mL IV/IO and reassess • smaller volume for CHF, renal/liver failure, or overload

↓

5. Persistent Hypotension
Paramedic
Norepinephrine 0.1-2 mcg/kg/min IV/IO • titrate to MAP ≥65 • alternatives require Medical Control

↓

6. 12-Lead ECG + Transmit
EMT / AEMT / Paramedic
Obtain as soon as feasible • repeat if immediate post-ROSC tracing or condition changes • open AC-04 when indicated

↓

7. Glucose + Neuro + Temperature
EMT / AEMT / Paramedic
Treat glucose <70 • document GCS/pupils/seizures • prevent fever • no rapid cold-fluid cooling

↓

8. Destination + Notify
Paramedic
PCI-capable center for STEMI, cardiogenic shock, recurrent ventricular arrhythmia, or ongoing ischemia • consider Covenant Health Air

Indications

- Sustained return of spontaneous circulation after adult cardiac arrest.
- Post-arrest patient with a pulse requiring stabilization, monitoring, cause identification, and destination selection.

Contraindications

- There is no contraindication to immediate post-ROSC stabilization.
- Do not continue AC-10 if the patient becomes pulseless; immediately resume CPR and open the rhythm-appropriate cardiac-arrest protocol.
- Do not give a rapid cold-IV-fluid infusion for prehospital temperature control after ROSC.

Assessment

Confirm ROSC and prepare for re-arrest

- Confirm pulse, blood pressure/MAP, organized rhythm, perfusion, respiratory effort, and clinical response.
- Keep pads attached and maintain immediate access to the defibrillator, suction, ventilation equipment, and arrest medications.
- Reassess pulse, rhythm, blood pressure, perfusion, SpO_2, $EtCO_2$, respiratory status, and neurologic status frequently and after every intervention.

Airway, oxygenation, and ventilation

- Avoid both hypoxemia and prolonged hyperoxia.
- Recognize that post-ROSC $EtCO_2$ may underestimate $PaCO_2$ and follow clinical response and blood-gas results when available.

Hemodynamics

- Assess peripheral perfusion, capillary refill, skin findings, mental status, lung sounds, and signs of cardiogenic shock or volume overload.
- Use cautious fluid boluses and transition promptly to norepinephrine when hypotension persists.

Cardiac, neurologic, glucose, and temperature assessment

- Document GCS, pupils, purposeful movement, focal findings, and seizure activity without making early field neurologic prognostication.
- Measure temperature when available, prevent fever, and avoid rapid warming of spontaneous hypothermia.

Treatment

1. Confirm any advanced airway with continuous waveform capnography and reassess breath sounds, chest rise, tube depth, and waveform after every movement.
2. Obtain frequent blood pressures and maintain MAP at least 65 mm Hg.
3. Obtain and transmit a 12-lead ECG as soon as feasible and repeat it when the initial tracing was obtained immediately after ROSC or the clinical condition changes.
4. Measure temperature when available, prevent fever, avoid rapid cold-IV-fluid prehospital cooling, and do not rapidly warm spontaneous hypothermia.
5. Transport patients with persistent STEMI/STEMI-equivalent, cardiogenic shock, recurrent ventricular arrhythmias, or significant ongoing ischemia to a PCI-capable hospital and consider Covenant Health Air when it shortens time to definitive care.
6. Otherwise transport to the closest appropriate receiving facility with early notification and continuous reassessment.

Medication orders

Normal saline

250-500 mL IV/IO bolus, then reassess

- AEMT and Paramedic standing order for post-ROSC MAP below 65 mm Hg.
- Use smaller volumes for CHF, renal failure, liver failure, or suspected volume overload.
- Reassess blood pressure/MAP, lung sounds, perfusion, and respiratory status after each bolus.

Norepinephrine

0.1-2 mcg/kg/min IV/IO; titrate to MAP at least 65 mm Hg

- Paramedic preferred vasopressor for persistent post-ROSC hypotension after cautious fluid resuscitation.
- Use continuous cardiac monitoring and frequent blood-pressure reassessment.
- Alternative post-ROSC vasopressors require Medical Control.

Dextrose

Treat glucose below 70 mg/dL through UP-04 Altered Mental Status

- Avoid hypoglycemia and recheck glucose after treatment.
- Do not treat isolated post-ROSC hyperglycemia routinely in the field unless another protocol indicates treatment.

Provider-level care

EMT Care Immediate post-ROSC stabilization, ventilation, oxygen titration, monitoring, neurologic assessment, and re-arrest preparation.

EMT: Confirm sustained ROSC using a palpable pulse, blood pressure, organized rhythm, perfusion, and clinical response.

- Maintain airway and ventilation, begin near 10 breaths/min, and avoid hyperventilation.

• Use 100% oxygen until SpO_2 can be measured reliably, then titrate oxygen to maintain SpO_2 90-98%.

• Keep defibrillation pads attached, obtain frequent complete vital signs, and prepare to resume CPR and defibrillation immediately.

• Check blood glucose, document GCS, pupils, purposeful movement and seizure activity, and measure temperature when available.

• Acquire and transmit a 12-lead ECG when trained and equipped without delaying stabilization or transport.

AEMT Care All EMT care plus vascular access, cautious fluid resuscitation, airway support, and medication preparation.

AEMT: Perform all EMT actions and establish IV/IO access without delaying transport.

• For MAP below 65 mm Hg, give normal saline 250-500 mL IV/IO and reassess blood pressure, lung sounds, perfusion, and respiratory status.

• Use smaller fluid volumes for CHF, renal failure, liver failure, or suspected volume overload.

• Assist with advanced-airway confirmation and continuous waveform capnography; reassess breath sounds, chest rise, tube depth, and waveform after every movement.

• Prepare norepinephrine and recurrent-dysrhythmia medications for paramedic use.

• Recheck glucose and complete vital signs after interventions or any clinical change.

Paramedic Care All AEMT care plus advanced airway/capnography, vasopressor support, rhythm management, ECG interpretation, and destination selection.

Paramedic: Perform all EMT and AEMT actions and identify the likely cause of arrest and immediate post-ROSC threats.

• Confirm an advanced airway with continuous waveform capnography and ventilate near 10 breaths/min, titrating toward $EtCO_2$ approximately 35-45 mm Hg while recognizing that $EtCO_2$ may underestimate $PaCO_2$ after ROSC.

• If MAP remains below 65 mm Hg after cautious fluid, start norepinephrine 0.1-2 mcg/kg/min IV/IO and titrate to MAP at least 65 mm Hg.

• Alternatives to norepinephrine for post-ROSC hypotension require Medical Control.

• Interpret and transmit the 12-lead ECG, repeat when indicated, and activate AC-04 for persistent STEMI/STEMI-equivalent or suspected ongoing ischemia.

• Treat recurrent ventricular dysrhythmia through the linked rhythm-specific protocol and prepare for immediate re-arrest.

• Choose the receiving destination based on STEMI, cardiogenic shock, recurrent ventricular arrhythmia, ongoing ischemia, transport time, and available cardiac-arrest/PCI capability.

Warnings

• Re-arrest is common; keep pads attached and be prepared to resume CPR and defibrillation immediately.

• Avoid hypoxemia, prolonged hyperoxia, and hyperventilation after ROSC.

• Do not rely on a single immediate post-ROSC ECG to exclude or confirm coronary occlusion; repeat the tracing when indicated.

• Coma alone does not exclude a patient from cardiac-center or catheterization-laboratory activation.

• Routine prophylactic amiodarone is not required after ROSC without recurrent ventricular dysrhythmia.

Clinical pearls

• A post-ROSC patient remains critically unstable even when the initial blood pressure and rhythm appear acceptable.

• AHA targets SpO_2 90-98%, normal physiologic ventilation, and MAP at least 65 mm Hg after ROSC.

• Post-ROSC $EtCO_2$ may underestimate $PaCO_2$; use it as a trend and ventilation guide rather than an exact substitute for arterial blood gas.

• Immediate post-ROSC ECG changes may evolve; repeat ECG can reduce both missed occlusion and false activation.

Special populations

Unresponsive after ROSC

• Prevent fever and communicate the need for protocolized hospital temperature control between 32°C and 37.5°C.

• Avoid early neurologic prognostication and document the examination before sedatives or paralytics when feasible.

Spontaneous hypothermia

• Remove wet clothing and prevent further heat loss.

• Do not rapidly warm a spontaneously hypothermic post-arrest patient.

Post-intubation care

- Open AR-08 for tube/BIAD confirmation, ventilation, sedation, analgesia, and ongoing airway management.
- Reassess tube position and waveform capnography after every movement or transfer.

Related pathways

- AC-04 ACS / STEMI (p. 80): Open for persistent STEMI/STEMI-equivalent or significant ongoing ischemia.
- AC-06 Narrow-Complex Tachycardia (p. 87): Open for recurrent narrow-complex tachycardia after ROSC.
- AC-07 Monomorphic Wide-Complex Tachycardia (p. 90): Open for recurrent monomorphic VT with a pulse.
- AC-08 Polymorphic Wide-Complex Tachycardia (p. 93): Open for recurrent polymorphic VT with a pulse.
- AC-09 VF / Pulseless VT (p. 96): Open immediately if the patient loses pulses in VF or VT.
- UP-04 Altered Mental Status (p. 14): Open for glucose below 70 mg/dL or another persistent altered-mental-status cause.
- AR-08 Post-Intubation / BIAD Management (p. 155): Open for advanced-airway confirmation, ventilation, sedation, and analgesia.
- AM-03 Dialysis / Renal Failure (p. 123): Open for suspected hyperkalemia, dialysis-related arrest, or renal failure.
- TE-07 Overdose / Toxic Ingestion (p. 211): Open for suspected toxicologic or overdose-related arrest.

AC-11
TEAM-Focused CPR

Revised October 16, 2026

Overview

Operational high-performance CPR pathway for coordinated adult cardiac-arrest care by Claiborne EMS.

Use AC-11 with AC-03 and the applicable rhythm-specific protocol; this page assigns team functions and CPR-quality expectations without replacing the clinical arrest algorithm.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Recognize Arrest
EMT / AEMT / Paramedic
Start CPR • apply pads/AED • BVM with oxygen • start scene timer

↓

2. Assign Team Functions
EMT / AEMT / Paramedic
Leader • compressor 1/2 • airway • monitor/defibrillator • IV/medications • recorder/timer

↓

3. Deliver High-Quality CPR
EMT / AEMT / Paramedic
100-120/min • depth 2-2.4 in • full recoil • compression fraction >80%

↓

4. Manage Ventilation
EMT / AEMT / Paramedic
30:2 before advanced airway • then continuous CPR with 1 breath every 6 sec • avoid hyperventilation

↓

5. Two-Minute Cycle
EMT / AEMT / Paramedic
Precharge when possible • rhythm/pulse check ≤10 sec • shock if indicated • immediately resume CPR

↓

6. Communicate + Record
EMT / AEMT / Paramedic
Closed-loop orders • announce rhythms, shocks, medications, airway, $EtCO_2$, and times

↓

7. ROSC or Termination
EMT / AEMT / Paramedic
ROSC: AC-10/AC-13 • persistent arrest: evaluate AC-12

Indications

- Any adult cardiac arrest managed by two or more Claiborne EMS responders.
- Training, simulation, and quality-improvement review of adult cardiac-arrest team performance.

Assessment

Team assignment

- Identify one team leader and assign compressor 1, compressor 2, airway, monitor/defibrillator, IV/medications, and recorder/timer functions as personnel arrive.
- Combine functions when staffing is limited, but do not combine tasks in a way that interrupts compressions, delays defibrillation, or prevents the leader from maintaining oversight.
- Use predetermined equipment positioning and role cards when available.

CPR quality

- Change compressors every two minutes or sooner for fatigue; complete the change during the planned rhythm check in 10 seconds or less.
- Use an audiovisual CPR feedback device when available and correct low or decreasing $EtCO_2$ by reassessing CPR quality, airway, and ventilation.

Airway and rhythm-cycle coordination

- With an advanced airway in place, provide continuous compressions and one breath every six seconds with continuous waveform capnography.
- Prepare the next intervention before each two-minute rhythm check and precharge the defibrillator during compressions when feasible.
- Keep rhythm and pulse checks at 10 seconds or less and immediately resume compressions after any shock or no-shock decision.

Treatment

1. Assign team functions as personnel arrive; the team leader states the working rhythm pathway and the next expected intervention.
2. Use closed-loop communication: address the assigned provider, give one clear instruction, require repeat-back, and announce completion.
3. The recorder announces the approaching two-minute rhythm check and tracks CPR cycles, rhythms, shocks, medications, airway events, $EtCO_2$, ROSC, transport, and termination times.
4. Continuously reassess compression quality, ventilation rate, pad contact, airway position, vascular access, medication timing, and reversible causes.
5. After the event, complete a brief team debrief, preserve monitor/CPR data when available, and restore equipment readiness.

Provider-level care

Provider-Level Team Actions Every provider may fill more than one team function when staffing is limited; lifesaving BLS care begins immediately.

EMT: Begin compressions, BVM ventilation, AED/monitor pad placement, scene timing, and event documentation without waiting for ALS.

- Rotate compressors every two minutes or sooner for fatigue and provide real-time feedback on rate, depth, recoil, and pauses.
- Repeat assigned tasks and treatment orders back to the team leader and announce when each is complete.

AEMT: Perform all EMT actions and assume vascular-access/medication or airway functions as assigned.

- Establish IV access first; use IO when IV access is unsuccessful, not feasible, or would delay required arrest medication.
- Administer treatments authorized by the active cardiac-arrest protocol and announce medication, dose, route, and time.

Paramedic: Serve as team leader when assigned, preserve global oversight, and avoid task fixation.

- Direct rhythm interpretation, manual defibrillation, advanced-airway decisions, medication timing, reversible-cause treatment, ROSC transition, and termination evaluation.
- Use waveform capnography and available CPR feedback to identify correctable ventilation or compression problems.

Warnings

- Do not interrupt compressions for IV/IO access, airway placement, medication preparation, or nonessential movement.
- Do not allow compressor rotation, rhythm analysis, pulse checks, charging, or shock delivery to create a pause longer than 10 seconds.
- Do not delay movement when transport is indicated by the active arrest, special-circumstance, or termination protocol.

Clinical pearls

- Predetermined positioning and role cards reduce duplicated work and delays.
- The next compressor should be in position before the two-minute cycle ends.

- Charging during compressions and resuming compressions immediately after shock delivery shortens the peri-shock pause.
- Structured post-event feedback and debriefing can improve CPR quality and team performance.

Special populations

Limited staffing

- Prioritize compressions, defibrillation, and effective ventilation before access, advanced airway, or documentation tasks.
- Request additional personnel early and reassign functions explicitly as responders arrive.

Arrest during movement or transport

- Stop vehicle movement when required to safely deliver manual CPR or defibrillation.
- Reconfirm team functions and equipment security before patient movement and after every transfer between surfaces.

Related pathways

- AC-03 Adult Cardiac Arrest (p. 77): Open for the primary adult cardiac-arrest algorithm.
- AC-09 VF / Pulseless VT (p. 96): Open for shockable VF/pVT arrest treatment.
- AC-10 Post-Resuscitation Care (p. 99): Open immediately after sustained ROSC.
- AC-12 Termination of Resuscitation (p. 104): Open when persistent adult OHCA meets evaluation criteria.
- AC-13 Post-Arrest Temperature Management (p. 106): Open after ROSC when the adult remains unresponsive to verbal commands.
- TB-10 Traumatic Arrest (p. 196): Open for cardiac arrest caused by trauma.

AC-12
Termination of Resuscitation

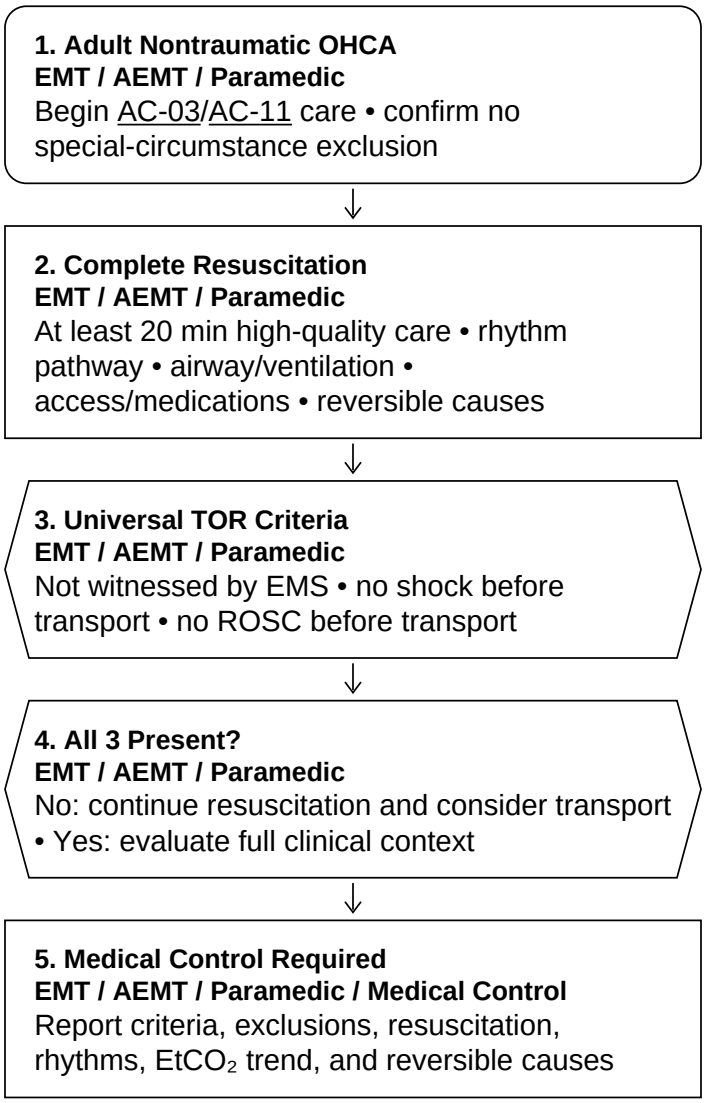
Revised October 16, 2026

Overview

Field termination pathway for an adult, nontraumatic, out-of-hospital cardiac arrest after a complete resuscitation and application of the Universal Termination of Resuscitation rule.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.



Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Termination Authorized?
EMT / AEMT / Paramedic
Yes: stop CPR, record time, support family, preserve scene • No/unavailable: continue and transport

↓

7. Document + Disposition
EMT / AEMT / Paramedic
Times • criteria • interventions • physician order • notifications • family support • scene disposition

Indications

- Adult, nontraumatic, out-of-hospital cardiac arrest with persistent pulselessness after at least 20 minutes of documented high-quality resuscitation.

Contraindications

- EMS-witnessed arrest, any shock delivered before transport, or any ROSC before transport.
- Pediatric cardiac arrest, traumatic arrest, overdose or suspected toxicologic arrest, pregnancy, hypothermia, drowning/submersion, lightning/electrocution, or in-hospital cardiac arrest.
- A reversible special circumstance that warrants continued treatment or transport.
- Medical Control declines termination or cannot be reached.

Assessment

Required resuscitation before consideration

- Provide appropriate defibrillation, ventilation, airway management, vascular access, arrest medications, and reversible-cause treatment without interrupting compressions.
- Review witness information, bystander care, downtime, all rhythms, shocks, transient pulse or movement, medication response, airway position, $EtCO_2$ trend, and possible reversible causes.
- The 20-minute safeguard does not independently establish futility and does not replace the Universal TOR criteria or Medical Control decision.

Exclusion screen

- Use separate obvious-death, valid DNR/advance-directive, traumatic-arrest, and special-circumstance protocols when applicable.
- When history or circumstances are uncertain, continue resuscitation, consult Medical Control, and transport as directed.

Treatment

1. Continue AC-03 and AC-11 resuscitation while termination eligibility is evaluated.
2. Contact Medical Control and report the patient, circumstances, Universal TOR criteria, exclusions, duration, rhythms, shocks, medications, airway, $EtCO_2$ trend, reversible causes, and response to treatment.
3. If termination is authorized, repeat the order back, stop resuscitation, confirm persistent apnea and pulselessness, and document the termination time.
4. If termination is declined or Medical Control is unavailable, continue resuscitation and transport unless a separate obvious-death or valid DNR policy applies.
5. Support family and witnesses, explain the process compassionately, preserve the scene when investigation is possible, and coordinate disposition and required notifications.

Provider-level care

Provider-Level Actions All levels continue resuscitation while criteria are evaluated; no provider stops CPR solely from the checklist.

EMT: Continue high-quality CPR, AED use, ventilation, timing, and documentation while determining EMS witness status, shocks delivered, and any ROSC.

- Identify special circumstances and immediately report any exclusion or possible reversible cause.
- Do not terminate resuscitation without a documented Medical Control order unless a separate obvious-death or valid DNR policy applies.

AEMT: Perform all EMT actions plus indicated vascular access, arrest medication, airway support, and reversible-cause treatment.

- Confirm the duration and adequacy of resuscitation and assist with the Medical Control report and order read-back.
- Continue resuscitation and transport when any Universal TOR criterion is absent, an exclusion is present, or Medical Control is unavailable.

Paramedic: Confirm that at least 20 minutes of documented resuscitation, rhythm-specific care, and reversible-cause evaluation have been completed.

- Confirm all three Universal TOR criteria and absence of exclusions before requesting termination authorization.
- Provide the complete resuscitation summary to Medical Control, repeat the order back, direct

termination when authorized, and coordinate disposition.

Medical Control: Review eligibility, Universal TOR criteria, exclusions, resuscitation course, rhythms, shocks, ROSC, $EtCO_2$ trend, reversible causes, and operational circumstances before authorizing termination.

Warnings

- $EtCO_2$ must never be the sole reason to terminate resuscitation.
- In an intubated adult, $EtCO_2$ below 10 mmHg after 20 minutes may contribute only as one component of a multimodal decision.
- Do not apply a specific $EtCO_2$ cutoff to a nonintubated patient or a patient ventilated through a supraglottic airway.
- Do not terminate when any Universal TOR criterion is absent, an exclusion is present, or Medical Control authorization has not been obtained.
- Do not remove tubes, vascular access, clothing, or other potential evidence when the death may require medical-examiner or law-enforcement investigation.

Clinical pearls

- The Universal TOR rule is intended for a mixed BLS/ALS or tiered EMS response system.
- Low $EtCO_2$ should first prompt reassessment of compression quality, ventilation rate, airway placement, and reversible causes.
- Use clear team communication and compassionate family support before and after the termination decision.

Special populations

Separate pathways required

- Use TB-10 for traumatic arrest, TE-07 for suspected overdose/toxicologic arrest, TE-05 for hypothermia, TE-03 for drowning/submersion, and AO-03 for pregnancy-associated emergencies.
- Consult Medical Control early for any uncertain cause or exceptional circumstance.

Obvious death or valid DNR

- Obvious-death criteria and valid DNR/advance-directive decisions are governed by separate policy and do not require completion of the Universal TOR pathway.

Related pathways

- AC-03 Adult Cardiac Arrest (p. 77): Continue the primary adult cardiac-arrest pathway while eligibility is evaluated.
- AC-09 VF / Pulseless VT (p. 96): Open for any shockable rhythm; a delivered shock excludes Universal TOR.
- AC-11 TEAM-Focused CPR (p. 102): Open for role assignment, CPR quality, timing, and documentation.
- TB-10 Traumatic Arrest (p. 196): Use instead for arrest caused by trauma.
- TE-03 Drowning / Submersion (p. 208): Use instead for drowning or submersion-related arrest.
- TE-05 Hypothermia (p. 210): Use instead for suspected hypothermic arrest.
- TE-07 Overdose / Toxic Ingestion (p. 211): Use instead for suspected overdose or toxicologic arrest.
- AO-03 OB-GYN Emergency (p. 140): Open for pregnancy-associated arrest or another obstetric emergency.

AC-13
Post-Arrest Temperature Management

Revised October 16, 2026

Overview

Field priorities are accurate temperature assessment, prevention of fever and further heat loss, avoidance of rapid cold-fluid cooling or rapid rewarming, and uninterrupted post-resuscitation stabilization and transport.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Sustained ROSC
EMT / AEMT / Paramedic
Open AC-10 • stabilize airway, ventilation, perfusion, rhythm, ECG, and glucose

↓

2. Follows Verbal Commands?
EMT / AEMT / Paramedic
Yes: continue AC-10 and prevent fever • No: enter temperature-management pathway

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. Measure Temperature
EMT / AEMT / Paramedic
Use a reliable core or approved temperature method when available • document site and trend

↓

4. Prevent Fever / Exposure
EMT / AEMT / Paramedic
Remove excessive heat sources • maintain normal environment • prevent further heat loss if cold

↓

5. No Routine Field Cooling
EMT / AEMT / Paramedic
Do not rapidly infuse cold IV fluids or initiate uncontrolled active cooling

↓

6. Spontaneously Hypothermic?
EMT / AEMT / Paramedic
Prevent additional heat loss • avoid rapid active rewarming

↓

7. Monitor + Transport
EMT / AEMT / Paramedic
ECG • SpO_2 • BP/MAP • temperature • $EtCO_2$ with advanced airway • early receiving notification

Indications

- Adult with sustained ROSC after cardiac arrest who remains unresponsive to verbal commands, regardless of arrest location or initial rhythm.

Contraindications

- No separate active prehospital temperature-control pathway is indicated for an adult who follows verbal commands after ROSC; continue AC-10 and prevent fever.
- Routine prehospital cooling with rapid cold-IV-fluid infusion is not indicated.

Assessment

Eligibility and neurologic assessment

- Assess whether the patient follows verbal commands and document the neurologic examination before sedatives or paralytics when feasible.

Temperature assessment

- Measure temperature as soon as practical using a reliable core or approved device when available.
- Document the measurement site, value, time, environmental exposure, wet clothing, and temperature trend.
- Assess for fever, spontaneous hypothermia, shivering, environmental heat or cold exposure, and treatments performed before EMS arrival.

Concurrent post-ROSC monitoring

- Continue cardiac monitoring, defibrillation-pad placement, SpO_2, serial blood pressure/MAP, glucose, and temperature trending.
- Use continuous waveform capnography when an advanced airway is present and reassess airway position after every movement.
- Obtain and transmit a 12-lead ECG, treat hypotension under AC-10, and remain prepared for immediate re-arrest.

Treatment

1. Open AC-10 and prioritize airway, physiologic ventilation, oxygenation, MAP at least 65 mmHg, glucose assessment, 12-lead ECG, and treatment of the arrest cause.
2. If the adult remains unresponsive to verbal commands, measure and trend temperature while preventing fever and avoidable environmental heat exposure.
3. If the patient is spontaneously hypothermic, remove wet clothing, insulate from further heat loss, handle gently, and avoid rapid active rewarming.
4. Continue continuous monitoring and frequent reassessment; treat re-arrest, dysrhythmia, hypotension, hypoglycemia, airway problems, or seizures under the applicable linked protocol.
5. Notify the receiving hospital early and report arrest details, ROSC time, neurologic responsiveness, temperature value/site/trend, airway, $EtCO_2$, ECG, blood pressure, glucose, and treatments.
6. Document that the receiving-hospital AHA temperature-control strategy for adults remaining unresponsive is 32°C-37.5°C for at least 36 hours; this is destination guidance and not a field target EMS must achieve.

Provider-level care

Provider-Level Actions Temperature management supplements AC-10 and must not delay post-ROSC stabilization or transport.

EMT: Assess response to verbal commands, obtain and document temperature when equipment is available, and prevent fever, excessive heat exposure, or further heat loss.

• Continue cardiac/AED monitoring, SpO_2, serial blood pressure, glucose assessment, airway support, and frequent reassessment.

• Do not initiate rapid cold-fluid cooling or uncontrolled active surface cooling.

AEMT: Perform all EMT actions and establish IV access without delaying transport.

• Administer fluids only for the hemodynamic indications and doses in AC-10; do not use chilled crystalloid for temperature control.

• Trend temperature, glucose, blood pressure, rhythm, oxygenation, and response to treatment.

Paramedic: Perform all prior actions and direct advanced airway, waveform capnography, ventilation, hemodynamic support, serial ECG, and re-arrest preparedness under AC-10.

• Document neurologic responsiveness before sedative or paralytic medication when feasible and report the temperature trend and neurologic findings to the receiving hospital.

• Communicate that an adult remaining unresponsive requires a deliberate receiving-hospital temperature-control strategy.

Warnings

• Do not let temperature measurement or intervention delay airway management, ventilation, perfusion support, ECG acquisition, destination notification, or transport.

• Do not rapidly infuse cold IV fluids for routine prehospital temperature control.

• Do not use uncontrolled ice packs, immersion, or another active-cooling method that cannot be continuously monitored and regulated.

• Do not make early neurologic prognostication in the field.

Clinical pearls

• AHA recommends a deliberate hospital temperature-control strategy for every adult who remains unresponsive to verbal commands after ROSC.

• Prehospital priorities are stabilization, fever prevention, accurate communication, and avoidance of harmful rapid cooling or rewarming.

• A normal initial temperature does not eliminate the need for ongoing fever prevention and hospital temperature control.

Special populations

Spontaneous hypothermia after ROSC

• Open TE-05 when hypothermia contributed to the arrest or remains a primary clinical problem.

Post-intubation care

• Open AR-08 for advanced-airway confirmation, ventilation, sedation, analgesia, and reassessment after movement.

Related pathways

• AC-10 Post-Resuscitation Care (p. 99): Primary pathway for all immediate post-ROSC stabilization.

• AC-09 VF / Pulseless VT (p. 96): Open immediately for re-arrest in VF or pulseless VT.

• AC-04 ACS / STEMI (p. 80): Open for persistent STEMI, acute coronary occlusion, or ongoing ischemia after ROSC.

• AR-08 Post-Intubation / BIAD Management (p. 155): Open for advanced-airway confirmation, ventilation, sedation, and analgesia.

• UP-04 Altered Mental Status (p. 14): Open for glucose below 70 mg/dL or another persistent altered-mental-status cause.

• TE-05 Hypothermia (p. 210): Open when hypothermia contributed to the arrest or remains clinically significant.

AC-14
LVAD Emergency

Revised October 16, 2026

Overview

Emergency assessment and treatment pathway for an adult with a durable continuous-flow left ventricular assist device outside the immediate implant hospitalization.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Unresponsive LVAD Patient
EMT / AEMT / Paramedic
Support ventilation as needed • do not diagnose arrest from pulse absence alone

↓

2. Assess Perfusion
EMT / AEMT / Paramedic
Skin color/temperature • capillary refill • Doppler MAP • waveform $EtCO_2$

↓

3. Adequate Perfusion?
EMT / AEMT / Paramedic
Any present: normal skin • normal capillary refill • MAP >50 mmHg • $EtCO_2$ >20 mmHg

↓

4. Yes - No Compressions
EMT / AEMT / Paramedic
Treat complaint and non-LVAD causes • monitor perfusion • contact LVAD center • transport

↓

5. No - Start CPR Now
EMT / AEMT / Paramedic
Begin compressions immediately • follow AC-03/AC-09 • do not delay for device troubleshooting

↓

6. Second Rescuer: Check LVAD
EMT / AEMT / Paramedic
Listen for hum • read alarms • inspect driveline/controller • restore connected power

↓

7. Transport + Coordinator
EMT / AEMT / Paramedic
LVAD-capable/implanting center when appropriate • bring caregiver and all backup equipment

Indications

- Adult with a durable continuous-flow LVAD who is ill, injured, unresponsive, poorly perfused, or experiencing a device alarm or suspected device malfunction.

Assessment

Perfusion assessment

- Assess responsiveness and breathing; support ventilation immediately when inadequate.
- Do not use absence of a palpable pulse alone to diagnose cardiac arrest because continuous-flow LVAD patients commonly have no palpable pulse.
- An unresponsive LVAD patient without any adequate-perfusion criterion is likely in cardiac arrest.
- Automated blood-pressure and pulse-oximetry measurements may be unreliable. Use Doppler with a manual cuff when available and trend waveform capnography when appropriate.

LVAD and equipment assessment

- Identify the device and implanting center from the emergency card, controller, equipment labels, caregiver, or medical alert information.
- Listen for a continuous mechanical hum over the left chest and left upper abdomen.
- Check the controller display and alarms, visible driveline connection and damage, battery charge, cable connections, and external power source.
- Reconnect an obvious disconnected driveline, battery, or power cable when it can be done without force or device disassembly.
- Ask the patient or trained caregiver what the alarm means, the patient's usual MAP and pump readings, and whether the current equipment differs from baseline.

Concurrent cause assessment

- Evaluate for dysrhythmia, hemorrhage or hypovolemia, hypoxia, hypoglycemia, overdose, right-ventricular failure, sepsis, stroke, trauma, and another time-sensitive complaint.
- Identify anticoagulant and antiplatelet medications, last dose when known, bleeding, melena or hematemesis, recent falls, and focal neurologic symptoms.
- A ventricular tachyarrhythmia may be tolerated while the patient remains conscious and perfused; treatment is determined by symptoms and perfusion rather than the monitor alone.

Treatment

1. Support ventilation as needed and determine whether adequate perfusion is present; do not delay immediate lifesaving care while locating device information.
2. When any adequate-perfusion criterion is present, do not perform chest compressions. Monitor closely and treat the complaint and non-LVAD causes under the applicable linked protocol.
3. When the patient is unresponsive with impaired perfusion, begin chest compressions immediately and follow AC-03 and the applicable rhythm pathway.
4. When a second rescuer is available, assess the LVAD concurrently: listen for hum, read alarms, inspect the driveline and controller, reconnect an obvious disconnection, and ensure adequate power.
5. Replace depleted batteries one at a time so that one power source remains connected at all times.
6. A controller exchange may be performed only by a trained patient or caregiver or under direct instruction from the LVAD coordinator.
7. Perform defibrillation or synchronized cardioversion when indicated without disconnecting the LVAD; position pads away from the implanted pump, controller, and driveline.
8. Control external hemorrhage aggressively. Give cautious isotonic-fluid aliquots only when impaired perfusion and suspected hypovolemia are present, reassessing for pulmonary edema, right-ventricular failure, or volume overload.
9. Contact the LVAD coordinator or implanting center early, but do not delay CPR, treatment, or transport. Consult Medical Control for destination or complex treatment decisions.
10. Transport to an LVAD-capable or implanting center when clinically and geographically appropriate. Bring the caregiver, backup controller, charged batteries, charger or power module, emergency card, device records, and medications.

Medication orders

Anticoagulant / antiplatelet consideration

No routine prehospital reversal under this protocol

- Expect chronic anticoagulant or antiplatelet therapy and control external hemorrhage aggressively.
- Use a separate approved reversal policy or direct Medical Control order when reversal is considered.

Isotonic crystalloid

Cautious aliquots for impaired perfusion with suspected hypovolemia

- Reassess MAP, lung sounds, work of breathing, skin perfusion, $EtCO_2$, and response after each aliquot.
- Use smaller aliquots when right-ventricular failure, pulmonary edema, or volume overload is suspected.

Provider-level care

Provider-Level Actions All provider levels assess perfusion and begin lifesaving BLS care immediately; device evaluation occurs concurrently when personnel permit.

EMT: Assess responsiveness, breathing, skin color and temperature, capillary refill, and other available signs of perfusion; pulse absence alone does not indicate arrest.

- Support ventilation, apply AED/monitor pads, and begin chest compressions immediately when the patient is unresponsive with impaired perfusion.
- Listen for the LVAD hum, read controller alarms, inspect visible connections, restore an obvious disconnected power or driveline connection, and gather all backup equipment.

AEMT: Perform all EMT actions plus IV/IO access, cautious isotonic-fluid aliquots for impaired perfusion with suspected hypovolemia, and frequent reassessment.

- Use smaller aliquots and reassess closely when right-ventricular failure, pulmonary edema, or volume overload is possible.
- Do not delay CPR, defibrillation, device troubleshooting by a second rescuer, coordinator contact, or transport for vascular access.

Paramedic: Perform all AEMT actions plus rhythm interpretation, manual defibrillation or synchronized cardioversion when indicated, advanced airway management, and waveform capnography.

- Use Doppler with a manual cuff when available, treat dysrhythmia and reversible causes under the linked protocols, and trend perfusion after each intervention.
- Coordinate the LVAD-center report, Medical Control consultation, destination decision, and any termination evaluation.

Warnings

- Do not cut, clamp, pull, or otherwise manipulate the driveline and do not force damaged connections.

- Do not withhold indicated defibrillation or synchronized cardioversion solely because an LVAD is present.
- Do not terminate resuscitation based solely on pulse absence, device alarm, rhythm, or $EtCO_2$.
- Destination selection must account for patient stability, transport time, LVAD-center capability, and coordinator or Medical Control direction.

Clinical pearls

- Bring the patient's trained caregiver whenever possible because the caregiver may know the alarm, normal readings, and emergency equipment better than responding clinicians.

Special populations

Perfusing ventricular tachyarrhythmia

- A conscious, adequately perfused LVAD patient may tolerate VT or VF; do not begin chest compressions solely from the ECG rhythm.
- Treat symptomatic or poorly perfused tachyarrhythmia under the applicable rhythm protocol and contact the LVAD center early.

Trauma or bleeding

- Assume anticoagulant or antiplatelet use until verified, control external hemorrhage immediately, and maintain a low threshold for serious occult bleeding or intracranial injury.
- Use the applicable trauma pathway without interrupting essential LVAD power or damaging the driveline.

Termination or advance directive

- Honor a valid DNR, MOST, or advance directive under agency policy and use separate obvious-death criteria when applicable.
- Otherwise, termination requires AC-12 criteria, Medical Control authorization, and LVAD-coordinator consultation when the coordinator is reachable.

Related pathways

- AC-03 Adult Cardiac Arrest (p. 77): Open immediately for an unresponsive LVAD patient with impaired perfusion.
- AC-09 VF / Pulseless VT (p. 96): Open for shockable VF/pVT arrest treatment when perfusion is absent.
- AC-10 Post-Resuscitation Care (p. 99): Open immediately after sustained ROSC or restoration of adequate perfusion.
- AC-11 TEAM-Focused CPR (p. 102): Open for concurrent CPR-team roles, timing, and quality management.
- AC-12 Termination of Resuscitation (p. 104): Open when termination is being considered; Medical Control and LVAD consultation are required as specified above.
- UP-04 Altered Mental Status (p. 14): Open for hypoglycemia or another non-LVAD altered-mental-status cause.
- TE-07 Overdose / Toxic Ingestion (p. 211): Open for suspected overdose or toxicologic cause.
- TB-10 Traumatic Arrest (p. 196): Use for LVAD patients in traumatic cardiac arrest.

AC-15
Total Artificial Heart Emergency

Revised October 16, 2026

Overview

Device-specific emergency pathway for a patient supported by a SynCardia Total Artificial Heart (TAH). This is not an LVAD; use AC-14 for a durable LVAD.

A TAH replaces both native ventricles and all four native valves. The external pneumatic driver produces pulsatile circulation, so a functioning system should generally produce a palpable pulse and measurable blood pressure.

Standard chest compressions, ECG-based rhythm treatment, defibrillation, cardioversion, pacing, antiarrhythmics, and vasopressors do not restore TAH output and must not be used for TAH pump failure.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Identify Total Artificial Heart
EMT / AEMT / Paramedic
Confirm TAH/Freedom or Companion driver • gather trained caregiver and backup equipment

↓

2. Assess Perfusion
EMT / AEMT / Paramedic
Mental status • ventilation • skin • capillary refill • palpable pulse • BP • $EtCO_2$ when available

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. Contact Transplant Center
EMT / AEMT / Paramedic
Call coordinator immediately • SynCardia clinical emergency support 866-771-9437 • do not delay treatment

↓

4. Pulse / Perfusion Absent?
EMT / AEMT / Paramedic
Check both air drivelines, kinks, connections, driver power, battery charge, and backup driver

↓

5. Low BP / Poor Perfusion
AEMT / Paramedic
SBP <90: isotonic fluid 500 mL IV/IO • may repeat once • maximum 1 L

↓

6. High BP + Pulmonary Edema
AEMT / Paramedic
SBP ≥150: oxygen/CPAP • nitroglycerin 0.4 mg SL every 5 min • maintain SBP ≥90

↓

7. Output Not Restored
EMT / AEMT / Paramedic / Medical Control
No CPR/defibrillation/pacing/ACLS drugs • coordinator + Medical Control • death-in-field policy

↓

8. Specialty Transport
EMT / AEMT / Paramedic
TAH-capable/implanting center • caregiver • primary and backup drivers • batteries • chargers • records

Indications

- Patient supported by a SynCardia Total Artificial Heart with illness, injury, poor perfusion, abnormal blood pressure, driver alarm, loss of pulse, or suspected system failure.

Assessment

Patient and perfusion assessment

- Assess responsiveness, airway, ventilation, skin color and temperature, capillary refill, palpable peripheral pulse, manual or automated blood pressure, and $EtCO_2$ when available.
- A functioning TAH produces pulsatile flow. Absent pulse or blood pressure indicates absent or severely impaired artificial-heart output until proven otherwise.
- Do not use ECG or 12-lead findings to determine TAH output. The native ventricles are absent, and the electrical tracing does not drive or represent artificial-heart circulation.
- Look for hypovolemia, hemorrhage, pulmonary edema, stroke, infection or sepsis, respiratory failure, trauma, driver failure, power loss, and air-driveline obstruction.

Driver and driveline assessment

- Expose the external driver, power source, batteries, and both right- and left-sided pneumatic drivelines.
- Ensure each air driveline is firmly connected and is not kinked, bent, compressed, cracked, or disconnected.
- Confirm the driver is operating, read available alarms, verify external power connections, and check each battery charge indicator.
- Use the patient's backup driver only with a trained caregiver or direct transplant-coordinator guidance whenever possible.

Treatment

1. Support airway and ventilation and contact the transplant coordinator immediately without delaying device assessment or treatment.
2. For absent pulse or poor perfusion, correct external air-driveline kinks or loose connections, restore power, replace depleted batteries as directed, and transition to the backup driver when available with trained assistance.
3. Do not give furosemide or administer the patient's home antihypertensive under this protocol.
4. Treat hemorrhage, stroke, infection, respiratory failure, trauma, or another concurrent emergency under the linked complaint-specific protocol without disrupting the TAH system.
5. If pulse and perfusion cannot be restored after troubleshooting, backup-driver exchange, ventilation, and appropriate fluid, contact the transplant center and Medical Control and follow the death-in-field policy.

6. Transport to the implanting or TAH-capable transplant center when feasible with the caregiver and all primary and backup equipment.

Medication orders

Isotonic crystalloid

500 mL IV/IO; may repeat once to maximum 1 L

- For SBP below 90 mmHg or poor perfusion after confirming driver function.
- Reassess pulse, blood pressure, skin perfusion, lung sounds, ventilation, and $EtCO_2$ after each bolus.

Nitroglycerin

0.4 mg SL every 5 minutes

- For SBP at least 150 mmHg with pulmonary edema or device-related symptoms.
- Maintain SBP at least 90 mmHg.

Provider-level care

Provider-Level Actions Restore pneumatic driver output, support ventilation and perfusion, and avoid standard cardiac-arrest interventions that cannot operate a TAH.

EMT: Confirm the device is a TAH rather than an LVAD; assess ventilation, pulse, blood pressure, skin perfusion, mental status, and driver function.

- Provide airway and ventilatory support, expose the driver and both air drivelines, correct an obvious kink or loose external connection, and confirm power and battery charge.
- Contact the transplant coordinator, keep the trained caregiver with the patient, gather all primary and backup equipment, and do not initiate CPR or AED use for TAH pump failure.

AEMT: Perform all EMT actions plus IV/IO access and isotonic fluid 500 mL for SBP below 90 mmHg or poor perfusion; may repeat once to a maximum of 1 liter.

- For SBP at least 150 mmHg with pulmonary edema or device-related symptoms, support oxygenation/CPAP and give nitroglycerin 0.4 mg SL every five minutes while maintaining SBP at least 90 mmHg.
- Assist a trained caregiver or coordinator-directed driver exchange without disconnecting both pumping lines simultaneously.

Paramedic: Perform all AEMT actions plus advanced airway and waveform-capnography support when indicated; trend blood pressure, pulse, perfusion, and $EtCO_2$ rather than ECG rhythm.

- Coordinate troubleshooting, Medical Control, transplant-center consultation, specialty destination, and the death-in-field decision when output cannot be restored.
- Do not give vasopressors or antiarrhythmics and do not perform defibrillation, cardioversion, pacing, or chest compressions for TAH pump failure.

Warnings

- Do not perform manual or mechanical chest compressions on a TAH patient.
- Do not use ECG rhythm, 12-lead ECG, AED, defibrillation, synchronized cardioversion, or transcutaneous pacing to treat TAH pump failure.
- Do not disconnect both pneumatic drivelines simultaneously, disassemble the external driver, or force a damaged connection.
- Do not confuse a Total Artificial Heart with an LVAD; the AC-14 perfusion and CPR pathway does not apply to a TAH.
- Do not add furosemide or assist with the patient's home antihypertensive under this protocol.

Clinical pearls

- The TAH replaces both ventricles and all four native valves while preserving the atria and great-vessel roots.
- The right and left artificial ventricles are powered pneumatically through two separate air drivelines; kinking either line can stop or reduce output.
- $EtCO_2$ can help trend perfusion when ventilation is controlled, but restoration of driver output is the essential intervention.
- The trained caregiver and transplant coordinator are critical clinical resources and may be the most experienced people present with the patient's exact equipment.

Special populations

Anticoagulation, bleeding, and stroke

- Expect anticoagulant or antiplatelet therapy until verified and control external hemorrhage immediately.
- Maintain a low threshold for gastrointestinal bleeding, intracranial hemorrhage, thromboembolic stroke, and device-associated infection.

DNR, MOST, or obvious death

- Honor a valid directive under agency policy and use the separate obvious-death or death-in-field policy when applicable.
- When pump output cannot be restored and no effective circulatory intervention remains, contact Medical Control and the transplant center before disposition whenever feasible.

Related pathways

- Call SynCardia Clinical Emergency Support - tel:8667719437: 24-hour clinical support for qualified healthcare providers: 866-771-9437.
- AC-14 LVAD Emergency (p. 108): Use instead when the patient has a durable left ventricular assist device rather than a total artificial heart.
- AR-01 Adult Airway (p. 143): Open for airway or ventilatory failure.
- AR-04 COPD / Asthma (p. 148): Open for respiratory distress or bronchospasm when appropriate.
- UP-04 Altered Mental Status (p. 14): Open for persistent altered mental status or a concurrent metabolic cause.
- UP-14 Stroke (p. 42): Open for focal neurologic deficit or suspected stroke.
- UP-15 Sepsis (p. 46): Open for suspected infection, sepsis, or shock.
- AM-05 Hypotension / Shock (p. 129): Open for concurrent non-device shock after driver function is restored.

AC-16 Wearable Cardioverter-Defibrillator Vest

Revised October 16, 2026

Overview

Emergency pathway for a patient wearing a ZOLL LifeVest or another wearable cardioverter-defibrillator who is symptomatic, has received a shock, has a device alarm, or is in cardiac arrest.

The alert tone and voice prompt determine whether rescuers may touch the patient. During an active treatment siren, clear the patient and allow the imminent device shock; resume assessment and CPR as soon as the warning ends.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Assess Patient + Alert
EMT / AEMT / Paramedic
Standard assessment • silent/single gong vs two-tone treatment siren • read/listen to prompt

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. Active Treatment Siren?
EMT / AEMT / Paramedic
"Press response buttons" or "Bystanders, do not interfere" means shock is likely

↓

3. Clear the Patient
EMT / AEMT / Paramedic
Stop contact/CPR temporarily • only patient may press response buttons • allow device treatment

↓

4. Siren Stops / CPR Prompt
EMT / AEMT / Paramedic
Immediately reassess • start or resume CPR when indicated

↓

5. EMS Takes Over
EMT / AEMT / Paramedic
Pull battery first • remove/disconnect vest • clean/dry chest • apply EMS pads • AC-03/AC-09

↓

6. Pulse Present / Shock Delivered
EMT / AEMT / Paramedic
Continuous monitor • 12-lead • assess ACS/dysrhythmia • document shock and symptoms

↓

7. Transport + Equipment
EMT / AEMT / Paramedic
Bring garment, monitor, battery, charger/hotspot, extra battery, and event information

Indications

- Patient wearing a wearable cardioverter-defibrillator vest who is symptomatic, shocked, or in cardiac arrest.

Assessment

Alert and patient assessment

- Determine whether the device is silent, producing a single-tone gong, announcing that treatment was delivered, or producing the two-tone treatment siren.

- Assess responsiveness, breathing, pulse, perfusion, chest pain, dyspnea, palpitations, syncope, injury, number and timing of shocks, and why the LifeVest was prescribed.
- If the device is silent, producing a gong, or announcing that treatment was delivered, proceed with standard assessment and treatment.

Post-shock and device assessment

- For a perfusing patient, obtain continuous monitoring and a 12-lead ECG, inspect the skin, document the shock and symptoms, and evaluate for ACS, dysrhythmia, heart failure, electrolyte disturbance, medication effect, or device malfunction.
- Identify blue conductive gel released by the LifeVest treatment electrodes and determine whether the vest will remain operational or be removed for EMS care.

Treatment

1. During an active two-tone treatment siren, stop touching the patient, temporarily stop CPR, clear everyone, and allow the device to complete its treatment sequence.
2. Only the patient may press the response buttons. EMS personnel, caregivers, and bystanders must never press them.
3. When EMS monitoring and defibrillation are available and EMS is taking over treatment, pull out the LifeVest battery first and remove or disconnect the vest without delaying CPR or an indicated EMS shock.
4. Before external defibrillation, cardioversion, or pacing, disconnect the LifeVest monitor from the electrode belt. Prefer complete battery and garment removal when urgent EMS electrical therapy is required.
5. Do not place EMS pads over LifeVest electrodes. Use anterior-lateral or anterior-posterior placement on a clean, dry chest.
6. Leave blue conductive gel in place while the LifeVest remains operational. If the vest is removed, rinse away the gel with water and dry the chest before applying EMS pads.
7. After any delivered shock, obtain continuous monitoring and a 12-lead ECG, treat ACS or dysrhythmia under the linked protocol, and transport.

Provider-level care

Provider-Level Actions Respect the active-shock warning, then transition immediately to standard resuscitation or complaint-specific care.

EMT: Identify the alert and voice prompt, clear the patient during an active treatment warning, and ensure that only the patient touches the response buttons.

- When the warning ends, assess pulse and breathing, begin or resume CPR, and use the AED after the LifeVest is safely disconnected or removed.
- Provide airway/ventilatory support, obtain vital signs, gather all device equipment, document device prompts and shocks, and transport.

AEMT: Perform all EMT actions plus IV/IO access, authorized medications, and treatment of associated hypoxia, hypotension, chest pain, or another complaint.

- Do not delay CPR or an indicated EMS shock for vascular access, device history, technical support, or equipment collection.
- Assist with battery removal, vest disconnection, chest preparation, and EMS pad placement.

Paramedic: Perform all AEMT actions plus rhythm interpretation, manual defibrillation, synchronized cardioversion, transcutaneous pacing, advanced airway management, and waveform capnography when indicated.

- Disconnect the LifeVest monitor from the electrode belt before EMS electrical therapy and place EMS pads clear of the vest electrodes.
- Lead post-shock assessment, 12-lead interpretation, destination, Medical Control, ZOLL technical-support contact, and handoff.

Warnings

- Do not touch the patient while the LifeVest announces that bystanders must not interfere or while a treatment shock is being delivered.
- Never press the response buttons for the patient; only the patient may use them as a consciousness test.
- Do not perform CPR during the brief active-shock warning; resume immediately when the warning ends or the device prompts CPR.
- Do not place EMS pads over LifeVest electrodes, conductive gel, medication patches, or damaged skin.
- Do not delay an indicated EMS shock while troubleshooting, collecting equipment, or contacting technical support.

Clinical pearls

- A LifeVest treatment shock is typically delivered 25-60 seconds after a treatable ventricular arrhythmia is detected.
- The two-tone siren and bystander warning indicate that a shock is likely; a silent device or single-tone gong does not require rescuers to clear the patient.
- Blue gel improves shock conduction and reduces burns; leave it in place if the LifeVest remains operational.
- The LifeVest does not automatically notify EMS or the patient's clinician when a shock occurs.
- Assess the patient and rhythm after every reported or witnessed device shock.

Special populations

Cardiac arrest

- Once EMS assumes monitoring and defibrillation, disconnect or remove the LifeVest and follow AC-03 and AC-09.

Perfusing patient after shock

- Obtain a 12-lead ECG, evaluate for ACS and recurrent ventricular dysrhythmia, inspect for skin injury, and transport with continuous monitoring.

Related pathways

- Call ZOLL LifeVest Technical Support - tel:8005433267: 24-hour technical support: 800-543-3267.
- AC-03 Adult Cardiac Arrest (p. 77): Open when the patient is pulseless after the active LifeVest warning ends.
- AC-04 ACS / STEMI (p. 80): Open for chest discomfort, anginal-equivalent symptoms, or ischemic ECG findings.
- AC-07 Monomorphic Wide-Complex Tachycardia (p. 90): Open for perfusing monomorphic VT.
- AC-08 Polymorphic Wide-Complex Tachycardia (p. 93): Open for perfusing polymorphic VT or recurrent unstable ventricular rhythm.
- AC-09 VF / Pulseless VT (p. 96): Open for VF or pulseless VT after EMS assumes defibrillation.
- AC-10 Post-Resuscitation Care (p. 99): Open after ROSC following a LifeVest or EMS-delivered shock.
- ZOLL LifeVest First-Responder Pocket Card - https://cardiac.zoll.com/hubfs/LifeVest_EMS_PocketCard.pdf?hsLang=en: Official manufacturer emergency-management instructions.

AM-01
Allergic Reaction / Anaphylaxis

Revised October 2026

Overview

Rapidly distinguish an isolated mild skin reaction from anaphylaxis involving the airway, breathing, circulation, or multiple body systems.

Anaphylaxis may occur without rash. Respiratory compromise or hypotension after a likely allergen exposure is sufficient to treat immediately.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Suspected Allergic Reaction
EMT / AEMT / Paramedic
Remove trigger when safe • assess airway, breathing, circulation, skin, GI symptoms, and perfusion

↓

2. Anaphylaxis?
EMT / AEMT / Paramedic
Respiratory compromise • hypotension/poor perfusion • or 2 or more body systems

↓

3. IM Epinephrine Now
EMT / AEMT / Paramedic
1 mg/mL: 0.5 mg IM lateral thigh • 0.3 mg autoinjector acceptable • repeat every 5 min

↓

4. Position + Support Airway
EMT / AEMT / Paramedic
Supine when tolerated • oxygen/suction/BVM • albuterol for persistent bronchospasm • early advanced airway

↓

5. Hypotension / Poor Perfusion?
AEMT / Paramedic
IV/IO isotonic crystalloid 500 mL aliquots • repeat with reassessment • maximum 2 L

Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Refractory Shock?
Paramedic / Medical Control
Continue IM epinephrine • epinephrine infusion 2-10 mcg/min • Medical Control required

↓

7. Bradykinin Angioedema?
Paramedic / Medical Control
Early difficult-airway plan • consider TXA 1 g IV/IO over 10 min with Medical Control

↓

8. Monitor + Transport
EMT / AEMT / Paramedic
Continuous cardiac/SpO_2/BP monitoring • 12-lead • $EtCO_2$ when indicated • early notification

Indications

- Acute allergic symptoms after a suspected exposure, including urticaria, pruritus, flushing, angioedema, wheezing, stridor, dyspnea, gastrointestinal symptoms, hypotension, collapse, or shock.
- Suspected anaphylaxis even when no rash or skin finding is present.
- Suspected ACE-inhibitor, hereditary, or other bradykinin-mediated angioedema.

Contraindications

- There is no absolute contraindication to IM epinephrine when anaphylaxis is present or strongly suspected.

Assessment

Severity classification

- Mild allergic reaction: isolated hives, itching, flushing, or localized swelling with a normal airway, normal breathing, and normal perfusion.
- Anaphylaxis: respiratory compromise, hypotension or poor perfusion, or acute involvement of two or more systems including skin/mucosa, respiratory, cardiovascular, or gastrointestinal symptoms.
- Acute hypotension, bronchospasm, or laryngeal involvement after a known or highly probable allergen exposure is anaphylaxis even without skin findings.

Focused assessment

- Identify the suspected trigger, route and time of exposure, symptom onset and progression, prior anaphylaxis, epinephrine given before EMS arrival, prescribed autoinjector, asthma, mast-cell disorder, and previous biphasic reaction.
- Review beta-blocker and ACE-inhibitor use and ask about hereditary or recurrent angioedema.
- Assess voice change, drooling, dysphagia, tongue or facial swelling, stridor, wheezing, work of breathing, oxygenation, skin findings, vomiting or abdominal pain, pulse, blood pressure, perfusion, and mental status.
- Reassess airway swelling, lung sounds, blood pressure, perfusion, mental status, and response after every epinephrine dose and fluid aliquot.

Treatment

1. Stop or remove the triggering exposure when safe, but do not delay epinephrine or transport to remove an embedded object or perform decontamination that can occur later.
2. Position most patients supine when tolerated. Use a position of comfort for severe respiratory distress, left-lateral positioning during pregnancy, and the recovery position for an unconscious patient with adequate breathing.
3. Repeat IM epinephrine every five minutes while significant airway, respiratory, cardiovascular, or progressive multisystem symptoms persist.
4. Diphenhydramine is limited to isolated or persistent skin symptoms and must never replace or delay epinephrine. Routine corticosteroids are not part of the acute anaphylaxis pathway.
5. For refractory anaphylactic shock, continue IM epinephrine and begin a Medical-Control-authorized epinephrine infusion rather than giving an IV epinephrine bolus.
6. Transport every patient who receives epinephrine or has airway, respiratory, cardiovascular, progressive, or multisystem symptoms with continuous reassessment and early destination notification.

Medication orders

Epinephrine 1 mg/mL - IM

0.5 mg IM lateral thigh; 0.3 mg autoinjector acceptable; repeat every 5 minutes

- EMT, AEMT, and Paramedic standing order for anaphylaxis.
- There is no absolute contraindication when anaphylaxis is present or strongly suspected.
- Do not delay administration for vascular access, cardiac monitoring, antihistamine, corticosteroid, or transport preparation.

Albuterol

2.5 mg nebulized; repeat every 5-15 minutes as needed, maximum 3 treatments

- For persistent bronchospasm after epinephrine.
- Albuterol does not treat upper-airway edema, hypotension, or shock and is not a substitute for repeated epinephrine.

Isotonic crystalloid

500 mL IV/IO aliquots; repeat with reassessment; maximum 2 L

- For hypotension or poor perfusion only.
- Use 250 mL aliquots for CHF, renal failure, liver failure, pulmonary edema, or volume-overload risk.

Diphenhydramine

25-50 mg slow IV/IO or deep IM

- Paramedic only for isolated or persistent skin symptoms; EMT/AEMT administration is not enabled without documented specific Tennessee authority.
- Do not administer for hypotension, poor perfusion, or decreased mental status.
- Adjunct only; never delay or replace epinephrine.

Epinephrine infusion

2-10 mcg/min IV/IO; titrate to SBP ≥90 mmHg or MAP ≥65 mmHg

- Paramedic with Medical Control for refractory shock after repeated IM epinephrine and appropriate fluid.
- Use an infusion pump and continuous cardiac, blood-pressure, and SpO_2 monitoring.
- Do not give an IV epinephrine bolus under this protocol.

Glucagon

1-2 mg IV/IO over 5 minutes

- Paramedic with Medical Control for refractory shock or bradycardia in a beta-blocked patient.
- Prepare for vomiting and protect the airway.

Tranexamic acid

1 g IV/IO over 10 minutes

- Paramedic with Medical Control for suspected ACE-inhibitor or hereditary bradykinin-mediated angioedema.
- Evidence is limited; do not delay difficult-airway preparation or transport.

Provider-level care

Provider-Level Actions Give epinephrine first, support airway and perfusion, and reassess continuously.

EMT: Remove or stop the suspected trigger when safe, perform the primary assessment, position the patient appropriately, and provide oxygen, suction, or BVM ventilation when indicated.

• For anaphylaxis, give epinephrine 1 mg/mL 0.5 mg IM in the lateral thigh; a 0.3 mg autoinjector is acceptable when that is the available device. Repeat every five minutes while significant symptoms persist.

• Give albuterol 2.5 mg by nebulizer for persistent bronchospasm after epinephrine; repeat every 5-15 minutes as needed to a maximum of three treatments.

• Apply available monitoring, request ALS early, initiate transport, and do not delay epinephrine for any other treatment.

AEMT: Perform all EMT actions and establish IV/IO access without delaying IM epinephrine or transport.

• For hypotension or poor perfusion, give isotonic crystalloid in 500 mL IV/IO aliquots with reassessment, maximum 2 liters. Use 250 mL aliquots when CHF, renal failure, liver failure, pulmonary edema, or volume overload is present or suspected.

• Continue IM epinephrine every five minutes for ongoing anaphylaxis and notify the paramedic immediately when shock, airway swelling, or refractory symptoms persist.

Paramedic: Perform all EMT and AEMT actions; establish an early difficult-airway plan for progressive tongue, pharyngeal, or laryngeal swelling and use waveform capnography when ventilatory support or an advanced airway is required.

• For isolated or persistent skin symptoms, give diphenhydramine 25-50 mg slow IV/IO or deep IM only when mental status and blood pressure are adequate.

• For refractory shock after repeated IM epinephrine and appropriate fluid, begin an epinephrine infusion at 2-10 mcg/min and titrate to SBP at least 90 mmHg or MAP at least 65 mmHg. Medical Control is required.

• For a beta-blocked patient with refractory shock or bradycardia, give glucagon 1-2 mg IV/IO over five minutes with Medical Control and prepare for vomiting.

• For suspected ACE-inhibitor or hereditary bradykinin-mediated angioedema, consider tranexamic acid 1 g IV/IO over ten minutes with Medical Control while prioritizing airway management and transport.

Warnings

• Do not delay initial or repeat IM epinephrine for IV/IO access, monitoring, albuterol, diphenhydramine, corticosteroids, or transport preparation.

• Do not give an IV epinephrine bolus for anaphylaxis under this protocol; dosing errors can cause severe hypertension, myocardial ischemia, and lethal dysrhythmia.

• Do not allow a hypotensive or poorly perfused patient to stand or walk because abrupt cardiovascular collapse may occur.

• ACE-inhibitor and hereditary angioedema may progress rapidly and may respond poorly to epinephrine, antihistamines, and corticosteroids.

• Routine corticosteroids are not recommended for acute anaphylaxis and have not been shown to reliably prevent a biphasic reaction.

Clinical pearls

• The shorter the interval from exposure to symptom onset, the greater the concern for rapid progression.

• A normal skin examination does not exclude anaphylaxis; prioritize airway, breathing, circulation, perfusion, and gastrointestinal findings.

• Obtain a 12-lead ECG and continuous cardiac monitoring in moderate or severe reactions without delaying epinephrine.

• Use continuous SpO_2 and serial blood pressure monitoring; add $EtCO_2$ when ventilation is supported or airway compromise is present.

• Document the suspected trigger, exact symptom-onset time, every epinephrine dose and route, response, recurrence, and all treatment given before EMS arrival.

Special populations

Pregnancy

• Use left-lateral positioning when feasible and treat anaphylaxis with the same prompt IM epinephrine dosing.

• Maternal oxygenation and perfusion are the immediate priorities for fetal resuscitation.

Beta-blocker therapy

• Expect possible resistance to epinephrine and increased risk of refractory hypotension or bradycardia.

• Glucagon may be used with Medical Control; give slowly and prepare for vomiting.

ACE-inhibitor or hereditary angioedema

- Prioritize early difficult-airway preparation and rapid transport.
- Give epinephrine when anaphylaxis cannot be excluded or airway compromise is developing, while recognizing that nonhistaminergic angioedema may respond poorly.
- TXA is a Medical-Control-directed formulary option with limited supporting evidence.

Related pathways

- AR-01 Adult Airway (p. 143): Open for progressive airway swelling, ventilatory failure, or advanced-airway management.
- AR-04 COPD / Asthma (p. 148): Open for persistent bronchospasm or a concurrent asthma/COPD exacerbation.
- AM-05 Hypotension / Shock (p. 129): Open for persistent hypotension or another shock etiology after anaphylaxis treatment.

AM-02
Diabetic Emergency - Adult

Revised October 2026

Overview

Treat symptomatic hypoglycemia promptly while protecting the airway and identifying stroke, seizure, overdose, trauma, infection, or another concurrent emergency.

Use titrated D10 rather than a fixed hypertonic dextrose bolus. The treatment target is clinical improvement and glucose at least 70 mg/dL-not routine hyperglycemia.

Marked hyperglycemia requires assessment for DKA, HHS, infection, dehydration, and shock. Claiborne EMS does not administer insulin under this protocol.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Check Glucose + ABCs
EMT / AEMT / Paramedic
Altered mental status • weakness • seizure • diaphoresis • vomiting • suspected diabetes

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. Glucose <70 mg/dL?
EMT / AEMT / Paramedic
Treat when symptomatic or when symptoms cannot be assessed reliably because of altered mental status

↓

3. Safe Swallow?
EMT / AEMT / Paramedic
Yes: oral glucose 15-25 g • No: AEMT/Paramedic D10 IV/IO or glucagon IM

↓

4. Titrate D10
AEMT / Paramedic
Up to 250 mL IV/IO • stop when mental status improves and glucose ≥70 • maximum 25 g

↓

5. Reassess
EMT / AEMT / Paramedic
Parenteral: glucose + neuro exam at 5 min • oral: repeat glucose within 10-15 min

↓

6. Hyperglycemia / DKA Features?
EMT / AEMT / Paramedic
Glucose ≥250 or Kussmaul breathing, vomiting, abdominal pain, dehydration, infection, shock, or SGLT2 use

↓

7. Hypotension / Poor Perfusion?
AEMT / Paramedic
Isotonic fluid 500 mL IV/IO aliquots • 250 mL if volume-overload risk • maximum 2 L

↓

8. Transport / Refusal Screen
EMT / AEMT / Paramedic
Transport high-risk medications, recurrence, seizure, pregnancy, overdose, uncertain cause, DKA/HHS, or persistent deficit

Indications

- Symptoms or altered mental status with suspected hypoglycemia or hyperglycemia.

- Known diabetes with abnormal glucose, medication complication, insulin-pump problem, or inability to maintain oral intake.
- Suspected DKA, HHS, euglycemic DKA, or another acute diabetic complication.

Contraindications

- Do not give oral glucose, food, drink, or oral medication to a patient who cannot swallow reliably or protect the airway.
- Do not administer insulin or independently alter an insulin-pump dose under this protocol.
- Do not administer routine IV fluid solely because the glucose is elevated; fluid is reserved for hypotension or poor perfusion.

Assessment

Immediate assessment

- Assess airway protection, ventilation, oxygenation, circulation, mental status, focal neurologic findings, seizure activity, trauma, temperature, and blood glucose.
- If a continuous glucose monitor value is inconsistent with the clinical presentation, confirm with the agency glucometer when feasible without delaying treatment of severe suspected hypoglycemia.
- Determine the diabetes type, usual glucose range, insulin and oral medication names, last medication dose, last meal, recent medication changes, pump or continuous-monitor use, and treatment given before EMS arrival.
- Ask about intentional overdose, alcohol use, prolonged fasting, vomiting, diarrhea, infection symptoms, pregnancy, renal or liver disease, and previous recurrent hypoglycemia.

Hyperglycemic emergency assessment

- A glucose at least 250 mg/dL should prompt evaluation for dehydration, Kussmaul respirations, abdominal pain, nausea or vomiting, infection, altered mental status, hypotension, poor perfusion, DKA, and HHS.
- Consider euglycemic DKA even when glucose is below 250 mg/dL, especially with SGLT2-inhibitor use, pregnancy, prolonged fasting, vomiting, alcohol use, or significant acute illness.
- Inspect insulin-pump tubing and the insertion site without manipulating or disconnecting the system. Ask about pump alarms, interruption, empty reservoir, dislodgement, or recent dose changes.

Treatment

1. If vascular access cannot be obtained promptly, give glucagon 1 mg IM. Position for aspiration protection and continue airway monitoring.
2. Repeat or continue titrated D10 within the 25 g maximum when hypoglycemia and symptoms persist. Do not create unnecessary hyperglycemia after the treatment target is reached.
3. For persistent altered mental status or focal neurologic findings after glucose correction, continue evaluation and treatment under the altered-mental-status, stroke, seizure, overdose, trauma, or sepsis pathway as appropriate.
4. For hyperglycemia, provide supportive care and treat hypotension or poor perfusion with isotonic crystalloid. Do not administer insulin or routine maintenance fluid.
5. A capable patient may stop or suspend their own insulin pump. EMS must not independently change the programmed dose; transport suspected pump failure or DKA.
6. Strongly recommend transport for oral hypoglycemic or sulfonylurea use, long-acting insulin, recurrent hypoglycemia, seizure, injury, intentional overdose, pregnancy, infection, renal or liver disease, uncertain cause, persistent neurologic abnormality, suspected DKA/HHS, or insulin-pump failure.

Medication orders

Oral glucose

15-25 g PO according to the packaged dose

- EMT, AEMT, and Paramedic when glucose is below 70 mg/dL with symptoms and the patient is awake, follows commands, swallows reliably, and protects the airway.
- Rapid-carbohydrate food or drink is acceptable when commercial oral glucose is unavailable and swallowing is safe.
- Reassess clinically and repeat glucose within 10-15 minutes.

Dextrose 10% (D10)

Up to 250 mL IV/IO, titrated to improving mental status and glucose ≥70 mg/dL; maximum 25 g

- AEMT and Paramedic when oral glucose is unsafe or ineffective.
- Recheck glucose and neurologic status after five minutes.
- Stop titration when the clinical and glucose targets are reached.

Glucagon

1 mg IM

- AEMT and Paramedic when oral glucose is unsafe and vascular access cannot be obtained promptly.
- Protect against aspiration because vomiting may occur.
- May be ineffective in malnutrition, prolonged fasting, chronic alcohol use, severe liver disease, or depleted glycogen states.

Isotonic crystalloid

500 mL IV/IO aliquots; repeat with reassessment; maximum 2 L

- For hypotension or poor perfusion only-not an elevated glucose value alone.
- Use 250 mL aliquots for CHF, renal failure, dialysis, liver failure, pulmonary edema, or other volume-overload risk.
- Do not begin a routine maintenance infusion under AM-02.

Provider-level care

Provider-Level Actions Correct hypoglycemia safely, identify high-risk recurrence, and provide supportive care for hyperglycemic emergencies.

EMT: Assess airway, breathing, circulation, mental status, and neurologic findings; obtain blood glucose early and repeat it whenever the patient's condition changes.

- When glucose is below 70 mg/dL and the patient is symptomatic, awake, follows commands, swallows reliably, and protects the airway, give oral glucose 15-25 g according to the packaged dose.
- Do not give oral glucose, food, fluid, or medication when swallowing or airway protection is impaired. Request ALS and provide aspiration protection and ventilatory support as indicated.
- Identify diabetes medications, last dose, last meal, treatment before EMS arrival, recurrent episodes, and high-risk transport features.

AEMT: Perform all EMT actions and establish IV access when parenteral treatment is required; use IO access when urgently needed and IV access cannot be obtained within the Universal Patient Care attempt limit.

- When oral treatment is unsafe or ineffective, give D10 up to 250 mL IV/IO, titrated to improving mental status and glucose at least 70 mg/dL, maximum 25 g.
- When oral treatment is unsafe and vascular access cannot be obtained promptly, give glucagon 1 mg IM and protect against aspiration because vomiting may occur.
- For hypotension or poor perfusion, give isotonic crystalloid in 500 mL aliquots with reassessment, maximum 2 liters; use 250 mL aliquots when volume-overload risk is present.

Paramedic: Perform all EMT and AEMT actions and evaluate persistent symptoms for stroke, seizure, overdose, trauma, sepsis, dysrhythmia, hypoxia, or another time-sensitive condition.

- Obtain continuous cardiac monitoring and a 12-lead ECG when dysrhythmia, ischemia, severe electrolyte disturbance, significant hyperglycemic illness, or another cardiac indication is suspected.
- Manage airway failure, seizure, shock, and associated emergencies under the corresponding linked protocol.
- Claiborne EMS does not administer insulin or independently change an insulin-pump dose under AM-02.

Warnings

- Sulfonylureas, other oral hypoglycemic agents, and long-acting insulin may cause delayed or recurrent hypoglycemia after an initially normal repeat glucose.
- A normal glucose after treatment does not explain persistent altered mental status, focal deficit, seizure, or shock; continue evaluation for another emergency.

Clinical pearls

- Persistent focal neurologic deficit after glucose correction remains a stroke alert until proven otherwise.
- D10 permits controlled titration and reduces the hypertonic exposure associated with concentrated D50 administration.
- Quality-control checks must be maintained for agency glucometers according to manufacturer and agency policy.
- For possible oral-agent overdose or uncertain medication effect, contact Poison Control early and provide the exact product, strength, amount, and ingestion time when available.
- Document the initial glucose, neurologic status, treatment dose and route, response, repeat glucose values, meal availability, medication risks, capacity assessment, and transport recommendation.

Special populations

Refusal after hypoglycemia

- Refusal may be considered only when the patient has known diabetes and a clear correctable cause; normal mental status, neurologic examination, and decision-making capacity; glucose at least 80 mg/dL and stable on repeat; ability to eat a meal containing longer-acting carbohydrate; and a responsible adult remaining with access to further glucose monitoring.
- Transport is strongly recommended and refusal is high risk when oral hypoglycemic or sulfonylurea use, long-acting insulin, overdose, recurrence, seizure, injury, pregnancy, infection, renal or liver disease, uncertain cause, persistent deficit, lack of food, or absence of a responsible adult is present.
- Contact Medical Control whenever any refusal criterion or recurrence risk is uncertain.

Pregnancy

- Treat maternal hypoglycemia promptly using the same adult dosing.
- Consider euglycemic DKA with vomiting, illness, fasting, SGLT2 exposure, or compatible symptoms even when glucose is below 250 mg/dL.

Renal, hepatic, or cardiac disease

- Use 250 mL fluid aliquots for hypotension or poor perfusion and reassess lungs, oxygenation, blood pressure, and perfusion after each aliquot.
- Recognize increased recurrence risk from impaired medication clearance and possible poor glucagon response with severe liver disease.

Related pathways

- Call Poison Control - tel:18002221222: National Poison Control Center: 1-800-222-1222.
- UP-04 Altered Mental Status (p. 14): Open for persistent altered mental status or another suspected metabolic, toxicologic, or neurologic cause.
- UP-13 Seizure (p. 39): Open for active seizure or persistent postictal findings.
- UP-14 Stroke (p. 42): Open when focal neurologic deficits persist after glucose correction.
- AM-05 Hypotension / Shock (p. 129): Open for persistent hypotension or poor perfusion.

AM-03
Dialysis / Renal Failure

Revised 2026-10-30

Overview

Renal-failure and dialysis patients may present with life-threatening hyperkalemia, pulmonary edema, sepsis, shock, hemorrhage, or dialysis-access complications.

Protect the dialysis access, recognize dangerous ECG changes early, and treat the immediate emergency without delaying transport for laboratory confirmation.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. ABCs + Dialysis History
EMT / AEMT / Paramedic
Last completed treatment • missed/shortened sessions • access type • dry weight • residual urine

↓

2. Hyperkalemia Suspected?
EMT / AEMT / Paramedic
Weakness • syncope • bradycardia • hypotension • peaked T waves • PR prolongation • P-wave loss • wide QRS

↓

3. ECG Changes / Instability?
Paramedic
Paramedic: formulation-specific calcium dose • sodium bicarbonate up to 50 mEq • albuterol 10 mg neb when pulse present

↓

4. Pulmonary Edema?
EMT / AEMT / Paramedic
Upright • oxygen when indicated • early CPAP • AC-05 pathway • no routine fluid

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Hypotension / Poor Perfusion?
AEMT / Paramedic
Normal saline 250 mL IV/IO • reassess after each aliquot • maximum 500 mL without Medical Control

⇩

6. Access Hemorrhage?
EMT / AEMT / Paramedic
Small gauze + focused fingertip pressure • tourniquet only for life-threatening bleeding uncontrolled by pressure

⇩

7. Infection / Peritonitis?
EMT / AEMT / Paramedic
Fever • chills • access redness/drainage • abdominal pain • cloudy peritoneal effluent → sepsis pathway

⇩

8. Transport + Early Alert
EMT / AEMT / Paramedic
Bring dialysis information and medication list • notify receiving facility of instability or urgent dialysis concern

Indications

- Hemodialysis or peritoneal-dialysis patient with acute symptoms, missed treatment, or treatment complication.
- Acute or chronic renal failure with suspected hyperkalemia, pulmonary edema, shock, infection, or altered mental status.
- Bleeding, infection, malfunction, displacement, or other complication involving a dialysis fistula, graft, catheter, or peritoneal-dialysis system.

Contraindications

- Do not access or use a dialysis catheter under AM-03. Existing dialysis-catheter use during interfacility transport is governed by the separate transfer policy.
- Do not administer routine crystalloid for pulmonary edema, asymptomatic missed dialysis, or an elevated potassium concern without hypotension or poor perfusion.
- Do not manipulate, disconnect, flush, drain, or obtain a specimen from a peritoneal-dialysis system in the field.

Assessment

Dialysis and renal history

- Determine the cause of renal disease, dialysis type, normal schedule, last fully completed treatment, missed or shortened treatments, dry weight, recent weight change, residual urine output, and known potassium problems.
- Identify prescribed medications, potassium supplements, potassium-sparing medications, recent medication changes, anticoagulants or antiplatelet drugs, and any treatment given before EMS arrival.
- Ask about weakness, paralysis, palpitations, syncope, chest discomfort, dyspnea, orthopnea, fever, chills, nausea, vomiting, diarrhea, decreased intake, bleeding, access pain, redness, drainage, swelling, loss of thrill, pump alarms, or catheter damage.

Suspected hyperkalemia

- Suspect hyperkalemia with renal failure, missed dialysis, known elevated potassium, crush injury, medication effect, or severe metabolic illness plus weakness, paralysis, syncope, bradycardia, hypotension, dysrhythmia, or compatible ECG changes.
- ECG findings may progress from peaked T waves to PR prolongation, diminished or absent P waves, QRS widening, bradycardia, ventricular dysrhythmia, or sine-wave morphology.
- A normal mental status or absence of classic ECG findings does not exclude dangerous hyperkalemia. Do not delay treatment of an unstable patient for laboratory confirmation.

Dialysis-access assessment

- Identify fistula, graft, tunneled catheter, or peritoneal-dialysis access and inspect for active bleeding, aneurysmal change, trauma, erythema, warmth, tenderness, swelling, drainage, displacement, disconnection, or exposed components.
- When bleeding is controlled, assess distal circulation and note whether the usual fistula or graft thrill is present without delaying transport for hemorrhage, ischemia, or suspected thrombosis.

Treatment

1. Protect the dialysis access and address airway failure, hypoxia, dysrhythmia, shock, pulmonary edema, sepsis, or hemorrhage immediately.

2. Use calcium chloride and calcium gluconate as alternative formulations; do not administer both as the initial calcium dose. Calcium chloride may be repeated once after 5-10 minutes only when life-threatening ECG changes persist. Do not repeat calcium gluconate without a specific Medical Control order confirming the cumulative dose.
3. Do not mix calcium and sodium bicarbonate in the same solution or administer them simultaneously through the same tubing. Flush the line thoroughly between medications.
4. For cardiac arrest strongly suspected to result from hyperkalemia, follow the linked arrest protocol and consider calcium and sodium bicarbonate as cause-specific treatment. Their effectiveness during hyperkalemic arrest remains uncertain under the 2025 AHA guideline.
5. For pulmonary edema, position upright when tolerated, provide oxygen when indicated, initiate CPAP early when appropriate, and follow AC-05. Do not give routine fluid.
6. For fistula or graft hemorrhage, use small gauze and firm focused fingertip pressure at the exact bleeding site. Avoid bulky or circumferential dressings. If immediately life-threatening extremity hemorrhage remains uncontrolled despite effective direct pressure, apply a commercial tourniquet proximal to the access, document the time, and transport emergently.
7. For suspected access infection or peritoneal-dialysis peritonitis, follow the sepsis pathway. Keep peritoneal-dialysis components capped and clamped as found and do not manipulate or drain the system.
8. Transport symptomatic patients and notify the receiving facility early when hyperkalemia, pulmonary edema, urgent dialysis need, uncontrolled bleeding, access infection, peritonitis, or hemodynamic instability is suspected.

Medication orders

Calcium gluconate 10%

1.5-3 g IV for hyperkalemia/arrhythmia; maximum 3 g per episode

- Paramedic for suspected hyperkalemia with significant ECG changes or hemodynamic instability.
- Preferred calcium formulation when administration is through a peripheral IV.
- For persistent life-threatening ECG changes, do not repeat calcium gluconate without a specific Medical Control order confirming the cumulative dose.
- Monitor the IV site continuously and stop for suspected extravasation.

Calcium chloride 10%

500-1,000 mg IV/IO for hyperkalemia/arrhythmia

- Paramedic alternative to calcium gluconate for suspected hyperkalemia with significant ECG changes or hemodynamic instability.
- Use a secure large proximal IV or IO when possible and monitor closely for extravasation.
- May repeat once after 5-10 minutes for persistent life-threatening ECG changes.

Sodium bicarbonate

1 mEq/kg IV/IO; maximum 50 mEq

- Paramedic for suspected hyperkalemia with significant ECG changes, suspected severe metabolic acidosis, or strongly suspected hyperkalemic arrest.
- Flush the line thoroughly before or after calcium administration; do not mix with calcium in the same solution or tubing.

Albuterol

10 mg by nebulizer

- Paramedic high-dose regimen for suspected hyperkalemia with a pulse.
- Do not administer nebulized albuterol during cardiac arrest.
- This dose is specific to AM-03 and is distinct from the routine bronchospasm dose.

Normal saline

250 mL IV/IO; repeat with reassessment; maximum 500 mL without Medical Control

- AEMT and Paramedic for hypotension or poor perfusion without pulmonary edema.
- Reassess lungs, oxygenation, blood pressure, and perfusion after every aliquot.
- Do not give routine fluid solely for renal failure, missed dialysis, or suspected hyperkalemia.

Provider-level care

Provider-Level Actions Protect dialysis access, recognize hyperkalemia and fluid overload, and treat immediately life-threatening complications.

EMT: Assess airway, breathing, circulation, mental status, oxygenation, perfusion, and blood glucose; provide oxygen or ventilation when clinically indicated.

- Identify the dialysis type and schedule, last completed treatment, missed or shortened sessions, access type and location, dry weight, residual urine output, medications, anticoagulants, and recent access problems.

• Protect a functioning fistula or graft extremity from blood-pressure cuffs, routine tourniquets, venipuncture, IV access, and constrictive dressings when another option exists.

• Control access hemorrhage with small gauze and focused fingertip pressure directly over the bleeding point. Request ALS early for instability, suspected hyperkalemia, pulmonary edema, or significant hemorrhage.

AEMT: Perform all EMT actions and establish peripheral IV access away from the fistula or graft extremity when vascular treatment is indicated; use IO access when urgently required and conventional IV access cannot be obtained within the approved attempt limit.

• Apply CPAP when indicated and tolerated for pulmonary edema or respiratory failure while following the linked acute-pulmonary-edema pathway.

• For hypotension or poor perfusion without pulmonary edema, give normal saline 250 mL IV/IO and reassess lungs, oxygenation, blood pressure, and perfusion after each aliquot; maximum 500 mL without Medical Control direction.

• Do not administer calcium, sodium bicarbonate, or the high-dose hyperkalemia albuterol regimen under AM-03.

Paramedic: Perform all EMT and AEMT actions; institute continuous cardiac monitoring and obtain a 12-lead ECG for suspected hyperkalemia, dysrhythmia, syncope, significant weakness, chest symptoms, pulmonary edema, or hemodynamic instability.

• For suspected hyperkalemia with significant ECG changes or hemodynamic instability, administer calcium and potassium-shifting treatment as specified in this protocol without delaying for laboratory confirmation.

• Treat bradycardia, cardiac arrest, pulmonary edema, shock, respiratory failure, or another concurrent emergency under the linked protocol while continuing cause-specific renal care.

• For persistent shock after cautious fluid, begin norepinephrine under the shock protocol rather than administering repeated large-volume crystalloid.

Warnings

• Calcium extravasation can cause severe tissue injury. Calcium gluconate is preferred through a peripheral IV; monitor the site continuously with either formulation.

• Do not mix calcium and sodium bicarbonate because precipitation may occur; flush the line thoroughly between administrations.

• Nebulized beta-2 agonists are not recommended as hyperkalemia treatment during cardiac arrest under the 2025 AHA guideline.

• Avoid routine or large-volume crystalloid in dialysis patients, anuric patients, pulmonary edema, or known volume overload.

• A dialysis-access hemorrhage can cause rapid exsanguination. Effective bleeding control takes priority over preservation of access patency.

• Claiborne EMS does not administer insulin for hyperkalemia under AM-03.

Clinical pearls

• Hyperkalemia may initially present with generalized weakness, paralysis, syncope, or bradycardia before a classic ECG pattern becomes obvious.

• Calcium stabilizes the myocardium but does not lower the serum potassium; definitive removal generally requires dialysis.

• Repeated large fluid boluses may worsen pulmonary edema in an anuric or volume-overloaded patient. Use cautious aliquots and early vasopressor support when shock persists.

• Small gauze and focused fingertip pressure are more effective for access bleeding than a bulky absorbent dressing that disperses pressure.

• Bring available dialysis paperwork, medication lists, recent laboratory results, and the patient's dialysis-center information with the patient.

• Document the last completed dialysis, missed time, access findings, ECG findings, medication doses, response, fluid volume, bleeding-control method, tourniquet time when applicable, and receiving-facility notification.

Special populations

Peritoneal dialysis

• Abdominal pain, fever, vomiting, guarding, or reported cloudy effluent may indicate peritonitis and sepsis.

• Keep the transfer set and catheter capped and clamped as found. Do not manipulate, flush, disconnect, drain, or collect effluent in the field.

Access hemorrhage

• Continue direct focused pressure during movement and transport; do not repeatedly lift the dressing to inspect the site.

• For immediately life-threatening hemorrhage uncontrolled by direct pressure, a commercial

tourniquet proximal to the access is a life-saving intervention even though the access may be lost.

Related pathways

- AC-01 Asystole / PEA (p. 72): Open for nonshockable cardiac arrest with suspected hyperkalemia.
- AC-09 Ventricular Fibrillation / Pulseless VT (p. 96): Open for shockable cardiac arrest with suspected hyperkalemia.
- AC-02 Bradycardia With a Pulse (p. 74): Open for symptomatic bradycardia while treating suspected hyperkalemia.
- AC-05 CHF / Acute Pulmonary Edema (p. 83): Open for pulmonary edema or acute decompensated heart failure.
- UP-15 Suspected Sepsis (p. 46): Open for suspected dialysis-access infection or peritoneal-dialysis peritonitis.
- AM-05 Hypotension / Shock (p. 129): Open for persistent hypotension or poor perfusion after cautious fluid.

AM-04 Hypertension / Suspected Hypertensive Emergency

Revised October 2026

Overview

Treat acute end-organ injury, not an isolated blood-pressure number.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Confirm Blood Pressure
EMT / AEMT / Paramedic
Correct cuff size • repeat an extreme or inconsistent value manually when feasible

↓

2. End-Organ Emergency?
EMT / AEMT / Paramedic
Stroke/CVA • ACS • pulmonary edema • aortic syndrome • encephalopathy • pregnancy

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. Undifferentiated CVA?
Paramedic
If SBP >220 or DBP >110: labetalol 10 mg slow IV over 2 min • reassess • repeat 10 mg once after at least 10 min only if still above target

↓

4. Stroke BP Target
Paramedic
Goal SBP ≤220 and DBP ≤110 • do not lower further without a condition-specific order

↓

5. Other Acute Injury
EMT / AEMT / Paramedic / Medical Control
Open the linked ACS, pulmonary-edema, aortic, obstetric, or facility-transfer pathway • Medical Control for non-stroke antihypertensive treatment

↓

6. No Acute Injury
EMT / AEMT / Paramedic
Address pain, anxiety, hypoxia, urinary retention, stimulant exposure, and missed medications • do not give an antihypertensive solely to normalize BP

↓

7. Transport + Reassess
EMT / AEMT / Paramedic
Serial BP • HR/rhythm • neurologic status • chest symptoms • perfusion

Indications

- Markedly elevated BP with symptoms or findings concerning for acute end-organ injury.
- Undifferentiated suspected CVA with manually confirmed SBP above 220 mm Hg or DBP above 110 mm Hg.

Contraindications

- Do not lower BP solely to normalize an isolated number without a protocol-defined indication.
- Do not use the undifferentiated-CVA labetalol branch for post-TNK transfer; follow AM-06 and written facility orders.
- Do not give labetalol for pulmonary edema, acute decompensated heart failure, cardiogenic shock,

bradycardia, advanced heart block, or active bronchospasm.

Assessment

End-organ screen

- New focal neurologic deficit, altered mental status, seizure, chest or back pain, pulmonary edema, visual symptoms, pregnancy-related symptoms, or reduced perfusion.
- Document antihypertensive adherence, stimulant or sympathomimetic exposure, baseline neurologic status, symptom onset, and last known well when CVA is suspected.
- For suspected CVA, obtain and document serial BP, HR, rhythm, glucose, focused neurologic findings, and response to any treatment.

Treatment

1. Verify the reading and address pain, anxiety, hypoxia, urinary retention, stimulant exposure, and the underlying emergency.
2. Use UP-14 for suspected stroke, AC-04 for ACS, AC-05 for pulmonary edema, the aortic pathway or Medical Control for suspected aortic syndrome, and AO-03 for pregnancy-related hypertension.
3. For undifferentiated suspected CVA with manually confirmed SBP above 220 mm Hg or DBP above 110 mm Hg, give labetalol 10 mg slow IV over 2 minutes. Reassess BP, HR, rhythm, perfusion, and neurologic status every 5 minutes.
4. If still above target after at least 10 minutes, repeat labetalol 10 mg slow IV once. Maximum total dose is 20 mg. The target is SBP 220 mm Hg or less and DBP 110 mm Hg or less; do not lower further without a condition-specific order.

Medication orders

Labetalol

Undifferentiated suspected CVA with SBP >220 or DBP >110: 10 mg slow IV over 2 min; may repeat 10 mg once after at least 10 min if still above target; maximum total 20 mg

- Paramedic standing order only for the defined undifferentiated-CVA branch after manual confirmation.
- Goal BP: SBP ≤220 and DBP ≤110. Do not use this target for possible thrombolysis/TNK or post-TNK transfer.
- For any other hypertensive emergency, Medical Control must define the indication, dose, and endpoint.

Provider-level care

Provider Actions Confirm the measurement, identify the condition causing the elevation, and avoid indiscriminate blood-pressure reduction.

EMT: Repeat BP when feasible, perform a stroke screen, and assess chest or back pain, dyspnea, pregnancy-related symptoms, seizure, altered mental status, visual symptoms, and perfusion.

- Do not administer a patient's home antihypertensive unless specifically authorized.

AEMT: Perform all EMT actions; establish IV access when clinically indicated without delaying transport, continue monitoring, and provide condition-specific care.

Paramedic: Perform all prior care and obtain a 12-lead ECG when it will not delay time-critical care or transport.

- For undifferentiated suspected CVA with manually confirmed SBP above 220 mm Hg or DBP above 110 mm Hg, give labetalol 10 mg slow IV over 2 minutes. Reassess every 5 minutes; if still above the target after at least 10 minutes, repeat 10 mg once. Maximum total dose is 20 mg. The target is SBP 220 mm Hg or less and DBP 110 mm Hg or less.
- For a non-stroke hypertensive emergency, use the linked condition-specific pathway. Any labetalol use outside the undifferentiated-CVA branch requires Medical Control to define the indication, dose, and endpoint.

Warnings

- Rapid uncontrolled BP reduction can worsen cerebral, coronary, or renal perfusion.

Clinical pearls

- Severe hypertension without acute end-organ injury is not usually a prehospital medication emergency.
- Accurate cuff size and manual confirmation prevent major treatment errors.
- A suspected stroke is undifferentiated until imaging; do not delay destination or imaging for BP normalization.

Related pathways

- UP-14 Suspected Stroke (p. 42): Open for time-critical stroke assessment, alert, destination, and the matching undifferentiated-CVA BP branch.
- AM-06 TNK Transfer (p. 133): Open for post-TNK transfer; the tighter BP target and written facility orders govern.
- AC-04 Acute Coronary Syndrome / STEMI (p. 80): Open for ischemic chest discomfort or ECG concern.

- AC-05 CHF / Acute Pulmonary Edema (p. 83): Open for pulmonary edema or acute decompensated heart failure; do not give labetalol under AM-04.
- AO-03 OB/GYN Emergency (p. 140): Open for pregnancy-related hypertension or eclampsia concern.

AM-05
Hypotension / Shock

Revised 2026-10-30

Overview

Identify and treat the cause while supporting perfusion, avoiding harmful fluid loading, and moving rapidly toward definitive care.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Recognize Shock
EMT / AEMT / Paramedic
SBP <90 mmHg or inadequate perfusion regardless of BP • confirm an abnormal automated reading manually when feasible

↓

2. Identify the Cause
EMT / AEMT / Paramedic
Hemorrhagic • septic • hypovolemic • cardiogenic • obstructive • anaphylactic • toxicologic • adrenal • obstetric

↓

3. Immediate Support
EMT / AEMT / Paramedic
Airway/ventilation • oxygen when indicated • control bleeding • prevent heat loss • position for condition

↓

4. Monitor + Obtain Access
AEMT / Paramedic
Cardiac monitor • glucose • repeat vitals • IV access; move rapidly to IO when critically unstable

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Cause-Directed Resuscitation
AEMT / Paramedic
Cause-specific reassessed fluids • hemorrhage: warmed isotonic crystalloid to perfusion when blood is unavailable • suspected TBI: adult SBP ≥110 mmHg, no permissive hypotension • treat reversible causes

↓

6. Persistent Shock?
Paramedic
Paramedic norepinephrine pathway • use epinephrine infusion when indicated • reassess continuously

↓

7. Rapid Transport
EMT / AEMT / Paramedic
Early notification • appropriate destination • consider Covenant Health Air when it provides meaningful advantage

Indications

- Adult with SBP below 90 mmHg.
- Adult with suspected TBI and SBP below the 110 mmHg target; prevent hypotension rather than waiting for SBP below 90 mmHg.
- Adult with clinical evidence of inadequate perfusion regardless of blood pressure, including altered mental status, weak pulses, cool or mottled skin, delayed capillary refill, or progressive deterioration.

Contraindications

- Do not give indiscriminate large-volume crystalloid in uncontrolled hemorrhage, pulmonary edema, cardiogenic shock, or suspected volume overload.
- Do not use permissive hypotension when traumatic brain injury is suspected.
- Do not treat a single automated blood-pressure value without considering the patient's complete perfusion assessment and BP trend.

Assessment

Confirm Shock + Trend Perfusion

- Repeat the blood pressure manually when feasible and use the correct cuff size.
- Trend mental status, pulse quality, skin temperature and color, capillary refill, blood pressure or MAP,

heart rate, respiratory pattern, SpO_2, and $EtCO_2$ when available.

Identify the Cause

- Assess for external or internal hemorrhage, infection or sepsis, anaphylaxis, dehydration or gastrointestinal loss, cardiac ischemia, dysrhythmia, cardiogenic shock, pulmonary embolism, tension pneumothorax, and cardiac tamponade.
- Consider medication effect or overdose, adrenal crisis, ectopic pregnancy, and obstetric or postpartum hemorrhage.
- Treat immediately reversible causes through the appropriate linked protocol.

Monitoring

- Use continuous cardiac monitoring, pulse oximetry, serial vital signs, blood glucose, and temperature when indicated.

Treatment

1. Control external hemorrhage immediately and prioritize rapid transport for suspected internal bleeding or another time-critical cause.
2. Establish IV access without delaying transport. In critical instability, move rapidly to IO when IV access cannot be obtained; the approved three-attempt vascular-access limitation remains applicable.
3. For nonhemorrhagic shock without fluid-overload risk, give isotonic crystalloid 500 mL IV/IO, then reassess mental status, lung sounds, work of breathing, peripheral perfusion, BP or MAP, heart rate, and $EtCO_2$ when available.
4. Repeat 500 mL aliquots only while hypotension or poor perfusion persists and the patient remains fluid responsive.
5. Outside the suspected-TBI branch, stop routine fluid boluses when perfusion adequately improves, SBP reaches approximately 90-100 mmHg, or MAP reaches at least 65 mmHg. The higher adult TBI SBP target of at least 110 mmHg takes precedence when TBI is suspected. In any branch, stop routine boluses for pulmonary edema, worsening respiratory status, or lack of fluid responsiveness and reassess the resuscitation strategy.
6. For suspected sepsis with hypotension or inadequate perfusion, use 500 mL isotonic crystalloid aliquots and target a cumulative initial volume up to 30 mL/kg only while the patient remains fluid responsive and free of pulmonary edema. Individualize volume in patients at risk of overload and do not delay transport to complete the target volume.
7. Use blood products only within an authorized, supported transfusion program with required training, monitoring, and destination coordination.
8. For pregnancy beyond approximately 20 weeks, use left uterine displacement or left lateral positioning when feasible, avoid prolonged supine positioning, treat hemorrhage or another cause, and follow AO-03.
9. For cardiogenic or obstructive shock, avoid repeated large-volume fluid boluses. Consider a cautious 250 mL trial only when volume depletion is possible and pulmonary edema is absent; treat tension pneumothorax or causative dysrhythmia immediately.
10. For anaphylactic shock, give IM epinephrine immediately under AM-01; do not delay epinephrine for IV access or fluid administration.
11. Use norepinephrine 0.05-0.5 mcg/kg/min IV/IO as the preferred vasopressor and titrate to adequate perfusion and MAP at least 65 mmHg.
12. Begin transport early, notify the receiving facility, and do not delay for repeated IV attempts, completion of a fluid target, or normalization of blood pressure.

Medication orders

Isotonic Crystalloid

500 mL IV/IO aliquots for nonhemorrhagic shock; use 250 mL aliquots when fluid-overload risk is present

- Reassess perfusion, BP or MAP, lung sounds, work of breathing, and response after every aliquot.
- For suspected sepsis, cumulative initial fluid may approach 30 mL/kg only while the patient remains fluid responsive and free of pulmonary edema.
- In hemorrhagic shock when blood is unavailable, titrate warmed isotonic crystalloid to perfusion rather than applying the nonhemorrhagic aliquot schedule; use the higher adult SBP target of at least 110 mmHg for suspected TBI. Minimize unnecessary crystalloid in uncontrolled hemorrhage and suspected cardiogenic shock.

Norepinephrine Infusion

0.05-0.5 mcg/kg/min IV/IO; titrate to adequate perfusion and MAP at least 65 mmHg

- Paramedic standing order for persistent shock after an appropriate fluid trial or when additional fluid is contraindicated.
- Use continuous ECG and frequent blood-pressure monitoring.

Epinephrine Infusion

2-10 mcg/min IV/IO; titrate to adequate perfusion and MAP at least 65 mmHg

- Paramedic alternative when norepinephrine is unavailable or significant bradycardia accompanies shock.
- Use continuous ECG and frequent blood-pressure monitoring.

Push-Dose Epinephrine

Temporary bridge only-open the dedicated Push-Dose Epinephrine reference for the approved indication, labeled 10 mcg/mL preparation, weight-based pediatric dose, and two-bolus limit

- Use the linked medication reference rather than preparing or dosing from memory.
- Begin a vasoactive infusion when indicated and contact Medical Control before any dose beyond the two-bolus limit.

Provider-level care

EMT Care Recognize shock, support airway and ventilation, control hemorrhage, prevent heat loss, and prioritize transport.

EMT: Follow Universal Patient Care; assess mental status, pulse quality, skin signs, capillary refill, blood-pressure trend, respiratory status, and evidence of inadequate perfusion.

- Repeat an abnormal automated blood pressure manually when feasible using the correct cuff size; do not delay treatment or transport when shock is clinically evident.
- Maintain the airway, assist ventilation when needed, and give oxygen for hypoxemia, respiratory distress, shock, or another clinical indication.
- Position supine when tolerated unless respiratory distress, pulmonary edema, pregnancy, or another condition requires alternate positioning.
- Control external hemorrhage immediately using direct pressure, wound packing, tourniquets, pelvic stabilization, and other approved measures when indicated.
- Prevent heat loss, obtain frequent vital signs, identify time-critical causes, and begin rapid transport with early receiving-facility notification.

AEMT Care All EMT care plus monitoring, vascular access, glucose assessment, and cause-appropriate crystalloid with frequent reassessment.

AEMT: Perform all EMT actions; apply continuous cardiac monitoring, obtain blood glucose, and obtain temperature when infection, sepsis, or environmental illness is suspected.

- Establish IV access without delaying transport. When the patient is critically unstable and IV access cannot be obtained rapidly, proceed to IO access under the approved vascular-access protocol.
- For nonhemorrhagic shock without fluid-overload risk, give isotonic crystalloid 500 mL IV/IO and reassess after every aliquot.
- For heart failure, renal failure, liver failure, suspected cardiogenic shock, or another risk of volume overload, use 250 mL aliquots with reassessment after each aliquot.
- For suspected hemorrhagic shock without traumatic brain injury when blood is unavailable, titrate warmed isotonic crystalloid only to maintain a palpable radial pulse, adequate mental status, or SBP approximately 80-90 mmHg; avoid unnecessary large-volume crystalloid. Reassess after treatment.
- For suspected TBI, prevent hypotension: target adult SBP at least 110 mmHg, including age 70. Do not use permissive hypotension; use normal saline when crystalloid is required and reassess blood pressure frequently and after treatment.
- Immediately report persistent hypotension, pulmonary edema, worsening respiratory status, altered mental status, dysrhythmia, or other deterioration to the Paramedic and receiving facility.

Paramedic Care All AEMT care plus advanced hemodynamic assessment, cause-specific intervention, vasopressor treatment, and critical destination decisions.

Paramedic: Perform all EMT and AEMT actions; obtain and interpret a 12-lead ECG when cardiac ischemia, dysrhythmia, pulmonary embolism, or cardiogenic shock is possible.

- Use waveform capnography when ventilation is assisted and when clinically useful for trending perfusion in severe shock.
- Treat immediately reversible causes, including tension pneumothorax and unstable dysrhythmia, using the applicable linked protocol.
- For persistent hypotension or inadequate perfusion after an appropriate fluid trial-or earlier when further fluid is contraindicated-begin norepinephrine 0.05-0.5 mcg/kg/min IV/IO and titrate to adequate perfusion and MAP at least 65 mmHg.
- When norepinephrine is unavailable or significant bradycardia accompanies shock, begin epinephrine infusion 2-10 mcg/min IV/IO and titrate to adequate perfusion and MAP at least 65 mmHg.
- Use continuous ECG monitoring and frequent blood-pressure reassessment during vasopressor treatment.
- Begin transport early, notify the receiving facility, choose the closest appropriate destination for the suspected cause, and consider Covenant Health Air

when it provides a meaningful time or capability advantage.

Warnings

- A normal mental status or GCS of 15 does not exclude shock.
- Hypotension may be a late sign of severe shock; use the complete perfusion assessment and trend.
- The treatment goal is adequate perfusion-not normalization of blood pressure.
- Fluids and vasopressors support perfusion but do not replace correction of the underlying cause.

Clinical pearls

- A falling $EtCO_2$ in a spontaneously breathing patient may indicate worsening perfusion.
- Reassess after every fluid aliquot and vasopressor adjustment; response to treatment is part of the diagnosis.
- Early vasopressor support may be preferable to repeated fluid in cardiogenic shock, pulmonary edema, or a patient who is no longer fluid responsive.
- Control hemorrhage, decompress tension pneumothorax, treat anaphylaxis, and correct causative dysrhythmia without waiting for blood-pressure normalization.

Special populations

Heart, Renal, or Liver Failure

- Use 250 mL crystalloid aliquots with frequent reassessment and stop for pulmonary edema, worsening respiratory status, or lack of fluid responsiveness.

Suspected Traumatic Brain Injury

- Prevent hypotension: target adult SBP at least 110 mmHg, including age 70. Do not use permissive hypotension. Avoid hypoxemia and excessive crystalloid; use normal saline when crystalloid is required. Reassess blood pressure frequently and after treatment.

Adrenal Insufficiency

- Ask about prescribed stress-dose steroids and follow the patient-specific emergency plan when authorized.

Related pathways

- Push-Dose Epinephrine - Medication Reference (p. 224): Open the dedicated reference for the approved indication, labeled preparation, concentration, dosing, and two-bolus limit.
- UP-01 Universal Patient Care (p. 4): Open universal assessment, monitoring, vascular-access, reassessment, and transport requirements.
- AM-01 Allergic Reaction / Anaphylaxis (p. 117): Open for anaphylactic shock and immediate IM epinephrine treatment.
- AM-03 Dialysis / Renal Failure (p. 123): Open for renal-failure considerations and fluid-risk guidance.
- AM-07 Crashing Patient (p. 135): Open for peri-arrest deterioration and resuscitation priorities.
- AC-04 Acute Coronary Syndrome / STEMI (p. 80): Open when myocardial ischemia or infarction may be causing shock.
- AC-05 CHF / Acute Pulmonary Edema (p. 83): Open for pulmonary edema or suspected acute heart failure.
- AC-02 Bradycardia (p. 74): Open for bradycardia causing cardiopulmonary compromise.
- AC-06 Narrow-Complex Tachycardia (p. 87): Open for unstable narrow-complex tachycardia.
- AC-07 Monomorphic Wide-Complex Tachycardia (p. 90): Open for unstable monomorphic wide-complex tachycardia.
- AO-03 OB/GYN Emergency (p. 140): Open for ectopic pregnancy, obstetric hemorrhage, or pregnancy-related shock.
- TB-05 Head Trauma (p. 188): Suspected TBI: higher adult perfusion target, serial reassessment, and controlled ventilation.
- TB-12 Abdominal / Pelvic Trauma (p. 201): Concealed traumatic hemorrhage and pelvic stabilization with the shared adult resuscitation strategy.
- TB-06 Multiple Trauma (p. 190): Open for traumatic shock, hemorrhage control, and trauma destination decisions.
- TE-07 Overdose / Toxic Ingestion (p. 211): Open for medication effect, overdose, or another toxicologic cause.
- AR-01 Adult Airway Management (p. 143): Open for advanced airway and ventilation support.

AM-06
Suspected Stroke - Tenecteplase (TNK) Transfer

Revised October 2026

Overview

TNK is a single IV bolus. There is no thrombolytic infusion to continue, titrate, or stop during transport.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Verify Transfer
Paramedic
Accepted sending/receiving plan • exact TNK bolus time known • dose/weight verified • last known well documented

↓

2. Stabilize Before Departure
Paramedic
Airway • glucose • baseline neuro exam • BP <180/105 • IV sites checked • written orders verified

↓

3. Monitor Every 15 Minutes
Paramedic
BP • GCS • pupils • focal findings • airway • SpO_2 • bleeding/angioedema symptoms

↓

4. Complication?
Paramedic
Severe headache • vomiting • acute hypertension • worsening deficit • seizure • bleeding • angioedema

↓

5. Support + Notify
Paramedic / Medical Control
ABCs • stop nonessential infusions • contact receiving stroke facility and Medical Control immediately

Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Direct Transport
EMT / AEMT / Paramedic
NPO • cardiac monitor • no unnecessary invasive procedures • minimize delay

Indications

- Interfacility transfer after hospital-administered Tenecteplase (TNK) for suspected acute ischemic stroke.

Contraindications

- Do not initiate transfer until the patient has an accepted sending/receiving plan and the exact TNK bolus-administration time is known and documented.
- Do not administer antiplatelet agents, anticoagulants, additional thrombolytic therapy, or unapproved IM medication during transfer unless specifically ordered by the sending stroke physician and confirmed with Medical Control.

Assessment

Before departure

- Document exact last known well and symptom-discovery times.
- Verify patient weight, total TNK dose, and exact TNK bolus administration time; an estimate is not acceptable.
- Obtain baseline BP, GCS, pupils, focal neurologic findings, glucose, airway assessment, IV-site assessment, and NIHSS or facility neurologic examination when available.
- Confirm written medication/infusion orders, receiving physician/facility acceptance, destination, and transfer plan.

En route

- Repeat BP, GCS, pupils, focused neurologic examination, airway assessment, and symptoms at least every 15 minutes; increase frequency for instability or active antihypertensive infusion.
- Watch for severe headache, nausea/vomiting, acute hypertension, declining consciousness, new or worsening deficit, seizure, angioedema, and external, GI, GU, or IV-site bleeding.

Treatment

1. Maintain oxygenation and ventilation; keep the patient NPO.

2. Maintain BP below 180/105 mmHg for the first 24 hours after thrombolysis. Do not use the pre-thrombolysis 185/110 threshold as the post-TNK target, and avoid intensive SBP lowering below 140 mmHg.
3. Use the limb without an infusion for BP readings when practical.
4. For neurologic deterioration, bleeding, angioedema, seizure, severe headache, or vomiting, support ABCs, stop nonessential infusions, recheck BP and glucose, and immediately contact the receiving stroke facility and Medical Control.
5. Transport directly without unnecessary delay. Consider Covenant Health Air only when it provides a meaningful time or capability advantage and does not delay definitive movement.

Medication orders

Tenecteplase (TNK)

Single IV bolus administered by the sending hospital

- There is no ongoing TNK infusion during transport.
- The exact bolus administration time must be known and documented before departure.

Antihypertensive infusion

Continue only under written sending-facility orders and receiving-facility acceptance

- Maintain BP below 180/105 mmHg after thrombolysis.
- Follow approved interfacility-transfer policy and pump parameters; do not begin or alter treatment without Medical Control or receiving-facility direction.

Provider-level care

Provider-Level Actions A Paramedic-led interfacility transport with focused post-thrombolysis monitoring.

EMT: Assist airway support, monitoring, movement, NPO precautions, bleeding surveillance, and documentation under the transport-team plan.

- Immediately report neurologic decline, vomiting, new bleeding, or airway symptoms.

AEMT: Assist with authorized monitoring and essential vascular access. Avoid unnecessary invasive procedures, IM medication, and additional vascular attempts after thrombolysis.

- Apply prolonged direct pressure after any necessary vascular access.

Paramedic: Verify the exact TNK bolus administration time before departure; do not depart without it.

- Verify patient weight, total TNK dose, last known well, baseline neurologic examination or NIHSS when available, sending/receiving acceptance, destination, and written medication/infusion orders.
- Perform and document BP and focused neurologic reassessment at least every 15 minutes; maintain continuous cardiac monitoring and pulse oximetry.
- Continue an antihypertensive infusion only under written sending-facility orders and accepted pump parameters. Do not initiate or titrate a new antihypertensive under AM-06 without Medical Control or receiving-facility direction.
- For deterioration, support ABCs, recheck BP and glucose, contact the receiving stroke facility and Medical Control immediately, and continue direct transport unless immediate lifesaving diversion is required.

Warnings

- Sudden neurologic deterioration, severe headache, vomiting, seizure, acute hypertension, or bleeding after TNK may indicate intracranial hemorrhage or edema.
- Angioedema can threaten the airway even without other allergic findings.
- TNK is a bolus medication; there is no thrombolytic pump or remaining drug volume to manage.
- Do not delay transport for nonessential procedures or attempt to normalize blood pressure beyond the ordered post-thrombolysis target.

Clinical pearls

- Precise times-not estimates such as 'about an hour ago'-are essential for stroke care and must be part of the handoff.
- A stable initial examination does not eliminate the risk of hemorrhage or angioedema.
- NPO status and avoidance of unnecessary invasive procedures reduce avoidable post-thrombolysis risk.

Special populations

Wake-up stroke

- Last known well is the last time the patient was known to be at neurologic baseline, not the time symptoms were discovered.

Airway deterioration

- Prioritize airway assessment and early airway preparation for orolingual angioedema, stridor, inability to protect the airway, or inadequate ventilation. Use the airway and anaphylaxis protocols while immediately notifying the receiving facility and Medical Control.

Related pathways

- UP-04 Altered Mental Status (p. 14): Open for ongoing neurologic and reversible-cause assessment.
- AM-05 Hypotension / Shock (p. 129): Open for hypotension or inadequate perfusion.
- AM-01 Allergic Reaction / Anaphylaxis (p. 117): Open for angioedema or another allergic airway emergency.
- AR-01 Adult Airway Management (p. 143): Open for airway protection or ventilatory support.
- UP-13 Seizure (p. 39): Open for new seizure or recurrent convulsive activity.
- AM-04 Hypertension (p. 127): Open for the defined undifferentiated-CVA BP branch only; post-TNK treatment remains governed by written facility orders.
- AM-07 Crashing Patient (p. 135): Open for peri-arrest deterioration.

AM-07
Crashing Medical Patient

Revised October 2026

Overview

Use for immediate or impending physiologic collapse requiring simultaneous resuscitation, rapid decisions, and expedited transport.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Crashing Patient?
EMT / AEMT / Paramedic
Hypoxia/ventilatory failure • severe distress • SBP <90/shock • declining mental status • peri-arrest dysrhythmia

↓

2. Assign Roles + Bring Equipment
EMT / AEMT / Paramedic
Leader • airway • monitor/defibrillator • IV/IO/medications • hemorrhage • documentation

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. Immediate Threats
EMT / AEMT / Paramedic
Airway/ventilation • hemorrhage • tension pneumothorax • hypoglycemia • anaphylaxis • dysrhythmia • shock

↓

4. Access + Cause-Directed Care
AEMT / Paramedic
Rapid IV/IO • reassessed fluids/vasopressors • pads early • use the linked definitive pathway

↓

5. Move + Reassess
EMT / AEMT / Paramedic
Move when lifesaving care is underway • reassess at least every 5 min during prolonged extrication

Indications

- Immediate or impending physiologic collapse: SpO_2 below 90% or inadequate ventilation, severe respiratory distress/exhaustion, SBP below 90 or shock, rapidly declining mental status, poor perfusion, extreme respiratory rate, or peri-arrest dysrhythmia.

Contraindications

- Do not remain on scene when safety is threatened, definitive treatment is unavailable, hemorrhage requires surgery or blood products unavailable to the crew, or transport/rendezvous is the fastest route to needed care.
- Do not use NIPPV for vomiting, facial trauma, inability to cooperate, inability to protect the airway, or impending arrest.

Assessment

Immediate assessment

- Assess airway, ventilation, oxygenation, pulse, BP/perfusion, severe hemorrhage, glucose, temperature when indicated, and reversible causes.
- A normal GCS or initially acceptable BP does not exclude a crashing patient; reassess after every intervention.

Treatment

1. Bring all anticipated resuscitation equipment to the patient and assign team functions.

2. Treat airway, ventilation, severe hemorrhage, tension pneumothorax, hypoglycemia, anaphylaxis, dysrhythmia, and shock before prolonged movement when safe.
3. Prioritize BVM ventilation and oxygenation. Use advanced airway management when the patient cannot protect the airway or cannot be adequately oxygenated/ventilated less invasively; confirm with waveform capnography.
4. Identify peri-intubation hypotension and use AM-05 fluid/vasopressor treatment before high-risk intubation when possible.
5. Use AM-05 for fluid and norepinephrine strategy; avoid indiscriminate crystalloid in uncontrolled hemorrhage, pulmonary edema, cardiogenic shock, or volume overload.
6. Apply pads early when instability is present or anticipated and use the linked cardiac pathway for dysrhythmia.
7. Move promptly once critical interventions are underway; plan reassessment stops during prolonged extrication and reassess at least every 5 minutes.

Medication orders

Push-Dose Epinephrine

Temporary bridge only-open the dedicated reference for the approved indication, labeled 10 mcg/mL preparation, weight-based pediatric dose, and two-bolus limit.

- Use the medication reference rather than preparing or dosing from memory.
- Begin a vasoactive infusion when indicated and contact Medical Control before any dose beyond the two-bolus limit.

Provider-level care

Provider-Level Actions Higher levels include preceding-level actions.

EMT: Assign team roles; obtain a full vital set within 5 minutes when feasible without delaying lifesaving care.

- Position, suction, use BVM ventilation and oxygen when indicated, control hemorrhage, apply AED/monitor support, prevent heat loss, request resources early, and prepare transport.

AEMT: Establish IV access or proceed rapidly to IO when urgent access is needed and IV is unsuccessful or unlikely.

- Give cause-appropriate isotonic fluid with reassessment and use a supraglottic airway when indicated.

Paramedic: Provide advanced airway/ventilation with waveform capnography, dysrhythmia treatment, vasopressors, and cause-specific resuscitation.

- Select the closest appropriate destination and consider Covenant Health Air only when it improves time or capability without delaying movement.

Warnings

- Do not allow procedures to delay definitive care.
- Fluids and vasopressors support perfusion but do not replace correction of the cause.

Clinical pearls

- The goal is not prolonged scene care; it is to avoid moving an unresuscitated patient through a difficult extrication when immediate EMS interventions can improve stability.

Related pathways

- Push-Dose Epinephrine - Medication Reference (p. 224): Open the dedicated reference for the approved preparation, dosing, and two-bolus limit.
- UP-01 Universal Patient Care (p. 4)
- AM-05 Hypotension / Shock (p. 129)
- AR-01 Adult Airway Management (p. 143)
- AC-03 Adult Cardiac Arrest (p. 77)
- AM-01 Allergic Reaction / Anaphylaxis (p. 117)
- TB-06 Multiple Trauma (p. 190)
- TE-07 Overdose / Toxic Ingestion (p. 211)

AO-01
Childbirth / Labor

Revised October 16, 2026

Overview

Determine whether transport is safe or delivery is imminent using gestational age, contraction pattern, urge to push, parity, prior labor duration, and perineal inspection.

Prioritize maternal positioning, preparation for neonatal care, recognition of abnormal presentation, and immediate treatment of postpartum hemorrhage.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Labor / Delivery Concern
EMT / AEMT / Paramedic
Due date • gravida/para • contractions • membrane rupture • bleeding • fetal movement.

↓

2. Maternal Emergency?
EMT / AEMT / Paramedic
Severe bleeding • hypertension/hypotension • seizure • severe pain • altered mental status.

↓

3. Inspect Perineum
EMT / AEMT / Paramedic
No digital vaginal examination. Look for crowning, cord, breech, limb, or meconium.

↓

4. Delivery Imminent?
EMT / AEMT / Paramedic
Crowning, urge to push, frequent contractions, advanced parity, or unsafe transport.

↓

5. Normal or Complicated?
EMT / AEMT / Paramedic
Normal delivery • prolapsed cord • breech • shoulder dystocia • multiple gestation.

Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Mother + Newly Born Care
EMT / AEMT / Paramedic
Control bleeding, maintain warmth, record times/APGAR, and notify destination.

Indications

- Pregnant patient with contractions, rupture of membranes, crowning, urge to push, vaginal discharge or bleeding, or concern for imminent delivery.
- Patient who has delivered immediately before EMS arrival and requires maternal or neonatal care.

Assessment

Priority findings

- Crowning or an uncontrollable urge to push.
- Gestation less than 36 weeks.
- Abnormal presentation, prolapsed cord, severe bleeding, seizure, or multiple gestation.
- History of rapid prior deliveries or very frequent contractions.

Required documentation

- Contraction onset, duration, and frequency; membrane rupture; delivery time; placenta delivery; maternal bleeding; and treatment.
- Infant APGAR at 1 and 5 minutes without interrupting resuscitation.

Treatment

1. Delay cord clamping for at least 60 seconds in a preterm infant under 37 weeks who does not require immediate resuscitation.
2. Clamp the cord approximately 10 cm from the infant and place the second clamp approximately 5 cm farther away.
3. After placental delivery, perform uterine massage for postpartum hemorrhage and continue maternal reassessment.

Medication orders

Tranexamic acid

1 g IV/IO over 10 minutes for postpartum hemorrhage with shock when within 3 hours of delivery.

- Use with AM-05 Hypotension/Shock and AO-03 OB/GYN Emergency.

Provider-level care

Provider-Level Actions Perform immediately indicated care at the highest authorized level present while protecting both maternal and neonatal physiology.

EMT: Place the mother left lateral when delivery is not imminent; document contraction frequency/duration and continuously reassess.

• For field delivery, support the head and body without pulling, check for a nuchal cord, dry and warm the infant, and move immediately to the Newly Born protocol.

• For prolapsed cord, elevate hips/knees to chest, relieve pressure from the presenting part with a gloved hand, cover exposed cord with moist saline dressing, and transport emergently.

• For breech presentation, support the presenting parts without pulling and create an airway space at the infant's nose if the head does not deliver.

AEMT: Perform all EMT care plus IV/IO access and protocol-directed treatment of maternal hypotension or hemorrhage without delaying delivery or transport.

• Prepare neonatal resuscitation equipment and obtain additional resources for prematurity or multiple gestation.

Paramedic: Perform all prior care plus advanced management of maternal shock, seizure, airway compromise, and complicated delivery.

• After field delivery, administer tranexamic acid for postpartum hemorrhage with shock when indicated; oxytocin is not carried on the Claiborne formulary.

Warnings

• Do not perform a digital vaginal examination.

• Do not pull on the infant, umbilical cord, breech-presenting parts, or placenta.

• Uterine massage is performed only after delivery of the placenta.

• Large-volume or free vaginal bleeding is abnormal and requires urgent hemorrhage management.

Clinical pearls

• Transport is generally preferred, but crowning, urge to push, frequent contractions, parity, and prior rapid labor may make field delivery safer.

• Twins are commonly premature and at higher risk for hypothermia; request additional resources early.

Special populations

Postpartum hemorrhage

• After placental delivery, perform uterine massage for excessive bleeding. For shock, use AM-05 and tranexamic acid 1 g IV/IO over 10 minutes when within 3 hours of delivery.

AO-02
Newly Born

Revised 2026-10-31

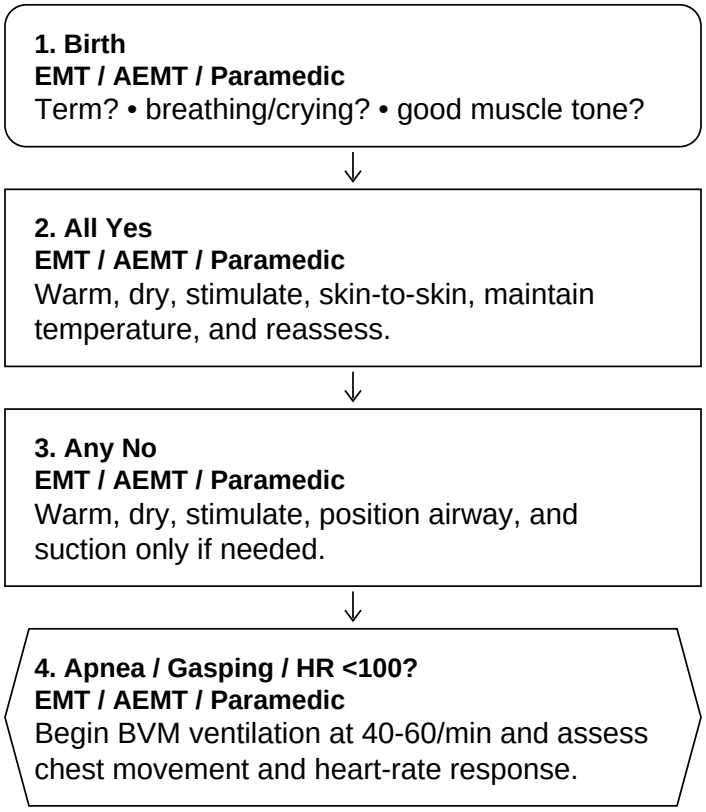
Overview

Most newly born infants require only warmth, drying, stimulation, and appropriate cord management.

Effective positive-pressure ventilation is the most important intervention when respirations are inadequate or heart rate remains below 100/min.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.



Continue at step 5 in the next column/page

Quick flow continued at step 5

5. HR Remains <60?
EMT / AEMT / Paramedic
After effective ventilation, begin two-thumb encircling 3:1 compressions, use 100% oxygen, and escalate care.

↓

6. Reassess + Transport
EMT / AEMT / Paramedic
Maintain warmth, right-hand SpO_2, cardiac monitoring, glucose, and early notification.

Indications

- Infant in the immediate newborn period following field, home, or facility delivery.
- Newly born infant with apnea, gasping, respiratory distress, persistent central cyanosis, poor tone, or bradycardia.

Assessment

Initial newborn assessment

- Term gestation, spontaneous breathing or crying, and good muscle tone.
- Heart rate, respiratory effort, central color, temperature, and responsiveness.
- Gestational age, meconium, delivery complications, multiple gestation, congenital disease, and maternal medications or substance exposure.

Expected preductal oxygen saturation

- 2 minutes: 65-70%; 3 minutes: 70-75%; 4 minutes: 75-80%.
- 5 minutes: 80-85%; 10 minutes: 85-95%.
- Use right-hand/wrist/upper-arm preductal monitoring. These expected values apply to an infant not requiring active resuscitation.

Treatment

1. If a non-vigorous infant does not improve after warming, drying, and stimulation within 30 seconds, move quickly to positive-pressure ventilation.
2. Add oxygen when the infant is not responding with an increasing heart rate or adequate breathing.
3. Do not interrupt resuscitation to obtain APGAR scores; document APGAR at 1 and 5 minutes.

Medication orders

Epinephrine 1:10,000

0.01 mg/kg IV/IO every 3-5 minutes as needed

- For heart rate below 60 despite effective ventilation and coordinated compressions. Paramedic-only when a pulse is present; AEMT/Paramedic for confirmed pulseless arrest within approved arrest orders. Use UVC/IO access when available.

Normal saline

10 mL/kg IV/IO; may repeat once when not responding to epinephrine

- Consider suspected hypovolemia.

Provider-level care

Provider-Level Actions Perform immediately indicated care at the highest authorized level present while protecting both maternal and neonatal physiology.

EMT: Warm, dry, and stimulate; position the airway and use bulb suction only when secretions obstruct breathing.

- If apnea/gasping or heart rate below 100 persists, provide BVM ventilation at 40-60 breaths/min and correct mask seal, airway position, and technique until chest movement is present.
- If heart rate remains below 60 after 30-60 seconds of effective ventilation, begin two-thumb encircling chest compressions coordinated at a 3:1 ratio.
- Apply pulse oximetry to the right hand, wrist, or upper arm and prevent hypothermia throughout care.

AEMT: Perform all EMT care plus authorized supraglottic airway support, IV/IO access, fluid and glucose treatment within current neonatal standing orders. Epinephrine while a pulse is present is Paramedic-only; AEMT epinephrine remains available for confirmed pulseless arrest within the approved arrest orders.

- Use cardiac monitoring to determine heart rate when available without interrupting ventilation or compressions.

Paramedic: Perform all prior care plus advanced airway management when ventilation remains ineffective and protocol-directed neonatal resuscitation medications. Paramedic: give the specified epinephrine dose for persistent HR below 60 with a pulse despite effective ventilation and coordinated compressions.

- Evaluate persistent failure to respond for hypovolemia, pneumothorax, hypoglycemia, congenital disease, infection, or maternal medication effect.

Warnings

- Do not routinely suction the airway; suction only when needed for obstruction.
- Do not perform routine endotracheal suctioning for meconium; provide supportive ventilation.
- Naloxone is not recommended for neonatal respiratory depression from maternal narcotics; provide ventilatory support.
- Ventilation must be effective before chest compressions are started.
- AO-02 does not specify a neonatal D10 dose. After immediate newborn stabilization, use PM-02 for the approved pediatric hypoglycemia pathway.

Clinical pearls

- A rising heart rate is the best indicator that newborn ventilation and resuscitation are effective.
- Skin-to-skin contact with the mother is the preferred warming method for a vigorous infant.
- Delay cord clamping for approximately 60 seconds unless immediate resuscitation is required.

Special populations

Premature infant

- Premature infants lose heat rapidly; use a cap, plastic wrap, thermal mattress, radiant heat, or other approved warming adjunct.
- Anticipate respiratory support, hypoglycemia, and the need for specialized neonatal destination care.

Related pathways

- PM-02 Pediatric Diabetic / Hypoglycemia (p. 177): After immediate newborn stabilization, open for the approved pediatric hypoglycemia and dextrose pathway.

AO-03
OB-GYN Emergency

Revised 2026-10-31

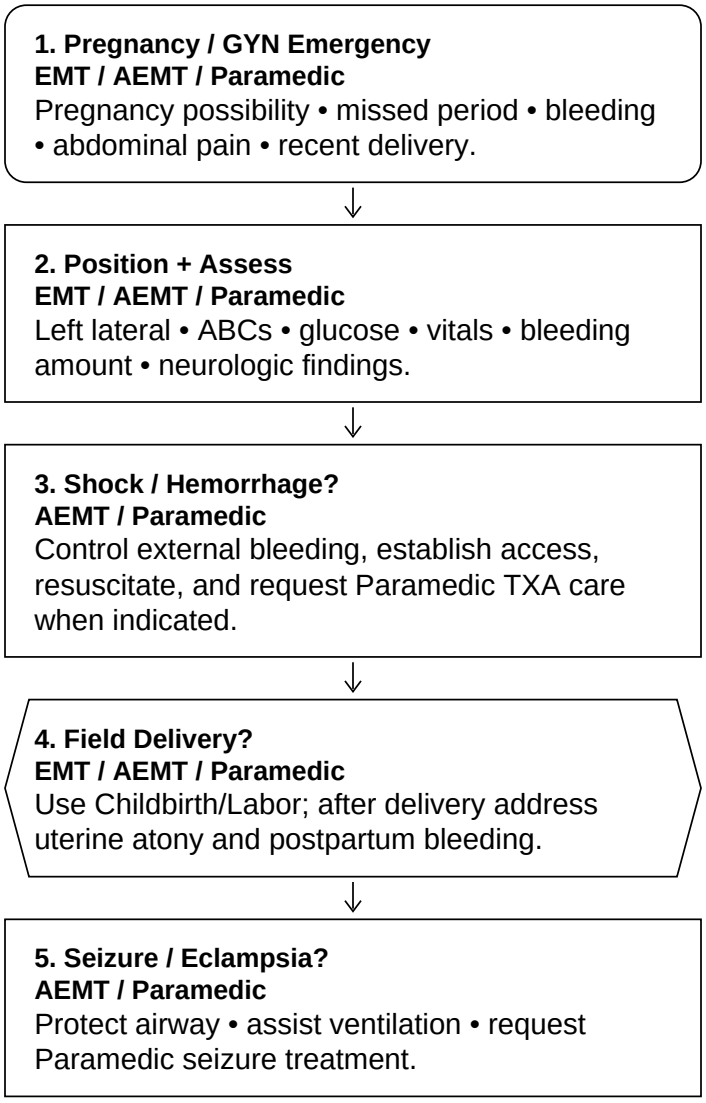
Overview

Evaluate pregnancy-related and gynecologic emergencies including hemorrhage, ectopic pregnancy, miscarriage, preeclampsia, eclampsia, shock, and trauma.

Use left lateral positioning, early hemorrhage recognition, seizure control, and rapid destination notification.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.



Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Paramedic - Seizure / Hemorrhage Paramedic
Magnesium for eclampsia • midazolam for ongoing convulsion • TXA for qualifying postpartum hemorrhage; use doses and safeguards below.

↓

7. Rapid Transport
EMT / AEMT / Paramedic
Continuous reassessment and early OB/trauma-capable destination notification.

Indications

- Known or suspected pregnancy with abdominal pain, vaginal bleeding, hypertension, seizure, syncope, or shock.
- Postpartum hemorrhage or uncontrolled vaginal hemorrhage with signs of shock regardless of pregnancy status.
- Pregnant patient involved in trauma or motor-vehicle collision.

Assessment

Preeclampsia / eclampsia findings

- Blood pressure above 140 systolic or 90 diastolic, or a significant rise from the patient's prepregnancy baseline.
- Severe headache, visual disturbance, right-upper-quadrant or epigastric pain, nausea/vomiting, hyperreflexia, or edema of the hands and face.
- Seizure in pregnancy or the postpartum period may represent eclampsia even without a prior diagnosis.

Hemorrhage and shock

- Quantify bleeding by pads per hour, visible blood loss, clots, free bleeding, and associated syncope or abdominal pain.
- Consider occult ectopic rupture or placental abruption when shock is disproportionate to visible bleeding.

Treatment

1. Paramedic: for eclampsia or suspected eclampsia, start magnesium sulfate promptly as the cause-directed therapy. For an ongoing convulsion, use the standardized UP-13 / medication-reference midazolam regimen when immediate seizure control is required.
2. After placental delivery, perform uterine massage and follow postpartum hemorrhage treatment.
3. A pregnant patient after motor-vehicle collision requires immediate physician evaluation; pregnancy beyond 20 weeks commonly requires 4-6 hours of fetal monitoring.

Medication orders

Tranexamic acid - Paramedic only

1 g IV/IO over 10 minutes for postpartum hemorrhage with shock when within 3 hours of delivery

- For uncontrolled vaginal or postpartum hemorrhage with signs of shock; follow the indication and timing limits above.
- Postpartum use is contraindicated when birth occurred more than 3 hours before EMS arrival.

Midazolam - Paramedic only

Use the standardized adult-seizure regimen: 10 mg IM when IV/IO is not established, 10 mg IN as an alternative, or 5 mg IV/IO; repeat 5 mg IV/IO once after 5 minutes if seizure continues (maximum total 10 mg).

- Do not delay IM treatment while attempting vascular access during active seizure.
- Magnesium sulfate remains the cause-directed eclampsia therapy.

Magnesium sulfate

4-6 g IV over 20-30 minutes, then 1-2 g/hr infusion for eclampsia or suspected eclampsia

- Primary cause-directed therapy for eclampsia or suspected eclampsia.
- Continuously monitor ECG, BP, respiratory status, SpO_2, and $EtCO_2$.

Provider-level care

Provider-Level Actions Perform immediately indicated care at the highest authorized level present while protecting both maternal and neonatal physiology.

EMT: Place the patient left lateral with the right side elevated 10-20 degrees when pregnancy is advanced or supine hypotension is possible.

- Assess airway, breathing, circulation, mental status, abdominal pain, bleeding amount, headache, vision changes, edema, and seizure activity.
- For seizure, protect from injury, support oxygenation and ventilation, check glucose, and request ALS immediately.
- Treat all pregnant trauma patients as requiring physician evaluation and transport.

AEMT: Perform all EMT care plus IV/IO access and AM-05 shock resuscitation. Support airway and

ventilation during seizure and request Paramedic care; midazolam, magnesium and TXA under AO-03 are Paramedic-only.

• Do not delay ventilation, CPR when indicated, or transport for IV/IO attempts; assist the Paramedic with medication preparation within scope.

Paramedic: Perform all prior care plus cardiac monitoring, midazolam for ongoing convulsion without delaying IM treatment for vascular access, magnesium treatment, TXA for qualifying postpartum hemorrhage, calcium-gluconate rescue for magnesium toxicity, and advanced resuscitation.

• Identify ectopic pregnancy, placental abruption/previa, preeclampsia/eclampsia, and non-obstetric causes while minimizing transport delay.

Warnings

• For suspected eclampsia, magnesium sulfate is the cause-directed therapy; give midazolam promptly for an ongoing convulsive seizure when needed.

• Magnesium sulfate may cause hypotension and respiratory depression; stop it for clinically significant toxicity and give calcium gluconate 1 g IV slowly with Medical Control.

• Do not use AM-04's undifferentiated-CVA labetalol standing order for pregnancy-related hypertension. Obtain Medical Control for the obstetric indication, agent, dose, and blood-pressure endpoint.

• Do not suggest that a normal ultrasound alone would exclude injury after maternal trauma; appropriate fetal monitoring and physician evaluation are required.

• Vaginal bleeding may be absent in ectopic pregnancy or placental abruption.

• Oxytocin is not on the Claiborne formulary and is not authorized under this protocol.

Clinical pearls

• Maintain a high index of suspicion for ectopic pregnancy in any patient of childbearing age with abdominal pain, syncope, or shock.

• Supine hypotensive syndrome is reduced by left lateral positioning with the right side elevated.

Special populations

Transport

• Use the closest appropriate emergency/OB-capable facility with early notification for hemorrhage, ectopic rupture, eclampsia, severe hypertension, trauma, or shock.

• Pregnancy with any chance of delivery should remain on ground transport and not be placed on an aircraft.

Related pathways

• AO-01 Childbirth / Labor (p. 137)
• AO-02 Newly Born (p. 138)
• AM-05 Hypotension / Shock (p. 129)
• UP-13 Seizure (p. 39)

AR-01
Adult Airway

Revised 2026-10-30

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Airway / Breathing Adequate?
EMT / AEMT / Paramedic
Assess rate, effort, patency, oxygenation, ventilation, and mental status.

↓

2. Basic Maneuvers First
EMT / AEMT / Paramedic
Position • suction • OPA/NPA • two-person BVM ± PEEP.

↓

3. Foreign-Body Obstruction?
EMT / AEMT / Paramedic
Use age-appropriate obstruction procedure and direct visualization when authorized.

↓

4. Support Still Needed?
EMT / AEMT / Paramedic
All levels: oxygen/BVM. AEMT/Paramedic: authorized NIPPV or BIAD. Paramedic: intubation when clinically indicated.

↓

5. Oxygenation / Ventilation Effective?
EMT / AEMT / Paramedic
Maintain SpO_2 at least 92% unless a condition-specific target applies; suspected TBI requires at least 94%. If ineffective, move to the failed-airway pathway.

Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Suspected TBI?
EMT / AEMT / Paramedic
SpO_2 ≥94%; urgently correct <90% or inadequate ventilation. Initial/routine assisted $EtCO_2$ 35-45 mmHg. No routine hyperventilation. Clear active/imminent herniation only: brief $EtCO_2$ 30-35 with capnography, avoid <30, and expedite transport. Open TB-05.

↓

7. Confirm + Reassess
EMT / AEMT / Paramedic
Waveform $EtCO_2$ after ETT/BIAD, secure device, monitor, and notify destination.

Indications

- Adult with actual or threatened airway obstruction, inadequate oxygenation, inadequate ventilation, or inability to protect the airway.

Treatment

1. Target adult ventilation at approximately 10-12 breaths/min and use $EtCO_2$ and clinical response to avoid hyperventilation.
2. Before an advanced-airway attempt, assess for difficult BVM ventilation, difficult laryngoscopy, inability to place a BIAD, and potential surgical-airway need. Prepare backup equipment and a rescue plan before beginning.
3. For suspected traumatic brain injury, continuously monitor oxygenation and ventilation. Aim for SpO_2 at least 94%; urgently correct SpO_2 below 90% or inadequate ventilation. The condition-specific TBI target takes priority over a lower general airway or post-ROSC oxygenation target.
4. For suspected traumatic brain injury requiring assisted ventilation, use an initial/routine $EtCO_2$ target of 35-45 mmHg. Avoid routine hyperventilation. Only for clear active or imminent cerebral herniation, use brief controlled ventilation targeting $EtCO_2$ 30-35 mmHg with continuous waveform capnography while expediting definitive care; avoid overshooting below 30 mmHg.
5. Prevent hypotension in suspected traumatic brain injury; do not use permissive hypotension. For adults, target SBP at least 110 mmHg, including age 70, with frequent reassessment and after treatment. Pediatric TBI blood-pressure targets require age-specific direction from Medical Control

until the separately approved pediatric TBI target table is published; do not substitute adult targets or generic pediatric shock thresholds.

Provider-level care

Provider-Level Actions Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT: Position and open the airway, suction, insert basic adjuncts, provide oxygen, and perform effective two-person BVM ventilation with PEEP when indicated.

• Continue basic airway care when it maintains adequate oxygenation and ventilation; do not delay transport solely to place an advanced airway.

AEMT: Perform all EMT care plus authorized BIAD placement and continuous waveform capnography once available. Endotracheal intubation is not included as an AEMT procedure in this protocol.

• After an unsuccessful BIAD attempt, do not repeat blind placement; continue effective BVM and request Paramedic airway support.

Paramedic: Perform all prior care plus ETT, advanced airway selection, DAI when authorized, chest decompression when indicated, and surgical airway for cannot-oxygenate/cannot-ventilate failure.

• Use waveform capnography continuously and transition immediately to the failed-airway protocol when criteria are met.

Warnings

• Failed airway includes inability to oxygenate at least 90% after an unsuccessful attempt, anatomy unsuitable for further attempts, or three total unsuccessful attempts.

• No more than three total intubation attempts by the most experienced Paramedic.

• Waveform capnography is mandatory after ETT and BIAD placement once available.

• Effective BVM is an acceptable definitive field airway; do not delay oxygenation or ventilation for intubation or repeated laryngoscopy.

Clinical pearls

• DOPE: displacement, obstruction, pneumothorax, and equipment failure.

• Manually stabilize the ETT during every move or transfer.

Related pathways

• TB-05 Head Trauma (p. 188): Open the shared TBI oxygenation, ventilation, herniation-rescue, and perfusion targets.

AR-02
Adult Failed Airway

Revised 2026-10-30

Overview

Recognize failure early, stop repeated laryngoscopy, and return immediately to the most effective oxygenation method.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Declare Failed Airway
EMT / AEMT / Paramedic
Cannot oxygenate/ventilate • unsuitable anatomy • or three total failed attempts.

↓

2. Call for Resources
EMT / AEMT / Paramedic
Bring the most experienced airway provider and rescue devices.

↓

3. BVM + Adjuncts
EMT / AEMT / Paramedic
Optimize position, suction, mask seal, OPA/NPA, PEEP, and two-person technique.

↓

4. Oxygenation Target Met?
EMT / AEMT / Paramedic
SpO_2 at least 92%, or at least 94% for suspected TBI: continue effective BVM. If oxygenation or ventilation is ineffective, place an authorized rescue airway.

↓

5. Rescue Airway
AEMT / Paramedic
AEMT/Paramedic: authorized BIAD. Paramedic: changed video-laryngoscopy rescue or surgical airway only when required.

Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Suspected TBI?
EMT / AEMT / Paramedic
SpO_2 ≥94%; urgently correct <90% or inadequate ventilation. Initial/routine assisted $EtCO_2$ 35-45 mmHg. No routine hyperventilation. Clear active/imminent herniation only: brief $EtCO_2$ 30-35 with capnography, avoid <30, and expedite transport. Open TB-05.

↓

7. Confirm + Transport
EMT / AEMT / Paramedic
Waveform $EtCO_2$, secure device, post-airway care, and early notification.

Indications

- Unable to ventilate and maintain oxygen saturation at least 90% during or after one or more unsuccessful attempts.
- Anatomy inconsistent with continued attempts or three unsuccessful attempts by the most experienced Paramedic.

Treatment

1. Optimize two-person BVM with positioning, suction, OPA/NPA as appropriate, high-flow oxygen, and PEEP when indicated.
2. If BVM maintains SpO_2 at least 92% (at least 94% for suspected TBI) and adequate ventilation, continue it as the definitive field airway and begin transport.
3. Place an authorized BIAD when BVM is ineffective or prolonged ventilation is required; confirm and secure it before movement.
4. Proceed to surgical cricothyrotomy only for a Paramedic-managed cannot-oxygenate/cannot-ventilate patient after less-invasive rescue measures fail or cannot be used.
5. Ventilate approximately 10-12 breaths/min with visible chest rise; use $EtCO_2$ and clinical response to avoid hyperventilation.
6. Suspected TBI oxygenation, ventilation and blood-pressure safeguards: see AR-01 (pp. 143-144).

Provider-level care

Provider-Level Actions Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT: Stop repeated attempts, optimize two-person BVM with airway adjuncts, suction, oxygen, and PEEP when indicated.

- Continue BVM when it maintains oxygen saturation at least 92% (at least 94% for suspected TBI) and adequate ventilation.

AEMT: Perform all EMT care plus authorized BIAD placement; do not perform ETT placement or video laryngoscopy under this protocol.

- Confirm a BIAD with waveform capnography and secure it during all movement.

Paramedic: Perform all prior care and proceed to surgical cricothyrotomy only when less invasive methods cannot oxygenate or ventilate.

- Avoid additional laryngoscopy after three total attempts or when anatomy makes success unlikely.

Warnings

- No more than three total intubation attempts; every repeated attempt must change the provider, approach, positioning, or equipment.
- Do not abandon effective BVM merely to obtain an advanced airway.
- Do not delay transport for repeated advanced-airway attempts once oxygenation and ventilation are effective.

Clinical pearls

- Use DOPE for sudden deterioration after airway placement.

Related pathways

- TB-05 Head Trauma (p. 188): Open the shared TBI oxygenation, ventilation, herniation-rescue, and perfusion targets.

AR-03
Airway, Drug Assisted

Revised 2026-10-30

Overview

Drug-assisted airway (DAI) is an optional, high-risk procedure requiring specific agency authorization, documented competency, performance improvement, and at least two Paramedics on scene.

Correct hypoxia and hypotension before administering a sedative-paralytic sequence whenever possible; resuscitation takes priority over intubation.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. DAI Indicated + Authorized?
Paramedic
Airway protection failure • oxygenation/ventilation failure • impending compromise; not for convenience.

↓

2. Prepare Completely
Paramedic
Two Paramedics • monitor • suction • BVM/PEEP • primary and rescue airway • surgical-airway plan.

↓

3. Preoxygenate + Resuscitate
Paramedic
High-flow oxygen, NIPPV or BVM/PEEP; correct hypoxia and hypotension before paralysis.

↓

4. Induction + Paralysis
Paramedic
Select one induction agent, use the appropriate paralytic, and proceed immediately to airway placement.

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Airway Successful?
Paramedic
Waveform $EtCO_2$, bilateral exam, tube depth, SpO_2, and securement. If not: BVM/BIAD and AR-02.

↓

6. Suspected TBI?
EMT / AEMT / Paramedic
SpO_2 ≥94%; urgently correct <90% or inadequate ventilation. Initial/routine assisted $EtCO_2$ 35-45 mmHg. No routine hyperventilation. Clear active/imminent herniation only: brief $EtCO_2$ 30-35 with capnography, avoid <30, and expedite transport. Open TB-05.

↓

7. Post-Airway Management
Paramedic
Pain-first analgesia/sedation • ventilate 10-12/min • continuous monitoring • transport.

Indications

- Failure to protect the airway, inability to oxygenate, inability to ventilate, or impending airway compromise when less-invasive care is inadequate and DAI is authorized.

Treatment

1. Do not perform DAI solely for convenience, because BVM is effective, or for a short transport when BVM or BIAD safely maintains oxygenation and ventilation.
2. Preoxygenate with high-flow oxygen, NIPPV when appropriate, or two-person BVM with PEEP. Prepare suction, an appropriately sized primary airway device, BIAD, and surgical-airway rescue before medication.
3. Use the AM-05 or AM-07 resuscitation pathway for peri-intubation shock. Treat hypotension before paralysis; use norepinephrine when indicated and open the dedicated Push-Dose Epinephrine reference only for its approved temporary-bridge use.
4. Limit laryngoscopy to three total attempts. Each subsequent attempt must change the provider, approach, positioning, or equipment; move immediately to AR-02 when oxygenation or ventilation is ineffective.

5. Confirm every advanced airway with continuous waveform $EtCO_2$, bilateral chest assessment, SpO_2, tube depth, and securement. Reassess after each movement.
6. After airway placement, treat pain first, then sedation. Ventilate an adult at approximately 10-12 breaths/min with visible chest rise and use $EtCO_2$ and clinical response to avoid hyperventilation.
7. Document indication, authorization, staffing, preoxygenation, medications and doses, oxygenation/hemodynamics before and after medication, attempt count, airway confirmation, ventilatory support, sedation/analgesia, and required quality-improvement form.
8. Suspected TBI oxygenation, ventilation and blood-pressure safeguards: see AR-01 (pp. 143-144).

Medication orders

Ketamine

1-2 mg/kg IV/IO

- Select one induction agent. Ketamine is generally preferred when hypotension or bronchospasm is present.
- If vascular access is unavailable: 4 mg/kg IM, maximum 400 mg; establish IV/IO access and proceed with airway management.
- Pediatric use requires a direct online order from the Claiborne EMS Medical Director or Assistant Medical Director.

Etomidate

0.3 mg/kg IV/IO

- Select one induction agent when hemodynamic response is uncertain.

Succinylcholine

1-2 mg/kg IV for RSI

Rocuronium

1 mg/kg IV/IO when succinylcholine is contraindicated or a longer duration is needed

- Do not routinely repeat before post-intubation analgesia and sedation are established.

Fentanyl

25-50 mcg IV/IO; repeat cautiously as needed, maximum 100 mcg

- Use lower or fractionated doses with shock, older age, or other hemodynamic risk.

Midazolam

2-5 mg IV/IO; repeat cautiously as needed

- Use after analgesia when sedation is needed.

Provider-level care

Provider-Level Actions Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT: Assist with preoxygenation, suction, BVM, equipment preparation, physiologic monitoring, and backup-airway readiness.

- Continue oxygenation and ventilation throughout preparation and between attempts.

AEMT: Perform all EMT care plus authorized vascular access, rescue BIAD support, monitoring, and assistance with the failed-airway plan.

Paramedic: Confirm indications, authorization, two-Paramedic staffing, monitoring, preoxygenation, suction, primary device, rescue BIAD, surgical-airway equipment, and a failed-airway plan before medication.

- Obtain two IV/IO access points when feasible, but do not delay life-saving oxygenation or ventilation for a second access attempt.
- Correct hypoxia and hypotension before paralytic administration, limit laryngoscopy attempts, verify with waveform capnography, and begin pain-first post-intubation care immediately.

Warnings

- This procedure removes protective airway reflexes and spontaneous breathing; do not begin without complete rescue capability.
- At least two Paramedics must be present under the approved protocol.
- DAI during unresolved hypoxia or hypotension substantially increases cardiac-arrest risk.
- No more than three total laryngoscopy attempts. Effective BVM is an acceptable definitive field airway.
- A paralyzed patient may be awake and in pain without visible movement; establish analgesia and sedation promptly after airway placement.
- Complete the required airway evaluation and performance-improvement documentation.

Clinical pearls

- When BVM maintains saturation at least 90%, continued basic airway management may be safer than DAI during a short transport.
- BIAD is preferred while resuscitating severe hypoxia or hypotension.

Related pathways

- AM-05 Hypotension / Shock (p. 129): Open for peri-intubation hypotension or shock.

- AM-07 Crashing Medical Patient (p. 135): Open for immediate physiologic resuscitation when the patient is critically unstable.
- AR-02 Adult Failed Airway (p. 144): Open immediately when oxygenation or ventilation is ineffective or further laryngoscopy is unsafe.
- AR-08 Post-Intubation / BIAD Management (p. 155): Open for continued analgesia, sedation, ventilation, and monitoring after airway placement.
- Push-Dose Epinephrine - Medication Reference (p. 224): Open the approved labeled preparation, dosing, and two-bolus limit.
- TB-05 Head Trauma (p. 188): Open the shared TBI oxygenation, ventilation, herniation-rescue, and perfusion targets.

AR-04
Adult COPD / Asthma

Revised October 17, 2026

Overview

Differentiate lower-airway bronchospasm from stridor and other causes of respiratory distress while supporting oxygenation and ventilation.

Escalate early for silent chest, fatigue, altered mental status, poor air movement, or impending respiratory failure.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Respiratory Distress
EMT / AEMT / Paramedic
Assess work of breathing, speech, air movement, SpO_2, lung sounds, mental status, and fatigue.

↓

2. Wheeze, Stridor, or Another Cause?
EMT / AEMT / Paramedic
Treat anaphylaxis immediately when suspected; assess for CHF, ACS, PE, pneumonia, pneumothorax, obstruction, or foreign body.

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. Oxygen + Bronchodilator
EMT / AEMT / Paramedic
Position upright • target oxygen appropriately • albuterol ± ipratropium.

↓

4. NIPPV Appropriate?
AEMT / Paramedic
AEMT/Paramedic: use early if cooperative and stable; do not use with emesis, inability to protect airway, or shock.

↓

5. Severe / Not Improving?
AEMT / Paramedic
Epinephrine only under the indication- and provider-specific orders below • BVM support • request Paramedic escalation.

↓

6. Paramedic - Severe Bronchospasm
Paramedic
Methylprednisolone • magnesium when indicated • advanced-airway preparation; follow doses and safeguards below.

↓

7. Reassess + Transport
EMT / AEMT / Paramedic
Continuous SpO_2; $EtCO_2$ when indicated; reassess after each treatment and notify early.

Indications

- Adult with respiratory distress from suspected asthma, COPD, reactive airway disease, bronchospasm, or stridor.

Treatment

1. Target SpO_2 93-95% for asthma or undifferentiated hypoxemia. For known COPD with hypercapnia risk, usual lower saturation, or an individualized oxygen plan, target 88-92%; do not withhold oxygen from a critically hypoxemic patient.
2. Give albuterol promptly; add ipratropium to the initial treatment for moderate or severe bronchospasm. Repeat bronchodilator treatment according to clinical response.

3. Use NIPPV early for a cooperative patient with significant distress, fatigue, or hypercapnic COPD physiology. Stop NIPPV and move to BVM/airway-resuscitation management for hypotension, worsening mental status, emesis, intolerance, or deteriorating ventilation.
4. For severe asthma or COPD exacerbation, give methylprednisolone early; it is adjunctive and must not delay bronchodilator treatment.
5. Use magnesium sulfate for severe asthma or bronchospasm not improving after initial bronchodilator therapy; monitor for hypotension.
6. Use IM epinephrine for anaphylaxis, severe upper-airway edema, or life-threatening asthma with poor air movement or impending respiratory failure. Paramedic only: use nebulized epinephrine for stridor or upper-airway edema, not routine lower-airway COPD/asthma treatment.
7. Reassess respiratory effort, ability to speak, air movement, SpO_2, $EtCO_2$ when used, heart rate, and blood pressure after each intervention. Transport all severe, recurrent, or incompletely responsive patients and notify early for NIPPV, epinephrine, magnesium, or impending airway failure.

Medication orders

Albuterol

2.5-5 mg nebulized; repeat as needed ×3 and continue when clinically indicated

Ipratropium

0.5 mg nebulized with the initial albuterol treatment for moderate or severe bronchospasm

Epinephrine 1 mg/mL (1:1,000)

0.3-0.5 mg IM for anaphylaxis, severe upper-airway edema, or life-threatening asthma

Nebulized epinephrine - Paramedic only

1 mg of 1 mg/mL (1:1,000) in 2 mL normal saline; may repeat once for stridor or upper-airway edema

Methylprednisolone

125 mg IV/IO/IM

Magnesium sulfate

1-2 g IV over 15-30 minutes for severe asthma/bronchospasm not improving after initial bronchodilators

Provider-level care

Provider-Level Actions Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT: Position for comfort, provide oxygen, assist ventilation when tiring, and administer authorized albuterol.

• Use epinephrine immediately for anaphylaxis within EMT scope and request ALS for severe distress or poor response.

• Monitor work of breathing, speech, mental status, air movement, SpO_2, and clinical response after every intervention.

AEMT: Perform all EMT care plus NIPPV for an appropriate cooperative, hemodynamically stable patient; obtain vascular access and give repeated albuterol/ipratropium within scope.

• Use authorized IM epinephrine for anaphylaxis or life-threatening asthma. Request Paramedic care for nebulized epinephrine for stridor/upper-airway edema; the nebulized route is Paramedic-only.

Paramedic: Perform all prior care plus nebulized epinephrine for stridor/upper-airway edema, cardiac and $EtCO_2$ monitoring when indicated, steroid and magnesium therapy, advanced-airway management, and treatment of competing diagnoses.

• Prepare BVM with PEEP and follow the airway pathway promptly for silent chest, exhaustion, declining mental status, or ineffective ventilation.

Warnings

- A silent chest with respiratory distress is a pre-arrest finding.
- Do not persist with NIPPV in a patient with hypotension, worsening mental status, emesis, inability to protect the airway, intolerance, or deteriorating ventilation.
- Do not delay assisted ventilation and airway preparation for repeated nebulizer treatments in a tiring patient.
- Medication administration must follow current Claiborne authorization and standing orders.

Clinical pearls

- A patient who can no longer speak, is tiring, or has decreasing air movement may be worsening even when wheezing is quieter.

Related pathways

- AR-01 Adult Airway (p. 143): Open for progressive respiratory failure or advanced-airway support.

- AR-03 Drug-Assisted Airway (p. 146): Open only when DAI is indicated and authorized for impending or actual airway failure.
- AM-01 Allergic Reaction / Anaphylaxis (p. 117): Open for suspected anaphylaxis or airway edema.
- AM-05 Hypotension / Shock (p. 129): Open for persistent hypotension or shock.
- AM-07 Crashing Medical Patient (p. 135): Open for immediate physiologic resuscitation in a critically unstable patient.

AR-05
Pediatric Airway

Revised 2026-10-30

Overview

For infants and children after the newly born period, begin with positioning, suction, basic adjuncts, and effective BVM; BVM is an acceptable definitive field airway when it provides adequate oxygenation and ventilation.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Airway / Breathing Adequate?
EMT / AEMT / Paramedic
Assess pediatric appearance, work of breathing, patency, oxygenation, ventilation, mental status, and perfusion.

↓

2. Basic Maneuvers First
EMT / AEMT / Paramedic
Neutral position • suction • OPA/NPA when appropriate • two-person age-sized BVM ± PEEP.

↓

3. Foreign-Body Obstruction?
EMT / AEMT / Paramedic
Use age-appropriate obstruction procedure; direct visualization only when authorized.

Continue at step 4 in the next column/page

Quick flow continued at step 4

4. Support Still Needed?
EMT / AEMT / Paramedic
Oxygen • BVM • authorized BIAD • paramedic ETT only when clinically necessary.

↓

5. Effective?
EMT / AEMT / Paramedic
If oxygenation or ventilation remains ineffective, move immediately to pediatric failed-airway rescue.

↓

6. Suspected TBI?
EMT / AEMT / Paramedic
SpO_2 ≥94%; urgently correct <90% or inadequate ventilation. Initial/routine assisted $EtCO_2$ 35-45 mmHg. No routine hyperventilation. Clear active/imminent herniation only: brief $EtCO_2$ 30-35 with capnography, avoid <30, and expedite transport. Open TB-05.

↓

7. Confirm + Reassess
EMT / AEMT / Paramedic
Waveform $EtCO_2$ after ETT/BIAD, secure device, monitor, maintain warmth, and transport.

Indications

- Pediatric patient with actual or threatened airway obstruction, inadequate oxygenation, inadequate ventilation, or inability to protect the airway.

Treatment

1. Use neutral airway positioning. For infants, consider a small shoulder roll to avoid flexion. Suction early and use age-appropriate OPA/NPA when clinically appropriate.
2. Use the length-based pediatric system as the primary source for equipment, medication, and tube sizing. Estimated ETT depth is approximately 3 × tube diameter; estimated uncuffed tube size is (16 + age in years) ÷ 4 only as a secondary estimate and not for infants.
3. For inadequate breathing with a pulse, ventilate 1 breath every 2-3 seconds (20-30/min) with visible chest rise. Use clinical response and $EtCO_2$ when available to avoid hyperventilation.

4. For severe foreign-body obstruction: infants-repeat 5 back blows and 5 chest thrusts; children-repeat 5 back blows and 5 abdominal thrusts. Do not perform blind finger sweeps.
5. Escalate early for apnea, exhaustion, altered mental status, worsening cyanosis, inadequate chest rise, or persistent hypoxemia despite optimized BVM.
6. Limit laryngoscopy to three total attempts. Each further attempt must change provider, approach, positioning, or equipment; move immediately to AR-06 when oxygenation or ventilation is ineffective.
7. Confirm ETT or BIAD with continuous waveform $EtCO_2$, chest examination, tube depth, SpO_2, and securement. Recheck after every move.
8. If heart rate remains below 60/min with poor perfusion despite effective oxygenation and ventilation, start CPR and follow PC-02 Pediatric Bradycardia / Poor Perfusion.
9. Pediatric cricothyrotomy is not routine. Consider it only for persistent cannot-oxygenate/cannot-ventilate failure when age-appropriate, specifically authorized, and within provider training and equipment capability.
10. Suspected TBI oxygenation, ventilation and blood-pressure safeguards: see AR-01 (pp. 143-144).

Provider-level care

Provider-Level Actions Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT: Position, suction, use basic adjuncts, and provide effective age-appropriate BVM ventilation; avoid unnecessary advanced-airway attempts.

• Maintain warmth and use the pediatric length-based resuscitation system for equipment sizing and dosing support.

• Continue effective BVM when it provides adequate oxygenation and ventilation; do not delay transport solely to place an advanced airway.

AEMT: Perform all EMT care plus authorized pediatric BIAD, vascular access, and continuous waveform capnography. Endotracheal intubation is not included as an AEMT procedure in this protocol.

• After a failed attempt, change approach/equipment and move promptly toward a rescue airway.

Paramedic: Perform all prior care plus ETT when clinically necessary and a skilled attempt is likely to succeed, chest decompression when indicated, and age-appropriate cricothyrotomy as an absolute last resort.

• Pediatric DAI requires a direct online Claiborne EMS Medical Director or Assistant Medical Director order.

• Secure the tube carefully and manually stabilize it during every movement.

Warnings

• Effective BVM ventilation is often safer than a difficult pediatric intubation.

• No more than three total intubation attempts; every further attempt must change the provider, approach, positioning, or equipment.

• Pediatric ketamine/DAI requires a direct online Claiborne EMS Medical Director or Assistant Medical Director order.

• Pediatric cricothyrotomy is a rare, age-appropriate, specifically authorized last-resort procedure for persistent cannot-oxygenate/cannot-ventilate failure.

Clinical pearls

• Respiratory failure is a common cause of pediatric cardiac arrest; support oxygenation and ventilation promptly.

• An improving SpO_2 alone does not prove adequate ventilation-reassess chest rise, mental status, perfusion, and waveform $EtCO_2$ when available.

Related pathways

• AO-02 Newly Born (p. 138): Open for neonatal resuscitation immediately after birth.

• AR-06 Pediatric Failed Airway (p. 152): Open immediately when oxygenation or ventilation is ineffective or further laryngoscopy is unsafe.

• AR-08 Post-Intubation / BIAD Management (p. 155): Open for airway confirmation, ventilation, monitoring, and continued post-airway management.

• PC-02 Pediatric Bradycardia / Poor Perfusion (p. 164): Open for HR below 60/min with poor perfusion despite oxygenation and ventilation.

• TB-05 Head Trauma (p. 188): Open the shared TBI oxygenation, ventilation, herniation-rescue, and perfusion targets.

AR-06
Pediatric Failed Airway

Revised 2026-10-30

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Declare Failed Airway
EMT / AEMT / Paramedic
Cannot oxygenate/ventilate • unsafe anatomy • deterioration • or three failed laryngoscopy attempts.

↓

2. Call + Optimize BVM
EMT / AEMT / Paramedic
Bring most experienced provider/rescue equipment; position • suction • adjuncts • two-person seal • oxygen • PEEP.

↓

3. BVM Effective?
EMT / AEMT / Paramedic
Judge chest rise, SpO_2 trend, mental status/perfusion, and $EtCO_2$ when available.

↓

4. Rescue Airway Needed?
AEMT / Paramedic
If BVM ineffective or prolonged ventilation required: authorized age/size-appropriate BIAD or deliberate paramedic rescue.

↓

5. Cannot Oxygenate / Ventilate?
Paramedic
Paramedic considers age-appropriate, specifically authorized surgical rescue only after less-invasive rescue fails or cannot be used.

Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Suspected TBI?
EMT / AEMT / Paramedic
SpO_2 ≥94%; urgently correct <90% or inadequate ventilation. Initial/routine assisted $EtCO_2$ 35-45 mmHg. No routine hyperventilation. Clear active/imminent herniation only: brief $EtCO_2$ 30-35 with capnography, avoid <30, and expedite transport. Open TB-05.

↓

7. Confirm + Transport
EMT / AEMT / Paramedic
Waveform $EtCO_2$ • secure device • maintain warmth • post-airway care • immediate notification.

Indications

- Unable to oxygenate or ventilate, anatomy unsafe for further attempts, or three total unsuccessful laryngoscopy attempts.
- Pediatric deterioration during or after an unsuccessful advanced-airway attempt when continued laryngoscopy could worsen hypoxemia or interrupt ventilation.

Treatment

1. Declare failure early. Do not wait for three attempts when BVM is ineffective, oxygenation/ventilation is deteriorating, or anatomy makes further attempts unsafe.
2. Stop laryngoscopy, bring the most experienced available airway provider and rescue equipment, and optimize BVM with neutral positioning, suction, OPA/NPA when appropriate, correct mask size/seal, oxygen, and PEEP when indicated.
3. Judge BVM success by chest rise, SpO_2 trend, mental status/perfusion, and waveform $EtCO_2$ when available-not SpO_2 alone.
4. If BVM is effective, continue it as the definitive field airway and begin transport. Do not pursue an advanced airway solely because BVM is not ideal.
5. Confirm every BIAD/ETT with continuous waveform $EtCO_2$, chest examination, SpO_2, depth, and securement; reassess after every movement.
6. If heart rate remains below 60/min with poor perfusion despite effective oxygenation and ventilation, start CPR and follow PC-02. If pulseless, follow PC-04 Pediatric Pulseless Arrest.

7. Maintain warmth, frequent reassessment, rapid transport, and early notification for any rescue airway, persistent hypoxemia, or respiratory failure.
8. Suspected TBI oxygenation, ventilation and blood-pressure safeguards: see AR-01 (pp. 143-144).

Provider-level care

Provider-Level Actions Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT: Stop further laryngoscopy attempts, call for resources, optimize two-person BVM with positioning, adjuncts, suction, oxygen, and PEEP when indicated.

• Continue BVM as the definitive field airway when oxygenation and ventilation are effective; begin transport without delay.

AEMT: Perform all EMT care plus authorized pediatric BIAD and continuous waveform capnography. Endotracheal intubation and video laryngoscopy are not included as AEMT procedures in this protocol.

• Use an age/size-appropriate BIAD only when BVM is ineffective or prolonged ventilation requires it; do not make repeated blind rescue-device attempts.

Paramedic: Perform all prior care plus an age/size-appropriate BIAD or a changed, deliberate video-laryngoscopy rescue approach only when BVM is ineffective or prolonged ventilation requires it.

• Consider age-appropriate cricothyrotomy only for persistent cannot-oxygenate/cannot-ventilate failure after less-invasive rescue fails or cannot be used, and only when specifically authorized.

Warnings

• Effective BVM ventilation is an acceptable definitive field airway; avoid repeated attempts that interrupt oxygenation or worsen hypoxemia.
• No more than three total laryngoscopy attempts; every further attempt must change the provider, approach, positioning, or equipment.
• Do not use repeated blind BIAD or laryngoscopy attempts.
• Pediatric cricothyrotomy is a rare, age-appropriate, specifically authorized last-resort procedure for persistent cannot-oxygenate/cannot-ventilate failure.

Clinical pearls

• Waveform $EtCO_2$ confirms continuous respiratory gas exchange after an advanced airway and helps identify deterioration promptly.

Related pathways

• AR-05 Pediatric Airway (p. 150): Open for the primary pediatric airway and BVM pathway.
• AR-08 Post-Intubation / BIAD Management (p. 155): Open for airway confirmation, ventilation, monitoring, and continued post-airway management.
• PC-02 Pediatric Bradycardia / Poor Perfusion (p. 164): Open for HR below 60/min with poor perfusion despite oxygenation and ventilation.
• PC-04 Pediatric Pulseless Arrest (p. 167): Open if the child is pulseless.
• TB-05 Head Trauma (p. 188): Open the shared TBI oxygenation, ventilation, herniation-rescue, and perfusion targets.

AR-07
Pediatric Respiratory Distress

Revised October 17, 2026

Overview

Assess pediatric respiratory distress by appearance, work of breathing, air movement, SpO_2, mental status, hydration/perfusion, and fatigue.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Pediatric Distress
EMT / AEMT / Paramedic
Position of comfort • appearance • work • air movement • SpO_2 • mental status • perfusion.

↓

2. Wheeze, Stridor, or Another Cause?
EMT / AEMT / Paramedic
Consider asthma, croup, bronchiolitis, epiglottitis, anaphylaxis, foreign body, or respiratory failure.

↓

3. Initial Support
EMT / AEMT / Paramedic
Least distressing effective oxygen method • suction when needed • pathway-specific bronchodilator/epinephrine.

Continue at step 4 in the next column/page

Quick flow continued at step 4

4. Improving?
EMT / AEMT / Paramedic
Reassess after each treatment; obtain access when severe or persistent.

↓

5. Severe / Impending Failure?
AEMT / Paramedic
PM-01 epinephrine when indicated and within provider scope • BVM support • request Paramedic airway escalation.

↓

6. Paramedic - Severe Pediatric Asthma
Paramedic
Medical-Control-directed corticosteroid; no standing pediatric steroid dose • magnesium when indicated • follow doses and safeguards below.

↓

7. Reassess + Transport
EMT / AEMT / Paramedic
Continuous SpO_2; $EtCO_2$ when indicated; maintain warmth and notify early.

Indications

- Pediatric patient with asthma, reactive airway disease, wheezing, bronchospasm, croup, stridor, bronchiolitis, or other respiratory distress.

Treatment

1. Do not force a mask on an agitated child who is maintaining oxygenation. Use blow-by or the least distressing effective oxygen method; support ventilation immediately when oxygenation or ventilation is inadequate.
2. For wheeze/bronchospasm, give albuterol and reassess after each treatment. Add ipratropium to the initial treatment only for moderate or severe bronchospasm.
3. For croup/stridor, keep the child calm and avoid unnecessary airway manipulation. Paramedic-only nebulized epinephrine is indicated for stridor at rest or moderate/severe upper-airway obstruction. Transport after treatment because symptoms can recur.
4. For suspected epiglottitis or bacterial tracheitis-drooling, tripod position, toxic appearance, or rapidly progressive upper-airway symptoms-do not force the child supine or examine the tongue/airway; prepare for a controlled airway response and notify the receiving facility early.
5. For anaphylaxis or life-threatening asthma, use the approved PM-01 Pediatric Allergic Reaction epinephrine pathway; do not use a conflicting separate epinephrine dose in this protocol.
6. For moderate or severe asthma/reactive-airway disease, contact Medical Control for corticosteroid consideration; it is adjunctive and must not delay bronchodilator therapy.
7. Use magnesium sulfate for severe asthma not improving after initial bronchodilator therapy; monitor blood pressure.
8. If heart rate remains below 60/min with poor perfusion despite effective oxygenation and ventilation, start CPR and follow PC-02.

Medication orders

Albuterol

Under 20 kg: 2.5 mg nebulized; 20 kg or greater: 5 mg nebulized for bronchospasm

Ipratropium

Under 20 kg: 0.25 mg nebulized; 20 kg or greater: 0.5 mg nebulized, with initial albuterol for moderate or severe bronchospasm

Nebulized epinephrine - Paramedic only

0.5 mL/kg of 1 mg/mL (1:1,000) epinephrine, maximum 5 mL (5 mg), nebulized for croup/stridor

Epinephrine 1 mg/mL (1:1,000)

Use the approved PM-01 Pediatric Allergic Reaction pathway for anaphylaxis or life-threatening asthma dosing

Methylprednisolone

Medical Control only; no standing pediatric dose, for moderate or severe asthma/reactive-airway disease

Magnesium sulfate

40-50 mg/kg IV once over 15-30 minutes; maximum 2 g, for severe asthma not improving after initial bronchodilators

Provider-level care

Provider-Level Actions Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT: Allow position of comfort, minimize agitation, provide oxygen by the least distressing effective method, assist ventilation when tiring, and administer authorized albuterol.

• Treat anaphylaxis immediately through PM-01 and avoid airway manipulation when epiglottitis is suspected.

• For clear bronchiolitis, provide supportive care-suction, oxygen when needed, hydration assessment, and ventilation support for failure-rather than routine bronchodilator or epinephrine treatment.

AEMT: Perform all EMT care plus vascular access, repeated albuterol and weight-appropriate initial ipratropium for moderate or severe bronchospasm, and PM-01-directed IM epinephrine. Request Paramedic care for nebulized epinephrine for croup/stridor; the nebulized route is Paramedic-only.

Paramedic: Perform all prior care plus nebulized epinephrine for croup/stridor, Medical-Control-directed corticosteroid consideration, magnesium for severe asthma, cardiac/$EtCO_2$ monitoring, and BVM/advanced-airway support for impending failure.

• Move promptly to AR-05/AR-06 for silent chest, cyanosis, exhaustion, declining mental status, bradycardia, or ineffective ventilation; do not continue repeated medications in a tiring child.

Warnings

• Do not force a child supine; allow the position that best maintains the airway.

• Drooling, rapid onset, fever, and tripod positioning suggest epiglottitis or bacterial tracheitis; airway manipulation may cause complete obstruction.

• A silent chest is a pre-arrest finding.

• Do not routinely give albuterol or epinephrine for clear bronchiolitis.

• Do not delay airway support for repeated medications in a tiring child.

• Do not perform blind finger sweeps.

Clinical pearls

• A child with decreasing wheeze, worsening fatigue, or declining mental status may be deteriorating rather than improving.

• Stridor is an upper-airway finding; albuterol does not treat croup or fixed upper-airway obstruction.

Related pathways

• PM-01 Pediatric Allergic Reaction (p. 176): Open for anaphylaxis or life-threatening asthma requiring IM epinephrine.

• AR-05 Pediatric Airway (p. 150): Open for BVM and primary pediatric airway support.

• AR-06 Pediatric Failed Airway (p. 152): Open for ineffective oxygenation/ventilation or unsafe further airway attempts.

• PC-02 Pediatric Bradycardia / Poor Perfusion (p. 164): Open for HR below 60/min with poor perfusion despite oxygenation and ventilation.

• AO-02 Newly Born (p. 138): Open for neonatal respiratory support immediately after birth.

AR-08 Post-Intubation / BIAD Management

Revised 2026-10-30

Overview

After every ETT or BIAD placement, continuously confirm airway position, oxygenation, ventilation, securement, analgesia, and sedation.

Treat pain first; paralysis does not provide analgesia or sedation. Newly born patients remain under AO-02.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. ETT / BIAD Placed
EMT / AEMT / Paramedic
Confirm success with continuous waveform $EtCO_2$, bilateral exam, SpO_2, tube depth, and securement.

↓

2. Secure + Ventilate
EMT / AEMT / Paramedic
Manual stabilization during movement • appropriate rate • visible chest rise • $EtCO_2$-guided reassessment.

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. Monitor Continuously
EMT / AEMT / Paramedic
Waveform $EtCO_2$ • SpO_2 • cardiac monitor • tube depth • breath sounds • frequent vital signs.

↓

4. Pain / Anxiety / Movement?
Paramedic
Paramedic: treat pain first, then add sedation when needed. Pediatric medication requires pediatric pathway/online order.

↓

5. Sudden Deterioration?
EMT / AEMT / Paramedic
Evaluate DOPE; disconnect ventilator and manually ventilate if cause is not immediately clear.

↓

6. Suspected TBI?
EMT / AEMT / Paramedic
SpO_2 ≥94%; urgently correct <90% or inadequate ventilation. Initial/routine assisted $EtCO_2$ 35-45 mmHg. No routine hyperventilation. Clear active/imminent herniation only: brief $EtCO_2$ 30-35 with capnography, avoid <30, and expedite transport. Open TB-05.

↓

7. Transport + Reconfirm
EMT / AEMT / Paramedic
Recheck after every move, document airway status/medications, and notify early.

Indications

- Any patient after successful endotracheal intubation or BIAD placement, excluding neonatal resuscitation immediately after birth.

Treatment

1. Ventilate adults at approximately 10-12 breaths/min and pediatric patients at 20-30 breaths/min (1 breath every 2-3 seconds), with visible chest rise. Use $EtCO_2$ and clinical response to avoid hyperventilation.
2. Avoid hypoxemia. After ROSC, target adult SpO_2 90-98% and pediatric SpO_2 94-99% once reliable monitoring is available.
3. For sudden deterioration, use DOPE: displacement, obstruction, pneumothorax, equipment failure. Disconnect from the ventilator and manually ventilate if the cause is not immediately clear.
4. Suction when indicated, elevate the head 10-20 degrees when feasible, and reassess breath sounds, tube depth, and waveform after suctioning or movement.
5. Document airway confirmation, tube depth, waveform $EtCO_2$, oxygenation, ventilation rate/settings, analgesia/sedation, reassessments, and all movement-related rechecks.
6. Suspected TBI oxygenation, ventilation and blood-pressure safeguards: see AR-01 (pp. 143-144).

Medication orders

Ketamine - adult

1 mg/kg IV/IO

- Adult alternative for post-airway sedation when clinically appropriate. Pediatric post-airway medication requires the pediatric system and direct online Medical Control or receiving-facility order.

Fentanyl - adult

25-50 mcg IV/IO; repeat cautiously as needed, maximum 100 mcg

- Treat pain first. Do not stack with morphine.

Morphine - adult alternative

4 mg IV/IO; repeat 2 mg every 5 minutes as needed, maximum 10 mg

- Select one analgesic strategy; do not stack with fentanyl.

Midazolam - adult

2-5 mg IV/IO; repeat cautiously as needed

- Use after analgesia when sedation is needed.

Provider-level care

Provider-Level Actions Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT: Manually stabilize and secure the device, ventilate at the appropriate rate, monitor SpO_2, and reassess after every move.

- Immediately report loss of waveform, resistance, reduced breath sounds, hypoxia, or changing tube depth.

AEMT: Perform all EMT care plus waveform capnography, IV/IO access, authorized ventilation support, suctioning, and assistance with DOPE management.

• This protocol does not grant AEMT authority for post-intubation analgesics, sedatives, or paralytics.

Paramedic: Perform all prior care plus pain-first adult analgesia/sedation, mechanical ventilation, and rescue management for DOPE deterioration.

• For pediatric patients, use the length-based pediatric medication system and direct online Medical Control or receiving-facility orders for post-airway analgesia/sedation; do not apply adult doses.

• Use paralysis only as a last resort after adequate analgesia and sedation. If movement remains unsafe, use paramedic judgment with continuous sedation and Medical Control or an established interfacility order.

Warnings

• Waveform capnography is mandatory after ETT and after BIAD placement once available.

• A paralyzed patient may be awake and in pain without visible movement; maintain analgesia and sedation.

• Loss of waveform or sudden deterioration requires immediate DOPE evaluation; never delay manual ventilation for ventilator troubleshooting.

• Propofol may not be initiated or bolused by field Paramedics; an already ordered interfacility infusion may continue when appropriate.

• Do not apply adult post-airway medication doses to pediatric patients.

• This protocol does not authorize routine repeat paralysis. Pediatric post-airway medication requires the pediatric pathway and a direct online Medical Control or receiving-facility order.

Clinical pearls

• Elevate the head 10-20 degrees when possible to reduce aspiration risk.

• For sudden deterioration, disconnect the ventilator, manually ventilate, and evaluate DOPE.

Related pathways

• AR-03 Drug-Assisted Airway (p. 146): Open for the approved DAI preparation, induction, and post-airway sequence.

• AR-05 Pediatric Airway (p. 150): Open for pediatric airway support and the length-based medication system.

• AR-09 Ventilator Emergencies (p. 157): Open for ventilator alarm, equipment failure, or unresolved ventilator deterioration.

• AR-11 Mechanical Ventilation, Adult (p. 160): Open for adult ventilator setup and management.

• TB-05 Head Trauma (p. 188): Open the shared TBI oxygenation, ventilation, herniation-rescue, and perfusion targets.

AR-09
Ventilator Emergencies

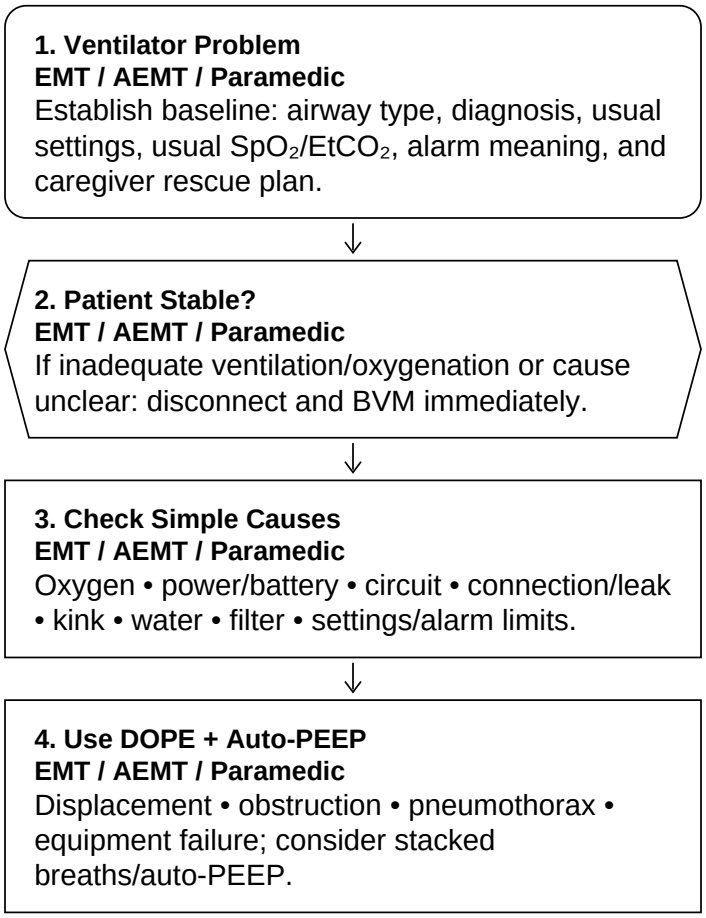
Revised 2026-10-30

Overview

Treat the patient, not the alarm. Use the caregiver and the patient's functioning equipment whenever possible, but never allow ventilator troubleshooting to delay manual ventilation.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.



Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Cause Corrected?
EMT / AEMT / Paramedic
Yes: resume established settings when safe. No: continue BVM and transport.

↓

6. Suspected TBI?
EMT / AEMT / Paramedic
SpO_2 ≥94%; urgently correct <90% or inadequate ventilation. Initial/routine assisted $EtCO_2$ 35-45 mmHg. No routine hyperventilation. Clear active/imminent herniation only: brief $EtCO_2$ 30-35 with capnography, avoid <30, and expedite transport. Open TB-05.

↓

7. Transport
EMT / AEMT / Paramedic
Ventilator • power • battery • oxygen reserve • BVM • suction • spare trach equipment • caregiver when feasible.

Indications

- Ventilator-dependent patient with alarm, equipment failure, hypoxia, abnormal $EtCO_2$, respiratory distress, or suspected airway/circuit problem.

Treatment

1. If the patient is stable, use the functioning ventilator and established settings. If oxygenation/ventilation is inadequate or the cause is not immediately clear, disconnect the ventilator and manually ventilate with BVM.
2. Check oxygen source/pressure, battery or AC power, circuit connection, leaks, kinks, water, filter, settings, and alarm limits.
3. Use DOPE for deterioration: displacement, obstruction, pneumothorax, equipment failure. For asthma/COPD or prolonged exhalation, consider stacked breaths/auto-PEEP; disconnect, manually ventilate slowly, and allow full exhalation before restarting.
4. For high-pressure alarm, assess kinking, biting, secretions/plugging, bronchospasm, pneumothorax, and patient-ventilator dyssynchrony. For low-pressure alarm, assess disconnection, circuit leak, cuff leak, loose tubing, or displaced airway.
5. Suction when secretions or obstruction are suspected; preoxygenate when possible and never force the catheter.
6. During BVM rescue, use age-appropriate rate, visible chest rise, and $EtCO_2$/clinical response; avoid hyperventilation.
7. Maintain continuous SpO_2 and waveform $EtCO_2$ when available. Reassess color, chest rise, breath sounds, airway depth, and perfusion after every intervention.
8. Transport with the ventilator, power cord, backup battery, adequate oxygen reserve, BVM, suction, and patient-specific spare tracheostomy equipment when available. Bring the knowledgeable caregiver and ventilator settings/information when feasible.
9. Suspected TBI oxygenation, ventilation and blood-pressure safeguards: see AR-01 (pp. 143-144).

Provider-level care

Provider-Level Actions Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT: Ask the caregiver for the baseline, check power/oxygen/circuit, assess the tracheostomy or ETT, suction when indicated, and immediately use BVM if correction is not rapid.

- Bring the patient's ventilator to the hospital even when malfunctioning.

AEMT: Perform all EMT care plus advanced suctioning, tracheostomy/BIAD support, capnography, vascular access when indicated, and assistance with ventilator troubleshooting.

Paramedic: Perform all prior care, maintain functioning home settings when appropriate, and manage airway, pneumothorax, bronchospasm, auto-PEEP, or other physiologic cause.

- Transport on the patient's functioning ventilator with current settings when safe; do not make unneeded setting changes during transport.

Warnings

- Never delay manual ventilation for troubleshooting. If the patient improves after disconnection, reassess for ventilator/circuit malfunction, obstruction, or auto-PEEP before reconnecting.
- Do not make unneeded setting changes when the patient is stable on a functioning home ventilator.
- Continuous SpO_2 and waveform $EtCO_2$ are required during assessment and transport once available.

Clinical pearls

- High-pressure and low-pressure alarms suggest different causes; assess the patient and airway before focusing on the machine.
- A ventilator malfunction does not replace the need to assess airway displacement, secretions, pneumothorax, or worsening underlying physiology.

Related pathways

- AR-08 Post-Intubation / BIAD Management (p. 155): Open for recently placed ETT/BIAD confirmation, monitoring, and DOPE response.
- AR-10 Tracheostomy Tube Emergencies (p. 159): Open for tracheostomy obstruction, displacement, or decannulation.
- AR-11 Mechanical Ventilation, Adult (p. 160): Open for adult ventilator setup and management after the emergency is stabilized.
- TB-05 Head Trauma (p. 188): Open the shared TBI oxygenation, ventilation, herniation-rescue, and perfusion targets.

AR-10 Tracheostomy Tube Emergencies

Revised October 17, 2026

Overview

First determine whether the patient has a tracheostomy or total laryngectomy; use caregiver knowledge, emergency information, and the patient's equipment.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Tracheostomy or Laryngectomy?
EMT / AEMT / Paramedic
Ask caregiver and inspect stoma/tube. If uncertain, apply oxygen to both face and stoma.

↓

2. Remove Simple Obstruction
AEMT / Paramedic
Remove speaking valve, cap, HME/filter, and inner cannula if present. Do not remove voice prosthesis.

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. Suction + Reassess
AEMT / Paramedic
Use patient-specific catheter/depth when known. Never force resistance.

↓

4. Catheter Will Not Pass / Tube Displaced?
AEMT / Paramedic
Deflate cuff if present, reassess face/stoma airflow, and remove the tracheostomy tube when obstruction/displacement is established.

↓

5. Ventilate Through Available Airway
EMT / AEMT / Paramedic
Trach: face and stoma as appropriate. Laryngectomy: stoma only with pediatric mask/stoma interface.

↓

6. Confirm + Transport
EMT / AEMT / Paramedic
Waveform $EtCO_2$, SpO_2, secure any rescue tube, bring spare equipment/caregiver, and notify destination.

Indications

- Tracheostomy or laryngectomy patient with distress, cyanosis, altered mental status, copious secretions, obstruction, displacement, decannulation, or bleeding.

Treatment

1. First determine tracheostomy versus total laryngectomy. If uncertain, apply oxygen to both face and stoma. For confirmed laryngectomy, the stoma is the only airway; do not attempt oral/nasal ventilation or intubation.
2. For suspected tracheostomy obstruction, immediately remove external attachments: speaking valve, cap, HME/filter, and inner cannula if present.
3. Then attempt suction with the patient's usual catheter/depth when known. If unknown, use the smallest appropriate catheter and stop at resistance; do not use a fixed depth.
4. If the suction catheter will not pass, treat the tube as obstructed or displaced: deflate the cuff when

present, reassess face and stoma airflow, and remove the tracheostomy tube rather than repeatedly attempting to ventilate through it.

5. After tracheostomy tube removal, provide oxygen to face and stoma. Use oral/nasal BVM if the upper airway is patent; occlude the stoma if necessary to obtain chest rise.
6. For laryngectomy rescue ventilation, use a pediatric mask or appropriate stoma interface over the stoma; confirm chest rise and waveform $EtCO_2$.
7. After any stoma tube/ETT placement, confirm with waveform $EtCO_2$, chest examination, SpO_2, depth, and securement.
8. Significant tracheostomy bleeding is potentially life-threatening: provide oxygen/ventilation, activate rapid transport, and contact Medical Control early.

Provider-level care

Provider-Level Actions Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT: Use caregiver and patient equipment, determine tracheostomy versus laryngectomy, provide oxygen, assist ventilation at the stoma, and prepare suction/replacement equipment.

• If uncertain whether a patient has a laryngectomy, apply oxygen to both face and stoma until anatomy is confirmed.

• A confirmed total laryngectomy disconnects the upper airway from the lungs; oxygenate and ventilate only through the stoma.

AEMT: Perform all EMT care plus remove obstructing valves/caps/HME/inner cannula, perform authorized suctioning, and assist with authorized tracheostomy replacement.

• Do not make repeated attempts to ventilate through a potentially displaced or obstructed tracheostomy tube. An endotracheal tube through the stoma is not an AEMT procedure in this protocol.

Paramedic: Perform all prior care and manage persistent obstruction, displacement, pneumothorax, respiratory failure, or rescue ventilation with advanced airway support.

• For a mature tract when replacement is appropriate, use the patient's spare tube or one size smaller; an appropriately sized cuffed ETT through the stoma is a paramedic rescue option when necessary.

• For a fresh/immature tract, do not blindly replace the tube because of false-passage risk; prioritize oxygenation/ventilation and rapid transport.

Warnings

• Do not force a suction catheter; inability to pass suggests obstruction or displacement.

• Deflate the cuff before removing a cuffed tracheostomy tube.

• Do not attempt oral or nasal ventilation after total laryngectomy; ventilate through the stoma.

• Do not remove a tracheoesophageal voice prosthesis during an emergency.

• Significant tracheostomy bleeding is potentially life-threatening; expedite transport and contact Medical Control.

Related pathways

• AR-01 Adult Airway (p. 143): Open for progressive respiratory failure or rescue airway support.

• AR-08 Post-Intubation / BIAD Management (p. 155): Open after a rescue airway is placed for confirmation, monitoring, and transport care.

• AR-09 Ventilator Emergencies (p. 157): Open for ventilator alarm, equipment failure, or home-ventilator troubleshooting.

AR-11
Mechanical Ventilation, Adult

Revised 2026-10-30

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Advanced Airway Confirmed
Paramedic
Continuous waveform EtCO2 - tube depth - bilateral exam - securement - analgesia and sedation through AR-08.

↓

2. Functioning Home / Transfer Ventilator?
Paramedic
Maintain established settings when safe; verify baseline, reserve oxygen/power, BVM, suction, and caregiver plan.

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. New Controlled Setup
Paramedic
Volume assist-control; tidal volume 6-8 mL/kg IBW. Default to 6 mL/kg for ARDS/restrictive physiology or elevated pressures.

↓

4. Choose Rate by Physiology
Paramedic
Usual adult: 10-12/min. COPD/asthma: 8-10/min, high inspiratory flow, and prolonged exhalation.

↓

5. Oxygenation + Pressure Check
Paramedic
PEEP 5 - FiO2 100% only until reliable oximetry - SpO2 goal 92-98% - assess synchrony, exhalation, and plateau pressure.

↓

6. Suspected TBI?
EMT / AEMT / Paramedic
SpO_2 ≥94%; urgently correct <90% or inadequate ventilation. Initial/routine assisted $EtCO_2$ 35-45 mmHg. No routine hyperventilation. Clear active/imminent herniation only: brief $EtCO_2$ 30-35 with capnography, avoid <30, and expedite transport. Open TB-05.

↓

7. Alarm / Deterioration?
EMT / AEMT / Paramedic
If cause is not immediately clear: disconnect, BVM, evaluate DOPES/DOTT, correct cause, and notify destination.

Indications

- Adult or protocol-authorized patient with advanced airway requiring mechanical ventilation.
- Interfacility or home-ventilator patient whose current settings are known and functioning.

Treatment

1. Initial new controlled setup: volume assist-control, tidal volume 6-8 mL/kg IBW, PEEP 5 cm H2O, and FiO2 100% only until a reliable oxygen-saturation measurement is available. Then use the lowest effective FiO2 and small, reassessed PEEP changes to maintain SpO2 92-98% unless a condition-specific target applies.
2. After return of spontaneous circulation, use the post-resuscitation target SpO2 90-98% and avoid using EtCO2 as a direct substitute for PaCO2. When blood-gas data are available, target PaCO2 35-45 mm Hg unless directed otherwise.
3. For suspected traumatic brain injury, follow the TB-05 targets below rather than an unspecified normal-ventilation goal. In severe metabolic acidosis, do not abruptly remove compensatory minute ventilation; reassess closely and seek Medical Control or receiving-facility direction.
4. For high PIP with high plateau pressure, suspect reduced compliance (lung, chest wall, or abdominal); for high PIP with normal plateau pressure, suspect airway/circuit resistance or auto-PEEP.
5. Suspected TBI oxygenation, ventilation and blood-pressure safeguards: see AR-01 (pp. 143-144).

Provider-level care

Provider-Level Actions Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT: Continuously monitor the airway, SpO_2, patient color, chest movement, and ventilator connection; manually ventilate immediately if failure occurs.

• Report alarms, loss of waveform, increased resistance, or clinical deterioration.

AEMT: Perform all EMT care plus capnography, authorized airway care, suctioning, and assistance with ventilator troubleshooting.

Paramedic: For a new controlled setup, use volume assist-control and calculate tidal volume from Ideal Body Weight (IBW) using height and sex; do not use actual scale weight.

• Set tidal volume at 6-8 mL/kg IBW. Start at 6 mL/kg for ARDS, restrictive physiology, or elevated pressures, and only increase when clinically appropriate while maintaining plateau pressure below 30 cm H2O.

• For usual adult physiology, begin 10-12/min and reassess ventilation from waveform EtCO2 trend, pH/PaCO2 information when available, chest

excursion, synchrony, and clinical response. For COPD/asthma, begin 8-10/min with high inspiratory flow and I:E about 1:4-1:5; ensure expiratory flow returns to baseline.

• Measure plateau pressure only during an inspiratory hold in a passive patient. For plateau pressure above 30 cm H2O, reduce tidal volume in 1 mL/kg IBW steps to no lower than 4 mL/kg IBW while correcting the underlying cause and obtaining additional direction as needed.

Warnings

- Tidal volume is based on Ideal Body Weight (IBW) from height and sex, never actual body weight.
- Do not persist with an alarming ventilator when the cause is unclear; disconnect and manually ventilate. In obstructive lung disease with suspected auto-PEEP, allow full exhalation and ventilate slowly after disconnecting.
- Do not force EtCO2 to a numeric target in severe obstructive lung disease when it worsens air trapping. EtCO2 may not equal PaCO2, especially after ROSC or poor perfusion.
- Do not use this new-setup pathway to overwrite stable home or transfer settings without a clinical reason.

Clinical pearls

- DOPES: displacement, obstruction, pneumothorax/pulmonary problem, equipment failure, stacked breaths/auto-PEEP.
- DOTT response: disconnect or decompress when indicated, oxygen/BVM, tube check, then adjust settings only after the cause is addressed.
- Document height, sex used for IBW, all initial and changed settings, oxygen/EtCO2 trends, pressure values, waveform findings, sedation/analgesia, alarms, troubleshooting, and patient response.

Related pathways

- AR-08 Post-Intubation / BIAD Management (p. 155): Open for analgesia, sedation, continuous airway confirmation, and post-airway reassessment.
- AR-09 Ventilator Emergencies (p. 157): Open for home-ventilator failure, equipment troubleshooting, transport preparation, and DOPES response.
- AC-10 Post-Resuscitation Care (p. 99): Open after ROSC for oxygenation, ventilation, blood-pressure, and destination priorities.
- TB-05 Head Trauma (p. 188): Open the shared TBI oxygenation, ventilation, herniation-rescue, and perfusion targets.

PC-01
Pediatric Asystole / PEA

Revised 2025-09-01

Overview

For infants and children after the newly born period, asystole and PEA are commonly the end result of hypoxia or respiratory failure; ventilation and high-quality CPR are central interventions.

Search aggressively for reversible causes while minimizing interruptions in compressions.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Pulseless Arrest
EMT / AEMT / Paramedic
Confirm apnea/pulselessness, DNR/MOST, and obvious-death criteria; apply monitor/AED and check leads before treating apparent asystole.

↓

2. High-Quality CPR
EMT / AEMT / Paramedic
100-120/min • depth at least ⅓ chest AP diameter • full recoil • pauses under 10 sec • change compressor every 2 min.

↓

3. Oxygenate + Ventilate
EMT / AEMT / Paramedic
BVM first; 15:2 with two rescuers or 30:2 with one rescuer when no advanced airway; avoid hyperventilation.

↓

4. IV/IO + Early Epinephrine
AEMT / Paramedic
Obtain rapid vascular access without delaying CPR; give 0.01 mg/kg IV/IO every 3-5 min.

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Reversible Cause?
EMT / AEMT / Paramedic
Hypoxia • hypovolemia • acidosis • temperature • metabolic/electrolyte cause • tension • toxins • thrombosis.

↓

6. ROSC?
EMT / AEMT / Paramedic
Yes: PC-08 Pediatric Post-Resuscitation. No: continue 2-min cycles, reassessment, and treatment of cause.

Indications

- Pediatric patient in pulseless arrest with asystole or pulseless electrical activity.

Treatment

1. Asystole/PEA is nonshockable. Continue CPR and treat reversible causes; defibrillate only if the rhythm changes to VF/pulseless VT.
2. With an advanced airway, use continuous compressions and ventilate 20-30/min (1 breath every 2-3 seconds). Avoid hyperventilation.
3. Check glucose, temperature, and likely reversible causes during CPR: hypoxia, hypovolemia, acidosis, hypo/hyperkalemia or metabolic disturbance, hypothermia, tension pneumothorax, tamponade, toxins, and thrombosis.
4. Use continuous waveform $EtCO_2$ when available. Use trends to improve CPR; an unexpected sustained increase may indicate ROSC, but do not use a single $EtCO_2$ value alone to make termination decisions.
5. Perform rhythm/pulse checks no more than every 2 minutes and keep each pause under 10 seconds.
6. After ROSC, move immediately to PC-08 Pediatric Post-Resuscitation Care; maintain airway support, oxygenation, ventilation, glucose/temperature management, blood pressure support, and early notification.

Medication orders

Epinephrine 0.1 mg/mL (1:10,000)

0.01 mg/kg IV/IO as early as possible; maximum single dose 1 mg; repeat every 3-5 minutes

Normal saline

10-20 mL/kg IV/IO only when hypovolemia is suspected; reassess perfusion, lungs, and suspected cause after each bolus

Provider-level care

Provider-Level Actions Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT: Begin continuous high-quality CPR, apply AED/monitor, provide oxygenation/ventilation, and use 15:2 with two rescuers or 30:2 with one rescuer when no advanced airway is present.

• Use a pediatric length-based resuscitation system, change compressors every 2 minutes, and limit rhythm/pulse checks to less than 10 seconds.

AEMT: Perform all EMT care plus IV/IO access, weight-based epinephrine, glucose analysis, crystalloid when hypovolemia is suspected, and authorized advanced airway support.

• Do not delay CPR or ventilation for repeated IV attempts; use IO promptly when IV access is not rapidly available.

Paramedic: Perform all prior care plus rhythm confirmation, continuous waveform $EtCO_2$, advanced reversible-cause treatment, and post-ROSC transition.

• Do not interrupt compressions to intubate; use effective BVM or an authorized BIAD when it reduces pauses.

Warnings

• Do not hyperventilate or pause compressions for endotracheal intubation.
• IV/IO access and medications are secondary to CPR and ventilation.
• Do not routinely give calcium or sodium bicarbonate unless a specific indication exists.
• Do not delay transport for repeated airway attempts.

Related pathways

• AR-05 Pediatric Airway (p. 150): Open for pediatric BVM and airway support.
• AR-06 Pediatric Failed Airway (p. 152): Open for ineffective oxygenation/ventilation or unsafe further airway attempts.
• PC-02 Pediatric Bradycardia / Poor Perfusion (p. 164): Open if a pulse returns but bradycardia with poor perfusion persists.
• PC-08 Pediatric Post-Resuscitation Care (p. 173): Open immediately after ROSC.
• AO-02 Newly Born (p. 138): Open for neonatal resuscitation immediately after birth.

PC-02
Pediatric Bradycardia With Poor Perfusion

Revised 2026-10-31

Overview

For infants and children after the newly born period, pediatric bradycardia is usually caused by hypoxia; correct airway, oxygenation, ventilation, and reversible causes before medication.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Bradycardia + Poor Perfusion
EMT / AEMT / Paramedic
HR <60 with hypotension, AMS, shock, weak pulses, or other poor perfusion.

↓

2. Airway / Oxygenation First
EMT / AEMT / Paramedic
Correct respiratory failure, obstruction, hypoxia, apnea/hypoventilation, hypoglycemia, hypothermia, acidosis, toxins, and other reversible causes.

↓

3. HR Still <60 + Poor Perfusion?
EMT / AEMT / Paramedic
After effective oxygenation and ventilation: begin CPR immediately, even with a pulse.

↓

4. Vascular Access
AEMT / Paramedic
Obtain rapid IV/IO without delaying CPR or effective ventilation; request Paramedic epinephrine for bradycardia with a pulse.

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Paramedic - Epinephrine With Pulse
Paramedic
Give 0.01 mg/kg IV/IO every 3-5 minutes while bradycardia with poor perfusion persists; maximum single dose 1 mg. Do not delay ventilation or CPR.

↓

6. Vagal / AV-Block Cause?
Paramedic
Atropine only for increased vagal tone or primary AV block; consider pacing for selected refractory conduction disease.

↓

7. Reassess + Transport
EMT / AEMT / Paramedic
Treat cause, monitor rhythm/perfusion, transition to arrest/ROSC pathway as needed, and notify pediatric-capable destination.

Indications

- Pediatric patient with bradycardia and hypotension, altered mental status, shock, poor perfusion, or impending arrest.

Treatment

1. Treat the child, not the number: bradycardia is an emergency when associated with hypotension, altered mental status, shock, weak pulses, or other poor perfusion.
2. If HR remains below 60/min with poor perfusion despite effective oxygenation and ventilation, start high-quality CPR immediately even if a pulse is still present.
3. Use continuous ECG, SpO_2, blood pressure, and waveform $EtCO_2$ after an advanced airway. Obtain a 12-lead if it does not delay resuscitation.
4. If the child becomes pulseless, transition immediately to PC-01/PC-04 Pediatric Pulseless Arrest. After sustained ROSC, move to PC-08 Pediatric Post-Resuscitation Care.

Medication orders

Epinephrine 0.1 mg/mL (1:10,000) - Paramedic only while pulse present

0.01 mg/kg IV/IO; maximum single dose 1 mg; repeat every 3-5 minutes while bradycardia with poor perfusion persists

Normal saline

10-20 mL/kg IV/IO only when hypovolemia is suspected; reassess perfusion, lungs, and cause after each bolus

Atropine

0.02 mg/kg IV/IO for increased vagal tone or primary AV block; minimum 0.1 mg; maximum single dose 0.5 mg child / 1 mg adolescent; may repeat once; maximum total 1 mg child / 2 mg adolescent

Provider-level care

Provider-Level Actions Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT: Open and support the airway, oxygenate and ventilate effectively, assess glucose when authorized, and begin CPR for persistent HR below 60 with poor perfusion.

- Check pulse and response approximately every 2 minutes and transition immediately to the arrest pathway if pulseless.

AEMT: Perform all EMT care plus IV/IO access, glucose analysis, crystalloid when hypovolemia is suspected, and authorized advanced airway support. Epinephrine for bradycardia with a pulse is Paramedic-only; transition to the AEMT/Paramedic pulseless-arrest epinephrine pathway only if the patient becomes pulseless.

- Do not delay CPR or ventilation for repeated IV attempts; use IO promptly when IV access is not rapidly available.

Paramedic: Perform all prior care plus cardiac monitoring, weight-based epinephrine for bradycardia with a pulse, atropine for increased vagal tone or primary AV block, and transcutaneous pacing for selected refractory conduction disease.

- Consider immediate pacing for complete heart block or sinus-node dysfunction with persistent cardiopulmonary compromise when vascular access is not immediately available.
- If pacing is used in a conscious child, provide age-appropriate analgesia/sedation with Medical Control when time permits; do not delay life-saving pacing in extremis.

Warnings

- Do not delay oxygenation and ventilation for vascular access or medication.
- Atropine is not a treatment for hypoxic bradycardia.
- Atropine may be ineffective or harmful after cardiac transplantation and may cause paradoxical bradycardia.

- Interpret the rate in clinical context; asymptomatic bradycardia requires monitoring and treatment of the cause, not automatic medication.

Clinical pearls

- Bradycardia associated with poor perfusion may be a harbinger of pediatric cardiac arrest; ventilation and timely CPR are central interventions.
- Transcutaneous pacing is for selected refractory conduction disease, not routine hypoxic bradycardia.

Related pathways

- AR-05 Pediatric Airway (p. 150): Open for pediatric BVM and airway support.
- PC-01 Pediatric Asystole / PEA (p. 163): Open for nonshockable pulseless arrest.
- PC-04 Pediatric Pulseless Arrest (p. 167): Open immediately if the child becomes pulseless.
- PC-08 Pediatric Post-Resuscitation Care (p. 173): Open after sustained ROSC.
- AO-02 Newly Born (p. 138): Open for neonatal resuscitation immediately after birth.

PC-03
Pediatric Pulmonary Edema / CHF

Revised 2026-10-26

Overview

For infants and children after the newly born period with suspected acute heart failure, pulmonary edema, or cardiogenic shock. Congenital anatomy and baseline physiology can make usual findings and oxygen targets misleading.

Support airway, breathing, and perfusion; avoid routine fluids, bronchodilators, and vasoactive medications without a specific indication and early Medical Control consultation.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Suspected Cardiac Cause
EMT / AEMT / Paramedic
Assess respiratory distress, perfusion, mental status, cardiac history/surgeries, baseline SpO_2, home medications, fluid restriction, feeding/urine output, fever, and caregiver-reported baseline.

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. Airway / Breathing Support
EMT / AEMT / Paramedic
Use position of comfort, oxygen and ventilation support for distress. Use gentle positive pressure and contact Medical Control early for known complex congenital heart disease.

↓

3. Known Baseline Different?
EMT / AEMT / Paramedic
Do not force a normal SpO_2 in cyanotic or single-ventricle disease; use the child's documented/caregiver-reported baseline and Medical Control guidance.

↓

4. Monitor + Identify Cause
EMT / AEMT / Paramedic
Continuous ECG, SpO_2, frequent BP, respiratory reassessment, and $EtCO_2$ when ventilatory support is used. Consider respiratory, infectious, cardiac, and anaphylactic causes.

↓

5. Volume Overload / Shock?
AEMT / Paramedic
No routine fluid bolus. If hypovolemia is genuinely suspected, use small reassessed aliquots with Medical Control. Vasodilator, analgesia, or vasoactive treatment requires Medical Control.

↓

6. Pediatric-Cardiac Destination
EMT / AEMT / Paramedic
Notify early, bring the caregiver's cardiac emergency plan/records when available, and transport without delaying for a medication trial.

Indications

- Infant or child with known or suspected cardiac disease plus respiratory distress, crackles/rales, poor feeding, cyanosis, edema, orthopnea, shock, hepatomegaly, or signs of poor perfusion.

Treatment

1. Pediatric acute heart failure may reflect congenital heart disease, myocarditis/cardiomyopathy, dysrhythmia, pulmonary hypertension, infection, or another cause; tailor treatment to physiology rather than to wheeze or crackles alone.
2. Avoid routine albuterol or fluid therapy without an identified indication. Excess fluid can worsen pulmonary edema, and usual saturation targets may be inappropriate for cyanotic or single-ventricle disease.
3. If anaphylaxis is clinically suspected, open PM-01 Pediatric Anaphylaxis. If severe bradycardia or pulseless arrest develops, move immediately to the appropriate pediatric cardiac pathway.

Medication orders

Fentanyl

1 mcg/kg IV/IO; maximum single dose 50 mcg only with Medical Control for pain/anxiety after airway and perfusion are stabilized

Nitroglycerin / vasoactive medication

Agent, dose, and indication determined by Medical Control

Provider-level care

Provider-Level Actions Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT: Position for comfort, support airway and ventilation, monitor SpO_2 and frequent vital signs, and obtain a focused cardiac history from the caregiver.

• Do not assume wheeze is asthma or treat it with albuterol unless bronchospasm is strongly supported.

• Identify and communicate the child's usual oxygen saturation, anatomy/surgeries, home medication schedule, and recent feeding/urine output.

AEMT: Perform all EMT care plus IV/IO access when indicated, ECG/$EtCO_2$ monitoring when available, and early Medical Control consultation.

• Do not give a routine fluid bolus. If hypovolemia is genuinely suspected, use cautious small aliquots with reassessment and Medical Control guidance.

Paramedic: Perform all prior care plus cardiac/12-lead monitoring when it does not delay care, careful ventilatory support, and Medical-Control-directed therapy based on the child's anatomy and physiology.

• Use vasodilator, vasoactive, or analgesic therapy only with Medical Control; treat shock or dysrhythmia under the applicable pediatric protocol.

Warnings

• Do not force a normal SpO_2 in cyanotic or single-ventricle disease without Medical Control guidance.

• Do not give routine albuterol, fluid, vasodilator, or vasoactive treatment without an identified indication and Medical Control direction.

• Do not delay transport for a medication trial or repeated airway attempts.

• Pediatric CHF treatment varies with the underlying congenital lesion; obtain Medical Control early.

Related pathways

• AR-05 Pediatric Airway (p. 150): Open for pediatric BVM and airway support.

• AR-07 Pediatric Respiratory Distress (p. 153): Open when respiratory pathology or bronchospasm is suspected.

• PM-01 Pediatric Anaphylaxis (p. 176): Open immediately when anaphylaxis is suspected.

• PC-02 Pediatric Bradycardia / Poor Perfusion (p. 164): Open for bradycardia with poor perfusion.

• PC-04 Pediatric Pulseless Arrest (p. 167): Open immediately if the child becomes pulseless.

PC-04
Pediatric Pulseless Arrest

Revised 2026-08-14

Overview

High-quality CPR and effective ventilation are central interventions because most pediatric arrests are respiratory/asphyxial in origin.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Pediatric Arrest
EMT / AEMT / Paramedic
Confirm pulselessness, age pathway, DNR/MOST, and obvious-death criteria. Use AO-02 through 3 days of age; use adult cardiac arrest at age 16 years or older.

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. High-Quality CPR + BVM
EMT / AEMT / Paramedic
100-120/min • depth at least ⅓ chest AP diameter • full recoil • pauses under 10 sec. Use 30:2 with one rescuer or 15:2 with two rescuers.

↓

3. AED / Monitor
EMT / AEMT / Paramedic
Apply immediately. Use pediatric pads/attenuator when available, but do not delay defibrillation if standard equipment is immediately available.

↓

4. Rhythm-Specific Pathway
EMT / AEMT / Paramedic
VF/pulseless VT: PC-07. Asystole/PEA: PC-01. Obtain IV/IO promptly without delaying CPR or ventilation.

↓

5. Suspected Opioid Cause?
AEMT / Paramedic
Give naloxone while continuing uninterrupted CPR, oxygenation, and ventilation. Naloxone is not a substitute for resuscitation.

↓

6. ROSC?
EMT / AEMT / Paramedic
Move immediately to PC-08 Pediatric Post-Resuscitation Care; maintain airway support and notify the pediatric-capable destination early.

Indications

• Pulseless patient age 3 days through 15 years.

Treatment

1. With an advanced airway, use continuous compressions and ventilate 20-30/min. Avoid excessive ventilation.
2. Perform rhythm and pulse checks approximately every 2 minutes; keep each pause under 10 seconds.
3. Assess glucose, temperature, and reversible causes during resuscitation. Follow PC-01 for asystole/PEA and PC-07 for VF/pulseless VT.
4. Use waveform $EtCO_2$ trends to improve CPR quality and identify possible ROSC; do not use a single $EtCO_2$ value as a termination decision.
5. After sustained ROSC, move immediately to PC-08; continue controlled oxygenation/ventilation, frequent BP and glucose reassessment, and early destination notification.

Medication orders

Naloxone

For suspected opioid-associated arrest, follow UP-04: parenteral naloxone 0.01 mg/kg IV/IO/IM initially; if needed, 0.1 mg/kg, maximum 2 mg per dose and 8 mg cumulative. This is not an intranasal order. Pediatric IN administration requires a product/device-specific Medical Control order pending stocked-product selection. Continue CPR and ventilation without interruption; do not delay resuscitation for naloxone.

Provider-level care

Provider-Level Actions Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT: Begin high-quality CPR with effective BVM ventilation, apply AED/monitor, and use the applicable shockable or nonshockable rhythm pathway.

• Change compressors about every 2 minutes, minimize all interruptions, and do not delay CPR, ventilation, or defibrillation for advanced procedures.

• Use a pediatric length- or weight-based resuscitation system and obtain caregiver history when it does not interfere with resuscitation.

AEMT: Perform all EMT care plus rapid IV/IO access, glucose and temperature assessment, authorized naloxone, and advanced airway support within current standing orders.

• Use IO promptly if IV access is not quickly available. Do not repeat IV attempts while delaying CPR or ventilation.

Paramedic: Perform all prior care plus rhythm-specific manual treatment, continuous waveform $EtCO_2$ after an advanced airway, reversible-cause management, and post-ROSC transition.

• Do not interrupt compressions to intubate; use effective BVM or an authorized BIAD when it reduces pauses.

Warnings

- Do not delay CPR, ventilation, or defibrillation for IV access, medication, or endotracheal intubation.
- Naloxone is only for suspected opioid-associated arrest and does not replace high-quality CPR and ventilation.
- Do not pause CPR after a shock to check a pulse; resume immediately until the next planned rhythm check.
- Do not use a single $EtCO_2$ value alone to make termination decisions.

Related pathways

- AR-05 Pediatric Airway (p. 150): Open for pediatric BVM and airway support.
- AR-06 Pediatric Failed Airway (p. 152): Open for ineffective oxygenation/ventilation or unsafe further airway attempts.
- PC-01 Pediatric Asystole / PEA (p. 163): Open for nonshockable pulseless arrest.
- PC-07 Pediatric VF / Pulseless VT (p. 172): Open for shockable pulseless arrest.
- PC-08 Pediatric Post-Resuscitation Care (p. 173): Open immediately after ROSC.

PC-05 Pediatric Narrow-Complex Tachycardia

Revised 2026-08-14

Overview

Differentiate sinus tachycardia from SVT. Sinus tachycardia is treated by correcting its cause; SVT is typically abrupt, regular, and fixed-rate.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Narrow Tachycardia
EMT / AEMT / Paramedic
Assess perfusion, mental status, BP, respiratory status, onset, P waves, R-R variability, and reversible causes.

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. Cardiopulmonary Compromise?
EMT / AEMT / Paramedic
Hypotension • acutely altered mental status • signs of shock • sudden collapse. If yes, prepare synchronized cardioversion immediately.

↓

3. Synchronized Cardioversion
Paramedic
0.5-1 J/kg; if unsuccessful, increase to 2 J/kg. Confirm sync mode before every shock. Sedate only if immediately available and it does not delay treatment.

↓

4. Stable + Regular SVT?
EMT / AEMT / Paramedic
If regular narrow-complex SVT is likely, attempt vagal maneuvers first. Irregular rhythm or uncertain/wide-complex tachycardia: Medical Control and appropriate alternate pathway.

↓

5. Adenosine
Paramedic
If IV/IO is readily available: rapid push followed by rapid saline flush, then document rhythm response.

↓

6. Reassess + Transport
EMT / AEMT / Paramedic
Treat the cause, monitor continuously, obtain 12-lead when it does not delay care, and notify destination.

Indications

- Pediatric narrow-complex tachycardia (QRS at or below 0.09 seconds) with suspected SVT or serious symptoms.

Treatment

1. Sinus tachycardia usually has visible P waves and variable R-R intervals; likely triggers must be addressed rather than treated with adenosine or cardioversion.

2. SVT is usually abrupt and regular with absent/abnormal P waves; infant rate is often above 220/min and child rate above 180/min.
3. Do not give adenosine for an irregular rhythm, polymorphic rhythm, or uncertain/wide-complex tachycardia. Use PC-06 when the QRS is wide or diagnosis is uncertain.

Medication orders

Adenosine

0.1 mg/kg rapid IV/IO push followed immediately by rapid saline flush; maximum 6 mg. If needed, repeat 0.2 mg/kg; maximum 12 mg. Only for regular narrow-complex SVT.

Midazolam

0.1-0.2 mg/kg IV/IO/IN for cardioversion; maximum single dose 2 mg, maximum total 5 mg

- Use only if immediately available; do not delay synchronized cardioversion. Verify against the current Claiborne pediatric medication reference.

Provider-level care

Provider-Level Actions Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT: Support airway and oxygenation as clinically indicated, obtain vital signs and rhythm strip when available, and identify cardiopulmonary compromise.

- Treat likely causes of sinus tachycardia-fever, pain, dehydration, hypoxia, sepsis, stimulant/toxin exposure, or anemia-within current standing orders.
- Keep the caregiver with the child when possible and prepare for immediate cardioversion if deterioration occurs.

AEMT: Perform all EMT care plus IV/IO access and supportive treatment of the underlying cause within current standing orders.

- Do not delay transport or cardioversion for repeated vascular-access attempts.

Paramedic: Perform all prior care plus synchronized cardioversion for cardiopulmonary compromise and vagal maneuvers/adenosine for stable, regular narrow-complex SVT.

- Use the most proximal practical IV/IO access for adenosine. Document rhythm before and after each intervention; obtain a 12-lead when stable without delaying treatment.
- For refractory stable SVT after vagal maneuvers and adenosine, contact Medical Control. Do not add antiarrhythmics not carried by the agency.

Warnings

- Do not delay synchronized cardioversion for sedation in a child with cardiopulmonary compromise.
- Do not give adenosine for an irregular rhythm, polymorphic rhythm, or uncertain/wide-complex tachycardia.
- If pulseless at any time, begin CPR and move immediately to Pediatric Pulseless Arrest.
- Do not delay treatment for a 12-lead ECG or repeated vascular-access attempts.

Related pathways

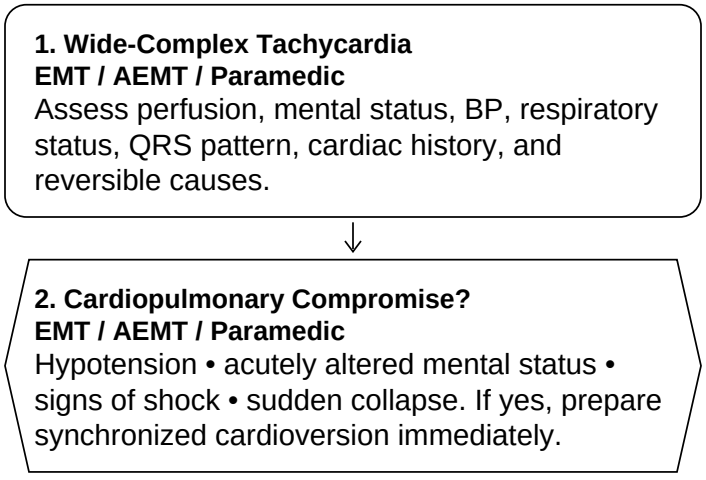
- AR-05 Pediatric Airway (p. 150): Open for airway and oxygenation support.
- PC-04 Pediatric Pulseless Arrest (p. 167): Open immediately if the child becomes pulseless.
- PC-06 Pediatric Wide-Complex Tachycardia (p. 170): Open for wide-complex or uncertain tachycardia.
- Medication Reference (p. 218): Open the current Claiborne medication reference.

PC-06
Pediatric Wide-Complex Tachycardia

Revised 2026-08-14

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.



Continue at step 3 in the next column/page

Quick flow continued at step 3

3. Synchronized Cardioversion
Paramedic
0.5-1 J/kg; if unsuccessful, increase to 2 J/kg. Confirm sync mode before every shock. Sedate only if immediately available and it does not delay treatment.

↓

4. Stable Rhythm Pattern?
EMT / AEMT / Paramedic
Obtain early Medical Control/expert consultation and Paramedic rhythm evaluation. Continue monitoring and transport support.

↓

5. Medication Strategy
Paramedic
For stable regular monomorphic rhythm, adenosine only with Medical Control/expert consultation. All other antiarrhythmic decisions are Medical-Control-directed.

↓

6. Reassess + Transport
EMT / AEMT / Paramedic
Continuous monitoring, rhythm strips, cause-directed treatment, and early pediatric-capable destination notification.

Indications

- Pediatric tachycardia with QRS greater than 0.09 seconds and suspected VT, SVT with aberrancy, or polymorphic tachycardia.

Treatment

1. Treat a wide-complex rhythm as potentially ventricular until proven otherwise. Obtain early Medical Control/expert consultation before antiarrhythmic treatment in a stable child.
2. Assess and treat reversible causes: hypoxia, electrolyte/metabolic abnormality, toxin/stimulant exposure, myocarditis/cardiomyopathy, and congenital heart disease.
3. Use continuous ECG, SpO_2, blood pressure, and rhythm strips before and after interventions. Obtain a 12-lead only when it does not delay treatment.
4. If the child becomes pulseless, begin CPR and move immediately to PC-04. If the rhythm is VF/pulseless VT, use PC-07.

Medication orders

Adenosine

0.1 mg/kg rapid IV/IO push followed immediately by rapid saline flush; maximum 6 mg. If needed, repeat 0.2 mg/kg; maximum 12 mg. Only for stable, regular, monomorphic wide-complex tachycardia with Medical Control/expert consultation.

Midazolam

0.1-0.2 mg/kg IV/IO/IN for cardioversion; maximum single dose 2 mg, maximum total 5 mg

- Use only if immediately available; do not delay synchronized cardioversion. Verify against the current Claiborne pediatric medication reference.

Antiarrhythmic therapy

Agent and dose determined by Medical Control; do not administer amiodarone and procainamide together.

Provider-level care

Provider-Level Actions Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT: Support airway and oxygenation as clinically indicated, identify cardiopulmonary compromise, obtain rhythm strip when available, and prepare for synchronized cardioversion.

- Assess reversible causes including hypoxia, toxin/stimulant exposure, fever, cardiac history, and symptoms of poor perfusion.

AEMT: Perform all EMT care plus IV/IO access and supportive treatment within current standing orders.

- Do not delay cardioversion or transport for repeated vascular-access attempts.

Paramedic: Perform all prior care plus synchronized cardioversion for cardiopulmonary compromise and early Medical Control/expert consultation for stable wide-complex tachycardia.

- Use adenosine only for a stable, regular, monomorphic wide-complex rhythm and only with Medical Control/expert consultation.
- Document rhythm before and after each intervention; obtain a 12-lead when stable without delaying treatment.

Warnings

- Do not delay synchronized cardioversion for sedation, 12-lead acquisition, or repeated IV attempts.

- Do not give adenosine for unstable, irregular, or polymorphic wide-complex tachycardia.
- Do not administer amiodarone and procainamide together.
- If pulseless at any time, begin CPR and move immediately to Pediatric Pulseless Arrest.

Clinical pearls

- Adenosine can be diagnostic or therapeutic only in the stable, regular, monomorphic wide-complex rhythm; expert consultation is part of that decision.
- Stable pediatric wide-complex tachycardia still warrants early specialty-capable destination planning because myocarditis, cardiomyopathy, toxins, and congenital disease may be causes.

Related pathways

- AR-05 Pediatric Airway (p. 150): Open for airway and oxygenation support.
- PC-04 Pediatric Pulseless Arrest (p. 167): Open immediately if the child becomes pulseless.
- PC-05 Pediatric Narrow-Complex Tachycardia (p. 169): Open for narrow-complex tachycardia.
- PC-07 Pediatric VF / Pulseless VT (p. 172): Open for shockable pulseless arrest.
- Medication Reference (p. 218): Open the current Claiborne medication reference.

PC-07
Pediatric VF / Pulseless VT

Revised 2026-08-14

Overview

High-quality CPR and early defibrillation are the central treatments for pediatric VF/pulseless VT.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. VF / Pulseless VT
EMT / AEMT / Paramedic
Begin continuous CPR, effective BVM ventilation, and defibrillator/AED application. Minimize all interruptions.

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. Shock 1: 2 J/kg
AEMT / Paramedic
Defibrillate, then immediately resume CPR for 2 minutes while preparing vascular access.

↓

3. Shock 2: 4 J/kg
AEMT / Paramedic
Defibrillate, resume CPR immediately, obtain IV/IO, and give epinephrine 0.01 mg/kg IV/IO.

↓

4. Refractory Shockable Rhythm
AEMT / Paramedic
Subsequent shocks: at least 4 J/kg, maximum 10 J/kg or adult dose. Give epinephrine every 3-5 minutes and consider one authorized antiarrhythmic.

↓

5. Torsades / Reversible Cause?
Paramedic
Paramedic: give magnesium only for torsades/polymorphic VT or suspected hypomagnesemia; identify and treat reversible causes.

↓

6. ROSC?
EMT / AEMT / Paramedic
Yes: PC-08 Pediatric Post-Resuscitation. No: continue 2-minute CPR, rhythm, shock, medication, and cause-treatment cycles.

Indications

- Pediatric pulseless arrest with ventricular fibrillation or pulseless ventricular tachycardia.

Treatment

1. After an advanced airway is placed, use continuous compressions and ventilate 20-30/min. Use waveform $EtCO_2$ to improve CPR and identify possible ROSC, not as a stand-alone termination decision.
2. Assess reversible causes during CPR: hypoxia, hypovolemia, acidosis, hypo/hyperkalemia or metabolic cause, hypothermia, tension

pneumothorax, tamponade, toxins, and thrombosis.

3. After sustained ROSC, move immediately to PC-08 for controlled oxygenation/ventilation, BP and glucose reassessment, and destination notification.

Medication orders

Epinephrine 0.1 mg/mL (1:10,000)

0.01 mg/kg IV/IO after the second shock or sooner only if rapid defibrillation is not possible; maximum single dose 1 mg; repeat every 3-5 minutes

Amiodarone

5 mg/kg IV/IO for shock-refractory VF/pulseless VT; maximum 300 mg. Select one antiarrhythmic only unless Medical Control directs otherwise.

Lidocaine

1 mg/kg IV/IO for shock-refractory VF/pulseless VT; maximum 100 mg. Select one antiarrhythmic only unless Medical Control directs otherwise.

Magnesium sulfate

Paramedic only: 25-50 mg/kg IV/IO once; maximum 2 g; only for torsades/polymorphic VT or suspected hypomagnesemia

Provider-level care

Provider-Level Actions Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT: Begin high-quality CPR and effective BVM ventilation, apply AED/monitor, minimize pauses, and resume compressions immediately after every shock.

• Use the pediatric length- or weight-based resuscitation system. Do not delay defibrillation for advanced procedures.

AEMT: Perform all EMT care plus IV/IO access, epinephrine, and manual defibrillation only when authorized by current standing orders.

• Use IO promptly if IV access is not quickly available; do not delay CPR or defibrillation for repeated IV attempts.

Paramedic: Perform all prior care plus manual defibrillation, one authorized antiarrhythmic for shock-refractory VF/pulseless VT, magnesium when indicated for torsades/polymorphic VT or suspected hypomagnesemia, $EtCO_2$-guided CPR improvement, and reversible-cause management.

• Charge during compressions, minimize the peri-shock pause, and do not interrupt compressions for endotracheal intubation.

Warnings

• Do not pause CPR after a shock to check a pulse; resume immediately until the next planned rhythm check.

• Do not delay CPR or defibrillation for IV access, medication, or endotracheal intubation.

• Do not routinely administer calcium or sodium bicarbonate unless a specific indication exists.

• Use one antiarrhythmic only unless Medical Control directs otherwise.

Related pathways

• AR-05 Pediatric Airway (p. 150): Open for pediatric BVM and airway support.

• AR-06 Pediatric Failed Airway (p. 152): Open for ineffective oxygenation/ventilation or unsafe further airway attempts.

• PC-04 Pediatric Pulseless Arrest (p. 167): Open for overarching pediatric arrest routing.

• PC-08 Pediatric Post-Resuscitation Care (p. 173): Open immediately after ROSC.

• Medication Reference (p. 218): Open the current Claiborne medication reference.

PC-08
Pediatric Post-Resuscitation

Revised 2026-08-14

Overview

For sustained ROSC in patients age 3 days through 15 years. Prevent recurrent arrest and secondary injury through controlled oxygenation, ventilation, perfusion, glucose, temperature, and rhythm management.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Sustained ROSC
EMT / AEMT / Paramedic
Confirm pulse/perfusion and immediately stabilize airway, breathing, circulation, glucose, temperature, and rhythm.

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. Optimize Oxygenation / Ventilation
EMT / AEMT / Paramedic
Once reliable SpO_2 is available, titrate toward 94-99% unless an individualized baseline applies. Avoid both hypocapnia and hypercapnia.

↓

3. Support Perfusion
AEMT / Paramedic
Maintain systolic and mean arterial pressure above the 10th percentile for age and sex. Avoid routine fluid boluses; use cautious reassessed fluid or Medical-Control-directed vasopressor support when indicated.

↓

4. Temperature + Glucose
EMT / AEMT / Paramedic
Check temperature and glucose. Avoid core temperature above 37.5 °C in the comatose child; correct hypoglycemia within standing orders.

↓

5. Persistent Arrhythmia?
Paramedic
Return to the rhythm-specific pediatric pathway. Continuing antiarrhythmic therapy requires Medical Control.

↓

6. Pediatric-Capable Destination
EMT / AEMT / Paramedic
Notify early, reassess continuously, and transport to the most appropriate pediatric-capable destination.

Indications

- Patient age 3 days through 15 years with sustained return of spontaneous circulation after respiratory or cardiac arrest.

Treatment

1. Target normoxemia appropriate to the child's underlying condition. In most children with reliable pulse oximetry, titrate to SpO_2 94-99%; individualize for cyanotic congenital heart disease or another known baseline exception.
2. Target ventilation appropriate to the child's underlying condition and limit exposure to hypocapnia and hypercapnia. Avoid excessive ventilation.
3. Maintain both systolic and mean arterial blood pressure above the 10th percentile for age and sex; use the pediatric calculator and Medical Control for field targets.
4. Avoid core temperature above 37.5 °C in the comatose child. Do not begin field targeted-temperature management unless specifically directed by Medical Control.

Provider-level care

Provider-Level Actions Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT: Maintain airway and appropriate ventilation, titrate oxygen toward SpO_2 94-99% when reliable unless a known individualized target applies, and monitor vital signs frequently.

- Check glucose when authorized, prevent fever, observe neurologic response, and reassess continuously for recurrent arrest.
- Use AO-02 Newly Born for patients through 3 days of age and the adult post-arrest pathway at age 16 years or older.

AEMT: Perform all EMT care plus IV/IO access, glucose correction, cautious cause-directed fluid support when hypovolemia is suspected, and authorized post-airway analgesia/sedation.

- Do not administer a routine fluid bolus in post-arrest care; reassess perfusion and lungs after any fluid given.

Paramedic: Perform all prior care plus cardiac/12-lead monitoring when it does not delay care, continuous waveform $EtCO_2$ after an advanced airway, rhythm-specific treatment, and Medical-Control-directed vasopressor support.

- Confirm advanced-airway position continuously with waveform capnography. Treat pain first, then provide age-appropriate sedation; paralysis alone is not sedation.
- Select a destination with pediatric ICU, cardiology, neurology, and temperature-management capability when available.

Warnings

- Do not use an adult oxygen target or force a usual saturation target in cyanotic congenital heart disease without individualized guidance.

- Do not administer routine fluid boluses or begin field targeted-temperature management without a specific indication and Medical Control direction.
- A paralyzed or ventilated child may be awake and in pain; treat pain before anxiety and maintain sedation.
- Post-ROSC condition can change rapidly; reassess continuously and return to the arrest pathway immediately if pulseless.

Clinical pearls

- Avoiding post-arrest fever and hypotension are key neuroprotective priorities.
- Pulse oximetry can overestimate oxygenation in patients with darker skin pigmentation; interpret the reading with the overall clinical condition.

Related pathways

- AR-08 Post-Intubation / BIAD Management (p. 155): Open for post-airway ventilation, analgesia, and monitoring.
- PC-01 Pediatric Asystole / PEA (p. 163): Open if nonshockable arrest recurs.
- PC-04 Pediatric Pulseless Arrest (p. 167): Open if the child becomes pulseless again.
- PC-07 Pediatric VF / Pulseless VT (p. 172): Open for recurrent shockable arrest.
- Medication Reference (p. 218): Open the current Claiborne medication reference.

PM-01
Pediatric Anaphylaxis / Allergic Reaction

Revised 2026-10-26

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Suspected Allergic Reaction
EMT / AEMT / Paramedic
Assess airway, breathing, circulation, exposure, onset, skin, GI, and cardiovascular findings. Remove the trigger when safe.

↓

2. Anaphylaxis Suspected?
EMT / AEMT / Paramedic
Airway/breathing compromise, hypotension/poor perfusion, or multi-system symptoms after likely allergen: treat immediately. Do not wait for every criterion when concern is high.

↓

3. IM Epinephrine Now
EMT / AEMT / Paramedic
Epinephrine 1 mg/mL: 0.01 mg/kg IM in the mid-outer thigh; maximum 0.3 mg prepubertal child / 0.5 mg adolescent. Repeat every 5 minutes as needed.

↓

4. Support Airway + Breathing
EMT / AEMT / Paramedic
Oxygen/ventilation as indicated; albuterol only for lower-airway bronchospasm. Prepare early for rapid airway deterioration.

↓

5. Anaphylactic Shock?
AEMT / Paramedic
Normal saline 20 mL/kg IV/IO with reassessment after each bolus. Repeat IM epinephrine and contact Medical Control early for refractory symptoms.

Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Reassess + Transport
EMT / AEMT / Paramedic
Continuous monitoring, repeat epinephrine when indicated, and ED evaluation for every child treated with epinephrine.

Indications

- Pediatric allergic reaction or suspected anaphylaxis after a likely exposure.

Treatment

1. IM epinephrine is the time-critical definitive treatment. Do not delay it for IV/IO access, diphenhydramine, corticosteroids, nebulizers, or transport preparation.
2. Give albuterol only for lower-airway bronchospasm. It does not reverse upper-airway edema, hypotension, or shock.
3. All children treated with epinephrine for anaphylaxis require ED evaluation and monitoring for recurrence, even if symptoms resolve.

Medication orders

Epinephrine 1 mg/mL (1:1,000)

0.01 mg/kg IM in the mid-outer thigh; maximum 0.3 mg prepubertal child / 0.5 mg adolescent; repeat every 5 minutes as needed

Albuterol

Under 20 kg: 2.5 mg nebulized; 20 kg or greater: 5 mg nebulized for lower-airway bronchospasm

- Adjunct only; does not treat upper-airway edema, hypotension, or anaphylaxis itself.

Diphenhydramine

Paramedic: 1 mg/kg PO/IV/IO/IM; maximum 50 mg

- Adjunct for isolated or persistent cutaneous symptoms only after stabilization.

Normal saline

20 mL/kg IV/IO for anaphylactic shock; reassess perfusion and lungs after each bolus

Methylprednisolone

2 mg/kg IV; maximum 125 mg

- Adjunct only; do not delay or replace epinephrine.

Epinephrine infusion

Medical-Control-directed for refractory anaphylaxis after repeated IM epinephrine; do not use a routine IV/IO epinephrine bolus

Provider-level care

Provider-Level Actions Use a length- or weight-based pediatric system and reassess after every intervention.

EMT: Remove the trigger when safe, support airway/ventilation, give IM epinephrine for suspected anaphylaxis, and apply cardiac/SpO_2 monitoring when available.

• For mild isolated skin symptoms, provide supportive care and reassessment. Do not use oral medication for anaphylaxis or an unprotected airway.

AEMT: Perform all EMT care plus IV/IO access, normal saline for anaphylactic shock with reassessment after each bolus, nebulized albuterol for bronchospasm, and authorized repeat IM epinephrine.

• Do not delay epinephrine, airway support, or transport for antihistamine administration.

Paramedic: Perform all prior care plus advanced airway management, Paramedic-only diphenhydramine for isolated or persistent cutaneous symptoms after stabilization, and Medical-Control-directed epinephrine infusion for refractory anaphylaxis after repeated IM doses.

• Give methylprednisolone 2 mg/kg IV (maximum 125 mg) only as an adjunct after epinephrine. Do not use a routine IV/IO epinephrine bolus for refractory anaphylaxis.

Warnings

• Do not delay IM epinephrine for IV access, diphenhydramine, corticosteroids, or albuterol.
• Do not give oral medication to a child with decreased mental status, inability to swallow, or an unprotected airway.
• Do not use intranasal epinephrine as the EMS anaphylaxis treatment pathway.
• Do not use a routine IV/IO epinephrine bolus for refractory anaphylaxis.

Clinical pearls

• A falling blood pressure may be late; poor perfusion and respiratory compromise require action before hypotension develops.
• The lateral thigh is the preferred IM epinephrine site for reliable absorption.

Related pathways

• AR-05 Pediatric Airway (p. 150): Open for pediatric BVM and airway support.
• AR-06 Pediatric Failed Airway (p. 152): Open for rapidly worsening or ineffective airway management.
• PM-03 Pediatric Hypotension / Shock (p. 179): Open for persistent shock after epinephrine and fluid reassessment.
• Medication Reference (p. 218): Open the current Claiborne medication reference.

PM-02
Pediatric Hypoglycemia / Diabetic Emergency

Revised 2026-10-26

Overview

Check blood glucose early in any child with altered mental status, seizure, weakness, or suspected diabetic emergency.

Treat hypoglycemia promptly while protecting the airway. Evaluate hyperglycemia for dehydration, poor perfusion, and possible DKA without treating a glucose number alone.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Check Blood Glucose
EMT / AEMT / Paramedic
Assess airway, mental status, medications/insulin, intake, diabetes history, perfusion, and possible toxin exposure. Confirm unexpected CGM readings with capillary glucose when feasible.

↓

2. Glucose <70 mg/dL?
EMT / AEMT / Paramedic
Yes: treat hypoglycemia. Do not delay treatment of a symptomatic child for CGM confirmation. No: evaluate other causes and hyperglycemia/DKA features.

↓

3. Can Swallow Safely?
EMT / AEMT / Paramedic
Yes: oral glucose/food per local weight-based product. No: D10 IV/IO or IM glucagon.

Continue at step 4 in the next column/page

Quick flow continued at step 4

4. D10 for Pediatric Hypoglycemia
AEMT / Paramedic
D10 2 mL/kg IV/IO (0.2 g/kg); recheck glucose and neurologic status after 5 minutes; repeat once if hypoglycemia persists.

↓

5. Suspected DKA + Dehydration?
AEMT / Paramedic
Treat clinical dehydration/poor perfusion, not hyperglycemia alone: NS 10 mL/kg IV/IO, reassess, and repeat once only if needed; maximum 20 mL/kg before Medical Control.

↓

6. Recheck + Transport
EMT / AEMT / Paramedic
Repeat glucose and neurologic assessment. Monitor for DKA cerebral-edema signs; notify destination early and transport.

Indications

- Pediatric patient with glucose below 70 mg/dL, significant hyperglycemia, or symptoms suggesting a diabetic emergency.

Treatment

1. Treat glucose below 70 mg/dL when symptomatic or when symptom assessment is unreliable. Recheck glucose and neurologic status after every intervention.
2. Hyperglycemia alone does not diagnose DKA. Evaluate dehydration, perfusion, vomiting, abdominal pain, Kussmaul respirations, and mental-status change.
3. Watch for cerebral-edema warning signs: worsening headache, altered mental status, recurrent vomiting, bradycardia, hypertension, or abnormal respirations. Request rapid transport and Medical Control if present.
4. Refusal after hypoglycemia requires normal neurologic status/capacity, glucose at least 80 mg/dL, ability to eat, a reliable caregiver, a clear reversible cause, and no long-acting oral hypoglycemic-agent concern.

Medication orders

Oral glucose

Per local product and weight-based system; give only when the child can swallow and protect the airway

Dextrose 10% (D10)

2 mL/kg IV/IO (0.2 g/kg); recheck glucose and neurologic status after 5 minutes and repeat once for persistent hypoglycemia. Contact Medical Control for additional dextrose dosing or infusion.

Glucagon

<20 kg: 0.5 mg IM; ≥20 kg: 1 mg IM when oral glucose is unsafe and vascular access cannot be obtained promptly

Normal saline

For suspected DKA with dehydration/poor perfusion: 10 mL/kg IV/IO, reassess, and repeat once only if needed; maximum 20 mL/kg before Medical Control

Provider-level care

Provider-Level Actions Use a length- or weight-based pediatric system and reassess after every intervention.

EMT: Check glucose and give oral glucose only if the child is awake, follows commands, and can swallow/protect the airway.

- If oral treatment is unsafe, support airway/ventilation and request ALS promptly. Do not delay treatment of symptomatic hypoglycemia for a confirmatory CGM or capillary reading.
- Use AO-02 Newly Born for immediate newborn resuscitation. After immediate stabilization, use this pathway for the approved pediatric hypoglycemia treatment.

AEMT: Perform all EMT care plus IV/IO access and D10 2 mL/kg (0.2 g/kg); recheck after 5 minutes and repeat once for persistent hypoglycemia.

- Use glucagon when vascular access is unavailable and oral therapy is unsafe. For suspected DKA with dehydration/poor perfusion, give cautious normal saline as directed and reassess.

Paramedic: Perform all prior care plus cardiac monitoring, evaluation for alternate causes, and management of dysrhythmia, seizure, shock, or airway compromise.

- Do not give insulin or sodium bicarbonate in the field for suspected pediatric DKA. Contact Medical Control early for worsening neurologic status, shock, or suspected cerebral edema.

Warnings

- Do not administer oral glucose when swallowing or airway protection is impaired.

- Do not give insulin or sodium bicarbonate in the field for suspected pediatric DKA.
- Do not give large or repeated fluid boluses routinely in suspected pediatric DKA.
- Do not delay treatment of symptomatic hypoglycemia for a confirmatory glucose reading.

Clinical pearls

- Pediatric DKA can deteriorate neurologically; avoid over-resuscitation and communicate serial mental-status findings to the receiving team.

Related pathways

- AO-02 Newly Born (p. 138): Open for immediate newborn resuscitation before transitioning to the approved hypoglycemia pathway.
- AR-05 Pediatric Airway (p. 150): Open when hypoglycemia causes unsafe airway protection or ventilation.
- PM-03 Pediatric Hypotension / Shock (p. 179): Open for persistent poor perfusion or shock.
- Medication Reference (p. 218): Open the current Claiborne medication reference.

PM-03
Pediatric Hypotension / Shock

Revised 2026-10-30

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Recognize Shock
EMT / AEMT / Paramedic
Assess mental status, central/peripheral pulses, capillary refill, skin temperature/color, work of breathing, BP trends, history, and likely cause. Hypotension is a late sign.

↓

2. Airway + Oxygenation
EMT / AEMT / Paramedic
Support ventilation, control external hemorrhage, keep warm, check glucose, and monitor ECG/SpO_2. AEMT/Paramedic: obtain rapid IV/IO without delaying care.

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. Identify Shock Type
EMT / AEMT / Paramedic
Hypovolemic/hemorrhagic • distributive/septic • cardiogenic • obstructive. Treat the likely cause while supporting perfusion.

↓

4. Fluid Strategy
AEMT / Paramedic
Nonhemorrhagic hypovolemic/distributive: isotonic crystalloid 10-20 mL/kg aliquots with reassessment after every bolus. Cardiogenic: 5-10 mL/kg cautiously with early Medical Control. Suspected traumatic hemorrhage: control bleeding, expedite transport, and obtain Medical Control direction; do not apply the medical/septic bolus target indiscriminately.

↓

5. Fluid-Refractory / Obstructive?
Paramedic
For septic shock unresponsive to fluid, begin Medical-Control-directed epinephrine or norepinephrine infusion. Decompress tension pneumothorax and treat other obstructive causes when indicated.

↓

6. Rapid Transport
EMT / AEMT / Paramedic
Reassess perfusion, lungs, mental status, glucose, and BP after every intervention; notify a pediatric-capable destination early.

Indications

- Pediatric patient with poor perfusion, altered mental status, weak pulses, delayed capillary refill, cool/mottled skin, or age-defined hypotension.

Treatment

1. Do not use a single shock-index shortcut as the sole determinant of pediatric shock. Use age-adjusted vital signs, perfusion findings, and clinical trends.
2. Give isotonic crystalloid in 10- or 20-mL/kg aliquots for nonhemorrhagic hypovolemic/distributive shock and reassess after every bolus for fluid responsiveness and volume overload.

3. For suspected traumatic hemorrhage, control bleeding, prevent hypothermia, and expedite pediatric-capable trauma care. Use a separately approved pediatric traumatic-hemorrhage order or online Medical Control for fluid selection, aliquot, target, and reassessment limits; do not automatically apply medical/septic-shock boluses or cumulative targets. Blood products require an authorized, supported transfusion program.
4. For suspected TBI, prevent hypotension and do not use permissive hypotension. The generic hypotension recognition thresholds below are not TBI perfusion targets; obtain Medical Control direction until the pediatric TBI numeric target table and age boundaries are approved.
5. Treat the cause: control hemorrhage, give epinephrine for anaphylaxis within the indicated provider scope, obtain Paramedic-only decompression for tension pneumothorax when indicated, and use the appropriate cardiac or airway pathway.

Medication orders

Isotonic crystalloid-nonhemorrhagic hypovolemic/distributive shock

10-20 mL/kg IV/IO aliquots; reassess perfusion, lungs, hepatomegaly, and work of breathing after every bolus. Further fluid is individualized with Medical Control.

Isotonic crystalloid-cardiogenic shock

5-10 mL/kg IV/IO cautiously with early Medical Control; reassess after each aliquot

Epinephrine or norepinephrine infusion

Medical-Control-directed for fluid-refractory septic shock; verify pediatric formulation and dose in the current medication reference

Emergency corticosteroid

For known/suspected adrenal crisis: use the patient's prescribed emergency plan or
Medical-Control-directed protocol equivalent

Provider-level care

Provider-Level Actions Use a length- or weight-based pediatric system and reassess after every intervention.

EMT: Support airway/ventilation, control external hemorrhage, keep the child warm, assess glucose, apply monitoring when available, and expedite transport.

• Identify early compensated-shock findings: tachycardia, weak pulses, delayed refill, cool/mottled skin, altered behavior, reduced urine output/poor intake, and worsening work of breathing.

AEMT: Perform all EMT care plus rapid IV/IO access and shock-type-specific isotonic-crystalloid aliquots with reassessment after every dose. The medical/septic aliquot instructions do not establish a pediatric traumatic-hemorrhage order; obtain Medical Control direction for that branch.

• Do not delay treatment for repeated IV attempts; use IO promptly when vascular access is not quickly available in critical shock.

Paramedic: Perform all prior care plus cardiac monitoring, needle decompression for tension pneumothorax, and Medical-Control-directed vasoactive infusion for fluid-refractory septic shock.

• Use epinephrine or norepinephrine as the initial vasoactive infusion for fluid-refractory septic shock when authorized. For suspected adrenal crisis, give the patient's prescribed emergency steroid or Medical-Control-directed equivalent.

Warnings

- Do not use medical/septic-shock fluid targets indiscriminately for traumatic hemorrhage. Do not substitute generic pediatric hypotension recognition thresholds for TBI-specific perfusion targets.
- Hypotension is late and ominous in pediatric shock: neonate SBP <60, 1 month-<1 year <70, age 1-9 <70 + (2 × age), age ≥10 <90 mmHg.
- Avoid routine 20-mL/kg boluses in suspected cardiogenic shock; pulmonary edema may worsen rapidly.
- Reassess after every fluid bolus for fluid responsiveness and signs of volume overload.
- Do not use routine corticosteroids for septic shock unless adrenal crisis is suspected or Medical Control directs otherwise.

Related pathways

- TB-06 Multiple Trauma (p. 190): Open for hemorrhage control, trauma priorities, and destination guidance; pediatric traumatic-hemorrhage fluid parameters still require Medical Control.
- TB-05 Head Trauma (p. 188): Open for TBI oxygenation and ventilation safeguards; pediatric TBI blood-pressure targets are Medical Control-directed.
- AR-05 Pediatric Airway (p. 150): Open for airway and ventilation support.
- PC-03 Pediatric Pulmonary Edema / CHF (p. 166): Open for suspected cardiogenic shock or pulmonary edema.
- PM-01 Pediatric Anaphylaxis (p. 176): Open for anaphylactic shock.

• Medication Reference (p. 218): Open the current Claiborne medication reference.

PM-04
Pediatric Pain Control

Revised 2026-10-08

Overview

PM-04 is the pediatric pain pathway for patients age 2 through younger than 16 years. Use actual or length-based weight and treat the cause of pain without delaying time-critical care or transport.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Assess Pain + Cause
EMT / AEMT / Paramedic
Confirm age and actual or length-based weight. Assess pain, airway, breathing, perfusion, mental status, allergies, prior analgesics, and immediate threats.

↓

2. Treat Immediate Threats
EMT / AEMT / Paramedic
Provide airway care, hemorrhage control, wound care, splinting, positioning, ice or heat when appropriate, and reassurance. Do not delay transport.

↓

3. Oral Route Safe?
EMT / AEMT / Paramedic
If alert and able to protect the airway and swallow: acetaminophen OR ibuprofen using the approved weight-based dose and maximum.

↓

4. Severe Pain / Oral Route Unsafe or Inadequate?
Paramedic
Paramedic: fentanyl 1 mcg/kg slow IV/IO once, maximum 50 mcg. A second fentanyl dose requires Medical Control authorization.

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Reassess + Transport
EMT / AEMT / Paramedic
Reassess pain, airway, ventilation, perfusion, mental status, vital signs, and medication response; document weight, dose, route, time, and response.

Indications

• Acute pain in a patient age 2 through younger than 16 years when analgesia can be provided without delaying airway management, hemorrhage control, shock treatment, splinting, or transport.

Treatment

1. Use the child's actual weight when available; otherwise use an approved length-based weight estimate. Document the weight source.
2. Address the cause of pain with appropriate positioning, splinting, wound care, ice or heat when indicated, calm reassurance, and rapid transport for time-sensitive illness or injury.
3. Reassess pain, airway, respiratory rate and effort, oxygenation, perfusion, mental status, and vital signs after every intervention and at handoff.

Medication orders

Acetaminophen

EMT/AEMT/Paramedic: 15 mg/kg PO once; maximum 650 mg.

• Use only when the child is alert and can swallow/protect the airway.
• Do not use with known hypersensitivity, severe liver disease or liver failure, suspected acetaminophen overdose, a recent maximum dose, or uncertain prior dose/time.

Ibuprofen

EMT/AEMT/Paramedic: 10 mg/kg PO once; maximum 400 mg.

• Use only when the child is alert and can swallow/protect the airway.
• Do not use with NSAID hypersensitivity, renal disease, significant dehydration, active gastrointestinal bleeding, or recent NSAID administration.

Fentanyl

Paramedic: 1 mcg/kg slow IV/IO once; maximum 50 mcg.

- Medical Control authorization is required before any second fentanyl dose; a second dose is not a standing order.
- Monitor airway, ventilation, oxygenation, mental status, and blood pressure; be prepared to support ventilation.

Provider-level care

Provider-Level Actions Use a length- or weight-based pediatric system and reassess after every intervention.

EMT: Confirm age and actual or length-based weight; treat immediate threats, provide nonpharmacologic pain care, and obtain a developmentally appropriate pain assessment before and after intervention.

- When the child is alert, can swallow safely, and has no medication-specific contraindication, administer one oral nonopioid: acetaminophen or ibuprofen using the approved weight-based dose and maximum. Do not routinely give both.
- Do not give oral medication when airway protection or swallowing is impaired, active vomiting creates an aspiration risk, or immediate transport/intervention takes priority.

AEMT: Perform all EMT care. Establish IV/IO access only when clinically indicated; do not delay transport or make repeated access attempts solely to provide analgesia.

- No additional AEMT standing-order analgesic is authorized by PM-04. Request Paramedic support or contact Medical Control when nonopioid care is inadequate or the oral route is unsafe.

Paramedic: Perform all prior care. For severe pain or when the oral route is unsafe or inadequate, administer fentanyl 1 mcg/kg slow IV/IO once; maximum 50 mcg.

- Monitor airway, ventilation, oxygenation, mental status, and blood pressure after fentanyl; keep suction and ventilation equipment immediately available and reassess before and during transport.

Warnings

- Do not use UP-11 Adult Pain Control for patients younger than 16 years; use PM-04.
- Do not delay airway management, hemorrhage control, shock treatment, splinting, or time-sensitive transport to complete analgesia.
- Do not administer oral medication when swallowing or airway protection is impaired, active vomiting creates an aspiration risk, or prior dose/time is uncertain.
- Do not give a second fentanyl dose without Medical Control authorization.
- PM-04 does not authorize ketorolac, morphine, hydromorphone, or ketamine as standing-order pediatric analgesics.

Clinical pearls

- Pain assessment can use self-report when reliable or a developmentally appropriate observational scale. Reassess the same measure after treatment.
- Meaningful improvement in comfort and function is the goal; complete elimination of pain is not required before transport.

Related pathways

- UP-11 Adult Pain Control (p. 31): Open only for patients age 16 years or older.
- Medication Reference (p. 218): Open the current Claiborne medication reference.

TB-01
Blast Injury / Incident

Revised 2026-10-30

Overview

Treat every blast as a potentially unsafe, intentional event until authorities determine otherwise.

Expect combined primary, secondary, and tertiary blast injury, including barotrauma, penetrating trauma, burns, crush injury, and toxic exposure.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Scene Safe?
EMT / AEMT / Paramedic
Stage for hazards, structural collapse, fire, hazardous material, or secondary device; activate the agency MCI plan when indicated.

↓

2. Triage + Rapid Removal
EMT / AEMT / Paramedic
Quantify patients, request resources, apply the local MCI/triage process when indicated, and begin load-and-go care.

↓

3. Identify Injury Pattern
EMT / AEMT / Paramedic
Airway burn • blast lung • hemorrhage • penetrating trauma • crush; consider radiation only when the mechanism suggests a radiological or nuclear event.

↓

4. Treat Immediate Threats
EMT / AEMT / Paramedic
Airway and ventilation • hemorrhage control • assess chest injury • hypothermia prevention • reassessed shock care. Urgently request ALS for suspected tension pneumothorax; decompression is Paramedic-only. Ventilate when clinically necessary.

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Blast Lung Injury?
EMT / AEMT / Paramedic
Dyspnea, hypoxia, hemoptysis, wheeze, chest pain, or diminished breath sounds; perform serial respiratory assessment, SpO_2, lung sounds, and monitoring when available.

↓

6. Rapid Trauma Transport
EMT / AEMT / Paramedic
Use trauma/burn destination plan and notify early, especially for respiratory compromise, penetrating injury, major burn, crush injury, or multisystem trauma.

Indications

- Patient with suspected blast injury / incident requiring prehospital assessment or treatment.

Provider-level care

Provider-Level Actions Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT: Maintain scene awareness, perform triage, control major hemorrhage, assess chest injury, oxygenation, ventilation, and perfusion, urgently request ALS for suspected tension pneumothorax, support ventilation, and prevent hypothermia.

- For an open chest wound in a spontaneously breathing patient, consider an approved vented seal and reassess frequently. If deterioration follows sealing, assess/release an obstructing seal and obtain immediate Paramedic treatment for suspected tension pneumothorax. Do not probe the wound. Positive-pressure ventilation requires the chest-injury standing orders and immediate Paramedic/Medical Control guidance, not automatic extrapolation of this seal instruction.
- Assess for hearing loss, ocular injury, burns, penetrating injury, abdominal injury, amputation, and crush syndrome.

AEMT: Perform all EMT care plus IV access and reassessed crystalloid only when indicated for shock after hemorrhage control, without delaying transport.

- Use authorized airway adjuncts and monitor closely for worsening blast-lung physiology.

Paramedic: Perform all prior care plus advanced airway management, cardiac/$EtCO_2$ monitoring, and chest decompression when indicated.

• When positive-pressure ventilation is clinically necessary for suspected blast lung injury, use the lowest effective pressures/volumes and monitor closely for deterioration.

Warnings

- Do not approach a patient who may be attached to wires, a package, or a triggering mechanism; withdraw and notify law enforcement.
- Do not withhold needed ventilation. Avoid unnecessarily high positive pressures/volumes and excessive fluids in suspected blast lung injury; if air transport is used, notify the flight crew of the concern.

Clinical pearls

- Tympanic membrane rupture and hearing loss are common and may impair communication.
- Secondary blast injuries from debris and shrapnel are the most common cause of death.

TB-02
Chemical and Electrical Burn

Revised 2026-10-30

Overview

Stop the exposure while protecting responders, then identify chemical, electrical, lightning, or associated traumatic injury.

Chemical and electrical burns may cause severe internal injury despite limited external findings.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Scene / PPE
EMT / AEMT / Paramedic
Identify agent or electrical source; do not enter until hazards are controlled.

↓

2. Stop Exposure
EMT / AEMT / Paramedic
Disconnect power when safe • remove contaminated clothing and jewelry • brush off dry chemical.

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. Decontaminate
EMT / AEMT / Paramedic
Brush away dry particulate before irrigation. For most chemical skin/eye exposures, begin prompt copious irrigation without delaying ordinary eye irrigation for product identification. Known water-reactive contamination requires immediate HazMat/Poison Help direction; open TE-09 for agent-specific exceptions. Protect unaffected skin and responders.

↓

4. Airway or Trauma Threat?
EMT / AEMT / Paramedic
Treat inhalation injury, dysrhythmia, fall, blast, or multisystem trauma.

↓

5. Monitor + Treat
AEMT / Paramedic
For high-voltage/lightning exposure, loss of consciousness, chest pain, dysrhythmia, abnormal ECG, or significant trauma: cardiac monitoring, vascular access, burn care, pain control, and reassessment.

↓

6. Burn-Center Transport
EMT / AEMT / Paramedic
Consult a burn center for every chemical injury, high-voltage (≥1,000 V) injury, or lightning injury. Arrange consultation and consider follow-up after low-voltage injury, including delayed symptoms or visual complaints; use the regional destination plan and notify early.

Indications

- Patient with suspected chemical and electrical burn requiring prehospital assessment or treatment.

Provider-level care

Provider-Level Actions Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT: Protect the crew, stop exposure, remove contaminated clothing and jewelry, and brush off dry particulate before irrigation. For most chemical skin/eye exposures, begin prompt copious irrigation; do not delay ordinary chemical-eye irrigation while identifying an unknown agent. Remove contact lenses when easily possible. For known water-reactive contamination, follow immediate HazMat/Poison Help direction and TE-09.

• After decontamination, cover burns with clean dry dressings except when white phosphorus is suspected: keep affected areas wet with cool water or wet dressings to prevent reignition until particles are removed. Protect the rest of the patient from hypothermia, manage associated trauma, and transport promptly.

AEMT: Perform all EMT care plus vascular access, reassessed fluid only for shock or significant burn-resuscitation need, and analgesia within standing orders.

• Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing, and medical-director standing orders; training alone does not grant permission.

• Prioritize cardiac monitoring after high-voltage/lightning exposure, loss of consciousness, chest pain, dysrhythmia, abnormal ECG, or significant associated trauma; do not delay transport.

Paramedic: Perform all prior care plus advanced airway treatment, dysrhythmia management, and treatment of shock or compartment-threatening injury.

• Consult Poison Help or Medical Control for agent-specific care and a burn center for every chemical injury, high-voltage (≥1,000 V) injury, or lightning injury. Arrange burn-center consultation and consider follow-up after low-voltage injury, including delayed symptoms or visual complaints.

Warnings

- Do not neutralize a chemical on the patient unless specifically directed by Poison Control, Medical Control, or HazMat; chemical reactions may generate heat.
- Do not apply ice, creams, ointments, or routine wet sheets to significant burns. White phosphorus is an exception: keep the affected area wet until particles are removed while keeping uninjured areas warm.

Clinical pearls

- Apply the trauma field-triage and destination plan: high-risk patients to the highest-level appropriate trauma center within regional constraints; favor pediatric-capable trauma care for children. Use a closer stabilization hospital only for an immediate critical need that cannot safely be managed during longer transport, under the destination plan/Medical Control. Notify early, document the reason, and activate the predefined transfer process. Do not delay ground movement solely while awaiting aircraft.
- Electrical current may cause dysrhythmia, deep tissue injury, rhabdomyolysis, fractures, or traumatic falls.
- Assess entrance and exit wounds, distal perfusion, neurologic status, and circumferential injury.

Related pathways

- TE-09 Chemical / HazMat Exposure (p. 213): Agent-specific decontamination, water-reactive contamination, white phosphorus, and antidote restrictions.
- TB-11 Eye Trauma (p. 198): Ordinary chemical-eye irrigation and known or strongly suspected open-globe exception.
- TB-09 Thermal Burn (p. 194): Burn assessment, controlled fluid rates, consultation, and hypothermia prevention.

TB-03
Crush Syndrome

Revised 2026-10-30

Overview

Suspect crush syndrome after compression of a large muscle mass, especially an extremity or torso; it can occur in under an hour and risk rises with longer entrapment.

Begin resuscitation before release when feasible because reperfusion can cause sudden hyperkalemia, acidosis, shock, and cardiac arrest.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Entrapment / Compression
EMT / AEMT / Paramedic
Confirm scene safety, duration, body region, access, and rescue plan.

↓

2. Crush Syndrome Risk?
EMT / AEMT / Paramedic
Large muscle mass • any significant entrapment time • swelling • neurologic deficit • shock. Document the time compression began when known.

↓

3. Before Release
AEMT / Paramedic
Airway • continuous cardiac monitoring/12-lead when feasible • IV/IO • normal saline preferred • prepare defibrillator and request early Medical Control/rescue-physician input. ECG-directed calcium, bicarbonate, and high-dose albuterol are Paramedic-only. Do not delay unsafe extrication.

↓

4. Release + Reassess
EMT / AEMT / Paramedic
Coordinate release with rescue personnel; assign a clinician to monitor the patient. Watch immediately for hypotension, hyperkalemia/dysrhythmia, respiratory failure, arrest, and compartment syndrome.

↓

5. Treat Complications
AEMT / Paramedic
Reassess shock, pain, compartment syndrome, and associated trauma; supportive treatment within scope. Request Paramedic treatment for suspected hyperkalemia.

Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Paramedic Hyperkalemia Care
Paramedic
Prepare ECG-directed calcium, sodium bicarbonate, and high-dose albuterol treatment before release for ECG changes, dysrhythmia, or peri-arrest deterioration under the applicable standing order or Medical Control. No routine prophylactic calcium or bicarbonate.

↓

7. Rapid Trauma Transport
EMT / AEMT / Paramedic
Minimize scene delay once safely released and notify the trauma center early for high-risk entrapment.

Indications

- Patient with suspected crush syndrome requiring prehospital assessment or treatment.

Provider-level care

Provider-Level Actions Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT: Coordinate safe rescue, control bleeding, manage airway and ventilation, prevent hypothermia, splint as indicated, and document the compression time and body region.

- Do not delay extrication for nonessential procedures when the scene is unsafe.

AEMT: Perform all EMT care plus large-bore vascular access and 0.9% saline before release when feasible: adult initial 1 L, then continue reassessed fluids during extrication. Prefer normal saline, but do not withhold otherwise indicated isotonic crystalloid solely because saline is unavailable. Use Medical Control for pediatric fluid guidance and for additional fluid rate/reassessment limits not specified by standing order.

- Do not administer calcium, sodium bicarbonate, or high-dose albuterol for hyperkalemia as an AEMT; these interventions are Paramedic-only.
- Monitor rhythm and perfusion, obtain a 12-lead when feasible, and prepare for rapid deterioration at the moment of release.

Paramedic: Perform all prior care plus protocol-directed treatment for ECG changes, dysrhythmia, or peri-arrest hyperkalemia, severe

pain, and shock; do not give routine prophylactic calcium or bicarbonate solely for crush injury.

• Coordinate the timing of release with rescue personnel and ensure the defibrillator and ECG-directed hyperkalemia treatment are immediately available. Obtain early Medical Control/rescue-physician input for prolonged large-muscle compression and follow the high-risk crush-release plan.

• Pre-release empiric medication use, fluid rate, and reassessment limits require the specified standing order or Medical Control; do not improvise. Do not repeat calcium gluconate without a specific Medical Control order confirming the cumulative dose.

Warnings

- Do not apply a prophylactic tourniquet solely because crush syndrome is suspected. Consider one only for hemorrhage control or with Medical Control/an established rescue plan when pre-release treatment cannot be provided.
- Reperfusion-related deterioration may occur immediately after the compressive force is removed.

TB-04
Extremity Trauma

Revised 2026-10-30

Overview

Prioritize hemorrhage control and neurovascular preservation in extremity injury.

Treat amputations, open fractures, dislocations, crush injury, and compartment syndrome as time-sensitive trauma.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Extremity Injury
EMT / AEMT / Paramedic
Expose, identify life-threatening hemorrhage, and assess associated trauma.

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. Major Hemorrhage?
EMT / AEMT / Paramedic
Direct pressure • approved wound packing • tourniquet for life-threatening extremity bleeding when pressure is ineffective or impractical. Tighten until bleeding stops and distal pulse is absent; reassess after movement.

↓

3. Check CSM
EMT / AEMT / Paramedic
Document distal pulse, capillary refill, skin color/temperature, sensation, and movement before and after every alignment, splint, or tourniquet.

↓

4. Align / Splint
EMT / AEMT / Paramedic
Cover open wounds, stabilize joints above and below, and reassess CSM. Gently realign only for absent distal perfusion or when needed to splint; stop for resistance or worsening pain. Do not force reduction.

↓

5. Pain / Shock Care
AEMT / Paramedic
Prevent hypothermia and provide authorized analgesia and vascular access.

↓

6. Appropriate Destination
EMT / AEMT / Paramedic
Apply the trauma field-triage and destination plan: high-risk patients to the highest-level appropriate trauma center within regional constraints; favor pediatric-capable trauma care for children. Use a closer stabilization hospital only for an immediate critical need that cannot safely be managed during longer transport, under the destination plan/Medical Control. Notify early, document the reason, and activate the predefined transfer process. Do not delay ground movement solely while awaiting aircraft.

Indications

- Patient with suspected extremity trauma requiring prehospital assessment or treatment.

Provider-level care

Provider-Level Actions Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT: For life-threatening extremity hemorrhage, apply an approved tourniquet promptly when direct pressure is ineffective or impractical. Tighten until bleeding stops and the distal pulse is absent. If bleeding persists despite proper placement and tightening, add a second tourniquet immediately proximal to the first when feasible. Record time, leave visible, and reassess after movement and throughout transport. For suitable junctional wounds, pack with approved gauze and maintain direct pressure per product instructions. Document distal pulse, capillary refill, skin color/temperature, sensation, and movement before and after every intervention.

- For open fractures, cover with a sterile dressing; do not probe the wound or push exposed bone back in. Wrap amputated tissue in moist sterile gauze, seal it in a labeled bag with patient/time, send it with the patient, and keep cool without direct contact with ice or water.

AEMT: Perform all EMT care plus vascular access, fluid therapy for shock, and authorized analgesia.

- Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing, and medical-director standing orders; training alone does not grant permission.
- Trend distal perfusion and report any deterioration immediately.

Paramedic: Perform all prior care plus advanced pain management and resuscitation of hemorrhagic shock within protocol.

- Coordinate direct trauma-center transport when limb or life is threatened.

Warnings

- Do not delay transport attempting perfect anatomic alignment. Do not force fracture or dislocation reduction; attempt gentle realignment only for absent distal perfusion or when needed to splint, and stop for resistance or worsening pain.
- Record tourniquet application time. Do not periodically loosen or remove a tourniquet. A pulseless extremity, expanding hematoma, severe pain out of proportion, or tense swelling requires urgent specialty care.

TB-05
Head Trauma

Revised 2026-10-30

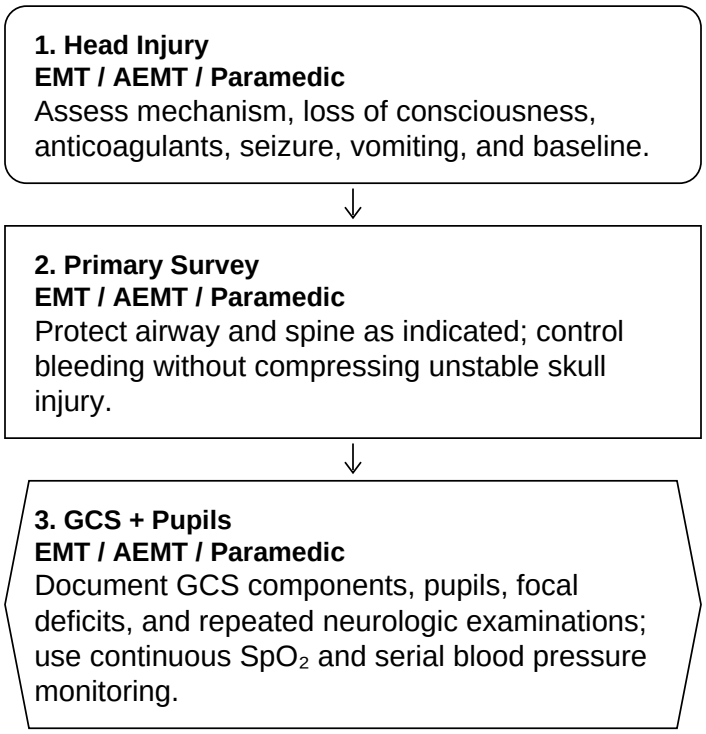
Overview

Prevent secondary brain injury by avoiding hypoxia, hypotension, hyperventilation, and transport delay.

Anticoagulant or antiplatelet use, age, mechanism, loss of consciousness, vomiting, seizure, and neurologic change increase risk.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.



Continue at step 4 in the next column/page

Quick flow continued at step 4

4. Prevent Secondary Injury
EMT / AEMT / Paramedic
Continuously monitor oxygenation and ventilation; aim for SpO_2 ≥94% and urgently correct SpO_2 <90% or inadequate ventilation. During initial and routine assisted ventilation, target $EtCO_2$ 35-45 mmHg; do not routinely hyperventilate. Maintain age-appropriate perfusion, glucose, and temperature. For suspected TBI, prevent hypotension: target adult SBP at least 110 mmHg, including age 70; do not use permissive hypotension. Reassess blood pressure frequently and after treatment; for children, obtain Medical Control for age-specific TBI resuscitation guidance rather than substituting an adult target.

↓

5. Herniation Signs?
AEMT / Paramedic
Falling GCS • unilateral dilated pupil • abnormal posturing • Cushing response. Position the head midline and elevate about 20-30° when this does not worsen hypotension or interfere with spinal precautions. Only for clear active or imminent herniation, use brief controlled rescue ventilation to $EtCO_2$ 30-35 mmHg with capnography while expediting transport; avoid overshooting below 30 mmHg; no routine hyperventilation.

↓

6. Appropriate Destination
EMT / AEMT / Paramedic
Minimize scene time, provide serial GCS and early notification, especially for neurologic deterioration or anticoagulant/antiplatelet use. Apply the trauma field-triage and destination plan: high-risk patients to the highest-level appropriate trauma center within regional constraints; favor pediatric-capable trauma care for children. Use a closer stabilization hospital only for an immediate critical need that cannot safely be managed during longer transport, under the destination plan/Medical Control. Notify early, document the reason, and activate the predefined transfer process. Do not delay ground movement solely while awaiting aircraft.

Indications

- Patient with suspected head trauma requiring prehospital assessment or treatment.

Provider-level care

Provider-Level Actions Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT: Maintain airway; continuously monitor oxygenation and ventilation, aim for SpO_2 ≥94%, and urgently correct SpO_2 <90% or inadequate ventilation. Use spinal motion restriction when indicated, control bleeding, obtain serial GCS/pupils, and transport promptly.

- Identify anticoagulants, antiplatelets, seizure, repeated vomiting, skull signs, focal deficit, abnormal pupils, and neurologic deterioration.

AEMT: Perform all EMT care plus vascular access and age-appropriate treatment of hypotension without delaying transport; use AM-05 for adults. For suspected TBI, target adult SBP at least 110 mmHg, including age 70; do not use permissive hypotension. Reassess blood pressure frequently and after treatment; for children, obtain Medical Control for age-specific TBI resuscitation guidance rather than substituting an adult target.

- Use authorized airway adjuncts and $EtCO_2$ monitoring when available. For initial and routine assisted ventilation, target $EtCO_2$ 35-45 mmHg; avoid routine hyperventilation.

Paramedic: Perform all prior care plus advanced airway management and controlled ventilation when the patient cannot protect the airway; for initial and routine assisted ventilation, including after advanced airway placement, target $EtCO_2$ 35-45 mmHg.

- Only for clear active or imminent herniation, use brief controlled rescue ventilation to $EtCO_2$ 30-35 mmHg with capnography while expediting transport; avoid overshooting below 30 mmHg; avoid routine hyperventilation. Treat active seizure under the seizure/medication pathway; do not provide prophylactic antiseizure medication in the field.

Warnings

- For pediatric TBI, obtain Medical Control guidance on age-specific perfusion targets. Do not substitute an adult target or a generic medical-shock threshold.
- A single normal examination does not exclude intracranial hemorrhage, especially in anticoagulated or older patients.
- Do not aggressively control hypertension in suspected traumatic brain injury unless another

approved protocol requires it. Initial and routine assisted-ventilation target: $EtCO_2$ 35-45 mmHg. Brief rescue target: $EtCO_2$ 30-35 mmHg only for clear active or imminent herniation, with capnography and expedited transport; avoid overshooting below 30 mmHg and do not routinely hyperventilate.

Clinical pearls

- Hypotension in isolated head injury should prompt a search for extracranial hemorrhage.
- Trend the individual eye, verbal, and motor GCS components rather than documenting only a total score.

Related pathways

- AM-05 Hypotension / Shock (p. 129): Adult shock care with the higher suspected-TBI perfusion target.

TB-06
Multiple Trauma

Revised 2026-10-30

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Major Trauma
EMT / AEMT / Paramedic
Scene safety • mechanism • resources • request air medical early when it meaningfully improves time to needed care or provides unavailable capability; never delay ground transport.

↓

2. XABCDE
EMT / AEMT / Paramedic
Exsanguination • airway • breathing • circulation • disability • exposure.

↓

3. High-Risk Trauma?
EMT / AEMT / Paramedic
Physiologic abnormality • high-risk anatomy • special considerations (age, anticoagulants, pregnancy, burns with trauma, suspected abuse) • concerning mechanism.

Continue at step 4 in the next column/page

Quick flow continued at step 4

4. Treat Immediate Threats
EMT / AEMT / Paramedic
Catastrophic hemorrhage control • airway/ventilation • assess chest injury and urgently request ALS for suspected tension pneumothorax • pelvic stabilization only when indicated • warming • reassessment.

↓

5. Chest Decompression
Paramedic
Paramedic-only chest decompression for suspected tension pneumothorax under current standing orders; reassess ventilation and perfusion after intervention.

↓

6. Resuscitate En Route
AEMT / Paramedic
Access • hemorrhage control • warmed isotonic crystalloid titrated to perfusion when blood is unavailable • higher adult TBI SBP target ≥110 mmHg • analgesia • warming • serial reassessment. Follow AM-05; avoid unnecessary large-volume crystalloid.

↓

7. Trauma TXA Eligibility
Paramedic
Paramedic only: after immediate lifesaving interventions, consider IV/IO TXA for traumatic hemorrhage with clinical hemorrhagic shock within 3 hours of injury. Adult dose: 1-2 g IV/IO: give 1 g over 10 minutes or 2 g over 20 minutes; do not exceed 100 mg/min. Verify the stocked concentration and dilution against the current medication-preparation instructions. Do not administer an automatic repeat dose; any additional dose requires a separate authorized order.

Continue at step 8 in the next column/page

Quick flow continued at step 8

8. Appropriate Destination
EMT / AEMT / Paramedic
Apply the trauma field-triage and destination plan: high-risk patients to the highest-level appropriate trauma center within regional constraints; favor pediatric-capable trauma care for children. Use a closer stabilization hospital only for an immediate critical need that cannot safely be managed during longer transport, under the destination plan/Medical Control. Notify early, document the reason, and activate the predefined transfer process. Do not delay ground movement solely while awaiting aircraft.

Indications

- Patient with suspected multiple trauma requiring prehospital assessment or treatment.

Provider-level care

Provider-Level Actions Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT: Perform rapid XABCDE, control hemorrhage, support airway/ventilation, assess chest injury, oxygenation and perfusion, and urgently request ALS for suspected tension pneumothorax. Stabilize the pelvis only when indicated, stabilize fractures, and prevent hypothermia.

- For life-threatening extremity hemorrhage, apply an approved tourniquet promptly when direct pressure is ineffective or impractical. Tighten until bleeding stops and the distal pulse is absent. If bleeding persists despite proper placement and tightening, add a second tourniquet immediately proximal to the first when feasible. Record time, leave visible, and reassess after movement and throughout transport. For suitable junctional wounds, pack with approved gauze and maintain direct pressure per product instructions. Do not periodically loosen or remove a tourniquet.
- For an open chest wound in a spontaneously breathing patient, consider an approved vented seal and reassess frequently. If deterioration follows sealing, assess/release an obstructing seal and obtain immediate Paramedic treatment for suspected tension pneumothorax. Do not probe the wound. Positive-pressure ventilation requires the chest-injury standing orders and immediate Paramedic/Medical Control guidance, not automatic extrapolation of this seal instruction.
- Package promptly; complete the secondary examination during transport when possible.

AEMT: Perform all EMT care plus IV/IO access, age-appropriate authorized shock resuscitation, and airway support during transport. For adult hemorrhagic shock when blood is unavailable, follow AM-05: titrate warmed isotonic crystalloid to perfusion; avoid unnecessary large-volume crystalloid. If TBI is suspected, prevent hypotension and target adult SBP at least 110 mmHg; do not use permissive hypotension.

- Trend mental status, perfusion, respiratory status, and response to treatment.

Paramedic: Perform all prior care plus advanced airway treatment, chest decompression, analgesia, and protocol-directed hemorrhagic-shock resuscitation.

- Paramedic only: after immediate lifesaving interventions, consider IV/IO TXA for traumatic hemorrhage with clinical hemorrhagic shock no more than 3 hours after injury. Adult dose: 1-2 g IV/IO: give 1 g over 10 minutes or 2 g over 20 minutes; do not exceed 100 mg/min. Verify the stocked concentration and dilution against the current medication-preparation instructions. Do not administer an automatic repeat dose; any additional dose requires a separate authorized order. Do not infer a systemic trauma regimen from topical indications. Record injury time and any TXA dose/time, and notify the receiving team to prevent duplicate dosing.
- Pediatric trauma TXA: local dose reference 15 mg/kg, Paramedic-only. Before administration, confirm pediatric eligibility, maximum dose, administration rate, and preparation with Medical Control within provider scope. Do not extrapolate the adult dose or infusion rate to children.
- Use blood products only within an authorized, supported transfusion program with required training, monitoring, and destination coordination.
- Lead destination, air-medical, and Medical Control decisions without extending scene time.

Warnings

- Do not delay transport for IV attempts, detailed examination, splinting, or procedures that can be completed en route.
- Prevent hypothermia throughout trauma resuscitation.
- Trauma TXA is Paramedic-only, not an AEMT intervention. Adult dose: 1-2 g IV/IO: give 1 g over 10 minutes or 2 g over 20 minutes; do not exceed 100 mg/min. Verify the stocked concentration and

dilution against the current medication-preparation instructions. Do not administer an automatic repeat dose; any additional dose requires a separate authorized order. Pediatric administration requires confirmation of the patient-specific maximum and rate with Medical Control within provider scope. Do not infer a trauma regimen from topical bleeding instructions.

- For pediatric traumatic hemorrhage, obtain Medical Control guidance and use only the authorized pediatric trauma resuscitation order; do not apply medical or septic-shock fluid targets indiscriminately.

Related pathways

- AM-05 Hypotension / Shock (p. 129): Adult hemorrhagic-shock strategy and suspected-TBI perfusion exception.
- TB-12 Abdominal / Pelvic Trauma (p. 201): Concealed hemorrhage, pelvic stabilization, and trauma TXA eligibility, the authorized adult dose range, and outstanding administration parameters.
- TB-08 Spinal Motion (p. 193): Open when selective spinal motion restriction is indicated.
- TB-10 Traumatic Arrest (p. 196): Open for traumatic pulseless arrest.

TB-07
Radiation Incident

Revised 2026-10-30

Overview

Radiation exposure, external contamination, and internal contamination are distinct problems; exposure alone does not make the patient or ambulance contaminated.

Treat life threats first while using time, distance, shielding, PPE, and contamination control.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Radiation Hazard
EMT / AEMT / Paramedic
Stage until Incident Command identifies a safe zone and the appropriate PPE and radiation monitoring/dosimetry when required.

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. Life Threat?
EMT / AEMT / Paramedic
Do not delay lifesaving airway, breathing, hemorrhage care, or transport for decontamination.

↓

3. Contaminated?
EMT / AEMT / Paramedic
If radioactive material is on clothing or skin, carefully remove outer clothing, bag and label belongings, then gently wash with soap and lukewarm water. Do not decontaminate an exposure-only patient.

↓

4. Assess Exposure
EMT / AEMT / Paramedic
Source • time • distance • shielding • symptoms • associated blast/burn/trauma • survey or dose information provided by radiation authorities.

↓

5. Coordinate Experts
AEMT / Paramedic
Incident Command/HazMat • local radiation safety • receiving facility • Medical Control.

↓

6. Controlled Transport
EMT / AEMT / Paramedic
Notify the destination early. If unstable, transport promptly with contamination precautions rather than delaying for completed decontamination.

Indications

- Patient with suspected radiation incident requiring prehospital assessment or treatment.

Provider-level care

Provider-Level Actions Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT: Use appropriate PPE and contamination control, treat immediate threats, and prevent spread. Exposure without contamination does not require patient or ambulance decontamination.

• For known or suspected external contamination, carefully remove outer clothing, bag and label belongings, and gently wash exposed skin or hair with soap and lukewarm water; do not scrub or injure the skin.

• Record source information, exposure time, distance, shielding, symptoms, decontamination performed, and any survey or dose information provided by radiation authorities.

AEMT: Perform all EMT care plus vascular access and supportive treatment within standing orders.

• Coordinate monitoring, ambulance preparation, and transport precautions with Incident Command.

Paramedic: Perform all prior care plus advanced supportive care and consultation with Medical Control, HazMat, local radiation safety, and the receiving facility.

• Manage associated trauma, burns, airway injury, shock, or dysrhythmia under the appropriate protocols. Do not give potassium iodide or other radiation countermeasures unless specifically directed by public-health or radiation authorities.

Warnings

• Do not enter a radiation hot zone without authorization, training, appropriate PPE, and required monitoring/dosimetry.

• Contamination alone is not a reason to withhold lifesaving care or delay transport of a seriously injured patient.

TB-08
Selective Spinal Immobilization

Revised 2026-10-30

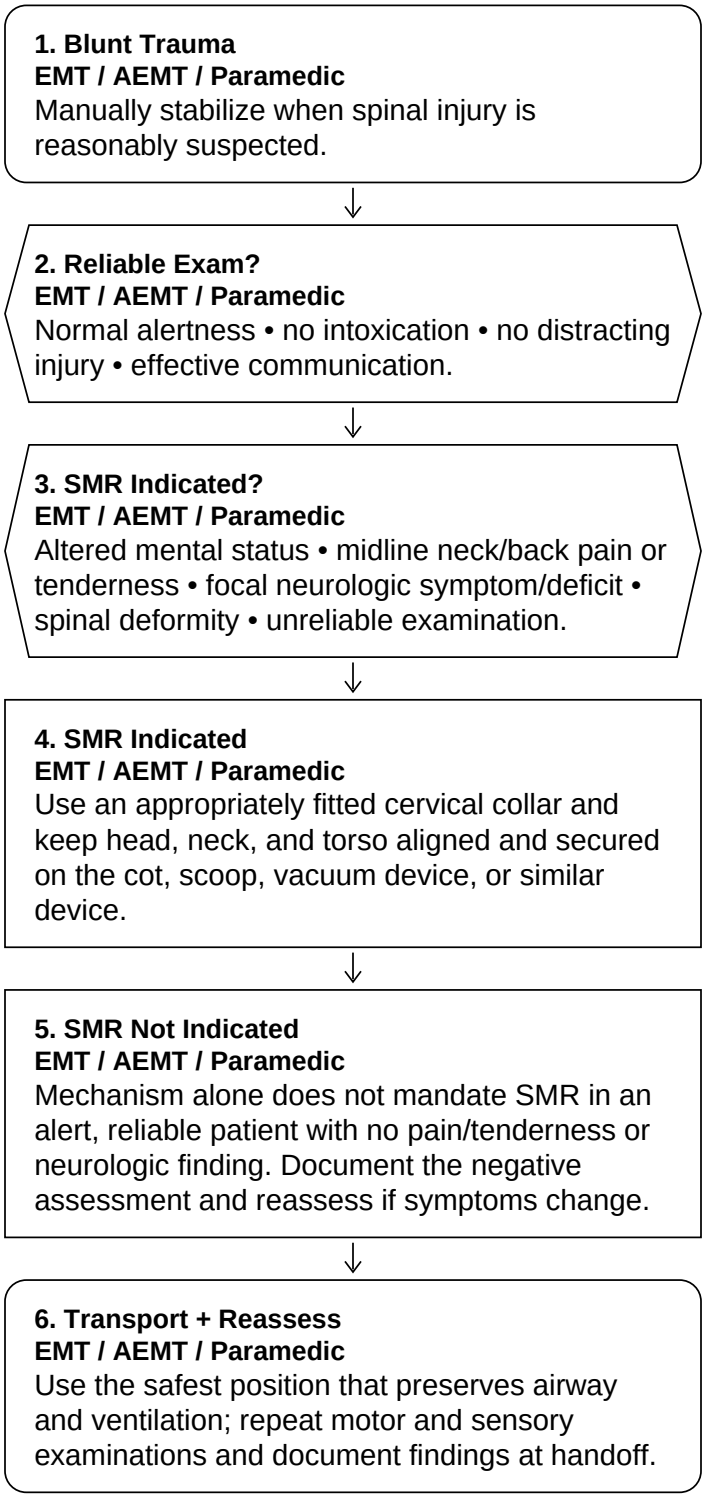
Overview

Apply spinal motion restriction (SMR) selectively after blunt trauma based on a reliable clinical examination and high-risk findings.

The goal is to minimize unwanted spinal movement and provide safe handling, not routine prolonged placement on a rigid long board.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.



Indications

• Patient with suspected selective spinal immobilization requiring prehospital assessment or treatment.

Provider-level care

Provider-Level Actions Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT: Assess alertness, reliability, midline neck/back pain or tenderness, neurologic findings, deformity, distracting injury, intoxication, and communication barriers.

• For children, include neck pain/torticollis, neurologic abnormality, altered mental status, high-risk motor-vehicle collision, high-impact diving, and substantial torso injury in the age-appropriate assessment. Age alone is not an indication, and mechanism alone does not mandate SMR in an alert, reliable, asymptomatic patient.

• Use appropriately sized equipment and padding to avoid excessive neck flexion; maintain airway access, do not force neutral alignment, and minimize time on rigid transfer devices.

• Apply SMR when indicated using an appropriately fitted cervical collar and aligned securement; pad voids, protect pressure points, and reassess motor/sensory findings and distal perfusion before and after movement.

AEMT: Perform all EMT care and support safe positioning, airway care, analgesia, and serial neurologic reassessment within scope.

• Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing, and medical-director standing orders; training alone does not grant permission.

Paramedic: Perform all prior care, resolve equivocal examinations, and manage airway or ventilation while minimizing cervical motion.

• Document the clinical basis for applying or clearing SMR and repeat neurologic findings before transfer of care.

Warnings

- Do not force the head or neck into neutral alignment when pain, resistance, neurologic change, airway compromise, or a position that cannot be tolerated occurs.
- A rigid long board is primarily an extrication or transfer device, not routine transport packaging; remove it when safely feasible.
- Routine SMR is not indicated for penetrating trauma without neurologic deficit and should not delay transport.
- For pediatric clearance, use only the approved pediatric decision aid and locally authorized clinician. If these requirements cannot be met, obtain Medical Control guidance; do not substitute an emergency-department imaging rule or treat these risk features as a stand-alone field-clearance rule.

TB-09
Thermal Burn

Revised 2026-10-30

Overview

Stop the burning process, identify airway or associated trauma, estimate partial- and full-thickness TBSA, and prevent hypothermia.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Stop Burning
EMT / AEMT / Paramedic
Remove the heat source and constricting items; protect responders. For a small thermal burn (<10% TBSA), use cool running water for approximately 20 minutes when feasible without hypothermia or transport delay. Monitor temperature and keep uninjured areas warm; no ice.

↓

2. Airway / Trauma Threat?
EMT / AEMT / Paramedic
Assess inhalation injury, progressive respiratory compromise, CO/cyanide risk, and multisystem trauma; plan early airway care when clinically indicated.

↓

3. Classify + Estimate TBSA
EMT / AEMT / Paramedic
Count only partial- and full-thickness burns. For scattered burns, the patient's palm plus fingers (whole palmar hand surface) is approximately 1% TBSA; use an age-adjusted validated chart for larger pediatric burns.

Continue at step 4 in the next column/page

Quick flow continued at step 4

4. Burn-Center Consultation?
EMT / AEMT / Paramedic
Full thickness • partial thickness ≥10% TBSA • potentially deep burn of any size • critical-area burn • inhalation injury • associated trauma • uncontrolled pain • complicating comorbidity • pediatric burn.

↓

5. Dry Cover + Resuscitate
AEMT / Paramedic
Prevent hypothermia • IV/IO when indicated • controlled warmed LR infusion using the burn-fluid table below (NS if LR unavailable) • pain control • frequent reassessment. Do not give an automatic 1 L bolus solely for burn size; shock requires assessment for hemorrhage or another cause.

↓

6. Appropriate Destination
EMT / AEMT / Paramedic
Consult a burn center for indicated injuries, including potentially deep burns of any size and complicating comorbidity. Consultation does not automatically require bypassing needed stabilization; follow the regional destination plan. Use a closer stabilization hospital for an immediate critical need that cannot safely be managed during longer transport; notify early, document the reason, and coordinate transfer.

Indications

- Patient with suspected thermal burn requiring prehospital assessment or treatment.

Assessment

Initial burn-fluid rates when resuscitation is indicated

- Record injury time, weight, and partial-/full-thickness TBSA. Use warmed lactated Ringer's when available; normal saline is an alternative, changing to LR when available.
- TBSA below 20%: 0.25 mL/kg × TBSA percentage per hour. Enter TBSA as a whole percentage, not a decimal fraction; for 70 kg and 15% TBSA, the starting rate is approximately 263 mL/hr.
- TBSA 20% or greater: age ≤5: 125 mL/hr; age 6-14: 250 mL/hr; age ≥15: 500 mL/hr.
- Reassess perfusion, respiratory status, and response frequently. Do not give an automatic 1 L bolus solely because a burn is extensive; do not delay transport.
- For shock, search for hemorrhage or another cause and use the appropriate shock-resuscitation pathway rather than relying on the isolated-burn starting rate.

Burn-center consultation or transfer considerations

- Full-thickness burns; potentially deep burns of any size; or comorbidity that complicates care.
- Partial-thickness burns of 10% TBSA or greater.
- Deep partial- or full-thickness burns involving the face, hands, feet, genitalia, perineum, or over major joints.
- Suspected inhalation injury, chemical or electrical injury, significant associated trauma, poorly controlled pain, or pediatric burns requiring specialty support.

Treatment

1. For shock, search for associated hemorrhage or another cause and use the appropriate shock-resuscitation pathway; do not delay transport. Do not apply medical/septic-shock fluid targets indiscriminately to pediatric traumatic hemorrhage.

Provider-level care

Provider-Level Actions Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT: Stop the burning process and remove jewelry/constricting items. For a small thermal burn (<10% TBSA), use cool running water for approximately 20 minutes when feasible without hypothermia or transport delay; monitor temperature and keep uninjured areas warm. Do not cool when hypothermia risk is significant. Cover with clean dry dressings, prevent hypothermia, and estimate TBSA.

- Support airway and ventilation, screen for carbon monoxide/cyanide exposure, and treat associated trauma.

AEMT: Perform all EMT care plus IV/IO access and authorized non-IM analgesia within scope. When burn-fluid resuscitation is indicated, use a controlled warmed lactated Ringer's infusion (normal saline if LR is unavailable) at the starting rate in the burn-fluid table; reassess perfusion, respiratory status, and response frequently.

• Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing, and medical-director standing orders; training alone does not grant permission.

• Consider two IV sites for larger burns without delaying transport.

• For shock, search for associated hemorrhage or another cause and follow the appropriate shock-resuscitation pathway; the isolated-burn starting rate does not replace shock treatment.

Paramedic: Perform all prior care plus advanced airway management when clinically required, protocol-directed analgesia, and burn resuscitation.

• Coordinate burn-center consultation or direct transport when appropriate. Consultation does not automatically require bypassing needed stabilization; apply the regional destination plan, document any immediate critical need for closer stabilization, and arrange transfer.

Warnings

• Do not include superficial first-degree burns in TBSA.

• Do not apply ice, wet sheets, creams, or ointments to significant burns; burn patients are highly susceptible to hypothermia.

• Do not use intramuscular analgesics in burn patients.

Related pathways

• AM-05 Hypotension / Shock (p. 129): Adult shock care when hypotension suggests hemorrhage or another cause beyond an isolated burn.

• TB-06 Multiple Trauma (p. 190): Associated traumatic hemorrhage and trauma destination priorities.

TB-10
Traumatic Arrest

Revised 2026-10-30

Overview

Apply the approved traumatic-arrest withholding and resuscitation criteria while prioritizing immediately reversible traumatic causes.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Arrest With Trauma
EMT / AEMT / Paramedic
Confirm apnea, pulselessness, rhythm when appropriate, DNR/MOST, and obvious-death criteria. If trauma does not clearly explain the arrest, use the age-appropriate medical cardiac-arrest pathway: adult cardiac arrest or PC-04 Pediatric Pulseless Arrest, while evaluating reversible causes.

↓

2. Withhold Resuscitation?
EMT / AEMT / Paramedic
Withhold only for a valid DNR/MOST or unmistakable death: decomposition, rigor mortis, dependent lividity, decapitation, hemicorporectomy, or injury incompatible with life. Do not use downtime alone as a withholding rule.

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3. Correct Reversible Causes
EMT / AEMT / Paramedic
Catastrophic hemorrhage • oxygenation/ventilation • Paramedic chest decompression when indicated • pelvic stabilization only when indicated • hypothermia prevention.

Continue at step 4 in the next column/page

Quick flow continued at step 4

4. CPR / Defibrillation
EMT / AEMT / Paramedic
Perform high-quality CPR and defibrillation for a shockable rhythm without delaying catastrophic hemorrhage control, effective oxygenation/ventilation, or indicated Paramedic chest decompression.

↓

5. ROSC or Transport Candidate?
EMT / AEMT / Paramedic
Witnessed arrest, organized rhythm, signs of life, feasible definitive-care transport, or an immediately reversible cause.

↓

6. Appropriate Destination
EMT / AEMT / Paramedic
Follow Medical Control and deceased-person policy. Apply the trauma field-triage and destination plan: high-risk patients to the highest-level appropriate trauma center within regional constraints; favor pediatric-capable trauma care for children. Use a closer stabilization hospital only for an immediate critical need that cannot safely be managed during longer transport, under the destination plan/Medical Control. Notify early, document the reason, and activate the predefined transfer process. Do not delay ground movement solely while awaiting aircraft.

Indications

- Patient with suspected traumatic arrest requiring prehospital assessment or treatment.

Provider-level care

Provider-Level Actions Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT: Confirm arrest and withholding criteria, control catastrophic hemorrhage, support airway/ventilation, apply AED/monitor, stabilize the pelvis only when indicated, and prevent hypothermia.

- Begin rapid transport when resuscitation criteria are met; do not delay for procedures that can occur en route.

AEMT: Perform all EMT care plus IV/IO access during transport. Do not perform chest decompression; assist the Paramedic with the procedure when requested.

- Use authorized airway adjuncts and reassess for signs of life and ROSC.

Paramedic: Perform all prior care plus advanced airway management, rhythm treatment, and chest decompression when tension physiology is suspected within standing orders. Consider bilateral decompression when the mechanism, examination, or physiology supports it.

- For ADULT clearly traumatic arrest, do not routinely give epinephrine. If used under local orders, it must not precede indicated lifesaving trauma interventions. Pediatric medication and resuscitation rules require the separate age-appropriate director-approved pathway; do not extrapolate the adult epinephrine instruction to children.
- Continue resuscitation or terminate only under current Claiborne policy and Medical Control requirements; do not use an app-only downtime cutoff.

Warnings

- Do not apply traumatic-arrest withholding criteria to lightning strike, drowning, or hypothermic arrest; initiate resuscitation.
- Do not hyperventilate. Avoid delaying transport for endotracheal intubation or vascular access.
- Chest decompression is Paramedic-only under the current Tennessee statewide protocol guidelines. Do not extend this procedure to an AEMT without formal state authorization.

Clinical pearls

- Treat the trauma triad of death by controlling hemorrhage, limiting scene delay, and aggressively preventing hypothermia.
- Withholding is limited to a valid directive or unmistakable death; an initiated resuscitation requires Claiborne policy and Medical Control for termination decisions.

Related pathways

- AC-03 Adult Cardiac Arrest (p. 77): Adult medical-arrest pathway when trauma does not clearly explain arrest.
- PC-04 Pediatric Pulseless Arrest (p. 167): Age-appropriate pediatric medical-arrest pathway when trauma does not clearly explain arrest.

- TB-06 Multiple Trauma (p. 190): Hemorrhage control, provider-specific chest care, and destination priorities.

TB-11
Eye Trauma

Revised 2026-10-30

Overview

Any vision loss, irregular pupil, hyphema, extrusion of ocular contents, impaled object, marked proptosis, severe pain, or high-velocity mechanism requires urgent transport and early receiving-facility notification.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Scene + Life Threats
EMT / AEMT / Paramedic
PPE - ABCs - associated head/facial trauma - spinal motion restriction only when indicated

↓

2. Chemical Exposure?
EMT / AEMT / Paramedic
Ordinary splash: start immediate gentle copious irrigation through handoff. Known/strongly suspected open globe with contamination: no irrigation into globe or pressure; noncontact rigid shield and urgent specialist/Medical Control guidance.

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3. Thermal Eye Injury?
EMT / AEMT / Paramedic
Briefly cool or gently irrigate with room-temperature water or saline - protect the eye - do not use ice

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4. Penetrating / Open Globe?
EMT / AEMT / Paramedic
No pressure, manipulation, irrigation, drops, or object removal - rigid shield - stabilize protruding object

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Blunt / Superficial?
EMT / AEMT / Paramedic
Assess vision and red flags - shield if globe injury suspected - irrigate only loose debris

↓

6. Vision Threat?
EMT / AEMT / Paramedic
Vision loss - irregular pupil - hyphema - proptosis - extrusion - severe pain - high-velocity mechanism

↓

7. Protect + Transport
EMT / AEMT / Paramedic
Control pain/nausea within scope - minimize Valsalva - serial vision check - early specialty notification

Indications

- Blunt or penetrating injury involving the eye or orbit.
- Chemical or thermal exposure involving the eye.
- High-velocity foreign body, impaled object, sudden visual deficit after trauma, or suspected globe rupture.
- Pain, photophobia, redness, swelling, inability to open the eye, abnormal pupil, hyphema, proptosis, diplopia, or loss of vision after injury.

Assessment

Focused eye assessment

- Identify the exact mechanism, time, eye involved, chemical or product when applicable, contact-lens use, protective eyewear, high-velocity metal/glass exposure, and associated head or facial trauma.
- Assess visual acuity in each eye separately before treatment when feasible and after treatment when it will not delay care. If a chart is unavailable, document the ability to read, count fingers, detect hand motion, or perceive light.
- Inspect without pressure for pupil shape and reactivity, redness, bleeding, hyphema, extrusion of tissue or fluid, corneal clouding, foreign body, lid injury, proptosis, and peri-orbital swelling.
- Assess extraocular movement and diplopia only when it can be done gently and without manipulating a suspected open globe. Do not force the eyelids open.

• Reassess pain, vision, pupils, mental status, and associated trauma throughout care.

Vision-threatening findings

• Decreased or lost vision, irregular or teardrop pupil, visible penetration, extruded ocular contents, hyphema, or an impaled object.

• Marked proptosis, rapidly increasing orbital swelling, severe pain, decreased vision, or an afferent pupillary abnormality after blunt trauma.

• Chemical burns, especially alkali or hydrofluoric-acid exposure; high-velocity metal or glass; explosion; severe lid laceration; or inability to adequately examine the eye.

Treatment

1. Treat immediate life threats and associated head, face, cervical-spine, burn, or multisystem trauma. Eye care must not delay critical trauma transport.
2. Ordinary chemical splash: begin immediate gentle copious water or normal-saline irrigation and continue during transport through handoff; do not wait for product identification or ideal fluid. Remove contact lenses when easily possible, irrigate from the inner corner outward, and prevent runoff into the unaffected eye or onto responders.
3. When trained and equipped to use an agency-approved ocular-pH endpoint, follow that endpoint and reassess pH after a pause. When those resources are unavailable, continue ordinary chemical-splash irrigation through handoff; do not delay irrigation to obtain equipment.
4. Brush away a dry chemical before irrigation when safe and appropriate. Do not attempt acid-base neutralization. Follow TE-09 Chemical / HazMat Exposure and Poison Help guidance for unusual or water-reactive agents.
5. Thermal injury: gently cool or irrigate with room-temperature water or saline, protect the eye, and avoid ice, ointment, or pressure.
6. Suspected penetrating/open-globe injury: do not remove an impaled object, manipulate the eye, apply pressure, irrigate, or instill medication. Stabilize a protruding object with bulky material around it without pressure, then protect with a rigid shield.
7. Blunt trauma: protect with a rigid shield when globe injury is possible. Do not place a pressure dressing. Evaluate for hyphema, irregular pupil, vision change, diplopia, restricted movement, proptosis, and associated facial/head injury.
8. Loose superficial debris may be irrigated only when penetrating injury or globe rupture is not suspected. Do not remove an embedded foreign body or use cotton swabs, magnets, or instruments on the eye.
9. Provide authorized analgesia and antiemetic care, minimize coughing, vomiting, agitation, and Valsalva, and position with the head elevated when safe for the overall trauma condition.
10. Transport promptly to the most appropriate facility under the trauma and specialty-destination plan. Notify early for open globe, impaled object, vision loss, orbital-compartment warning signs, significant chemical burn, or severe associated trauma.
11. Document baseline and repeat vision, pupil findings, laterality, mechanism, product/exposure time, irrigation start and duration, eye protection, medication, response, consultation, and destination rationale.

Medication orders

Analgesic

Use the current Claiborne County pain-management protocol and provider-level standing orders.

• Titrate carefully and monitor airway, ventilation, mental status, and blood pressure.

• Do not use ketamine for suspected penetrating or open-globe eye trauma under this protocol unless Medical Control expressly authorizes it.

Antiemetic

Use the current age-appropriate Claiborne County standing order when nausea or vomiting threatens the injured eye.

• Preventing vomiting and Valsalva is especially important when open-globe injury is suspected.

Topical ophthalmic anesthetic

Only if specifically stocked and adopted in a finalized Claiborne County standing order or ordered by Medical Control.

• May facilitate irrigation when authorized; never dispense for patient home use.

• Do not instill into an eye with suspected penetration or globe rupture.

Provider-level care

Provider-Level Actions Protect the injured eye, preserve baseline findings, treat pain and nausea, and avoid transport delay.

EMT: Treat immediate airway, breathing, circulation, hemorrhage, and associated trauma threats. Use spinal motion restriction only when indicated by mechanism and examination.

• For an ordinary chemical splash, begin immediate gentle copious water or normal-saline irrigation and

continue during transport through handoff; do not wait for product identification or ideal fluid. Remove contact lenses only when easily accomplished without delaying irrigation. Gently cool or irrigate thermal exposure as directed below.

• For suspected penetrating or open-globe injury, do not apply pressure, remove an object, manipulate the eye, irrigate, or instill medication. Stabilize a protruding object without pressure and place a rigid eye shield when available.

• If an open globe is known or strongly suspected with chemical contamination, do not irrigate into the globe or apply pressure. Protect with a noncontact rigid shield and urgently obtain ophthalmology, Poison Help, and Medical Control guidance. A hypothetical concern for rupture must not delay ordinary splash irrigation.

• Assess and document visual acuity in each eye separately when feasible, pupils, visible injury, mechanism, pain, and changes during care. Do not delay time-sensitive transport to complete testing.

AEMT: Perform all EMT actions plus vascular access when clinically indicated and authorized analgesia or antiemetic treatment under the current standing orders.

• Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing, and medical-director standing orders; training alone does not grant permission.

• Monitor for deterioration from associated head/facial trauma, hemorrhage, chemical exposure, or systemic injury and support transport without delaying irrigation or eye protection.

• Avoid medications or interventions that increase agitation, vomiting, coughing, or pressure on a suspected open globe.

Paramedic: Perform all prior actions; lead advanced airway, analgesia, antiemetic, monitoring, Medical Control, and destination decisions when the injury or associated trauma is severe.

• Use early Medical Control and receiving-facility consultation for vision loss, suspected open globe, impaled object, orbital compartment warning signs, hydrofluoric-acid exposure, or uncertain destination.

• Do not use ketamine for analgesia in suspected penetrating or open-globe eye trauma under this protocol unless Medical Control expressly authorizes it.

Warnings

• Never apply pressure to an eye with trauma, suspected penetration, irregular pupil, hyphema, extrusion, or possible globe rupture.

• Do not remove an impaled or embedded object and do not attempt field repair of an eyelid or globe injury.

• Do not delay ordinary chemical-splash irrigation to identify the product, obtain visual acuity, locate saline, or contact Poison Help. The known/strongly suspected contaminated open-globe exception requires protection and urgent expert guidance, not irrigation into the globe.

• Do not neutralize a chemical on the eye and do not use ice, ointment, cotton swabs, magnets, or instruments.

• Do not patch either eye. For suspected open-globe injury, protect the affected eye with a rigid shield without pressure.

• Sudden painless non-traumatic vision loss is not eye trauma. Contact Medical Control and use the appropriate medical pathway; do not apply ocular pressure when trauma or globe rupture is possible.

Clinical pearls

• A normal external appearance does not exclude an open globe or intraocular foreign body after a high-velocity mechanism.

• Pediatric patients may describe vision poorly and may resist examination; use calm, age-appropriate assessment and avoid forced eyelid opening.

Special populations

Pediatric patient

• Use a parent or caregiver to help with reassurance and irrigation when safe.

• Measure and record weight with the length-based system when age/size appropriate before medication, but do not delay irrigation or transport.

• Suspect non-accidental trauma when the history, development, mechanism, or injury pattern is inconsistent; follow the Family Violence / Abuse protocol.

Related pathways

• Chemical / HazMat Exposure (p. 213): Responder safety, decontamination, agent cards, and Poison Help.

• Chemical and Electrical Burn (p. 184): Associated facial, skin, electrical, or thermal burn care.

• Head Trauma (p. 188): Associated head injury and serial neurologic assessment.

• Multiple Trauma (p. 190): Multisystem injury and trauma destination decisions.

TB-12
Abdominal / Pelvic Trauma

Revised 2026-10-30

Overview

A normal initial examination or blood pressure does not exclude major abdominal, retroperitoneal, pelvic, solid-organ, hollow-viscus, vascular, genitourinary, or pregnancy-related injury.

Prioritize hemorrhage control, pelvic stabilization when indicated, prevention of hypothermia, rapid transport to definitive or necessary stabilization care, and resuscitation that does not delay definitive hemorrhage control.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. XABCDE + Rapid Trauma Survey
EMT / AEMT / Paramedic
Control external hemorrhage - airway/ventilation - perfusion - neuro status - expose and warm

↓

2. Internal Hemorrhage?
EMT / AEMT / Paramedic
Shock - pain/tenderness - distention - seatbelt sign - flank bruising - high-risk mechanism

↓

3. Open Injury?
EMT / AEMT / Paramedic
Impaled object or evisceration - do not remove or reduce - stabilize and cover without pressure

↓

4. Pelvic Injury Suspected?
EMT / AEMT / Paramedic
Significant mechanism plus pelvic pain, deformity, shock, or unreliable examination - binder at greater trochanters

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Resuscitate En Route
AEMT / Paramedic
IV/IO - warmed isotonic crystalloid titrated to adult perfusion when blood is unavailable (AM-05) - TBI: adult SBP ≥110 mmHg - analgesia/antiemetic - reassess

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6. Trauma TXA Eligibility
Paramedic
Paramedic only: after immediate lifesaving interventions, consider IV/IO TXA for traumatic hemorrhage with clinical hemorrhagic shock within 3 hours. Adult dose: 1-2 g IV/IO: give 1 g over 10 minutes or 2 g over 20 minutes; do not exceed 100 mg/min. Verify the stocked concentration and dilution against the current medication-preparation instructions. Do not administer an automatic repeat dose; any additional dose requires a separate authorized order.

↓

7. Trauma Center + Early Alert
EMT / AEMT / Paramedic
Minimize scene time - serial exam/vitals - pregnancy and pediatric considerations - destination plan

Indications

- Blunt or penetrating injury to the abdomen, flank, pelvis, groin, perineum, lower chest, or lower back.
- Abdominal or pelvic pain, tenderness, guarding, rigidity, distention, bruising, abrasion, seatbelt sign, penetrating wound, impalement, evisceration, or unstable vital signs after trauma.
- Pelvic pain or deformity, suspected pelvic fracture, blood at the urinary meatus, hematuria, rectal/vaginal bleeding, inability to void, or neurologic deficit after trauma.
- High-energy mechanism, multisystem trauma, anticoagulant use, pregnancy, pediatric trauma, older adult trauma, or unexplained shock with possible abdominal/pelvic injury.

Assessment

Rapid abdominal and pelvic assessment

- Determine mechanism, time, energy transfer, penetration trajectory, impaled object, restraint/seatbelt use, airbag deployment, vehicle intrusion, fall height, crush, blast, anticoagulants, pregnancy possibility/gestation, and associated chest, spine, head, or extremity injury.
- Inspect and gently palpate the abdomen and flanks once for pain, tenderness, guarding, rigidity, distention, abrasions, ecchymosis, penetrating wounds, evisceration, or exposed organs. Do not repeatedly examine a painful abdomen.
- Inspect the pelvis, groin, and perineum for wounds, bruising, deformity, swelling, blood at the urinary meatus, hematuria, rectal or vaginal bleeding, or neurologic deficit while preserving privacy and preventing heat loss.
- Assess the pelvis by history, mechanism, visual findings, and gentle palpation only when needed. Do not repeatedly rock or compress the iliac crests to test stability.
- Trend mental status, pulse, skin, capillary refill, blood pressure, pulse pressure, respiratory status, and shock index when useful. A normal initial pressure does not exclude compensated hemorrhagic shock.

High-risk findings

- Hypotension, altered mental status, tachycardia, weak pulses, cool or mottled skin, narrowing pulse pressure, or worsening shock.
- Penetrating torso injury, evisceration, impalement, expanding abdomen/flank/groin hematoma, pelvic deformity, significant pelvic pain with high-energy mechanism, or suspected open pelvic fracture.
- Seatbelt sign, lower-rib injury, severe pain, guarding/rigidity, distention, hematemesis, hematuria, blood at the meatus, bloody stool, vaginal bleeding, or inability to void.
- Pregnancy, pediatric or older patient, anticoagulant/antiplatelet use, bleeding disorder, or unreliable examination because of altered mental status, distracting injury, intoxication, or communication barrier.

Treatment

1. Perform XABCDE, control compressible external hemorrhage, support oxygenation and ventilation as indicated, expose only as needed, and actively prevent hypothermia.
2. Treat suspected abdominal or pelvic hemorrhage as time-critical trauma. Minimize scene time, begin destination coordination early, and complete nonessential assessment or procedures during transport.
3. Suspected pelvic fracture: apply the agency-approved commercial pelvic binder centered over the greater trochanters, not the abdomen, waist, or iliac crests. Secure the legs together when appropriate, minimize movement, and reassess distal neurovascular status.
4. Do not repeatedly manipulate, rock, or compress the pelvis and do not loosen or remove a correctly positioned binder in the field unless a separately approved protocol or Medical Control requires it.
5. Evisceration: do not replace organs into the abdomen. Cover loosely with sterile saline-moistened dressings and an occlusive layer when available, protect from heat loss, and avoid direct pressure.
6. Impaled object: do not remove. Stabilize in place with bulky dressings without pressure on exposed organs. Modify or remove only when essential for an immediate lifesaving intervention or safe transport and under rescue/Medical Control coordination.
7. Control external or junctional hemorrhage with direct pressure, wound packing, hemostatic gauze, or tourniquet when anatomically appropriate. Do not blindly pack material into the abdominal cavity.
8. Provide authorized analgesia and antiemetic treatment without masking deterioration or delaying transport. Keep the patient NPO and reassess frequently.
9. Apply the trauma field-triage and destination plan: high-risk patients to the highest-level appropriate trauma center within regional constraints; favor pediatric-capable trauma care for children. Use a closer stabilization hospital only for an immediate critical need that cannot safely be managed during longer transport, under the destination plan/Medical Control. Notify early, document the reason, and activate the predefined transfer process. Do not delay ground movement solely while awaiting aircraft.
10. Document mechanism, examination, serial vital signs, suspected bleeding source, pelvic-binder indication/position/time, distal neurovascular findings, open-injury care, medications/fluids/blood, response, pregnancy status, consultation, and destination rationale.

Medication orders

Analgesic

Use the current Claiborne County pain-management protocol and provider-level standing orders.

- Titrate to comfort while preserving ventilation, perfusion, mental-status reassessment, and rapid transport.

Antiemetic

Use the current age-appropriate Claiborne County standing order when indicated.

- Vomiting increases aspiration risk and can complicate evisceration, impalement, and pain control.

Warmed isotonic crystalloid

Adult hemorrhagic shock when blood is unavailable: titrate to perfusion under AM-05; avoid unnecessary large-volume crystalloid. Suspected TBI requires adult SBP at least 110 mmHg and normal saline when crystalloid is required.

- Reassess perfusion, respiratory status, and blood pressure frequently and after treatment. Definitive hemorrhage control takes priority; use blood products only within an authorized, supported transfusion program.

Tranexamic acid (TXA)

Paramedic only: consider IV/IO TXA after immediate lifesaving interventions for traumatic hemorrhage with clinical hemorrhagic shock within 3 hours of injury. Adult dose: 1-2 g IV/IO: give 1 g over 10 minutes or 2 g over 20 minutes; do not exceed 100 mg/min. Verify the stocked concentration and dilution against the current medication-preparation instructions. Do not administer an automatic repeat dose; any additional dose requires a separate authorized order.

- Do not delay transport or definitive hemorrhage control for TXA administration.
- Record injury time and any TXA dose/time; notify the receiving team to prevent duplicate dosing. Do not infer a systemic trauma regimen from topical bleeding indications.
- Pediatric trauma TXA: local dose reference 15 mg/kg, Paramedic-only. Before administration, confirm pediatric eligibility, maximum dose, administration rate, and preparation with Medical Control within provider scope. Do not extrapolate the adult dose or infusion rate to children.

Blood products

Use only within an authorized, supported transfusion program with required training, monitoring, and destination coordination.

Provider-level care

Provider-Level Actions Recognize concealed hemorrhage, stabilize the pelvis when indicated, protect open injuries, and move rapidly toward definitive care.

EMT: Perform XABCDE, control external hemorrhage, support airway and ventilation, prevent hypothermia, and rapidly assess the abdomen, flanks, pelvis, groin, and perineum for injury.

- Do not repeatedly compress or rock the pelvis. When pelvic fracture with significant mechanism, pain, deformity, or shock is suspected, apply the agency-approved pelvic binder centered over the greater trochanters and reassess distal circulation, sensation, and movement.
- Do not remove an impaled object or reduce eviscerated organs. Stabilize protruding objects without pressure and cover evisceration with a sterile saline-moistened dressing protected from heat loss.
- Package for rapid trauma transport, keep the patient NPO, and provide early notification for shock, penetrating injury, evisceration, impalement, pregnancy, or suspected unstable pelvic injury.

AEMT: Perform all EMT actions plus IV/IO access. For adult hemorrhagic shock when blood is unavailable, follow AM-05: titrate warmed isotonic crystalloid to clinical perfusion and avoid unnecessary large-volume crystalloid. Use age-appropriate analgesia or antiemetic care within current standing orders.

- Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing, and medical-director standing orders; training alone does not grant permission.
- Do not delay transport for vascular access. Trend mental status, pulse quality, skin findings, blood pressure, pulse pressure, and response to resuscitation.
- Do not delay transport for vascular access or fluid. Without suspected TBI, use the AM-05 endpoint of a palpable radial pulse, adequate mental status, or SBP approximately 80-90 mmHg. With suspected TBI, prevent hypotension and target adult SBP at least 110 mmHg, including age 70; do not use permissive hypotension. Use normal saline when crystalloid is required for suspected TBI and reassess blood pressure frequently and after treatment.

Paramedic: Perform all prior actions; lead advanced airway/ventilation, cardiac and EtCO2 monitoring,

shock resuscitation, Medical Control, trauma-center destination, and air-medical decisions.

• Use blood products only within an authorized, supported transfusion program with required training, monitoring, and destination coordination. Other advanced hemorrhage interventions remain subject to current standing orders.

• For a pregnant patient beyond approximately 20 weeks, relieve aortocaval compression with manual left uterine displacement or left lateral tilt when compatible with spinal and trauma care.

Warnings

- Do not delay transport for a detailed abdominal examination, repeated pelvic assessment, IV attempts, pain medication, splinting, or procedures that can be completed en route.
- Do not repeatedly rock or compress the pelvis. A normal or minimally painful pelvic examination does not exclude fracture or retroperitoneal hemorrhage.
- Place a pelvic binder over the greater trochanters, not the abdomen, waist, or iliac crests. A high binder may fail to reduce pelvic volume and can obstruct abdominal assessment.
- Do not remove an impaled object, reduce eviscerated organs, apply direct pressure to exposed viscera, or blindly pack the abdominal cavity.
- Do not allow exposed organs, wet dressings, cold fluids, or unnecessary exposure to cause hypothermia.
- Blood at the urinary meatus or suspected urethral injury is a warning for the receiving team; do not insert a urinary catheter in the field.
- Trauma TXA is Paramedic-only, not an AEMT intervention. Adult dose: 1-2 g IV/IO: give 1 g over 10 minutes or 2 g over 20 minutes; do not exceed 100 mg/min. Verify the stocked concentration and dilution against the current medication-preparation instructions. Do not administer an automatic repeat dose; any additional dose requires a separate authorized order. Pediatric administration requires confirmation of the patient-specific maximum and rate with Medical Control within provider scope. Do not infer a trauma regimen from topical bleeding instructions.
- Maternal resuscitation is the best initial fetal resuscitation. Do not delay trauma-center care to obtain fetal assessment unavailable to the crew.

Clinical pearls

- The abdomen, retroperitoneum, and pelvis can conceal life-threatening blood loss before hypotension appears.
- Pelvic pain plus a high-energy mechanism or shock is enough to justify early stabilization under the local binder policy; repeated manual testing adds risk without improving field care.
- A seatbelt sign, lower-rib injury, flank bruising, hematuria, or unexplained shock should increase suspicion even when abdominal tenderness is mild.
- Pediatric patients may maintain blood pressure until late shock. Older adults and patients taking anticoagulants may deteriorate after an initially reassuring examination.
- A correctly positioned binder and rapid transport are complementary; neither should delay the other.

Special populations

Pregnancy

- Assess gestational age, abdominal pain, contractions, vaginal bleeding/fluid, fetal movement when known, and maternal shock while prioritizing maternal XABCDE.
- Beyond approximately 20 weeks, use manual left uterine displacement or left lateral tilt when compatible with spinal and trauma care.
- Transport significant trauma to a destination capable of both maternal trauma resuscitation and obstetric/neonatal care when feasible without delaying definitive trauma treatment.

Pediatric patient

- Use the length-based system for equipment and medication calculations when appropriate, but do not delay hemorrhage control or transport.
- Use age-appropriate signs of shock; hypotension is a late finding.
- For pediatric traumatic hemorrhage, obtain Medical Control guidance and use only the authorized pediatric trauma resuscitation order; do not apply medical or septic-shock fluid targets indiscriminately.
- Use only an appropriately sized, agency-approved pelvic stabilization device and pediatric application procedure.

Related pathways

- AM-05 Hypotension / Shock (p. 129): Adult hemorrhagic-shock fluid strategy, reassessment, and the higher suspected-TBI perfusion target.

- TB-05 Head Trauma (p. 188): Suspected TBI: prevent hypotension and use the higher adult SBP target.
- Multiple Trauma (p. 190): XABCDE, hemorrhage control, shock care, and trauma destination.
- Selective Spinal Immobilization (p. 193): Spinal motion restriction based on mechanism and examination.
- Traumatic Arrest (p. 196): Use when abdominal/pelvic trauma progresses to cardiac arrest.
- OB / GYN Emergency (p. 140): Pregnancy-related assessment and obstetric coordination.

TE-01
Bites and Envenomations

Revised 2026-10-28

Overview

Do not endanger responders to capture or identify an animal; use a description or photograph when safely available.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Scene Safe?
EMT / AEMT / Paramedic
Stage or request resources until animal, insect, or environmental hazard is controlled.

↓

2. ABCs / Anaphylaxis / Shock
EMT / AEMT / Paramedic
Treat immediate threats and route to age-appropriate allergy or shock protocol.

↓

3. Wound + Immobilize
EMT / AEMT / Paramedic
Clean/cover the wound when it will not delay care, remove constricting items, and immobilize the extremity in a neutral position of comfort.

↓

4. Snake Bite?
EMT / AEMT / Paramedic
Mark swelling/redness with time; keep the patient calm, keep the limb at heart level, call Poison Help, and transport promptly. Do not apply ice, tourniquet, suction, or incision.

↓

5. Spider / Sting?
EMT / AEMT / Paramedic
For a visible stinger, scrape it away; wash the area and use a cold pack for 10-20 minutes. Treat anaphylaxis immediately when present.

Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Consult + Transport
EMT / AEMT / Paramedic
For suspected envenomation, unknown toxic bite/sting, or concerning symptoms, call Poison Help (1-800-222-1222). For animal bites/scratches, follow the local public-health/animal-control reporting process and arrange evaluation for rabies risk assessment.

Indications

- Animal/human bite, insect sting, or suspected snake/spider envenomation.

Provider-level care

Provider-Level Actions Protect rescuers, stop exposure when safe, support ABCs, and involve Poison Control or Medical Control early.

EMT: Provide wound care when it will not delay care, remove rings/bands, immobilize the extremity in a neutral position of comfort, mark progression, and treat anaphylaxis with epinephrine when indicated.

- For a visible insect stinger, scrape it away; wash the area and use a cold pack for 10-20 minutes.
- For suspected envenomation, unknown toxic bite/sting, or concerning symptoms, contact Poison Help (1-800-222-1222). For possible rabies exposure, follow the local public-health/animal-control reporting process.

AEMT: Perform all EMT care plus IV/IO access, fluids, and authorized pain/anaphylaxis treatment.

Paramedic: Perform all prior care plus cardiac monitoring for systemic illness and treat severe venom-associated muscle spasm or seizure under the applicable medication and seizure protocol.

Warnings

- Do not capture or transport a snake for identification.
- Do not apply ice to a snake bite.
- Do not delay transport for snake-bite interventions or identification.

Related pathways

- Medication Reference (p. 218): Open for approved midazolam indications, dosing, and monitoring.

TE-02
Carbon Monoxide / Cyanide

Revised 2026-10-28

Overview

Consider carbon monoxide and cyanide in any combustion exposure; conventional pulse oximetry and environmental readings do not exclude poisoning.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Scene Safety + Decontaminate
EMT / AEMT / Paramedic
Stage or request PPE/HAZMAT support, remove the patient from exposure, and decontaminate liquid or industrial cyanide contamination before ambulance transport when indicated.

↓

2. 100% Oxygen
EMT / AEMT / Paramedic
Give regardless of standard pulse-ox reading; support airway and ventilation.

↓

3. Assess Toxicity
EMT / AEMT / Paramedic
Assess for altered mental status, dyspnea, weakness, nausea/vomiting, syncope, seizure, chest pain, shock, neurologic deficit, or respiratory failure.

↓

4. Monitor + Access
AEMT / Paramedic
Obtain glucose, 12-lead, IV/IO, and cardiac monitoring as indicated. Use pulse CO-oximetry when available, but do not delay care or rely on one field SpCO value to rule out poisoning.

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. High Cyanide Suspicion?
EMT / AEMT / Paramedic
Enclosed-space fire, known industrial cyanide exposure, or ingestion with severe AMS, seizure, hypotension/collapse, or respiratory failure.

↓

6. Consult + Early Alert
EMT / AEMT / Paramedic
Call Poison Help (1-800-222-1222) and Medical Control for severe exposure; notify the receiving facility early. Claiborne ground units do not stock a cyanide antidote.

Indications

- Smoke/combustion exposure, industrial cyanide exposure, cyanide ingestion, or compatible unexplained illness.

Provider-level care

Provider-Level Actions Protect rescuers, stop exposure when safe, support ABCs, and involve Poison Control or Medical Control early.

EMT: Remove from exposure, provide 100% oxygen and ventilation support, obtain glucose, and rapidly transport.

AEMT: Perform all EMT care plus IV/IO access, 12-lead acquisition, and treatment of glucose abnormality or shock.

Paramedic: Perform all prior care plus cardiac and pulse CO-oximetry monitoring when available, Poison Help/Medical Control consultation, and early destination notification for severe exposure.

Warnings

- Provide 100% oxygen regardless of conventional pulse-oximetry reading; standard SpO_2 is unreliable in carbon monoxide exposure.
- Do not place a patient with liquid or industrial cyanide contamination in the ambulance before indicated decontamination and HAZMAT direction.
- For severe carbon monoxide exposure-loss of consciousness, neurologic deficit, dysrhythmia/ischemia, severe acidosis, or pregnancy-notify the receiving facility early for hyperbaric-consultation consideration.
- A cyanide antidote is not stocked on Claiborne ground units. Do not delay supportive care, Poison

Help/Medical Control consultation, or transport while seeking an unavailable antidote.

- Pregnant patients, children, and older adults may become symptomatic at lower CO levels.

TE-03
Drowning / Submersion Injury

Revised 2026-10-30

Overview

Drowning is respiratory impairment after submersion or immersion; reversal of hypoxia is the priority.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Safe Water Rescue
EMT / AEMT / Paramedic
Only trained/equipped rescuers enter hazardous water. Trained responders may provide in-water breaths only when safe and without delaying removal.

↓

2. Two Initial Breaths
EMT / AEMT / Paramedic
For apnea or unresponsiveness, provide two initial breaths/ventilations by the first effective means available; do not delay for routine suctioning.

↓

3. Pulse?
EMT / AEMT / Paramedic
No: begin CPR with breaths and compressions, then use the adult or pediatric arrest pathway.
Yes: continue oxygenation and airway support.

↓

4. Trauma Indication?
EMT / AEMT / Paramedic
Use spinal-motion restriction only when indicated and never at the expense of ventilation/CPR.

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Dry + Warm + Monitor
EMT / AEMT / Paramedic
Remove wet clothing; assess glucose, ECG, hypothermia, and pulmonary status.

↓

6. Transport / Observe
EMT / AEMT / Paramedic
Encourage evaluation, especially for cough, foam, dyspnea, abnormal sounds, or hypoxia.

Indications

- Any submersion/immersion event with respiratory symptoms, altered mental status, apnea, or arrest.

Provider-level care

Provider-Level Actions Protect rescuers, stop exposure when safe, support ABCs, and involve Poison Control or Medical Control early.

EMT: Prioritize rescue safety, give two initial breaths/ventilations when indicated, ventilate through foam without routine suctioning, begin CPR with breaths and compressions if pulseless, and dry/warm the patient.

AEMT: Perform all EMT care plus IV/IO access and treatment of glucose abnormality, shock, or associated hypothermia.

Paramedic: Perform all prior care plus advanced airway/cardiac monitoring and consultation for dive-related barotrauma or decompression illness.

Warnings

- Do not waste time suctioning airway foam; ventilate through it and suction water/vomit only when needed.
- Spinal-motion restriction should never delay oxygenation, ventilation, or CPR.

Clinical pearls

- Do not use a single submersion-time cutoff alone to determine whether to begin or continue resuscitation; use the applicable cardiac-arrest and termination pathway with the full clinical context.

TE-04
Hyperthermia

Revised 2026-10-30

Overview

Rapid cooling is initiated concurrently with standard resuscitation and takes precedence over transport in exertional heat stroke when effective immersion is available.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Remove From Heat
EMT / AEMT / Paramedic
Move to cool area, remove tight/excess clothing, obtain glucose; temperature must not delay care.

↓

2. Classify Severity
EMT / AEMT / Paramedic
Cramps • exhaustion • suspected heat stroke (heat exposure plus altered mental status; a temperature below 104°F does not exclude it).

↓

3. Heat Stroke?
EMT / AEMT / Paramedic
Begin immediate active cooling concurrent with resuscitation. Ice-water immersion is preferred; use tepid-water immersion or tarp-assisted ice-water cooling if unavailable.

↓

4. Cool to below 102°F
EMT / AEMT / Paramedic
Use rectal core temperature when feasible, without interrupting care. Stop active cooling once core temperature is below 102°F (39°C).

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Support Perfusion
AEMT / Paramedic
For hypotension, dehydration, or poor perfusion, give isotonic crystalloid in reassessed aliquots: adult 500 mL; pediatric 20 mL/kg. Do not use a routine fixed maximum.

↓

6. Reassess + Transport
EMT / AEMT / Paramedic
Airway, ECG, neurologic status, BP, and core temperature; notify destination.

Indications

- Heat cramps, heat exhaustion, heat stroke, or other hyperthermic emergency.

Medication orders

Isotonic crystalloid

Adult 500 mL IV/IO aliquots; pediatric 20 mL/kg IV/IO aliquots; repeat only after reassessment for hypotension, dehydration, or poor perfusion.

Provider-level care

Provider-Level Actions Protect rescuers, stop exposure when safe, support ABCs, and involve Poison Control or Medical Control early.

EMT: Move to a cool environment, remove excess clothing, check glucose, begin active cooling, and give oral fluid only for mild illness with intact mentation/swallowing.

AEMT: Perform all EMT care plus IV/IO isotonic crystalloid for hypotension, dehydration, or poor perfusion, with reassessment after each aliquot.

Paramedic: Perform all prior care plus cardiac monitoring and protocol treatment of seizures, dysrhythmia, shock, or airway failure.

Warnings

- Do not delay cooling to obtain a temperature; heat exposure with altered mental status is heat stroke until proven otherwise.
- Use rectal core temperature when feasible, but do not interrupt active cooling, airway management, or transport to obtain it.
- Do not overcool; stop active cooling once core temperature is below 102°F (39°C).

- Antipyretics and dantrolene are not treatments for environmental or exertional heat stroke; use active external cooling.

TE-05
Hypothermia / Frostbite

Revised 2026-10-30

Overview

Remove wet clothing, protect from further heat loss, and handle severe hypothermia carefully while assessing for signs of life for at least 60 seconds.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Remove Wet Clothing
EMT / AEMT / Paramedic
Move to warm shelter, dry patient, insulate, and measure temperature without delaying treatment.

↓

2. Localized or Systemic?
EMT / AEMT / Paramedic
Frostbite: no rubbing; if refreezing can be prevented, rapidly rewarm in 37-40°C (99-104°F) water for 20-30 minutes. Systemic: airway, glucose, warming, ECG.

↓

3. Pulse Check ≥60 Seconds
EMT / AEMT / Paramedic
If no pulse, begin the age-appropriate arrest pathway with hypothermia modifications and concurrent rewarming.

↓

4. Active Core Warming
EMT / AEMT / Paramedic
Warm blankets/packs to axillae and groin; protect skin and keep chemical heat away from frostbitten tissue; warmed fluids when available.

Continue at step 5 in the next column/page

Quick flow continued at step 5

5. Support Perfusion
AEMT / Paramedic
For hypotension or poor perfusion, give warmed isotonic crystalloid in reassessed aliquots: adult 500 mL; pediatric 20 mL/kg. Do not use a routine fixed maximum.

↓

6. Hypothermic Arrest / Transport
EMT / AEMT / Paramedic
Below 30°C (86°F): full resuscitation with rewarming. For VF/pVT, defibrillate once; if unsuccessful, defer further shocks and epinephrine until ≥30°C. Contact Medical Control early and consider ECLS-capable destination when feasible.

Indications

- Systemic hypothermia or localized cold injury/frostbite.

Medication orders

Warmed isotonic crystalloid

Adult 500 mL IV/IO aliquots; pediatric 20 mL/kg IV/IO aliquots; repeat only after reassessment for hypotension or poor perfusion.

Provider-level care

Provider-Level Actions Protect rescuers, stop exposure when safe, support ABCs, and involve Poison Control or Medical Control early.

EMT: Remove wet clothing, dry/insulate, apply protected heat packs to axillae/groin, check glucose, and provide gentle handling.

AEMT: Perform all EMT care plus IV/IO access and warmed isotonic crystalloid when available for hypotension or poor perfusion, with reassessment after each aliquot.

Paramedic: Perform all prior care plus cardiac monitoring and the hypothermic-arrest sequence below 30°C (86°F): full resuscitation with rewarming; one VF/pVT defibrillation attempt, then defer further shocks and epinephrine until ≥30°C.

Warnings

- Do not rub or massage frostbitten tissue, debride frostbite blisters, or apply direct chemical heat to frostbitten tissue.

- Do not rewarm frostbite if refreezing is possible. When sustained thaw can be assured, use 37-40°C (99-104°F) water for 20-30 minutes.
- Below 30°C (86°F), provide full resuscitation with rewarming. For VF/pVT, defibrillate once; if unsuccessful, defer further shocks and epinephrine until ≥30°C.

Clinical pearls

- Do not use a fixed temperature alone to stop resuscitation. Continue unless there is obvious lethal injury or the patient is frozen solid; seek early Medical Control and consider ECLS-capable destination when feasible.

TE-07
Overdose / Toxic Ingestion

Revised 2026-10-30

Overview

Stabilize ventilation and perfusion first, identify time/substance/amount/co-ingestants, and consult Poison Control.

Naloxone is titrated to adequate ventilation-not restoration of consciousness-and toxin-specific treatment follows the clinical toxidrome and ECG.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. ABCs + Exposure History
EMT / AEMT / Paramedic
Substance, route, amount, time, co-ingestants, intent; bring containers when safe. For skin/eye exposures, remove contaminated clothing and irrigate; for inhalation, move to fresh air only when the scene is safe.

↓

2. Inadequate Ventilation?
EMT / AEMT / Paramedic
Ventilate first; suspected opioid: naloxone titrated to breathing.

Continue at step 3 in the next column/page

Quick flow continued at step 3

3. Shock / AMS / Seizure?
EMT / AEMT / Paramedic
Treat by age-appropriate shock, diabetic, AMS, behavioral, and seizure pathways.

↓

4. Poison Control
EMT / AEMT / Paramedic
Call 1-800-222-1222; obtain 12-lead, IV/IO, and cardiac monitor as indicated.

↓

5. Specific Toxin?
EMT / AEMT / Paramedic
APAP/salicylate • beta/CCB • CO/cyanide • organophosphate • TCA. Use the current medication reference and Poison Help/Medical Control; do not use unapproved antidote doses.

↓

6. Treat + Disposition
EMT / AEMT / Paramedic
Patients treated with naloxone require healthcare observation for recurrent respiratory depression. Other low-risk exposures may follow Poison Help home-care advice only when agency non-transport criteria are met. Intentional, pediatric, pregnant, polysubstance, or symptomatic exposures require direct Poison Help consultation.

Indications

- Known or suspected overdose, poisoning, or toxic ingestion/exposure.

Provider-level care

Provider-Level Actions Protect rescuers, stop exposure when safe, support ABCs, and involve Poison Control or Medical Control early.

EMT: Support ventilation, administer naloxone for suspected opioid-induced respiratory depression only per the current Medication Reference, check glucose, and contact Poison Help.

- If activated charcoal is stocked and authorized, use only after Poison Control or Medical Control recommendation and only when airway protection and swallowing are intact.

AEMT: Perform all EMT care plus IV/IO access, broader-route naloxone per the current Medication Reference, reassessed fluids, and toxin-specific treatment only with Poison Help or Medical Control guidance.

Paramedic: Perform all prior care plus cardiac monitoring, pacing when indicated, and toxidrome-specific advanced care using the current Medication Reference with Poison Help or Medical Control guidance.

Warnings

- Ventilation has priority over naloxone; do not dose solely to restore consciousness. Use the Medication Reference for approved naloxone dosing.
- Charcoal is not routine and should not be given to a patient unable to protect the airway; use only when stocked, authorized, and recommended by Poison Control or Medical Control.
- Do not clear a patient only because they awaken after naloxone; recurrent respiratory depression can occur and requires healthcare observation.
- Do not rely only on history in a suspected intentional ingestion; assess for hidden medications and weapons.

Related pathways

- Call Poison Help - tel:+18002221222: National Poison Control line
- Medication Reference (p. 218): Open for approved naloxone and antidote dosing.
- TE-08 WMD / Nerve Agent (p. 212): Open for scene safety, decontamination, recognition, and escalation of suspected cholinergic exposure.

TE-08 WMD / Nerve-Agent Exposure

Revised 2026-10-30

Overview

Claiborne ground units do not carry a CHEMPACK, Mark I, or equivalent atropine/pralidoxime field kit. This protocol does not authorize routine field autoinjector or antidote administration.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Do Not Enter the Hot Zone
EMT / AEMT / Paramedic
Stage uphill/upwind when possible. Request HazMat, fire, law enforcement, and incident command. Only trained and properly protected personnel may enter the hazard area.

↓

2. Remove + Decontaminate
EMT / AEMT / Paramedic
Protected personnel remove the patient from the source, remove and bag contaminated clothing, brush dry material, and irrigate as directed. Prevent secondary contamination of responders, the ambulance, and the receiving facility.

↓

3. Airway / Ventilation / Seizure
EMT / AEMT / Paramedic
Suction secretions, provide oxygen and BVM ventilation, and use the applicable seizure pathway. Do not allow antidote retrieval to delay resuscitation.

↓

4. Cholinergic Syndrome?
EMT / AEMT / Paramedic
Miosis, salivation, lacrimation, sweating, bronchorrhea, bronchospasm, vomiting, diarrhea, urination, bradycardia, fasciculations, weakness, apnea, or seizure after a compatible exposure.

↓

5. Escalate
EMT / AEMT / Paramedic
Call Poison Help (1-800-222-1222) and Medical Control. Request HazMat/mutual-aid support and early receiving-facility coordination. Claiborne ground units have no CHEMPACK/Mark I or equivalent atropine/pralidoxime kit.

Continue at step 6 in the next column/page

Quick flow continued at step 6

6. Transport + Handoff
EMT / AEMT / Paramedic
Transport after indicated decontamination. Communicate agent, route, scene zone, PPE, symptoms, decontamination, airway/ventilatory care, and Poison Help/Medical Control guidance.

Indications

- Known or suspected nerve-agent release, intentional chemical event, or organophosphate/carbamate exposure with cholinergic findings.

Provider-level care

Provider-Level Actions Protect rescuers, stop exposure when safe, support ABCs, and involve Poison Control or Medical Control early.

EMT: Remain outside the hazard until assigned PPE and training level are appropriate; support removal, decontamination, suction, oxygenation, BVM ventilation, glucose assessment, and rapid transport.

- Recognize the cholinergic syndrome and notify Poison Help, Medical Control, HazMat, and the receiving facility early.

AEMT: Perform all EMT care plus IV/IO access, authorized fluids, reassessment, and preparation for rapid respiratory deterioration. Do not administer unstocked atropine/pralidoxime or use non-existent autoinjectors under this protocol.

Paramedic: Perform all prior care plus advanced airway/ventilation, cardiac monitoring, and seizure or shock care under the applicable approved pathway. Coordinate agent-specific therapy with Medical Control and the receiving facility; no CHEMPACK/Mark I field antidote pathway is authorized.

Warnings

- Do not enter a hot zone, vapor cloud, confined space, or oxygen-deficient atmosphere without required training, monitoring, PPE, and respiratory protection.
- Claiborne ground units do not carry a CHEMPACK, Mark I, or equivalent atropine/pralidoxime kit. Do not use external kit dosing or administer an unstocked antidote.
- Secondary contamination can injure responders and contaminate the ambulance or receiving facility; decontaminate before entry into the clean treatment area.
- Airway suction, ventilation, seizure care, decontamination, Poison Help, Medical Control, and rapid transport take priority over unavailable antidote retrieval.

Clinical pearls

- Atropine and pralidoxime are recognized antidotes for life-threatening cholinergic poisoning, but this Claiborne ground protocol does not create an unstocked field-antidote pathway. Future adoption requires documented kit availability, training, and standing orders.

Related pathways

- Call Poison Help - tel:+18002221222: National Poison Control line
- Chemical / HazMat Exposure (p. 213): Open for zones, PPE, decontamination, and agent cards.
- Overdose / Toxic Ingestion (p. 211): Open for airway, ventilation, and poison-exposure support.

TE-09
Chemical / HazMat Exposure

Revised October 30, 2026

Overview

Use for known or suspected hazardous-chemical exposure involving inhalation, skin or eye contact, ingestion, injection, or an unidentified substance.

Responder safety and contamination control come before patient contact. Do not enter a hot zone or confined space without the required training, monitoring, PPE, respiratory protection, and HazMat coordination.

Stop exposure, perform appropriate decontamination, support airway and ventilation, identify the substance and route when safe, and involve Poison Help and Medical Control early.

Quick flow

Apply this pathway together with the full provider-specific instructions and clinical precautions below.

1. Establish Zones + PPE
EMT / AEMT / Paramedic
Hot • warm • cold zones • identify wind/terrain/confined-space risk • request HazMat

Continue at step 2 in the next column/page

Quick flow continued at step 2

2. Safe Rescue Only
EMT / AEMT / Paramedic
Removal from the source by properly trained and protected personnel; prevent additional victims

↓

3. Decontaminate
EMT / AEMT / Paramedic
Remove clothing/jewelry • brush dry agents • blot liquids • irrigate when indicated • contain runoff

↓

4. ABCs + Immediate Threats
EMT / AEMT / Paramedic
Airway edema • bronchospasm • respiratory failure • shock • dysrhythmia • seizure • burns

↓

5. Identify Agent + Route
EMT / AEMT / Paramedic
SDS/ERG • label/placard • inhaled, skin, eye, ingestion • duration • concentration • symptoms

↓

6. Poison Help + Medical Control
EMT / AEMT / Paramedic
Call 1-800-222-1222 • obtain agent-specific guidance • verify antidote and destination

↓

7. Agent Card + Transport
EMT / AEMT / Paramedic
Use chlorine, ammonia, organophosphate, H2S, HF, or unknown/corrosive pathway

Indications

- Known or suspected exposure to an industrial, agricultural, household, transportation, combustion, laboratory, or intentionally released chemical.
- Multiple patients with similar unexplained respiratory, neurologic, ocular, gastrointestinal, or dermatologic findings.
- Contamination or symptoms after a spill, vapor release, confined-space event, pesticide application, fertilizer or pool-chemical incident, fumigation, sewer or manure-pit exposure, or unknown substance.

Assessment

General chemical-exposure assessment

- Confirm the scene has been evaluated for atmosphere, oxygen concentration, flammability, explosion, wind direction, vapor density, confined-space hazards, and the need for specialized chemical PPE or SCBA.
- Determine the agent or product, physical form, concentration, route, duration, time since exposure, decontamination already completed, number of exposed people, symptoms, and coexisting trauma or thermal injury.
- Assess airway voice and patency, work of breathing, lung sounds, oxygenation, ventilation, mental status, pupils, secretions, skin moisture and color, temperature, ECG, perfusion, pain, and exposed eyes and skin.
- Do not rely on pulse oximetry alone when a cellular asphyxiant, carbon monoxide, cyanide, hydrogen sulfide, or methemoglobinemia is possible.
- Repeatedly reassess because airway edema, pulmonary edema, chemical pneumonitis, dysrhythmia, seizures, electrolyte toxicity, and neurologic symptoms may be delayed.

Quick card: Chlorine / chloramine gas

- Common sources include pool products, industrial chlorine, and mixing bleach with acids or ammonia. Liquid-soaked clothing can continue to off-gas and contaminate responders.
- Expect burning eyes and throat, cough, chest tightness, wheeze, stridor, hypoxemia, and possible delayed pulmonary edema. Liquefied chlorine can cause frostbite.
- Move to fresh air through protected rescue, remove contaminated clothing and jewelry, wash exposed skin with mild soap and water, irrigate eyes, support ventilation, and treat bronchospasm.
- Do not routinely neutralize chemicals on the patient. Nebulized sodium bicarbonate for chlorine is permitted only if specifically adopted by Claiborne County, available in an approved preparation, and ordered by the finalized protocol or Medical Control.

Quick card: Ammonia

- Common sources include fertilizer, agricultural operations, industrial refrigeration, and concentrated cleaning products. Ammonia is highly water soluble and causes alkaline injury to moist tissues.
- Expect severe eye and upper-airway irritation, cough, bronchospasm, stridor, laryngeal edema, chemical burns, frostbite from liquefied product, hypotension, and pulmonary edema.

- Protected rescue and rapid removal from the source are critical. Remove clothing and jewelry, brush/blot visible material, wash skin, irrigate eyes continuously, and prepare early for a difficult or progressively swollen airway.

Quick card: Organophosphate / carbamate pesticide

- Consider after agricultural or pesticide exposure with miosis, sweating, salivation, lacrimation, vomiting, diarrhea, urination, fasciculations, weakness, bradycardia, bronchospasm, or profuse pulmonary secretions.
- Avoid secondary dermal contamination. Use appropriate chemical PPE, remove clothing, bag contaminated items, brush dry agents, wash skin and hair with soap and water, and protect the airway during decontamination.
- Airway suction, oxygenation, ventilation, and control of secretions are immediate priorities. Use TE-08 WMD / Nerve Agent for scene safety, decontamination, cholinergic recognition, seizure care, and escalation.
- Claiborne ground units do not carry a CHEMPACK, Mark I, or equivalent atropine/pralidoxime kit. Do not use legacy kit counts, autoinjector directions, or unstocked antidote doses.

Quick card: Hydrogen sulfide

- Common sources include sewers, septic systems, manure pits, petroleum operations, and decaying organic matter. It is heavier than air and can rapidly incapacitate rescuers in low or confined spaces.
- The rotten-egg odor is unreliable because olfactory fatigue occurs. Assume a dangerous atmosphere when unexplained collapse occurs near a confined space or organic-waste source.
- Expect eye irritation, cough, dyspnea, pulmonary edema, headache, confusion, seizure, coma, dysrhythmia, circulatory collapse, and sudden respiratory arrest.
- Only trained personnel with appropriate respiratory protection may remove the patient. Provide immediate high-concentration oxygen and ventilation, cardiac monitoring, seizure care, and Poison Help/Medical Control consultation. Use antidotal nitrite therapy only if specifically approved.

Quick card: Hydrofluoric acid / hydrogen fluoride

- Sources include glass etching, rust removal, industrial processing, some wheel or metal cleaners, and laboratory products. Pain and systemic toxicity may be delayed, especially with dilute solutions.
- Small-appearing burns can cause deep tissue injury, hypocalcemia, hypomagnesemia, hyperkalemia, prolonged QT, ventricular dysrhythmia, shock, or death. Severe pain out of proportion is a major warning.
- Use chemical PPE, remove contaminated clothing and jewelry, brush dry material, irrigate immediately and copiously, establish access in an unexposed extremity when feasible, and begin continuous ECG monitoring.
- After irrigation, apply or administer calcium gluconate only through the finalized Claiborne HF kit/protocol and Poison Help or Medical Control guidance. Eye, inhalation, ingestion, hand, face, genital, circumferential, large-area, or symptomatic exposures require urgent specialty evaluation.

Quick card: Unknown chemical / corrosive

- Treat an unidentified substance as hazardous until identified. Look for placards, container shape, labels, SDS, ERG information, witness reports, occupational setting, and toxidrome without touching contaminated materials.
- For dry material, brush away visible particles before irrigation unless authoritative agent-specific guidance directs otherwise. For most liquid skin or eye exposures, remove contaminated items and irrigate with copious tepid water.
- Do not induce vomiting, give oral neutralizers, or attempt acid-base neutralization on the body. Do not place water-reactive material in contact with water until the agent is identified and HazMat/Poison Help gives direction.
- Transport symptomatic patients and those with significant eye injury, airway symptoms, abnormal vital signs, altered mental status, intentional exposure, concentrated corrosive ingestion, uncertain decontamination, or a substance with known delayed toxicity.

Treatment

1. Establish command, zones, responder accountability, decontamination, and a clean treatment area. Do not bring a contaminated patient, clothing, equipment, or packaging into the ambulance or emergency department.
2. Remove the patient from exposure only when the rescue can be performed by personnel with the required training and PPE. Never enter a confined space for rescue without the appropriate team and respiratory protection.
3. Remove and bag contaminated clothing and jewelry. Brush away dry agents and blot excess liquid before irrigation when appropriate. Follow

HazMat direction for runoff containment and water-reactive substances.

4. Provide airway positioning, suction, oxygen when indicated, BVM ventilation, bronchodilator therapy, CPAP when appropriate, and early advanced-airway preparation for progressive edema or respiratory failure.
5. Irrigate exposed eyes and skin continuously with copious tepid water or saline unless reliable agent-specific guidance contraindicates water. Remove contact lenses when easily possible without delaying irrigation.
6. Treat shock, seizures, dysrhythmias, burns, temperature injury, and pain using the appropriate linked protocol and provider-level standing orders.
7. Call Poison Help at 1-800-222-1222 and contact Medical Control early for severe or unknown exposure, unusual antidote, persistent symptoms, ingestion, intentional exposure, pediatric patient, pregnancy, or destination guidance.
8. Coordinate transport to the most appropriate receiving facility, burn center, trauma center, pediatric center, or facility with toxicology capability. Notify the destination before arrival and report decontamination status.
9. Document the identified or suspected agent, source, route, concentration, exposure duration, PPE, zone of contact, decontamination, product information, Poison Help/Medical Control recommendations, treatment, response, and contamination status at handoff.

Medication orders

Bronchodilator

Use the current age-appropriate respiratory-distress protocol for clinically significant bronchospasm.

- Bronchodilator treatment does not replace decontamination, oxygenation, ventilation, or monitoring for delayed pulmonary edema.

Atropine / pralidoxime

Not stocked as a CHEMPACK, Mark I, or equivalent field kit on Claiborne ground units. Use TE-08 for supportive care, escalation, and future adoption safeguards.

- Do not use unapproved autoinjector counts or antidote doses.

Calcium gluconate for hydrofluoric-acid exposure

Use only the finalized Claiborne County HF preparation, route, dose, and approval pathway.

- Potential routes include topical, nebulized, or other specialty administration depending on exposure and local adoption.
- Continuous ECG monitoring and Poison Help or Medical Control consultation are required for symptomatic or significant exposure.

Other specialty antidotes

Methylene blue, nitrite therapy, chelation, or other uncommon antidotes require a separately approved standing order or direct Medical Control.

Provider-level care

Provider-Level Actions Protect responders, stop exposure, decontaminate, support ABCs, and use agent-specific guidance.

EMT: Remain outside the hazard until the scene is declared safe for the assigned PPE and training level; request HazMat, fire, law enforcement, and additional EMS resources early.

- Assist removal of contaminated clothing and jewelry, brush away dry material before irrigation, provide copious eye/skin irrigation when indicated, and prevent contamination of the ambulance or receiving facility.
- Support airway and ventilation, administer oxygen as indicated, treat bronchospasm within scope, obtain glucose when indicated, and contact Poison Help.
- Obtain the product name, container, label, placard, Safety Data Sheet, exposure route, time, and symptoms without re-entering the hazard or delaying care.

AEMT: Perform all EMT actions plus IV/IO access, authorized fluids and medications, continuous reassessment, and preparation for rapid respiratory deterioration.

- Use an unexposed extremity for vascular access when feasible and protect equipment from contamination.
- For a cholinergic presentation, use the linked WMD / Nerve Agent pathway for recognition, decontamination, ventilation, and escalation. Claiborne ground units do not carry a CHEMPACK, Mark I, or equivalent atropine/pralidoxime kit.

Paramedic: Perform all prior actions; lead advanced airway, ventilation, CPAP when appropriate, cardiac monitoring, 12-lead ECG, seizure, dysrhythmia, and shock management.

- Contact Medical Control and Poison Help for unidentified agents, severe symptoms, delayed-risk exposures, specialty antidotes, or destination questions.
- Administer calcium gluconate for hydrofluoric-acid exposure or other uncommon antidotes only from an approved kit and under the finalized agent-specific standing order or direct Medical Control.

Warnings

- Do not enter a hot zone, vapor cloud, confined space, or oxygen-deficient atmosphere without the required training, monitoring, PPE, and respiratory protection.
- Do not allow contaminated clothing, bags, equipment, or runoff to create secondary contamination of EMS personnel, the ambulance, the public, or the receiving facility.
- Do not mix household or industrial chemicals to identify them. Mixing bleach with acids or ammonia can generate toxic gas.
- Do not routinely neutralize acids, alkalis, or other chemicals on or inside the patient; exothermic reactions can worsen injury.
- Do not induce vomiting after caustic, hydrocarbon, or unknown ingestion.
- A normal initial examination or pulse oximetry reading does not exclude delayed pulmonary edema, cellular asphyxiation, methemoglobinemia, electrolyte toxicity, or deep tissue injury.
- Specialty therapies require an adopted protocol, stocked preparation, appropriate training and an applicable standing order or Medical Control order. An online order does not expand the provider's authorized scope.

Clinical pearls

- Removing contaminated clothing can eliminate a large portion of external contamination and should occur before entry into the clean treatment area.
- For eye exposure, irrigation begins immediately; do not delay irrigation to identify the exact product or locate specialty fluid.
- Product labels, photographs, SDS documents, shipping papers, placard numbers, and the Emergency Response Guidebook can improve identification without transporting a contaminated container.
- Nitrogen oxides, nitrates/nitrites, aniline dyes, and similar oxidizers can cause methemoglobinemia with cyanosis that responds poorly to oxygen; obtain co-oximetry when available and contact Poison Help.
- Methyl bromide and other fumigants may cause delayed neurologic symptoms, seizures, and pulmonary edema even after an initially mild presentation.
- Suspected arsenic, lead, mercury, copper, or other heavy-metal exposure is primarily supportive in the field; decontaminate, monitor, preserve product information, and obtain toxicology guidance rather than delaying transport for chelation.
- Opioid overdose is already addressed in TE-07 Overdose / Toxic Ingestion; use that pathway for ventilation and naloxone rather than this chemical-decontamination pathway.

Special populations

Children

- Children may receive a greater dose relative to body mass, remain closer to heavier-than-air vapors, and deteriorate rapidly from airway edema or respiratory fatigue.
- Use age-appropriate equipment, medication dosing, temperature protection, and pediatric destination criteria.

Pregnancy

- Maternal oxygenation, ventilation, perfusion, and decontamination are the priorities. Do not delay indicated maternal treatment because of pregnancy.
- Consult Poison Help and Medical Control for fetal considerations, antidotes, and destination decisions.

Responder exposure

- Any responder with symptoms, PPE breach, splash, inhalation, needlestick, or uncertain exposure requires immediate removal from duty, decontamination, medical evaluation, and agency exposure documentation.

Related pathways

- Call Poison Help - tel:+18002221222: National Poison Control line
- Opioid / Overdose (p. 211): Ventilation and naloxone pathway
- CO / Cyanide (p. 207): Cellular asphyxiant pathway
- Nerve Agent (p. 212): Scene safety, decontamination, recognition, and escalation pathway
- Chemical Burn (p. 184): Burn and destination pathway

Medication reference

Use the indication-specific protocol and provider level. This reference does not grant additional scope or override Medical Control requirements, contraindications, age limits, or product-specific restrictions.

Claiborne County EMS RX-R1 Formulary - 42 verified stocked medications plus protocol-specific references

https://www.tn.gov/content/dam/tn/health/events/TN%20State%20Guidelines%202024-2025%2009.11.2025.pdf

Acetaminophen

Trade name: Tylenol

Trade name: Ofirmev

Uses

- Pain
- Fever

EMT

Indication: Pain - oral

Adult: UP-11, age 16 and older: 650-1,000 mg PO once.

Pediatric: PM-04, age 2 to <16: 15 mg/kg PO once; maximum 650 mg. Under age 2: Medical Control.

- UP-11 Adult Pain Control (p. 31)

EMT

Indication: Uncomfortable fever - oral

Adult: 650-1,000 mg PO.

Pediatric: Older than 3 months: 15 mg/kg PO; maximum 650 mg. Use UP-10 for assessment and exclusions.

- UP-10 Fever (p. 28)

Paramedic

Indication: Adult pain - IV/IO; AEMT authorization pending

Adult: Age 16 and older: weight ≥50 kg, 1,000 mg; weight <50 kg, 15 mg/kg (maximum 750 mg). Infuse over 15 minutes once; never rapid push. Allow at least 6 hours after the previous acetaminophen dose; no EMS repeat. Total from all routes: ≤4,000 mg/24 hours at ≥50 kg, or ≤75 mg/kg/24 hours (maximum 3,750 mg) below 50 kg.

Pediatric: IV/IO acetaminophen is not a standing-order pediatric analgesic under PM-04; use its oral/nonpharmacologic options.

Conditional AEMT access remains disabled until separately verified by the agency.

Clinical restriction: IO use requires separate verified authorization; never establish IO solely for pain. Expanded AEMT use remains disabled pending verification.

- UP-11 Adult Pain Control (p. 31)

Cautions

- Check severe liver disease, allergy, suspected overdose, prior dose/time, and all acetaminophen sources against the route-, weight- and indication-specific dose limits. Do not repeat when previous dose/time is uncertain.
- Do not give IV/IO acetaminophen solely for uncomplicated fever. Do not establish IO solely for pain in a stable patient.
- AEMT expanded analgesia is not enabled while authorization is unverified. Current Tennessee authority for the exact drug/route, documented applicable education, agency competency/credentialing, and medical-director orders for the drug, route, indication, and age group are all required; course completion or selecting AEMT view alone is not authorization.

Governing pathways

- Pain Control - Adult (p. 31)
- Pediatric Pain Control (p. 181)
- Fever (p. 28)

Adenosine

Trade name: Adenocard

Uses

- Narrow-complex tachycardia
- Wide-complex tachycardia

Paramedic

Indication: Narrow-tach; wide-tach

Adult: First dose 6 mg rapid IV push with immediate flush; then 12 mg rapid IV push after 1-2 minutes if needed; may repeat the 12 mg dose once

Pediatric: First dose 0.1 mg/kg rapid IV/IO push (maximum 6 mg); then 0.2 mg/kg after 1-2 minutes if needed (maximum 12 mg)

Cautions

- Use only for a stable regular rhythm under the applicable tachycardia protocol.
- Give as a rapid push through the most proximal practical access with an immediate flush and continuous rhythm recording.
- Do not use for an irregular rhythm, sinus tachycardia, or active severe bronchospasm.
- Contact Medical Control before reduced-dose administration in a heart-transplant patient or through central venous access.

Governing pathways

- Adult Narrow-Complex Tachycardia (p. 87)
- Pediatric Narrow-Complex Tachycardia (p. 169)

Albuterol

Trade name: Proventil

Trade name: Ventolin

Uses

- Asthma/COPD
- Allergic reaction
- Hyperkalemia

EMT

Indication: Bronchospasm - inhaled

Adult: Nebulized 2.5-5 mg; MDI 2 puffs inhaled.

Pediatric: Under 20 kg: 2.5 mg nebulized; 20 kg or greater: 5 mg nebulized.

- AR-04 Adult COPD / Asthma (p. 148)

Paramedic

Indication: AM-03 suspected hyperkalemia with a pulse - nebulized

Adult: 10 mg by nebulizer. Do not administer nebulized albuterol during cardiac arrest.

Pediatric: The prior pediatric formulary reference is under 15 kg: 10 mg; 15 kg or greater: 15 mg. This is not a PM-04 analgesic or a pediatric standing order under adult AM-03; obtain the separate pediatric order/Medical Control.

Clinical restriction: No automatic repeat interval is authorized here. Confirm pediatric eligibility and repeat/cumulative limits in a separately approved order before use; the former generic every-2-hour instruction is not carried into AM-03.

- AM-03 Hyperkalemia / Renal Failure (p. 123)

Cautions

- Reassess heart rate, work of breathing, air movement, and response.
- For anaphylaxis, albuterol is adjunctive and does not replace IM epinephrine.
- High-dose hyperkalemia treatment is Paramedic-only, distinct from EMT bronchospasm care. Do not give nebulized albuterol during cardiac arrest.

Governing pathways

- Adult COPD / Asthma (p. 148)
- Pediatric Respiratory Distress (p. 153)
- AM-03 Hyperkalemia / Renal Failure (p. 123)

Amiodarone

Trade name: Cordarone

Uses

- Shock-refractory VF/pulseless VT
- Stable monomorphic VT with a pulse

Paramedic

Indication: AC-09 / PC-07 shock-refractory VF/pulseless VT

Adult: First dose 300 mg IV/IO; second dose 150 mg IV/IO. Paramedic standing order for VF/pVT refractory to the third shock. Administer during CPR without interrupting compressions. Do not combine with lidocaine without Medical Control.

Pediatric: 5 mg/kg IV/IO for shock-refractory VF/pulseless VT; maximum 300 mg. Select one antiarrhythmic only unless Medical Control directs otherwise..

- AC-09 Ventricular Fibrillation / Pulseless VT (p. 96)

Paramedic

Indication: AC-07 stable adult monomorphic VT with a pulse

Adult: 150 mg IV/IO over 10 minutes. Paramedic standing order for stable monomorphic VT with a pulse. Stop the infusion and cardiovert immediately if instability develops. After conversion, if VT recurs, repeat 150 mg over 10 minutes once and begin maintenance infusion at 1 mg/min. Do not combine with another antiarrhythmic without Medical Control.

Pediatric: PC-06 antiarrhythmic agent and dose require Medical Control; do not extrapolate this adult order.

- AC-07 Adult Monomorphic Wide-Complex Tachycardia (p. 90)

Cautions

- Administration rate and repeat dosing depend on whether a pulse is present.

Governing pathways

- Adult VF / Pulseless VT (p. 96)
- Pediatric VF / Pulseless VT (p. 172)
- AC-07 Adult Monomorphic Wide-Complex Tachycardia (p. 90)

Aspirin

Trade name: ASA

Uses

- Chest pain
- Suspected acute coronary syndrome

EMT

Indication: Chest pain / suspected ACS

Adult: 324 mg chewed

Pediatric: N/A

Cautions

- Confirm allergy, active bleeding, and whether an adequate dose was already taken.

Governing pathways

- Acute Coronary Syndrome / STEMI (p. 80)

Atropine

Trade name: Atropine sulfate

Uses

- Symptomatic bradycardia

Paramedic

Indication: AC-02 adult / PC-02 pediatric symptomatic bradycardia

Adult: 1 mg IV/IO; repeat every 3-5 minutes to maximum total 3 mg. For bradycardia causing cardiopulmonary compromise; do not delay indicated pacing.

Pediatric: 0.02 mg/kg IV/IO for increased vagal tone or primary AV block; minimum 0.1 mg; maximum single dose 0.5 mg child / 1 mg adolescent; may repeat once; maximum total 1 mg child / 2 mg adolescent.

- AC-02 Bradycardia With a Pulse (p. 74)

Cautions

- Follow the cause-specific bradycardia pathway. Do not delay ventilation, CPR or indicated pacing.
- This cardiac dosing row does not establish a toxicologic antidote order.

Governing pathways

- AC-02 Bradycardia With a Pulse (p. 74)
- PC-02 Pediatric Bradycardia (p. 164)

Calcium Chloride / Gluconate

Trade name: Calcium chloride

Trade name: Calcium gluconate

Uses

- Hyperkalemia/arrhythmia
- Calcium-channel blocker toxicity
- Calcium replacement

Paramedic

Indication: Hyperkalemia/arrhythmia - calcium chloride IV/IO

Adult: 500-1,000 mg IV/IO. May repeat once after 5-10 minutes for persistent life-threatening ECG changes. Use AM-03 indication and monitoring requirements.

Pediatric: 20 mg/kg IV/IO reference; not a pediatric standing order under adult AM-03. Pediatric indication and limits require the separate order/Medical Control.

- AM-03 Hyperkalemia / Renal Failure (p. 123)

Paramedic

Indication: Hyperkalemia/arrhythmia - calcium gluconate IV

Adult: 1.5-3 g IV initial; maximum 3 g per episode. Preferred for peripheral IV administration. For persistent life-threatening ECG changes, do not repeat without a specific Medical Control order confirming cumulative dose.

Pediatric: 60-100 mg/kg IV/IO reference; pediatric eligibility and limits require the separate order/Medical Control.

Clinical restriction: For persistent life-threatening ECG changes, do not repeat calcium gluconate without a specific Medical Control order confirming cumulative dose. The AM-03 standing-order maximum is 3 g per episode.

- AM-03 Hyperkalemia / Renal Failure (p. 123)

Paramedic

Indication: Calcium-channel blocker toxicity / other calcium indications

Adult: Use the condition-specific order; do not transfer the hyperkalemia regimen to another indication.

Pediatric: Use the separately approved pediatric indication-specific order or Medical Control.

- TE-07 Overdose / Toxic Ingestion (p. 211)

Paramedic

Indication: AO-03 clinically significant magnesium toxicity - Medical Control

Adult: Stop magnesium sulfate; calcium gluconate 1 g IV slowly with Medical Control.

Pediatric: Pregnancy-related indication: follow AO-03 and Medical Control for age-specific care.

- AO-03 OB-GYN Emergency (p. 140)

Cautions

- Verify calcium salt, concentration, route, indication, and cumulative dose. Calcium chloride and gluconate are alternatives; do not give both as the initial calcium treatment.
- Calcium gluconate is preferred through a peripheral IV. Monitor continuously for extravasation; calcium chloride can cause severe tissue injury.

- Do not mix calcium and bicarbonate or infuse simultaneously through the same tubing; flush thoroughly between medications.

Governing pathways

- AM-03 Hyperkalemia / Renal Failure (p. 123)
- Overdose / Toxic Ingestion (p. 211)
- AO-03 OB-GYN Emergency (p. 140)

Dexamethasone

Trade name: Decadron

Uses

- Allergic reaction
- Asthma/COPD
- Croup/stridor

Paramedic

Indication: Allergic reaction; asthma/COPD; croup/stridor

Adult: 10 mg IV/IO or IM

Pediatric: 0.6 mg/kg

Cautions

- Steroid effect is delayed and does not replace epinephrine, bronchodilator, airway, or ventilation care.

Governing pathways

- Adult COPD / Asthma (p. 148)
- Pediatric Respiratory Distress (p. 153)

Dexmedetomidine

Trade name: Precedex

Uses

- Advanced sedation
- Pain

Paramedic

Indication: Advanced sedation / pain

Adult: 0.2-1.5 mcg/kg/hr

Pediatric: Same as adult formulary dose

Cautions

- Continuous airway, respiratory, cardiac, and blood-pressure monitoring is required.
- Watch for bradycardia and hypotension.

Governing pathways

- Post-Intubation / BIAD Management (p. 155)

Dextrose

Trade name: D10W

Trade name: D25W

Trade name: D50W

Uses

- Hypoglycemia

AEMT

Indication: Hypoglycemia

Adult: D10: up to 250 mL IV/IO, titrated to improving mental status and glucose at least 70 mg/dL; maximum 25 g

Pediatric: D10: 2 mL/kg IV/IO (0.2 g/kg); recheck glucose and neurologic status after 5 min; may repeat once

Cautions

- Pediatric D10 dosing is weight-based, not age-based.
- Verify concentration, patient weight, vascular patency, and maximum volume before administration.
- Recheck glucose and clinical response.

Governing pathways

- Pediatric Diabetic Emergency (p. 177)
- Altered Mental Status (p. 14)

Diazepam

Trade name: Valium

Trade name: Diastat

Uses

- Chemical restraint
- Seizures

Paramedic

Indication: REFERENCE ONLY - no active standing order established

Adult: No active adult standing-order regimen is established in this app. Medical Control must specify indication, dose, route, rate, repeat interval and maximum within verified Paramedic scope.

Pediatric: Historical formulary reference only, NOT an administration order: 0.2 mg/kg IV/IO; 0.5 mg/kg IM/PR. No active pediatric route-specific regimen, repeat interval or maximum is established here; obtain an explicit Medical Control order within verified scope.

Cautions

- Not an automatically authorized alternative to the active seizure or agitation pathway. This reference entry does not establish an administration order or expand scope.
- Monitor ventilation, oxygenation, blood pressure, and sedation depth.

Governing pathways

- UP-13 active seizure pathway (not an order for this drug) (p. 39)
- UP-18 active agitation pathway (not an order for this drug) (p. 57)

Diltiazem

Trade name: Cardizem

Uses

- Significantly symptomatic stable atrial fibrillation/flutter

Paramedic

Indication: AC-06 stable symptomatic AF/flutter with SBP ≥110 mmHg

Adult: 0.25 mg/kg IV/IO (maximum 20 mg); if needed, 0.35 mg/kg IV/IO after 15 minutes (maximum 20 mg); infusion 5-10 mg/hr. Paramedic standing order for significantly symptomatic stable atrial fibrillation/flutter when SBP is at least 110 mmHg. For prolonged transport, an infusion of 5-10 mg/hr requires Medical Control. Avoid with hypotension, acute decompensated heart failure, suspected systolic dysfunction, acute coronary syndrome, pre-excitation, or an undifferentiated wide-complex rhythm.

Pediatric: Medical Control; no pediatric standing order under AC-06.

- AC-06 Adult Narrow-Complex Tachycardia (p. 87)

Cautions

- Avoid in hypotension, pre-excited atrial fibrillation, or an undifferentiated wide-complex rhythm.
- Use only under the applicable rhythm protocol.

Governing pathways

- Adult Narrow-Complex Tachycardia (p. 87)

Diphenhydramine

Trade name: Benadryl

Uses

- Abdominal pain/vomiting
- Allergic reaction
- Acute dystonic / extrapyramidal reaction

Paramedic

Indication: Abdominal pain/vomiting; allergic reaction

Adult: 25-50 mg IV/IO or IM

Pediatric: 1 mg/kg

Paramedic

Indication: UP-17 acute dystonic / extrapyramidal reaction

Adult: 50 mg IV/IO/IM.

Pediatric: Younger than 16 years: 1 mg/kg IV/IO/IM; maximum 50 mg.

Clinical restriction: Paramedic-only for dystonia until specific current Tennessee AEMT authority for this indication/route, competency and signed local orders are documented. AEMT care is recognition, airway assessment/support and requesting Paramedic care.

- UP-17 Behavioral / Psychiatric Emergency (p. 53)

Cautions

- May cause sedation, anticholinergic effects, and hypotension.
- For dystonia, assess for airway involvement and monitor after administration. Allergy treatment authority does not establish dystonia authority; Medical Control cannot expand provider scope.

Governing pathways

- Behavioral / Psychiatric Emergency (p. 53)

Dopamine

Trade name: Intropin

Uses

- Bradycardia
- Medical shock

Paramedic

Indication: AC-02 persistent unstable adult bradycardia

Adult: 5-20 mcg/kg/min IV/IO infusion; titrate to clinical response. Paramedic standing order for persistent unstable bradycardia when atropine is ineffective. Taper slowly after clinical stabilization when appropriate.

Pediatric: No pediatric standing dose is established by AC-02; use the pediatric condition-specific protocol and Medical Control.

- AC-02 Bradycardia With a Pulse (p. 74)

Cautions

- Use an infusion pump, titrate to the protocol endpoint, and monitor for dysrhythmia or extravasation.

- The AC-02 regimen is indication-specific. Other shock indications require the applicable protocol or a specific Medical Control order, not extrapolation from this row.

Governing pathways

- Bradycardia With a Pulse (p. 74)

Epinephrine

Trade name: Adrenalin

Trade name: EpiPen

Uses

- Anaphylaxis
- Cardiac arrest
- Bradycardia
- Severe upper-airway edema / asthma

EMT

Indication: Anaphylaxis - IM

Adult: 0.5 mg of 1 mg/mL IM in the lateral thigh; 0.3 mg autoinjector acceptable; repeat every 5 min as needed

Pediatric: 0.01 mg/kg of 1 mg/mL IM in the mid-outer thigh; maximum 0.3 mg prepubertal child or 0.5 mg adolescent; repeat every 5 min as needed

AEMT

Indication: AC-01 / AC-03 / AC-09 adult pulseless cardiac arrest - IV/IO

Adult: 1 mg of 0.1 mg/mL IV/IO every 3-5 minutes. Follow the rhythm-specific AC-01/AC-03/AC-09 timing; in VF/pulseless VT give after the second shock as soon as possible without delaying CPR or defibrillation.

Pediatric: Use the separate pediatric pulseless-arrest row. Pulseless-arrest authorization does not establish neonatal resuscitation, pulsed bradycardia or vasopressor-infusion authority.

- AC-09 Adult VF / Pulseless VT (p. 96)

AEMT

Indication: Pediatric pulseless cardiac arrest - IV/IO

Adult: Use the separate adult pulseless-arrest row.

Pediatric: 0.01 mg/kg of 0.1 mg/mL IV/IO; maximum single dose 1 mg; repeat every 3-5 minutes. Follow PC-01/PC-04/PC-07 rhythm-specific timing; not a newborn resuscitation order.

- PC-07 Pediatric VF / Pulseless VT (p. 172)

Paramedic

Indication: PC-02 pediatric bradycardia with poor perfusion - IV/IO

Adult: Use AC-02 for adult bradycardia; this is not an adult IV bolus order.

Pediatric: 0.01 mg/kg of 0.1 mg/mL IV/IO; maximum single dose 1 mg; repeat every 3-5 minutes while bradycardia with poor perfusion persists. Follow PC-02 airway, ventilation and CPR priorities. Not a newborn resuscitation order.

- PC-02 Pediatric Bradycardia With Poor Perfusion (p. 164)

Paramedic

Indication: Persistent unstable bradycardia - infusion

Adult: 2-10 mcg/min IV/IO infusion; titrate to clinical response

Pediatric: Use the pediatric cardiac bradycardia pathway

Paramedic

Indication: AR-04 adult stridor/upper-airway edema or AR-07 pediatric croup/stridor - nebulized

Adult: AR-04: 1 mg of 1 mg/mL epinephrine in 2 mL normal saline, nebulized; may repeat once for stridor or upper-airway edema.

Pediatric: AR-07: 0.5 mL/kg of 1 mg/mL epinephrine, maximum 5 mL (5 mg), nebulized for croup/stridor.

- AR-07 Pediatric Respiratory Distress (p. 153)

Paramedic

Indication: AM-01 refractory anaphylactic shock - infusion with Medical Control

Adult: 2-10 mcg/min IV/IO; titrate to SBP ≥90 mmHg or MAP ≥65 mmHg. Paramedic with Medical Control for refractory shock after repeated IM epinephrine and appropriate fluid. Use an infusion pump and continuous cardiac, blood-pressure, and SpO_2 monitoring. Do not give an IV epinephrine bolus under this protocol.

Pediatric: This adult pathway is not a pediatric standing order; use the age-specific protocol and Medical Control.

- AM-01 Allergic Reaction / Anaphylaxis (p. 117)

Paramedic

Indication: AM-05 adult shock - infusion alternative

Adult: 2-10 mcg/min IV/IO; titrate to adequate perfusion and MAP at least 65 mmHg. Paramedic alternative when norepinephrine is unavailable or significant bradycardia accompanies shock. Use continuous ECG and frequent blood-pressure monitoring.

Pediatric: This adult pathway is not a pediatric standing order; use the age-specific protocol and Medical Control.

- AM-05 Hypotension / Shock (p. 129)

Paramedic

Indication: UP-15 pediatric septic shock - Medical Control selection

Adult: Use the separate adult UP-15 norepinephrine pathway.

Pediatric: Medical Control selects epinephrine or norepinephrine and specifies the initial rate (approved range 0.05-0.1 mcg/kg/min), titration increment/interval, ceiling and perfusion target. Do not select solely from a warm/cold label.

- UP-15 Suspected Sepsis (p. 46)

AEMT

Indication: AR-04 / AR-07 life-threatening asthma - IM

Adult: AR-04: 0.3-0.5 mg of 1 mg/mL epinephrine IM for life-threatening asthma with poor air movement or impending respiratory failure.

Pediatric: AR-07 uses PM-01: 0.01 mg/kg of 1 mg/mL IM; maximum 0.3 mg prepubertal child or 0.5 mg adolescent. Follow PM-01 reassessment and repeat limits; not a nebulized order.

- AR-04 Adult COPD / Asthma (p. 148)

Paramedic

Indication: AO-02 newly born resuscitation - HR below 60 with a pulse despite effective ventilation and coordinated compressions

Adult: Newly born resuscitation only; not an adult dosing order.

Pediatric: Epinephrine 0.1 mg/mL (1:10,000): 0.01 mg/kg IV/IO every 3-5 minutes as needed for heart rate below 60 despite effective ventilation and coordinated compressions. Paramedic-only while a pulse is present. Use UVC/IO access when available under AO-02.

- AO-02 Newly Born (p. 138)

Cautions

- Verify indication, concentration, route, dose, and repeat interval before every administration.
- Do not use a routine IV/IO epinephrine bolus for refractory anaphylaxis; follow the applicable anaphylaxis pathway.

Governing pathways

- Adult Allergic Reaction (p. 117)
- Pediatric Allergic Reaction (p. 176)
- Adult COPD / Asthma (p. 148)
- Pediatric Respiratory Distress (p. 153)
- AM-01 Allergic Reaction / Anaphylaxis (p. 117)
- AM-05 Hypotension / Shock (p. 129)
- UP-15 Suspected Sepsis (p. 46)
- AC-01 Adult Asystole / PEA (p. 72)
- AC-03 Adult Cardiac Arrest (p. 77)
- AC-09 Adult VF / Pulseless VT (p. 96)
- PC-07 Pediatric VF / Pulseless VT (p. 172)
- AR-04 Adult COPD / Asthma (p. 148)
- AR-07 Pediatric Respiratory Distress (p. 153)
- AO-02 Newly Born (p. 138)

Push-Dose Epinephrine

Uses

- Temporary bridge for severe hypotension
- Peri-intubation hypotension

Paramedic

Indication: Severe hypotension / peri-intubation shock - temporary bridge only

Adult: 10 mcg slow IV/IO (1 mL of a labeled 10 mcg/mL syringe) every 3-5 min as needed; maximum two total boluses, then Medical Control before any additional dose

Pediatric: 0.5-1 mcg/kg slow IV/IO (0.05-0.1 mL/kg of a labeled 10 mcg/mL syringe) every 3-5 min as needed; maximum 10 mcg (1 mL) per dose and two total boluses, then Medical Control before any additional dose

Cautions

- Use only a labeled 10 mcg/mL (0.01 mg/mL) syringe. To prepare when a standardized prepared syringe is unavailable: add 1 mL of epinephrine 0.1 mg/mL (1:10,000) to 9 mL normal saline in a 10 mL syringe, then label the syringe with drug and concentration.
- Do not administer undiluted epinephrine 1 mg/mL (1:1,000) or the 0.1 mg/mL (1:10,000) cardiac-arrest syringe as a push-dose bolus.
- Paramedic use only. Use only as a temporary bridge while a vasoactive infusion is prepared and the underlying cause of shock is treated; use continuous ECG and frequent blood-pressure reassessment.
- Do not use for cardiac arrest or under AC-02 Bradycardia With a Pulse. For anaphylaxis, give IM epinephrine immediately under AM-01 or PM-01; push-dose epinephrine does not replace that pathway.

Governing pathways

- Adult Hypotension / Shock (p. 129)
- Crashing Medical Patient (p. 135)
- Airway, Drug Assisted (p. 146)
- Pediatric Hypotension / Shock (p. 179)

Protocol-specific preparation/reference; not an additional stocked formulary item.

Etomidate

Trade name: Amidate

Uses

• Drug-assisted intubation/RSI

Paramedic

Indication: DAI/RSI

Adult: 0.3 mg/kg IV/IO

Pediatric: 0.3 mg/kg IV/IO; pediatric DAI requires direct online Medical Director or Assistant Medical Director order

Cautions

• DAI dosing is weight-based. Etomidate provides hypnosis without analgesia or ongoing sedation.
• Use only with complete airway, paralysis, and post-intubation plans.

Governing pathways

• Airway Drug-Assisted Intubation (p. 146)

Fentanyl

Trade name: Sublimaze

Uses

• Pain
• Procedures
• Advanced sedation

Paramedic

Indication: Pain - IV/IO

Adult: UP-11 age 16 and older: 50 mcg slow IV/IO; repeat 25-50 mcg every 5 minutes as needed; maximum total 100 mcg.

Pediatric: PM-04 age 2 to <16: 1 mcg/kg slow IV/IO once; maximum 50 mcg. Medical Control authorization is required before any second dose.

• UP-11 Adult Pain Control (p. 31)

Paramedic

Indication: DAI / post-intubation analgesia - IV/IO

Adult: AR-03: 25-50 mcg IV/IO; repeat cautiously as needed; maximum 100 mcg. Use the separate airway/sedation order, not the pain regimen.

Pediatric: Pediatric DAI requires a direct online Medical Director or Assistant Medical Director order; use the pediatric/airway pathway.

• AR-03 Drug-Assisted Airway (p. 146)

Paramedic

Indication: AC-04 persistent significant ischemic discomfort

Adult: 25-50 mcg IV/IO; repeat as needed to a maximum total dose of 100 mcg. Paramedic standing order for persistent significant ischemic discomfort after aspirin and appropriate nitroglycerin. Use continuous cardiac, respiratory, SpO_2, and blood-pressure monitoring. Avoid routine opioid administration; opioids may delay absorption of oral antiplatelet medications.

Pediatric: This adult pathway is not a pediatric standing order; use the age-specific protocol and Medical Control.

• AC-04 Acute Coronary Syndrome / STEMI (p. 80)

Paramedic

Indication: PC-03 pediatric cardiac pain/anxiety - Medical Control

Adult: Use the adult indication-specific pathway.

Pediatric: 1 mcg/kg IV/IO; maximum single dose 50 mcg only with Medical Control for pain/anxiety after airway and perfusion are stabilized.

• PC-03 Pediatric Pulmonary Edema / CHF (p. 166)

Paramedic

Indication: AR-08 adult post-airway analgesia

Adult: 25-50 mcg IV/IO; repeat cautiously as needed, maximum 100 mcg. Treat pain first. Do not stack with morphine.

Pediatric: Direct online Medical Control or receiving-facility order through the pediatric system.

• AR-08 Post-Intubation / BIAD Management (p. 155)

Cautions

• Select one opioid for pain; do not combine fentanyl, morphine and hydromorphone under UP-11.
• Use continuous SpO_2 and waveform $EtCO_2$ with opioids; monitor airway, ventilation, mental status, blood pressure, cumulative dose and response.
• Claiborne opioid analgesia remains Paramedic-only. AEMT expanded-analgesia coursework does not create a local opioid standing order.

Governing pathways

• Pain Control - Adult (p. 31)
• Pediatric Pain Control (p. 181)
• AC-04 Acute Coronary Syndrome / STEMI (p. 80)
• PC-03 Pediatric Pulmonary Edema / CHF (p. 166)
• AR-08 Post-Intubation / BIAD Management (p. 155)

Glucagon

Trade name: GlucaGen

Uses

• Hypoglycemia

AEMT

Indication: Hypoglycemia - IM when oral glucose is unsafe and vascular access cannot be obtained promptly

Adult: 1 mg IM.

Pediatric: Below 20 kg: 0.5 mg IM; 20 kg or above: 1 mg IM.

• UP-04 Altered Mental Status (p. 14)

Paramedic

Indication: AM-01 refractory shock/bradycardia in a beta-blocked patient - Medical Control

Adult: 1-2 mg IV/IO over 5 minutes. Paramedic with Medical Control for refractory shock or bradycardia in a beta-blocked patient. Prepare for vomiting and protect the airway.

Pediatric: This adult pathway is not a pediatric standing order; use the age-specific protocol and Medical Control.

• AM-01 Allergic Reaction / Anaphylaxis (p. 117)

Cautions

• Nausea and vomiting are common; protect the airway and reassess glucose.

• Follow UP-04/PM-02 eligibility and aspiration precautions; this hypoglycemia row is not a toxicologic overdose dosing order.

Governing pathways

• Altered Mental Status (p. 14)

• Pediatric Diabetic Emergency (p. 177)

• AM-01 Allergic Reaction / Anaphylaxis (p. 117)

Hydromorphone

Trade name: Dilaudid

Uses

• Pain

Paramedic

Indication: Adult pain - slow IV/IO

Adult: Age 16 and older: 0.5 mg slow IV/IO; repeat 0.5 mg after 10 minutes as needed; maximum total 1 mg.

Pediatric: Not authorized as a standing-order pediatric analgesic under PM-04. Use PM-04; contact Medical Control for patients younger than 2 years.

• UP-11 Adult Pain Control (p. 31)

Paramedic

Indication: Adult pain - IM when vascular access unavailable

Adult: Age 16 and older: 0.5-1 mg IM once.

Pediatric: Not authorized as a standing-order pediatric analgesic under PM-04. Use PM-04; contact Medical Control for patients younger than 2 years.

• UP-11 Adult Pain Control (p. 31)

Cautions

• Potent opioid: verify dose units, prior analgesics, cumulative dose and ventilation. Select only one opioid; additional dosing beyond UP-11 limits requires Medical Control.

• Maintain continuous SpO_2 and waveform $EtCO_2$, cardiac monitoring, and reassessment every 5-10 minutes.

Governing pathways

• UP-11 Adult Pain Control (p. 31)

• PM-04 Pediatric Pain Control (p. 181)

Ibuprofen

Trade name: Advil

Trade name: Motrin

Uses

• Pain

• Fever

EMT

Indication: Pain; fever

Adult: 400 mg PO

Pediatric: Pain (PM-04, age 2 to <16): 10 mg/kg PO once; maximum 400 mg. Fever: see UP-10.

Cautions

• Confirm age, renal disease, GI bleeding, NSAID allergy, pregnancy, and prior NSAID use.

Governing pathways

• Pain Control - Adult (p. 31)

• Pediatric Pain Control (p. 181)

• Fever (p. 28)

Ipratropium

Trade name: Atrovent

Uses

• Asthma/COPD

EMT

Indication: Moderate or severe bronchospasm

Adult: 0.5 mg nebulized with the initial albuterol treatment

Pediatric: Under 20 kg: 0.25 mg nebulized; 20 kg or greater: 0.5 mg nebulized, with initial albuterol

Cautions

- Use as an adjunct to albuterol under the respiratory protocol.

Governing pathways

- Adult COPD / Asthma (p. 148)
- Pediatric Respiratory Distress (p. 153)

Ketamine

Trade name: Ketalar

Uses

- RSI
- Chemical restraint
- Seizures
- Pain
- Procedures
- Sedation

Paramedic

Indication: DAI/RSI - IV/IO or IM

Adult: 1-2 mg/kg IV/IO; if vascular access is unavailable, 4 mg/kg IM (maximum 400 mg).

Pediatric: Direct online Medical Director or Assistant Medical Director order for pediatric DAI; use the applicable pediatric pathway.

- AR-03 Drug-Assisted Airway (p. 146)

Paramedic

Indication: Adult pain - IV/IO

Adult: Age 16 and older: 0.3 mg/kg IV/IO slowly over 10 minutes; maximum 30 mg per dose. May repeat every 20 minutes as needed; maximum three total doses. Never rapid IV push.

Pediatric: Not authorized as a standing-order pediatric analgesic under PM-04. Use PM-04; contact Medical Control for patients younger than 2 years.

- UP-11 Adult Pain Control (p. 31)

Paramedic

Indication: Adult pain - IN when vascular access unavailable

Adult: Age 16 and older: 0.5-1 mg/kg IN once; maximum 100 mg. Single-dose option under UP-11.

Pediatric: Not authorized as a standing-order pediatric analgesic under PM-04. Use PM-04; contact Medical Control for patients younger than 2 years.

- UP-11 Adult Pain Control (p. 31)

Paramedic

Indication: UP-18 adult severe agitation (BARS 7) - IM

Adult: Age 16 and older: 2-4 mg/kg IM; maximum 400 mg. Do not delay treatment attempting vascular access.

Pediatric: Younger than 16: direct Medical Control medication and dose order required before BARS 7 chemical sedation.

- UP-18 Behavioral Agitation / Severe Agitation (p. 57)

Paramedic

Indication: UP-18 adult severe agitation (BARS 7) - established IV/IO only

Adult: Age 16 and older: 1 mg/kg IV/IO slowly; maximum 100 mg. Only if access is already safely established; do not attempt IV access in a violently struggling patient solely for this route.

Pediatric: Younger than 16: direct Medical Control medication and dose order required before BARS 7 chemical sedation.

- UP-18 Behavioral Agitation / Severe Agitation (p. 57)

Paramedic

Indication: AR-08 adult post-airway sedation

Adult: 1 mg/kg IV/IO. Adult alternative for post-airway sedation when clinically appropriate. Pediatric post-airway medication requires the pediatric system and direct online Medical Control or receiving-facility order.

Pediatric: Direct online Medical Control or receiving-facility order through the pediatric system.

- AR-08 Post-Intubation / BIAD Management (p. 155)

Cautions

- For suspected penetrating eye injury or open globe, do not give ketamine unless expressly authorized by Medical Control under TB-11; preserve this restriction when using an analgesia or airway reference.
- Ketamine analgesia, DAI, agitation, seizures, and post-intubation sedation have separate indications, doses and permissions. Do not substitute DAI doses for analgesia.
- Continuous airway, ventilation, oxygenation, cardiac, blood-pressure and sedation monitoring are required. Do not routinely start ketamine and an opioid together.
- Claiborne ketamine analgesia remains Paramedic-only; AEMT coursework alone does not authorize it.
- For BARS 7, do not routinely combine ketamine and midazolam or give prophylactic midazolam after ketamine. If dangerous agitation persists 10 minutes after ketamine, contact Medical Control

before repeating ketamine or adding another sedative.

Governing pathways

- Drug-Assisted Airway (p. 146)
- UP-11 Adult Pain Control (p. 31)
- PM-04 Pediatric Pain Control (p. 181)
- Eye Trauma / Open Globe Safeguards (p. 198)
- Behavioral Agitation / Sedation (p. 57)
- Seizure (p. 39)
- AR-08 Post-Intubation / BIAD Management (p. 155)

Ketorolac

Trade name: Toradol

Uses

- Adult pain

Paramedic

Indication: Adult pain - IV; AEMT authorization pending

Adult: Age 16 and older: 15 mg IV once; maximum 15 mg IV. No EMS repeat.

Pediatric: Not authorized as a standing-order pediatric analgesic under PM-04. Use PM-04; contact Medical Control for patients younger than 2 years.

Conditional AEMT access remains disabled until separately verified by the agency.

- UP-11 Adult Pain Control (p. 31)

Paramedic

Indication: Adult pain - IM; AEMT authorization pending

Adult: Age 16 and older: 30 mg IM once; maximum 30 mg IM. No EMS repeat.

Pediatric: Not authorized as a standing-order pediatric analgesic under PM-04. Use PM-04; contact Medical Control for patients younger than 2 years.

Conditional AEMT access remains disabled until separately verified by the agency.

- UP-11 Adult Pain Control (p. 31)

Cautions

- Select one route only; do not combine IV and IM doses or exceed the route-specific single-dose maximum. Apply the same dose ceiling in older or low-weight adults.
- Avoid with renal disease/transplant, significant dehydration, hypotension, active GI bleeding, NSAID allergy, clinically significant anticoagulant-associated bleeding risk, pregnancy, or recent NSAID administration.
- Do not give solely for uncomplicated fever. PM-04 does not authorize pediatric ketorolac standing orders.
- AEMT expanded analgesia is not enabled while authorization is unverified. Current Tennessee authority for the exact drug/route, documented applicable education, agency competency/credentialing, and medical-director orders for the drug, route, indication, and age group are all required; course completion or selecting AEMT view alone is not authorization.

Governing pathways

- UP-11 Adult Pain Control (p. 31)
- PM-04 Pediatric Pain Control (p. 181)

Labetalol

Trade name: Trandate

Uses

- Undifferentiated suspected CVA with severe hypertension
- Medical-Control-directed non-stroke hypertensive emergency

Paramedic

Indication: AM-04 undifferentiated suspected CVA with SBP >220 or DBP >110; other use only under a condition-specific pathway or Medical Control

Adult: Age 16 and older only. Undifferentiated suspected CVA: 10 mg slow IV over 2 min; may repeat 10 mg once after at least 10 min if still SBP >220 or DBP >110; maximum total 20 mg

Pediatric: Not authorized under Adult Medical AM-04; use a pediatric condition-specific pathway or Medical Control

Cautions

- Use the standing order only for manually confirmed undifferentiated suspected CVA with SBP >220 or DBP >110; target SBP ≤220 and DBP ≤110. Do not use this target after thrombolysis/TNK.
- For any non-stroke hypertensive emergency, use a linked condition-specific pathway or Medical Control; avoid rapid or unmonitored reduction.
- Do not give with bradycardia, advanced heart block, acute decompensated heart failure/pulmonary edema, cardiogenic shock, active bronchospasm, bronchial asthma, or known drug hypersensitivity.

Governing pathways

- Hypertension (p. 127)
- Suspected Stroke (p. 42)

Lidocaine

Trade name: Xylocaine

Uses

- VF/pulseless VT cardiac arrest
- Wide-complex tachycardia
- Normal-QT polymorphic VT
- Conscious IO infusion pain

Paramedic

Indication: VF/pulseless VT cardiac arrest; other ventricular dysrhythmias per linked protocol

Adult: VF/pulseless VT cardiac arrest: 1-1.5 mg/kg IV/IO initial; then 0.5-0.75 mg/kg every 5-10 min; maximum total 3 mg/kg. For perfusing ventricular tachycardia, follow the linked protocol.

Pediatric: 1 mg/kg IV/IO for shock-refractory VF/pulseless VT; maximum 100 mg. Select one antiarrhythmic only unless Medical Control directs otherwise..

- AC-09 Ventricular Fibrillation / Pulseless VT (p. 96)

AEMT

Indication: UP-06 conscious IO infusion pain - IO only

Adult: 20-40 mg IO slowly before the initial forceful flush.

Pediatric: 0.5 mg/kg IO slowly; maximum 40 mg.

Clinical restriction: Use 2% preservative-free, epinephrine-free lidocaine suitable for IV/IO administration. Device training and local authorization are required.

- UP-06 Venous Access (p. 19)

Paramedic

Indication: AC-08 recurrent polymorphic VT with normal baseline QT

Adult: 100 mg IV/IO initial dose. Paramedic standing order for recurrent polymorphic VT only when the baseline QT interval is normal. Treat associated myocardial ischemia and keep pads attached. Do not substitute lidocaine for immediate defibrillation of a sustained episode.

Pediatric: This adult pathway is not a pediatric standing order; use the age-specific protocol and Medical Control.

- AC-08 Adult Polymorphic Wide-Complex Tachycardia (p. 93)

Cautions

- Track cumulative dose and monitor for neurologic or cardiovascular toxicity.
- Do not combine lidocaine with amiodarone for shock-refractory VF/pulseless VT without Medical Control direction.
- The IO-pain row is separate from Paramedic-only cardiac/dysrhythmia dosing. After slow administration before the initial forceful flush, flush and use the IO according to the stocked device instructions.

Governing pathways

- Adult VF / Pulseless VT (p. 96)
- Pediatric VF / Pulseless VT (p. 172)
- Venous Access (p. 19)
- AC-08 Adult Polymorphic Wide-Complex Tachycardia (p. 93)

Lorazepam

Trade name: Ativan

Uses

- Chemical restraint
- Seizures

Paramedic

Indication: REFERENCE ONLY - no active standing order established

Adult: No active adult standing-order regimen is established in this app. Medical Control must specify indication, dose, route, rate, repeat interval and maximum within verified Paramedic scope.

Pediatric: Historical formulary reference only, NOT an administration order: 0.1 mg/kg. No active pediatric route-specific regimen, repeat interval or maximum is established here; obtain an explicit Medical Control order within verified scope.

Cautions

- Not an automatically authorized alternative to the active seizure or agitation pathway. This reference entry does not establish an administration order or expand scope.
- Monitor ventilation, oxygenation, blood pressure, and sedation depth.

Governing pathways

- UP-13 active seizure pathway (not an order for this drug) (p. 39)
- UP-18 active agitation pathway (not an order for this drug) (p. 57)

Magnesium Sulfate

Trade name: Magnesium

Uses

- Hypomagnesemia
- Eclampsia/pre-eclampsia
- Tocolysis
- Ventricular arrhythmia/torsades
- Asthma/COPD

Paramedic

Indication: AO-03 eclampsia / suspected eclampsia

Adult: 4-6 g IV over 20-30 minutes, then 1-2 g/hr infusion for eclampsia or suspected eclampsia. Primary cause-directed therapy for eclampsia or suspected eclampsia. Continuously monitor ECG, BP, respiratory status, SpO_2, and $EtCO_2$.

Pediatric: Pregnancy-related indication: follow AO-03 and Medical Control for age-specific care. No unrelated pediatric standing order is created.

- AO-03 OB-GYN Emergency (p. 140)

Paramedic

Indication: AC-08 recurrent prolonged-QT polymorphic VT/torsades with a pulse

Adult: 1-2 g IV over 15 minutes; may repeat once. Paramedic standing order for recurrent polymorphic VT associated with a prolonged QT interval/torsades de pointes. Defibrillation takes priority for any sustained episode. Do not administer routinely when the QT interval is normal. Monitor continuously for hypotension, bradycardia, respiratory depression, and recurrent dysrhythmia.

Pediatric: This adult pathway is not a pediatric standing order; use the age-specific protocol and Medical Control.

- AC-08 Adult Polymorphic Wide-Complex Tachycardia (p. 93)

Paramedic

Indication: AC-09 / PC-07 pulseless torsades / specified pediatric arrest indication

Adult: 1-2 g IV/IO push; may repeat once. Paramedic standing order only for suspected pulseless torsades/long-QT polymorphic VT. Routine magnesium is not indicated for other VF/pVT cardiac arrests.

Pediatric: 25-50 mg/kg IV/IO once; maximum 2 g; only for torsades/polymorphic VT or suspected hypomagnesemia.

- AC-09 Ventricular Fibrillation / Pulseless VT (p. 96)

Paramedic

Indication: AR-04 / AR-07 severe asthma

Adult: 1-2 g IV over 15-30 minutes for severe asthma under AR-04.

Pediatric: 40-50 mg/kg IV once over 15-30 minutes; maximum 2 g, for severe asthma not improving after initial bronchodilators.

- AR-04 Adult COPD / Asthma (p. 148)

Cautions

- Dose, dilution, rate, and endpoint differ by indication.
- Monitor respiratory status, blood pressure, and cardiac rhythm.
- Do not infer independent tocolysis, prolonged infusion, or repeat hypomagnesemia standing orders from historical formulary reference dosing. Those indications require a specific applicable protocol/order.

Governing pathways

- OB / GYN Emergency (p. 140)
- Adult COPD / Asthma (p. 148)
- AC-08 Adult Polymorphic Wide-Complex Tachycardia (p. 93)
- AC-09 Ventricular Fibrillation / Pulseless VT (p. 96)
- PC-07 Pediatric VF / Pulseless VT (p. 172)
- AR-07 Pediatric Respiratory Distress (p. 153)

Methylprednisolone

Trade name: Solu-Medrol

Uses

- Allergic reaction
- Asthma/COPD

Paramedic

Indication: AR-04 adult moderate/severe asthma or COPD

Adult: 125 mg IV/IO or IM.

Pediatric: AR-07: Medical Control only; no standing pediatric dose for asthma/reactive-airway disease. Do not substitute the PM-01 allergy dose.

- AR-04 Adult COPD / Asthma (p. 148)

Paramedic

Indication: PM-01 pediatric allergic reaction - adjunct

Adult: Use the applicable adult pathway; no separate adult dose is added by this pediatric row.

Pediatric: 2 mg/kg IV; maximum 125 mg. Adjunct only; do not delay or replace epinephrine.

- PM-01 Pediatric Anaphylaxis / Allergic Reaction (p. 176)

Cautions

- Steroid effect is delayed and does not replace immediate airway or bronchodilator therapy.

Governing pathways

- Adult COPD / Asthma (p. 148)
- AR-07 Pediatric Respiratory Distress (p. 153)
- PM-01 Pediatric Anaphylaxis / Allergic Reaction (p. 176)

Metoprolol

Trade name: Lopressor

Uses

- AF/flutter with RVR during Cardizem shortage

Paramedic

Indication: Stable, symptomatic atrial fibrillation/flutter with RVR when diltiazem is unavailable under the approved shortage contingency

Adult: 2.5-5 mg IV push over 1 minute every 10 minutes; maximum 3 doses or 15 mg

Pediatric: Medical Control

Cautions

- Use only during the approved Cardizem shortage contingency; do not independently combine or sequence with diltiazem.
- Avoid with hypotension, bradycardia, advanced heart block, cardiogenic shock, pre-excited atrial fibrillation/flutter, undifferentiated wide-complex rhythm, acute decompensated heart failure, or suspected systolic dysfunction.
- Reassess blood pressure, heart rate, rhythm, perfusion, respiratory status, and symptoms before every dose; use caution with bronchospasm.

Governing pathways

- Adult Narrow-Complex Tachycardia (p. 87)

Midazolam

Trade name: Versed

Uses

- Seizures
- Chemical restraint
- Post-intubation sedation

Paramedic

Indication: Seizure - age-specific dosing

Adult: 10 mg IM preferred without IV/IO; 10 mg IN alternative; or 5 mg IV/IO when access is established. For IV/IO, repeat 5 mg once after 5 min if seizure continues; maximum total 10 mg

Pediatric: 0.2 mg/kg IM/IN, maximum 10 mg; or 0.1 mg/kg IV/IO, maximum 5 mg. May repeat once after 5 min; maximum total 0.4 mg/kg, not to exceed 10 mg Count prescribed prearrival rescue benzodiazepine toward the two-dose episode limit.

- UP-13 Seizure (p. 39)

Paramedic

Indication: Adult dangerous agitation (BARS 6)

Adult: Age 16 and older: 2.5 mg IV/IO or 5 mg IM/IN; repeat once after 5 min if dangerous agitation persists after complete airway/hemodynamic reassessment. Maximum: 5 mg IV/IO or 10 mg IM/IN. Age ≥65, frail, significant respiratory disease, or suspected CNS-depressant intoxication: 1-2.5 mg IV/IO or 2.5 mg IM/IN; repeat once only after reassessment; maximum cumulative dose 5 mg.

Pediatric: Younger than 16: 0.1 mg/kg IV/IO or 0.2 mg/kg IM/IN; maximum initial dose 5 mg. Contact Medical Control when feasible before administration; repeat requires Medical Control. Maximum cumulative dose 10 mg.

- UP-18 Behavioral Agitation / Severe Agitation (p. 57)

Paramedic

Indication: Adult severe agitation (BARS 7) - alternative when ketamine unavailable or contraindicated

Adult: Age 16 and older: 10 mg IM preferred; 10 mg IN only if IM not feasible. Age ≥65, frail, or increased respiratory risk: 5 mg IM. Follow UP-18 reassessment and Medical Control requirements before additional sedation.

Pediatric: Younger than 16: direct Medical Control medication and dose order required before BARS 7 chemical sedation.

- UP-18 Behavioral Agitation / Severe Agitation (p. 57)

Paramedic

Indication: Post-intubation sedation

Adult: 2-5 mg IV/IO; repeat cautiously as needed after analgesia

Pediatric: Direct online Medical Control or receiving-facility order

Paramedic

Indication: AC-06 adult electrical-therapy sedation

Adult: 2-5 mg IV/IO; use the smallest effective dose. Paramedic procedural sedation for synchronized cardioversion when feasible. Do not delay cardioversion in an unstable patient. Continuous airway, respiratory, SpO_2, cardiac, and blood-pressure monitoring is required.

Pediatric: This adult pathway is not a pediatric standing order; use the age-specific protocol and Medical Control.

• AC-06 Adult Narrow-Complex Tachycardia (p. 87)

Paramedic

Indication: AC-07 adult electrical-therapy sedation

Adult: 2-5 mg IV/IO; use the smallest effective dose. Paramedic sedation for synchronized cardioversion when feasible. Do not delay cardioversion in an unstable patient. Continuous airway, respiratory, SpO_2, cardiac, and blood-pressure monitoring is required.

Pediatric: This adult pathway is not a pediatric standing order; use the age-specific protocol and Medical Control.

• AC-07 Adult Monomorphic Wide-Complex Tachycardia (p. 90)

Paramedic

Indication: AC-08 adult electrical-therapy sedation

Adult: 2-5 mg IV/IO; use the smallest effective dose. Paramedic sedation when a conscious patient requires unsynchronized defibrillation. Do not delay shock delivery for sedation. Continuous airway, respiratory, SpO_2, cardiac, and blood-pressure monitoring is required.

Pediatric: This adult pathway is not a pediatric standing order; use the age-specific protocol and Medical Control.

• AC-08 Adult Polymorphic Wide-Complex Tachycardia (p. 93)

Paramedic

Indication: PC-05 pediatric cardioversion sedation

Adult: Use the adult electrical-therapy pathway.

Pediatric: 0.1-0.2 mg/kg IV/IO/IN for cardioversion; maximum single dose 2 mg, maximum total 5 mg. Use only if immediately available; do not delay synchronized cardioversion. Verify against the current Claiborne pediatric medication reference.

• PC-05 Pediatric Narrow-Complex Tachycardia (p. 169)

Paramedic

Indication: PC-06 pediatric cardioversion sedation

Adult: Use the adult electrical-therapy pathway.

Pediatric: 0.1-0.2 mg/kg IV/IO/IN for cardioversion; maximum single dose 2 mg, maximum total 5 mg. Use only if immediately available; do not delay synchronized cardioversion. Verify against the current Claiborne pediatric medication reference.

• PC-06 Pediatric Wide-Complex Tachycardia (p. 170)

Cautions

• For agitation, IM is preferred over IN when IM is feasible.
• Continuously monitor airway, ventilation, oxygenation, blood pressure, cardiac rhythm, and sedation depth.
• Do not routinely combine ketamine and midazolam or give prophylactic midazolam after ketamine. BARS 6, BARS 7, seizure and airway regimens are separate orders.

Governing pathways

• Seizure (p. 39)
• Behavioral Agitation / Sedation (p. 57)
• Airway Drug-Assisted Intubation (p. 146)
• Post-Intubation / BIAD Management (p. 155)
• AC-06 Adult Narrow-Complex Tachycardia (p. 87)
• AC-07 Adult Monomorphic Wide-Complex Tachycardia (p. 90)
• AC-08 Adult Polymorphic Wide-Complex Tachycardia (p. 93)
• PC-05 Pediatric Narrow-Complex Tachycardia (p. 169)
• PC-06 Pediatric Wide-Complex Tachycardia (p. 170)

Morphine

Trade name: Morphine sulfate

Uses

• Pain

Paramedic

Indication: Adult pain - slow IV/IO

Adult: Age 16 and older: 5 mg slow IV/IO; repeat 5 mg after 10 minutes as needed; maximum total 10 mg.

Pediatric: Not authorized as a standing-order pediatric analgesic under PM-04. Use PM-04; contact Medical Control for patients younger than 2 years.

• UP-11 Adult Pain Control (p. 31)

Paramedic

Indication: Adult pain - IM when vascular access unavailable

Adult: Age 16 and older: 5-10 mg IM once.

Pediatric: Not authorized as a standing-order pediatric analgesic under PM-04. Use PM-04; contact Medical Control for patients younger than 2 years.

• UP-11 Adult Pain Control (p. 31)

Paramedic

Indication: AR-08 adult post-airway analgesia - alternative

Adult: 4 mg IV/IO; repeat 2 mg every 5 minutes as needed, maximum 10 mg. Select one analgesic strategy; do not stack with fentanyl.

Pediatric: Direct online Medical Control or receiving-facility order through the pediatric system.

- AR-08 Post-Intubation / BIAD Management (p. 155)

Cautions

- Monitor airway, ventilation, oxygenation, mental status, blood pressure and cumulative opioid dose. Select only one opioid; additional dosing beyond UP-11 limits requires Medical Control.
- Maintain continuous SpO_2 and waveform $EtCO_2$, cardiac monitoring, and reassessment every 5-10 minutes.

Governing pathways

- UP-11 Adult Pain Control (p. 31)
- PM-04 Pediatric Pain Control (p. 181)
- AR-08 Post-Intubation / BIAD Management (p. 155)

Nitroglycerin

Trade name: Nitrostat

Trade name: NTG

Uses

- Chest pain
- CHF
- Hypertension

EMT

Indication: Patient's own prescribed medication

Adult: 0.4 mg SL every 5 min; maximum 3 doses

Pediatric: N/A

AEMT

Indication: Agency formulary medication

Adult: 0.4 mg SL every 5 min; maximum 3 doses

Pediatric: N/A

Cautions

- Verify blood-pressure threshold, PDE-5 inhibitor use, right-ventricular infarction concern, and protocol indication.

Governing pathways

- Acute Coronary Syndrome / STEMI (p. 80)

Naloxone

Trade name: Narcan

Uses

- Suspected opioid-related respiratory depression

EMT

Indication: UP-04 suspected opioid-related respiratory depression - IN only

Adult: 2 mg IN; repeat every 2-3 min to adequate ventilation; maximum cumulative dose 8 mg. Retain this local adult order pending confirmation of the stocked product; do not silently substitute a 4 mg spray.

Pediatric: No numeric pediatric IN standing order is established by this update. Confirm stocked product/strength or injectable concentration plus atomizer; obtain a product/device-specific Medical Control dose and administration plan. Do not apply the IV 0.01-to-0.1 mg/kg sequence to IN.

Clinical restriction: Pediatric IN product, dose, repeat interval, limits and technique require medical-director selection before release. For atomized injectable product, specify deliverable volume, nostril division and device dead-space handling.

- UP-04 Altered Mental Status (p. 14)

AEMT

Indication: UP-04 suspected opioid-related respiratory depression - IV/IO/IM only

Adult: 0.4-2 mg IV/IO/IM; repeat every 2-3 min and titrate to adequate ventilation; maximum cumulative dose 8 mg.

Pediatric: 0.01 mg/kg IV/IO/IM initially; if ventilation remains inadequate, escalate to 0.1 mg/kg; maximum 2 mg per dose. Repeat every 2-3 min to adequate ventilation; maximum cumulative dose 8 mg.

- UP-04 Altered Mental Status (p. 14)

Cautions

- Support ventilation immediately; do not delay BVM ventilation while preparing or awaiting naloxone. Titrate to adequate ventilation rather than complete arousal.
- If ventilation remains inadequate after 8 mg, continue airway support and reassess the diagnosis.
- For resuscitation immediately after birth, follow AO-02 Newly Born; naloxone does not replace neonatal ventilation. This is not an expanded-analgesia privilege.

Governing pathways

- Altered Mental Status (p. 14)
- Overdose / Toxic Ingestion (p. 211)

Norepinephrine

Trade name: Levophed

Uses

- Bradycardia
- Medical shock
- Septic shock

Paramedic

Indication: AM-05 adult medical shock

Adult: 0.05-0.5 mcg/kg/min IV/IO infusion; titrate to adequate perfusion and MAP at least 65 mm Hg; maximum 0.5 mcg/kg/min under AM-05.

Pediatric: Do not use the adult AM-05 regimen as a pediatric standing order. Follow the age-specific shock pathway and Medical Control.

- AM-05 Hypotension / Shock (p. 129)

Paramedic

Indication: UP-15 adult septic shock

Adult: Start 0.1 mcg/kg/min IV/IO; titrate to MAP ≥65 mm Hg or SBP ≥90 mm Hg; maximum 2 mcg/kg/min.

Pediatric: Persistent pediatric septic shock: Medical Control selects epinephrine or norepinephrine and specifies the initial rate (approved range 0.05-0.1 mcg/kg/min), titration increment/interval, ceiling and perfusion target. Do not select solely from a warm/cold label or use the adult ceiling.

- UP-15 Sepsis (p. 46)

Paramedic

Indication: AC-05 persistent adult cardiogenic shock

Adult: Start 0.1 mcg/kg/min IV/IO; titrate to MAP ≥65 mmHg or SBP ≥90 mmHg; maximum 2 mcg/kg/min. Paramedic standing order for persistent cardiogenic shock. Continuous cardiac monitoring, frequent blood-pressure assessment, and careful IV/IO-site surveillance are required. Use a proximal, well-functioning IV or IO and reassess perfusion frequently.

Pediatric: This adult pathway is not a pediatric standing order; use the age-specific protocol and Medical Control.

- AC-05 CHF / Acute Pulmonary Edema (p. 83)

Paramedic

Indication: AC-10 persistent adult post-ROSC hypotension

Adult: 0.1-2 mcg/kg/min IV/IO; titrate to MAP at least 65 mm Hg. Paramedic preferred vasopressor for persistent post-ROSC hypotension after cautious fluid resuscitation. Use continuous cardiac monitoring and frequent blood-pressure reassessment. Alternative post-ROSC vasopressors require Medical Control.

Pediatric: This adult pathway is not a pediatric standing order; use the age-specific protocol and Medical Control.

- AC-10 Post-Resuscitation Care (p. 99)

Cautions

- Use an infusion pump, titrate to the protocol endpoint, and monitor the access site and distal perfusion.
- Indication-specific ceilings are distinct: UP-15 adult septic shock 2 mcg/kg/min; AM-05 adult medical shock 0.5 mcg/kg/min. For other indications, follow the condition-specific protocol. Pediatric initiation and titration require Medical Control; never use "same as adult."

Governing pathways

- Hypotension / Shock (p. 129)
- Adult CHF / Acute Pulmonary Edema (p. 83)
- Sepsis (p. 46)
- AC-05 CHF / Acute Pulmonary Edema (p. 83)
- AC-10 Post-Resuscitation Care (p. 99)

Ondansetron

Trade name: Zofran

Uses

- Nausea and vomiting

AEMT

Indication: Abdominal pain / nausea / vomiting

Adult: 4 mg IV/IO/IM/PO/ODT; may repeat once after 15 min after reassessment

Pediatric: 0.15 mg/kg IV/IO/IM/PO/ODT; maximum 4 mg; may repeat once after 15 min after reassessment

Cautions

- Avoid with known congenital long-QT syndrome or clinically significant QT prolongation; use caution with electrolyte loss, bradyarrhythmia, or other QT-prolonging medications.

Governing pathways

- Abdominal Pain / Vomiting / Diarrhea (p. 10)

Promethazine

Trade name: Phenergan

Uses

- Persistent nausea/vomiting after ondansetron

Paramedic

Indication: UP-03 adult refractory nausea/vomiting after ondansetron

Adult: 12.5 mg deep IM once after ondansetron; never IV/IO.

Pediatric: Medical Control; no pediatric standing order.

• UP-03 Abdominal Pain / Vomiting / Diarrhea (p. 10)

Cautions

• Deep IM is the only authorized route. Never administer IV or IO.
• Avoid with significant CNS depression, inability to protect the airway, or known hypersensitivity.

Governing pathways

• Abdominal Pain / Vomiting / Diarrhea (p. 10)

Propofol

Trade name: Diprivan

Uses

• Interfacility continuation of an established sedative infusion

Paramedic

Indication: Continuation of a pre-existing interfacility infusion only

Adult: 5-50 mcg/kg/min infusion, continued only according to the sending order

Pediatric: Continuation only according to the sending order and Medical Control direction

Cautions

• Do not initiate or bolus propofol in the field.
• Requires continuous airway, ventilation, oxygenation, cardiac, and blood-pressure monitoring.
• Propofol has no analgesic effect and may cause profound hypotension or apnea.

Governing pathways

• Post-Intubation / BIAD Management (p. 155)

Rocuronium

Trade name: Zemuron

Uses

• Drug-assisted intubation/RSI

Paramedic

Indication: DAI/RSI

Adult: 1 mg/kg IV/IO when succinylcholine is contraindicated or a longer duration is needed

Pediatric: 1 mg/kg IV/IO; pediatric DAI requires direct online Medical Director or Assistant Medical Director order

Cautions

• DAI dosing is weight-based. Establish post-intubation analgesia and sedation promptly; do not routinely repeat before they are established.

Governing pathways

• Airway Drug-Assisted Intubation (p. 146)

Sodium Bicarbonate

Trade name: Sodium bicarbonate

Uses

• Hyperkalemia
• Protocol-specific toxicologic emergency

Paramedic

Indication: AM-03 suspected hyperkalemia / severe metabolic acidosis - IV/IO

Adult: 1 mEq/kg IV/IO; maximum 50 mEq. Use the AM-03 indication and monitoring safeguards.

Pediatric: No pediatric trauma/hyperkalemia standing order is supplied by adult AM-03. Use a separately approved pediatric order or Medical Control.

• AM-03 Hyperkalemia / Renal Failure (p. 123)

Paramedic

Indication: Toxicologic indication - IV/IO

Adult: 1 mEq/kg IV/IO reference; use the toxin-specific order for dose, repeat, maximum and clinical endpoint.

Pediatric: 1 mEq/kg reference; use the pediatric toxin-specific order for route, repeat, maximum and endpoint.

• TE-07 Overdose / Toxic Ingestion (p. 211)

Cautions

• Not routine in every cardiac arrest. Use only the protocol-defined indication; do not transfer a toxicology regimen to traumatic or medical arrest.
• Verify concentration and adequate ventilation. Do not mix with calcium or administer simultaneously through the same tubing; flush thoroughly.

Governing pathways

• AM-03 Hyperkalemia / Renal Failure (p. 123)
• Overdose / Toxic Ingestion (p. 211)

Succinylcholine

Trade name: Anectine

Uses

• Drug-assisted intubation/RSI

Paramedic

Indication: DAI/RSI

Adult: 1-2 mg/kg IV for RSI

Pediatric: 1-2 mg/kg IV for RSI; pediatric DAI requires direct online Medical Director or Assistant Medical Director order

Cautions

- DAI dosing is weight-based. Verify contraindications and prepare continuous post-paralysis ventilation, analgesia, and sedation.

Governing pathways

- Airway Drug-Assisted Intubation (p. 146)

Tranexamic Acid

Trade name: TXA

Trade name: Cyklokapron

Uses

- Traumatic hemorrhage with clinical shock
- Topical epistaxis
- Topical dental hemorrhage
- Medical-Control-directed bradykinin angioedema

Paramedic

Indication: Traumatic hemorrhage with clinical shock - IV/IO

Adult: 1-2 g IV/IO: 1 g over 10 minutes OR 2 g over 20 minutes; maximum administration rate 100 mg/min. Verify the stocked concentration and preparation instructions. Do not automatically repeat; further dosing requires a specific order.

Pediatric: 15 mg/kg reference. Pediatric systemic trauma TXA is Medical-Control-directed only: obtain the patient-specific eligibility, route, infusion rate, maximum and repeat instructions before administration; no independent pediatric standing-order regimen.

Clinical restriction: Adult dose/rate options are approved as shown. Verify stocked concentration and preparation instructions; no automatic repeat. Pediatric systemic trauma TXA requires a patient-specific Medical Control order for eligibility, route, infusion rate, maximum and repeat policy. The locally selected 15 mg/kg differs from the Tennessee 20 mg/kg IV reference; current state/route authority and credential verification still apply. Do not infer systemic dosing from topical orders.

- TB-12 Abdominal / Pelvic Trauma (p. 201)

AEMT

Indication: Persistent atraumatic anterior epistaxis - topical only

Adult: UP-09: 1 g/10 mL total applied to anterior gauze after 15 minutes uninterrupted compression; maintain external pressure for 20 minutes.

Pediatric: UP-09: 15 mg/kg topically; maximum 1 g total, on appropriately sized anterior gauze; maintain external pressure for 20 minutes.

- UP-09 Epistaxis (p. 25)

AEMT

Indication: Persistent dental hemorrhage - topical only

Adult: UP-07: TXA 1 g/10 mL to saturate gauze/cotton; pack the bleeding socket and maintain direct pressure for 20 minutes.

Pediatric: Use UP-07 for the pediatric dental indication and Medical Control when the pediatric dose or eligibility is not specified; do not infer a systemic 15 mg/kg regimen.

- UP-07 Dental Problems (p. 21)

Paramedic

Indication: Suspected ACE-inhibitor / hereditary bradykinin angioedema - IV/IO with Medical Control

Adult: AM-01: 1 g IV/IO over 10 minutes with Medical Control.

Pediatric: Use a separate pediatric order / Medical Control; the adult AM-01 order is not a pediatric standing order.

- AM-01 Allergic Reaction / Anaphylaxis (p. 117)

Paramedic

Indication: AO-03 postpartum hemorrhage with shock - within 3 hours of delivery

Adult: 1 g IV/IO over 10 minutes for postpartum hemorrhage with shock when within 3 hours of delivery. Do not use this postpartum order more than 3 hours after delivery.

Pediatric: Pregnancy/postpartum indication only; no non-obstetric pediatric TXA order is created. Follow AO-03 and Medical Control for age-specific care.

- AO-03 OB-GYN Emergency (p. 140)

Cautions

- Systemic trauma TXA is Paramedic-only: after immediate lifesaving interventions, consider for traumatic hemorrhage with clinical hemorrhagic shock and injury no more than 3 hours earlier. Do not delay transport or hemorrhage control.
- Record injury time and TXA dose/time; notify the receiving team to avoid duplicate dosing. Do not infer systemic trauma dosing from topical indications.
- Topical nasal/oral TXA is a separate existing AEMT/Paramedic order; EMT administration is not authorized under UP-07/UP-09. The adult trauma 1-2 g range is not a topical dose.
- Do not give IV/IO TXA solely for isolated dental bleeding or epistaxis. Do not atomize TXA, or perform deep/blind/posterior nasal packing under UP-09.

Governing pathways

- Multiple Trauma (p. 190)
- Abdominal / Pelvic Trauma (p. 201)
- Epistaxis (p. 25)
- Dental Problems (p. 21)
- Allergic Reaction / Anaphylaxis (p. 117)
- AO-03 OB-GYN Emergency (p. 140)

Vecuronium

Trade name: Norcuron

Uses

- Drug-assisted intubation/RSI

Paramedic

Indication: DAI/RSI

Adult: DAI: 80-100 mcg/kg IV push once; intermittent maintenance: 10-15 mcg/kg IV every 15 min as needed

Pediatric: DAI: 80-100 mcg/kg IV push once; intermittent maintenance: 50-100 mcg/kg IV every 1 hr as needed; pediatric DAI requires direct online Medical Director or Assistant Medical Director order

Cautions

- Paralysis provides no sedation, amnesia, or analgesia.
- Confirm ventilation capability and give continuous post-intubation sedation/analgesia.

Governing pathways

- Airway Drug-Assisted Intubation (p. 146)